

IBM Storage Scale System
7.0.1.0

Deployment Guide



Note

Before using this information and the product it supports, read the information in [“Notices” on page 227](#).

This edition applies to Version 7 release 0 modification 1 of the following product and to all subsequent releases and modifications until otherwise indicated in new editions:

- IBM Storage Scale Data Management Edition for IBM Storage Scale System (product number 5765-DME)
- IBM Storage Scale Data Access Edition for IBM Storage Scale System (product number 5765-DAE)

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About this information

Who should read this information

This information is intended for administrators of IBM Storage Scale System that includes IBM Storage Scale RAID.

IBM Storage Scale System information units

IBM Storage Scale System Utility Node documentation consists of the following information units.

Information unit	Type of information	Intended users
Hardware Planning and Installation Guide	This unit provides Utility Node information including technical overview, planning, installing, troubleshooting, and cabling.	System administrators and IBM support team
Service Guide	This unit provides Utility Node information including servicing and parts listings.	System administrators and IBM support team
Problem Determination Guide	This unit provides Utility Node information including events, replacing servers, issues, maintenance procedures, and troubleshooting.	System administrators and IBM support team
Command Reference	This unit provides information about IBM Storage Scale System Utility Node commands and scripts.	System administrators and IBM support team
IBM Storage Scale RAID: Administration	This unit provides IBM Storage Scale RAID information including administering, monitoring, commands, and scripts.	• System administrators of IBM Storage Scale systems • Application programmers who are experienced with IBM Storage Scale systems and familiar with the terminology and concepts in the XDSM standard

IBM Storage Scale System 5147-102 Storage Enclosure documentation consists of the following information units.

Information unit	Type of information	Intended users
Hardware Planning and Installation Guide	This unit provides IBM Storage Scale System 5147-102 Storage Enclosure information including technical overview, planning, installing, troubleshooting, and cabling.	System administrators and IBM support team
Service Guide	This unit provides IBM Storage Scale System information including servicing and parts listings.	System administrators and IBM support team
Problem Determination Guide	This unit provides IBM Storage Scale System information including events, replacing servers, issues, maintenance procedures, and troubleshooting.	System administrators and IBM support team

Information unit	Type of information	Intended users
Command Reference	This unit provides information about IBM Storage Scale System commands and scripts.	System administrators and IBM support team
IBM Storage Scale RAID: Administration	This unit provides IBM Storage Scale RAID information including administering, monitoring, commands, and scripts.	• System administrators of IBM Storage Scale systems • Application programmers who are experienced with IBM Storage Scale systems and familiar with the terminology and concepts in the XD SM standard

IBM Storage Scale System 3500 documentation consists of the following information units.

Information unit	Type of information	Intended users
Hardware Planning and Installation Guide	This unit provides IBM Storage Scale System 3500 information including technical overview, planning, installing, troubleshooting, and cabling.	System administrators and IBM support team
Service Guide	This unit provides IBM Storage Scale System 3500 information including servicing and parts listings.	System administrators and IBM support team
Problem Determination Guide	This unit provides IBM Storage Scale System information including events, replacing servers, issues, maintenance procedures, and troubleshooting.	System administrators and IBM support team
Command Reference	This unit provides information about IBM Storage Scale System commands and scripts.	System administrators and IBM support team
IBM Storage Scale RAID: Administration	This unit provides IBM Storage Scale RAID information including administering, monitoring, commands, and scripts.	• System administrators of IBM Storage Scale systems • Application programmers who are experienced with IBM Storage Scale systems and familiar with the terminology and concepts in the XD SM standard

IBM Storage Scale System 3200 documentation consists of the following information units.

Information unit	Type of information	Intended users
Hardware Planning and Installation Guide	This unit provides IBM Storage Scale System 3200 information including technical overview, planning, installing, troubleshooting, and cabling.	System administrators and IBM support team
Service Guide	This unit provides IBM Storage Scale System 3200 information including servicing and parts listings.	System administrators and IBM support team

Information unit	Type of information	Intended users
Problem Determination Guide	This unit provides IBM Storage Scale System 3200 information including events, replacing servers, issues, maintenance procedures, and troubleshooting.	System administrators and IBM support team
Command Reference	This unit provides information about IBM Storage Scale System commands and scripts.	System administrators and IBM support team
IBM Storage Scale RAID: Administration	This unit provides IBM Storage Scale RAID information including administering, monitoring, commands, and scripts.	• System administrators of IBM Storage Scale systems • Application programmers who are experienced with IBM Storage Scale systems and familiar with the terminology and concepts in the XD SM standard

IBM Storage Scale System 3000 documentation consists of the following information units.

Information unit	Type of information	Intended users
Hardware Planning and Installation Guide	This unit provides IBM Storage Scale System 3000 information including technical overview, planning, installing, troubleshooting, and cabling.	System administrators and IBM support team
Service Guide	This unit provides IBM Storage Scale System 3000 information including events, servicing, and parts listings.	System administrators and IBM support team
Problem Determination Guide	This unit provides IBM Storage Scale System 3000 information including setting up call home, replacing servers, issues, maintenance procedures, and troubleshooting.	System administrators and IBM support team
Command Reference	This unit provides information about IBM Storage Scale System commands and scripts.	System administrators and IBM support team
IBM Storage Scale RAID: Administration	This unit provides IBM Storage Scale RAID information including administering, monitoring, commands, and scripts.	• System administrators of IBM Storage Scale systems • Application programmers who are experienced with IBM Storage Scale systems and familiar with the terminology and concepts in the XD SM standard

IBM Storage Scale System 5000 documentation consists of the following information units.

Information unit	Type of information	Intended users
Hardware Guide	This unit provides IBM Storage Scale System 5000 information including system overview, installing, and troubleshooting.	System administrators and IBM support team

Information unit	Type of information	Intended users
Model 092 storage enclosures	This unit provides information including initial hardware installation and setup, and removal and installation of field-replaceable units (FRUs), customer-replaceable units (CRUs) for IBM Storage Scale System 5000 Expansion – Model 092, 5147-092.	System administrators and IBM support team
Model 106 storage enclosures	This unit provides information including hardware installation and maintenance for IBM Storage Scale System 5000 Expansion – Model 106.	System administrators and IBM support team
Problem Determination Guide	This unit provides IBM Storage Scale System 5000 information including setting up call home, replacing servers, issues, maintenance procedures, and troubleshooting.	System administrators and IBM support team
Command Reference	This unit provides information about IBM Storage Scale System commands and scripts.	System administrators and IBM support team
IBM Storage Scale RAID: Administration	This unit provides IBM Storage Scale RAID information including administering, monitoring, commands, and scripts.	• System administrators of IBM Storage Scale systems • Application programmers who are experienced with IBM Storage Scale systems and familiar with the terminology and concepts in the XDSM standard

Related information

Related information

For information about:

- IBM Storage Scale RAID, see [IBM Documentation](#).
- mmvdisk command, see [mmvdisk documentation](#).
- Mellanox OFED (MLNX_OFED 3.1.0) Release Notes, go to <https://docs.nvidia.com/doca/archive/3-1-0/doca+release+notes/index.html>.
- IBM Storage Scale System, see [IBM Documentation](#).
- IBM Storage Scale call home, see [Understanding call home](#).
- Installing IBM Storage Scale and CES protocols with the installation toolkit, see [Installing IBM Storage Scale on Linux® nodes with the installation toolkit](#).
- Detailed information about the IBM Storage Scale installation toolkit, see [Using the installation toolkit to perform installation tasks: Explanations and examples](#).
- CES HDFS, see [Adding CES HDFS nodes into the centralized file system](#).
- Installation toolkit support by IBM Storage Scale System, see [IBM Storage Scale System awareness with the installation toolkit](#).
- IBM Power9 servers, see <https://www.ibm.com/docs/en/storage-scale-system/5147-092?topic=guide-5105-22e-reference-information>.

For the latest support information about IBM Storage Scale RAID, see the IBM Storage Scale RAID FAQ in [IBM Documentation](#).

Conventions used in this information

Table 1 on page xv describes the typographic conventions used in this information. UNIX file name conventions are used throughout this information.

Table 1. Conventions

Convention	Usage
bold	Bold words or characters represent system elements that you must use literally, such as commands, flags, values, and selected menu options. Depending on the context, bold typeface sometimes represents path names, directories, or file names.
<u>bold underlined</u>	<u>bold underlined</u> keywords are defaults. These take effect if you do not specify a different keyword.
constant width	Examples and information that the system displays appear in constant-width typeface. Depending on the context, constant-width typeface sometimes represents path names, directories, or file names.
<i>italic</i>	<i>Italic</i> words or characters represent variable values that you must supply. <i>Italics</i> are also used for information unit titles, for the first use of a glossary term, and for general emphasis in text.
<key>	Angle brackets (less-than and greater-than) enclose the name of a key on the keyboard. For example, <Enter> refers to the key on your terminal or workstation that is labeled with the word <i>Enter</i> .
\	In command examples, a backslash indicates that the command or coding example continues on the next line. For example: <pre>mkcondition -r IBM.FileSystem -e "PercentTotUsed > 90" \ -E "PercentTotUsed < 85" -m p "FileSystem space used"</pre>
{item}	Braces enclose a list from which you must choose an item in format and syntax descriptions.
[item]	Brackets enclose optional items in format and syntax descriptions.
<Ctrl-x>	The notation <Ctrl-x> indicates a control character sequence. For example, <Ctrl-c> means that you hold down the control key while pressing <c>.
item...	Ellipses indicate that you can repeat the preceding item one or more times.
	In <i>synopsis</i> statements, vertical lines separate a list of choices. In other words, a vertical line means <i>Or</i> . In the left margin of the document, vertical lines indicate technical changes to the information.

How to submit your comments

To contact the IBM Storage Scale development organization, send your comments to the following email address:

scale@us.ibm.com

Summary of changes

This topic summarizes changes to the IBM Storage Scale Systems information. A vertical line (|) to the left of text and illustrations indicates technical changes or additions made to the previous edition of the information.

Summary of changes for IBM Storage Scale Systems 7.0.1.x. as updated, July 2026

The following new features and updates are available in IBM Storage Scale Systems 7.0.1:

Upgrade

Upgrade flowchart

Added for high-level overview of the upgrade process. For more information, see [“Upgrade workflow”](#) on page 144.

Upgrade support matrix.

Updated upgrade guidelines and the matrix table. For more information, see [Chapter 4, “Software upgrade and firmware update,”](#) on page 139.

Decoupled precheck tool

A stand-alone tool validates system health across all nodes before upgrade. Run this tool after the container is started and before the upgrade begins. See [“Checking the system health by using the decoupled precheck tool”](#) on page 147.

Protocol nodes upgrade

Only online upgrade is supported. For more information, see [“Upgrading the protocol nodes”](#) on page 156

CES deployment from the GUI

Cluster Export Services (CES), Protocol VMs, AFM Gateway VMs, I/O Building Blocks, and Client VMs can now be deployed from the GUI on IBM Storage Scale System Utility Node systems. See [Chapter 3, “Deployment from an x86 Utility Node by using the GUI,”](#) on page 119.

Minimum drives for RAID arrays

Declassified arrays require a minimum of 20 drives for 16+2p and 21 drives for 16+3p RAID configurations. See [“Prerequisites”](#) on page 39.

Commands

- Updated the **essrun protocol configure --enable** command.
- Removed the **--include-firmwares** and **--skip-firmwares** options from the **essrun** command. The new **--jbof-deps** option installs JBOF dependencies for MES scenarios. See the **essrun** command in IBM Storage Scale System: Command Reference.
- The **essrun audit** command enables, verifies, and rolls back STIG audit configuration on nodes. See [“Enabling STIG audit”](#) on page 215.

Security

STIG audit support

STIG-compliant audit logging is available on IBM Storage Scale System nodes that are running RHEL 8.10 or RHEL 9.6. See [“Security Technical Implementation Guide”](#) on page 214 and [“Enabling STIG audit”](#) on page 215.

FIPS mode management

FIPS mode can be enabled, verified, and disabled on nodes by using the **essrun fips** command. This mode is supported on RHEL 8.10 and RHEL 9.6. See [“Federal Information Processing mode”](#) on page 217, [“Enabling and managing FIPS mode on nodes”](#) on page 220, and [“FIPS mode issues”](#) on page 222.

New messages

No new messages have been added in this release.

Hardware updates

VLAN tags on canisters is removed. Use untagged traffic when you deploy IBM Storage Scale System 3500 nodes or add S54 switches. See [“Configuring the MTM 5149-S54 switch” on page 30.](#)

Chapter 1. Quick Deployment and Upgrade Guide

This quick deployment and upgrade guide gives only the essential steps, and assumes that you understand the deployment and the upgrade of an IBM Storage Scale System. You can deploy or upgrade an IBM system from a management server, such as Power9 or a management server virtual machine (MSVM) for x86 systems. The MSVM runs on IBM Storage Scale System Utility Node, which is a bare metal.

Prerequisites

Complete the following steps before the deployment and the upgrade:

1. Download the IBM Storage Scale System package from [IBM Fix Central](#).

```
Scale_System_UNIFIED-7.0.1.x-EMS.tgz
```

2. Populate the `/etc/hosts` file to the management server.
3. Extract the installation package.

```
cd /home/deploy
xz --decompress <xz package>
```

4. Start the container.

```
sh <untarred binary file>
```

5. Import the system configuration and define nodes.

```
essrun -N MS1,IOnode1,IOnode2,IOnoden,prt01,prt02 config load -p
```

where:

MS1

Specifies a management server, such as Power9, MSVM for x86 systems.

IOnode1,IOnode2,IOnoden

Specifies one or more I/O nodes.

prt01,prt02

Specifies one or more protocol nodes.

6. Check the defined nodes before the deployment and the upgrade.

```
essrun -N MS1,IOnode1,IOnode2,IOnoden,prt01,prt02 update --precheck
```

Deployment from the Power9 management server

1. Complete all steps in the [Prerequisites](#).
2. If the Power9 server needs to be updated, run the following command:

Note: You might need to run the following command again, if the kernel is changed.

```
essrun -N MS1 update --offline
```

3. If canisters, I/O nodes, or protocol nodes need to be updated, run the following command:

```
essrun -N IOnode1,IOnode2,IOnoden,prt01,prt02 update --offline
```

4. Create network bonds.

```
essrun -N MS1,IOnode1,IOnode2,IOnoden network --suffix=-hs
```

5. Create a cluster.

```
essrun -N IOnodeclasses cluster --suffix=-hs
```

6. Add the Power9 server to the cluster.

```
essrun -N IOnode1 cluster --add-ems MS1 --suffix=-hs
```

7. Create a file system.

```
essrun -N IOnodeclasses filesystem --name fs1 --suffix=-hs
```

Where the I/O node classes can be similar to the ones in the following examples:

- IBM Storage Scale System 3000: `ess30001a` and `ess30001b`
- IBM Storage Scale System 3500: `ess35001a` and `ess35001b`
- IBM Storage Scale System 6000: `ess60001a` and `ess60001b`

8. Add a protocol node to the cluster. For more information, see [“Creating a shared root file system for CES”](#) on page 92.

9. Log in to each node and check whether nodes are installed properly.

```
essinstallcheck -N localhost
```

10. From the Power9 management server do the final health check.

```
gnrhealthcheck
```

```
mmhealth node show -a
```

11. Set a time zone and set up Chrony. Refer to [“Setting a time zone”](#) on page 106.

12. Set up call home. For more information about the call home configuration, see [“Configuring call home”](#) on page 108.

13. Configure the GUI for hardware monitoring.

```
essrun -N MS1,IOnode1,IOnode2 gui --configure
```

Deployment from an MSVM for x86 systems

Complete the following steps to deploy a system from an MSVM that runs on the Utility Node.

1. Download the `Scale_System_VM-x.x.x.x-x86_64-UTILITY-EMS.tgz` package on the `/serv` directory and extract the package in the `/serv` directory.
2. Copy all downloaded software to the `/serv` directory on the Utility Node.
3. Extract the EMSVM script package in the `/serv` directory.
4. Check the Utility Node.

```
cd /serv/EMSVM-main ; ./emsvm --check-host
```

5. Start the Utility Node.

```
./emsvm --start-EMS
```

6. Connect to the Utility Node.

```
./emsvm --connect-EMS
```

7. Log in to the virtual machine by using the default `root/ibmesscluster` credentials and change the root password immediately for security.

8. Run the `ess_ssr_setup` tool and follow the instructions.

9. Set the fully qualified domain name (FQDN).

```
hostnamectl set-hostname FQDN
```

10. Complete all steps in the [Prerequisites](#).

11. If the MSVM needs to be updated, run following command:

Note: You might need to run the following command again, if the kernel is changed:

```
essrun -N EMSVM update --offline
```

Where, *EMSVM* specifies an MSVM that runs on Utility Node.

12. If canisters, I/O nodes, or protocol nodes need to be updated, run following command:

```
essrun -N IOnode1,IOnode2,IOnoden,prt01,prt02 update --offline
```

13. Create network bonds.

```
essrun -N EMSVM,IOnode1,IOnode2,IOnoden network --suffix=-hs
```

14. Create a cluster.

```
essrun -N IOnodeclasses cluster --suffix=-hs
```

15. Add the MSVM to the cluster.

```
essrun -N IOnode1 cluster --add-ems EMSVM --suffix=-hs
```

16. Create a file system.

```
essrun -N IOnodeclasses filesystem --name fs1 --suffix=-hs
```

where, *fs1* specifies a file system.

17. Create a cesSharedRoot file system.

```
essrun -N IOnode1,IOnode2 filesystem --ces
```

18. Set up the Cluster Export Services (CES) by following instructions in [“Adding and configuring protocols in an exiting cluster by using IBM Storage Scale installation toolkit”](#) on page 96.

19. Configure GUI for hardware monitoring.

```
essrun -N EMSVM,IOnode1,IOnode2 gui --configure
```

Precheck nodes

Check the system health before upgrade. You can use the precheck tool to check all nodes on a cluster for any issues. For more information, see [Checking system health](#).

For complete documentation, troubleshooting guidance, and timeout configuration, see the README.md file in the [GitHub repository \(https://github.com/IBM/Scale_System_PRECHECK_TOOL\)](https://github.com/IBM/Scale_System_PRECHECK_TOOL).

Upgrade from the Power9 management server

You can upgrade online or offline from the Power9 management server.

1. Complete all steps in the [Prerequisites](#).
2. Update the Power9 server from the container.

Note: You might be prompted to run the following command again, if the kernel is changed:

```
essrun -N MS1 update --offline
```

3. If a cluster exists in the environment, restart the performance monitoring and GUI.

```
systemctl restart pmsensors
```

```
systemctl restart gpfsd
```

4. If the Power9 firmware needs to be updated in the management server, run the following commands:

```
cd /install/ess/otherpkgs/rhels8/ppc64le/firmware/  
update_flash -v -f 01VL950_131_045.img  
update_flash -f 01VL950_131_045.img  
update_flash -c
```

5. After the management server node is updated from the container, run the following command:

- **Offline mode:**

```
essrun -N IOnode1,IOnode2,IOnoden update --offline
```

- **Online mode:**

```
essrun -N IOnode1,IOnode2,IOnoden update
```

6. Restart the performance monitoring service on each upgraded I/O node.

```
systemctl restart pmsensors
```

7. Update enclosures firmware.

- **Offline mode:**

```
mmchfirmware --type storage-enclosure --fast-offline
```

- **Online mode:**

```
mmchfirmware --type storage-enclosure
```

8. Update the drive firmware.

- **Offline mode:**

```
mmchfirmware -type drive --fast-offline
```

- **Online mode:**

```
mmchfirmware --type drive
```

9. If the Power9 firmware needs to be updated in the I/O nodes, run the following commands:

```
cd /install/ess/otherpkgs/rhels8/ppc64le/firmware/  
update_flash -v -f 01VL950_131_045.img  
update_flash -f 01VL950_131_045.img  
update_flash -c
```

10. Update IBM Storage Scale in the protocol nodes.

- **Offline mode:**

For more information about how to upgrade IBM Storage Scale in protocol nodes, see [Performing offline upgrade or excluding nodes from upgrade by using installation toolkit](#).

- **Online mode:**

For more information about how to upgrade IBM Storage Scale in protocol nodes, see [Performing online upgrade by using the installation toolkit](#).

11. Upgrade protocol nodes operating system and other components by using the container.

- **Offline mode:**

```
essrun -N prt1,prt2 update --offline
```

- **Online mode:**

```
essrun -N prt1,prt2 update
```

12. If the Power9 firmware needs to be updated in the protocol nodes, run the following commands:

```
cd /install/ess/otherpkgs/rhels8/ppc64le/firmware/  
update_flash -v -f 01VL950_131_045.img  
update_flash -f 01VL950_131_045.img  
update_flash -c
```

13. Run a health check on each node that was upgraded.

```
essinstallcheck
```

Upgrade from an MSVM for x86 systems

You can upgrade online or offline from an MSVM that runs on the Utility Node.

1. Complete all steps in the [Prerequisites](#).
2. Update the MSVM from the container.

Note: You might be prompted to run the following command again, if the kernel is changed.

```
essrun -N EMSVM update --offline
```

3. If a cluster exists in the environment, restart the performance monitoring and GUI.

```
systemctl restart pmsensors
```

```
systemctl restart gpfsGUI
```

4. After the MSVM is updated from the container, run the following command:

- **Offline mode:**

```
essrun -N IOnode1,IOnode2,IOnoden update --offline
```

- **Online mode:**

```
essrun -N IOnode1,IOnode2,IOnoden update
```

5. Restart the performance monitoring service on each upgraded I/O node.

```
systemctl restart pmsensors
```

6. Update firmware of enclosures.

- **Offline mode:**

```
mmchfirmware --type storage-enclosure --fast-offline
```

- **Online mode:**

```
mmchfirmware --type storage-enclosure
```

7. Update the drives firmware.

- **Offline mode:**

```
mmchfirmware --type drive --fast-offline
```

- **Online mode:**

```
mmchfirmware --type drive
```

8. If the Power9 firmware needs to be updated in the I/O nodes, run the following commands:

```
cd /install/ess/otherpkgs/rhels8/ppc64le/firmware/  
update_flash -v -f 01VL950_131_045.img  
update_flash -f 01VL950_131_045.img  
update_flash -c
```

9. Update IBM Storage Scale on the protocol nodes.

- For more information about how to upgrade IBM Storage Scale on protocol nodes offline, see [Performing offline upgrade or excluding nodes from upgrade by using installation toolkit](#).
- For more information about how to upgrade IBM Storage Scale on protocol nodes online, see [Performing online upgrade by using the installation toolkit](#).

10. Upgrade operating system of protocol nodes and other components by using the container.

• **Offline mode:**

```
essrun -N prtvm1,prt2 update --offline
```

where, *prtvm1* specifies a protocol node virtual machine.

• **Online mode:**

```
essrun -N prtvm1,prt2 update
```

11. If the Power9 firmware needs to be updated on the protocol nodes, run the following commands:

```
cd /install/ess/otherpkgs/rhels8/ppc64le/firmware/  
update_flash -v -f 01VL950_131_045.img  
update_flash -f 01VL950_131_045.img  
update_flash -c
```

12. Verify whether all upgraded nodes were installed properly.

```
essinstallcheck
```

Error handling

If you encounter any issues during the deployment, see [Troubleshooting](#).

Chapter 2. Deployment through the command-line interface

You can deploy the IBM Storage Scale System from different management servers by using command-line interface (CLI).

The software deployment is consistent across environments. However, deployment has slight variations based on a management server such as IBM Power9, x86 IBM Storage Scale System Utility Node, or bring-your-own management server (BYOMS), hardware models, or specific customer requirements.

Table 2. Software deployment methods		
IBM Storage Scale System	Management server	Deployment method
3000, 3200, 3500, 5000	IBM Power9 server	Deployment by using essrun command from a container
3000, 3200, 3500, 5000, 6000	IBM Storage Scale System Utility Node (MTM 5149-23E)	Deployment through a management virtual machine by using CLI Deployment by using GUI
3500	Customer-provided x86 server	Deployment through a management virtual machine on non-IBM hardware by using CLI

Planning and configuring network

Management server network planning is essential for the deployment of IBM Storage Scale Systems. You can ensure stable communication between nodes, support high-speed data transfers, and enable redundancy for high availability. The network configuration enhances security through segmentation and firewall policies, can ensure accurate time synchronization, and supports both IPv4 and IPv6 environments.

Management servers and switches

IBM Storage Scale System needs a management server node and a management switch.

- **One management server:** Power9, IBM Storage Scale System Utility Node, or any other management server.

For more information about Utility Node, see [IBM Storage Scale Utility Node documentation](#).

- **Dual socket Utility Node:**

Starting with IBM Storage Scale System 6.2.3.0, you can configure a dual-socket IBM Storage Scale System Utility Node as a management server.

The characteristics of this dual-socket management server are as follows:

- One more socket is available.
- Maximum memory of 1 TB.
- Default memory of 512 GB.
- Minimum memory of 256 GB.
- The key difference between a single-socket and a dual-socket management server is the management NIC placement. The NIC must be installed in slot 5.

- Slot 1 supports an NVIDIA ConnectX-6 or ConnectX-7 VPI adapter that allows the system to accommodate a third VPI card.

Note: Support for single-socket management server continues.

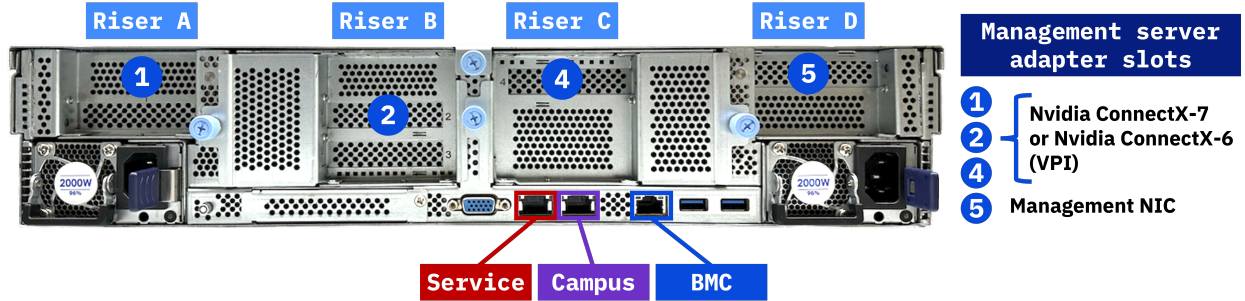


Figure 1. Slot placement rules (dual-socket management server)

- **1, 2, 4** NVIDIA Connect X-6 and ConnectX-7 (VPI)
- **5** Management NIC

For more information about the slot placement rules for an Utility Node, see Slot placement summary for the Utility Node (MTM 5149-23E) in the *IBM Storage Scale System Utility Node Hardware Planning and Installation Guide*.

- **Switch:** A high-speed switch, InfiniBand or Ethernet, for cluster communication. The switch must be split into management VLAN (VLAN 102) and FSP/BMC VLAN (VLAN 101).
 - To modify any existing management switches to the new VLAN configuration for IBM Storage Scale System 3200. For more information, see “Configuring a switch VLAN” on page 22.
 - For more information about supported switches, see [IBM Storage Scale Utility Node documentation](#).

The following table lists the ports that are connected to the management server VLAN:

IBM Storage Scale System	Management server	Port/Connection	VLAN	Notes
3000	Power9	C11-T1	Management VLAN	IPs and hostnames for all nodes
		C11-T2	FSP VLAN	
		C11-T3	Campus network or additional management connection	Recommended to access the management server and maintain a connection when the container starts.
		HMC1	FSP VLAN	
		Port 1 on each canister	Management VLAN	IPs and hostnames for all nodes
5000	Power9	C11-T1	Management VLAN	IPs and hostnames for all nodes
		C11-T2	FSP VLAN	
		C11-T3	Campus network	Recommended for remote access and GUI
		HMC1	FSP VLAN	

IBM Storage Scale System	Management server	Port/Connection	VLAN	Notes
3200	Power9	C11-T1	Management VLAN	IPs and hostnames for all nodes
			C11-T2	FSP VLAN
		C11-T3	Campus network	Recommended for remote access and GUI
		HMC1	FSP VLAN	
		Nodes (per canister)	Single management connection	Each connection is split into 2 MAC addresses: BMC and operating system. The BMC connection requires a VLAN tag that must be set for proper communication with the management server node.
		BMC connection	Shared management port	VLAN 101, BMC traffic routed through VLAN 101
		Management connection	Shared management port	VLAN 102, visible to OS
		High-speed connections	InfiniBand or Ethernet	

Examples of management server network

ESS Network (ESS 5000, ESS 3000, and ESS 3200)

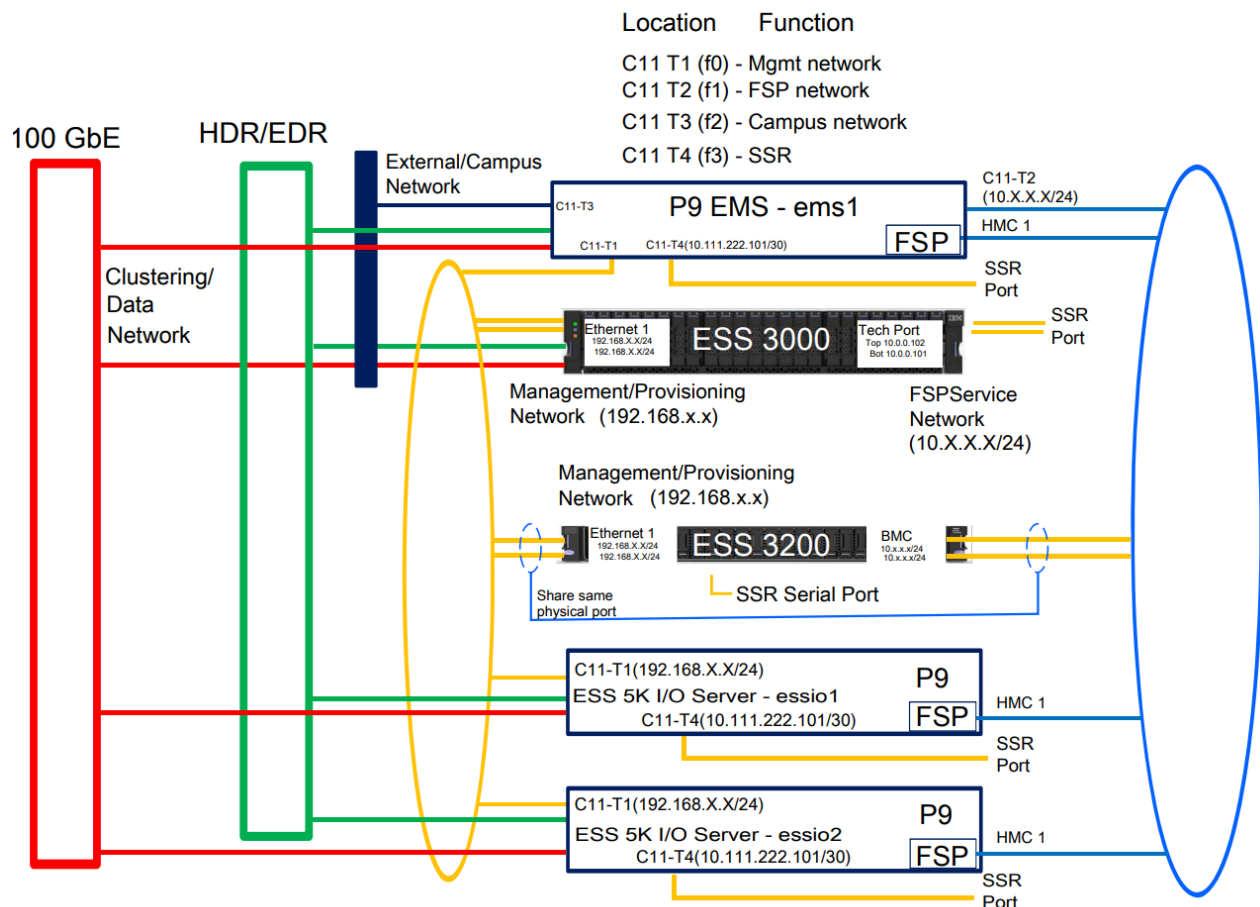


Figure 2. IBM Storage Scale Systems 3000, 3200, and 5000 management server network

The highlighted green ports are trunk ports of only IBM Storage Scale System 3200. These ports needed for each IBM Storage Scale System 3200 canister that has a single interface for the BMC and the OS but

unique MAC addresses. By using a VLAN tag, canister BMC MAC addresses are routed to the BMC/FSP/Service VLAN (Default is 101).

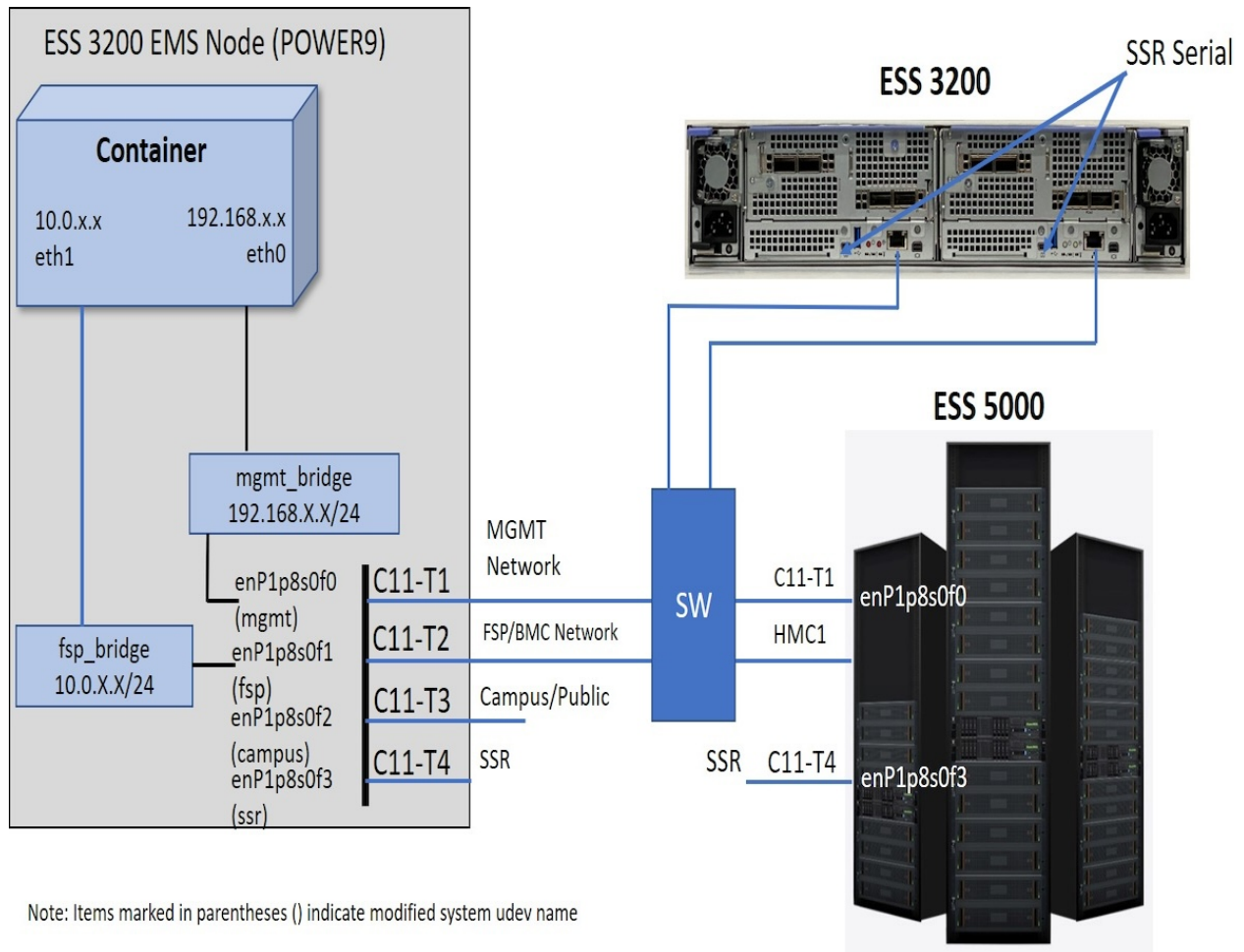
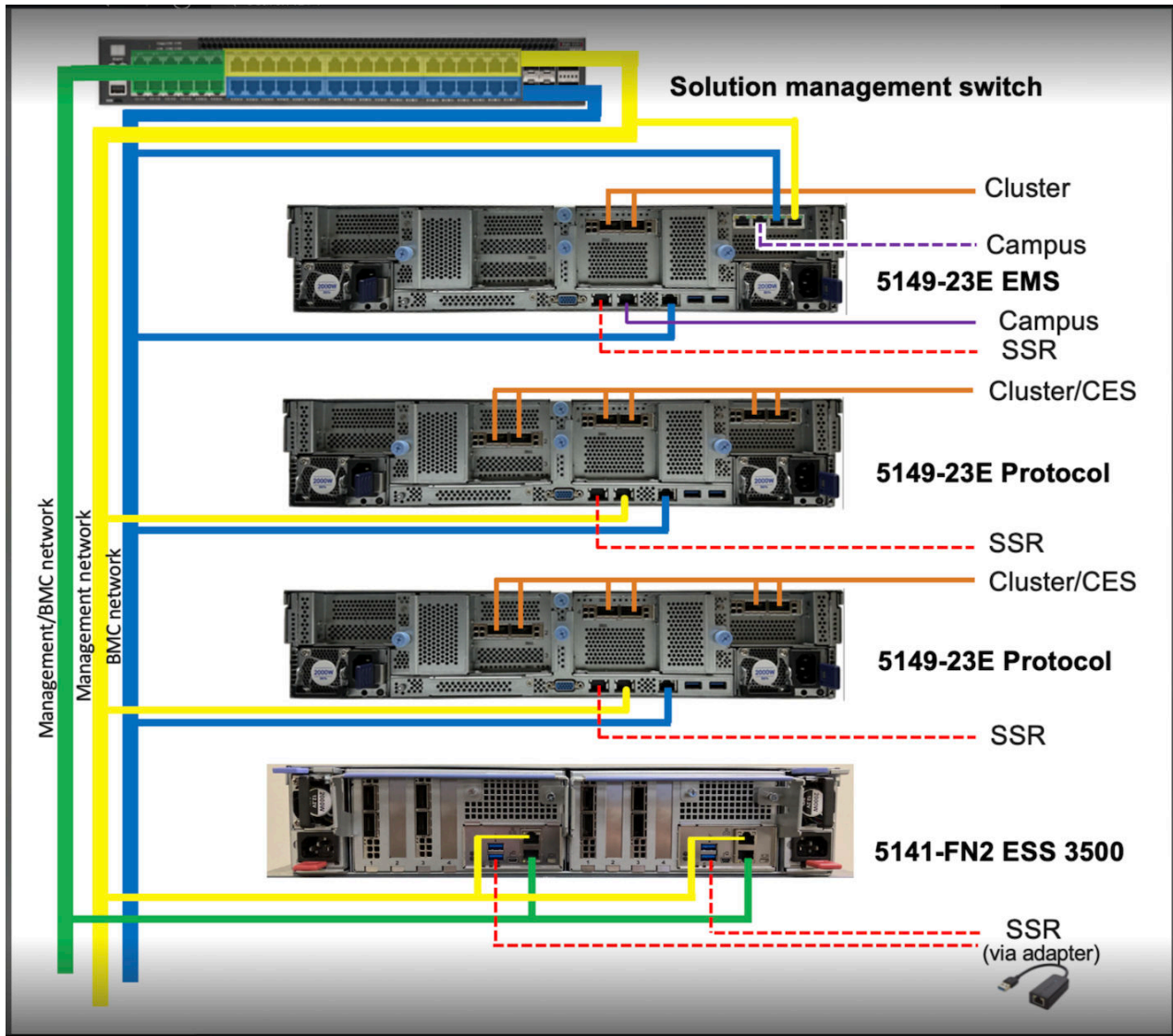


Figure 4. IBM Storage Scale Systems 3500 and 6000 management server network



Configuring multi-rail over TCP

The multi-rail over TCP (MROT) feature enables the concurrent use of multiple network interfaces and subnets. With MROT, the subnets attribute can be used to establish fault tolerance or automatic failover. All IP addresses that are defined in the subnets attribute are used to establish connections with the nodes within a cluster.

About this task

For more information, see the *Configuring multi-rail over TCP (MROT)* section in the *IBM Storage Scale: Administration Guide*.

Note: MROT and RDMA over Converged Ethernet (RoCE) coexist on the same IBM Storage Scale cluster at a specific time.

Procedure

Assign IP addresses to interfaces.

```
essgennetworks -N <NodeName> --suffix=-ce --interface <InterfaceName> --MROT <IP_Address> --
netmask /24 --mtu 9000
```

A sample output is as follows:

```
[INFO]Starting network generation...
[INFO]nodelist: <IOnden>
[INFO]suffix used for network hostname:-ce
[INFO]Interface(s) available on node <IOnden>-ce
[INFO]Considered interface(s) of node <IOnden>-ce are ['enP48p1s0f0'] with RDMA Port
['Inbuilt'] for this operation
[INFO]Checking for IPv4 address assignment on node <IOnden>

[INFO]Interface name enP48p1s0f0 consider for IP assignment with IP address 192.168.3.51 for
node <IOnden>-ce
[INFO]Successfully assigned IP Address to interface enP48p1s0f0 at node <IOnden>-ce
[INFO]IPv4 Assignment successful. Reloading connections.
[OK] essgennetworks passed successfully
[INFO]IP address assignment completed, now follow Scale MROT document to configure Multi Rail
over TCP.
```

Note: The **essgennetwork** command does not configure the MROT end-to-end. This command assigns an IP address to the specified interface.

Configure an IPv6 address on high-speed interfaces

From IBM Storage Scale System 6.1.6.0, by using the **essgennetwork** command you can configure an IPv6 address on high-speed interfaces. These high-speed interfaces can be used to create a future-ready IBM Storage Scale System cluster by using the IPv6 address over the high-speed network.

No updates are needed in the **essrun** command to use the IPv6 address during the IBM Storage Scale System cluster creation. The **essrun** command automatically selects an IPv6 address to create a cluster, if IPv6 addresses are present in the `/etc/hosts` file.

Before you issue the **essrun cluster** command to create an IBM Storage Scale System cluster by using an IPv6 address over the high-speed network, make sure that the following conditions are met:

- All necessary IPv6 addresses and a hostname must be present in the `/etc/hosts` file.
- IPv6 addresses must be assigned to all high-speed interfaces. You can use the bonded interface with the assigned IPv6 address as a part of the IBM Storage Scale System cluster.
- The high-speed interface must not be configured with an IPv6 link local address. For example, an IPv6 address that starts from `fe80::` must not be used. The interface must be assigned an IP address that can be pinged between nodes. An example of IPv6 addresses is as follows:

```
2001:700:570f:30::1 <MS1>-hs.gpfs.net <MS1>-hs
2001:700:570f:30::2 <IOnode1>-hs.gpfs.net <IOnode1>-hs
2001:700:570f:30::3 <IOnode2>-hs.gpfs.net <IOnode2>-hs
2001:700:570f:30::4 <protnode1>-hs.gpfs.net <protnode1>-hs
2001:700:570f:30::202 <IOnode1>-vm1-hs.gpfs.net <IOnode1>-vm1-hs
2001:700:570f:30::203 <IOnode2>-vm1-hs.gpfs.net <IOnode2>-vm1-hs
```

- The **essrun cluster** command must use the IPv6 hostname alias in the `-N` option during the cluster creation.

1. Assign IPv6 addresses to interfaces.

```
essgennetworks -N <IOnden> --assign-ip 2001:700:570f:30::1 --interface enP51p1s0f0
```

The system output is as follows:

```
[INFO] Starting network generation...
[INFO] nodelist: <IOnden>
```

An IPv6 address is assigned to the `enP51p1s0s0` interface.

2. For a bonded interface, during a bond creation an IP address can be configured by using the following command. By default, only IPv4 addresses are considered for the bond creation.

```
essgennetworks -N <IOnden> --suffix=-hs --create-bond --IPoIB --ip-version v6
```

The system output is as follows:

```
[INFO] Starting network generation...
[INFO] nodelist: <IOnodes>
```

The **essgennetwork --assign-ip** command can also be used on the bonded interface, if the IPv6 address was not used during the bond creation. This command assigns the IPv6 address to the created bonded interface.

3. When an IPv6 address is assigned to the interfaces or bond, issue the **essrun** command to create an IBM Storage Scale System cluster by using the IPv6 address over high-speed network. These hostnames must be the same that are used as IPv6 alias in the `/etc/hosts` file.

Restriction: The IPv6 support is available only for the GPFS admin or daemon network. The IPv6 support is not available on management, campus, and FSP networks. The IPv6 support for these interfaces needs to be configured manually.

Enabling RoCE

Remote Direct Memory Access (RDMA) over Converged Ethernet (RoCE) is a network protocol that allows direct access to the memory of another server without use of the CPU or operating system.

The current state of the OFED and firmware driver enables the optimization for performance and latency and simultaneously for better availability. It supports individual ports, multiple fabrics, and classical network bond to protect a service IP from network failures. With the current Mellanox OFED driver and adapter firmware, Mellanox supports RoCEv2 over bonded interface ports from one physical NIC.

RDMA provides faster access from one node data to another node. Direct memory access from the memory of one node to the memory of another node without involving the CPU or system resources of the kernel allows more efficient use of the environment. The network and host performance improve significantly because of lower latency, lower CPU load, and higher bandwidth. In contrast, TCP/IP communications require copy operations, which add latency and use CPU and memory resources.

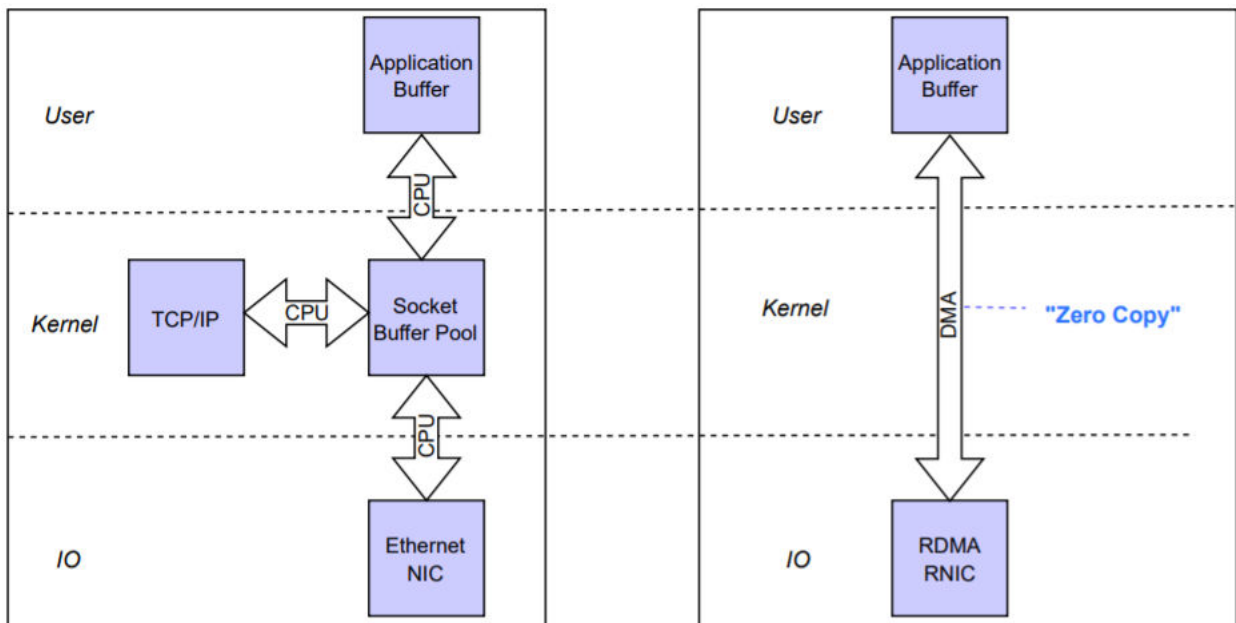


Figure 5. Comparison between TCP/IP and RoCE communications

The resources, which are needed for communication by using TCP/IP, are saved and can be used for real workload of the applications.

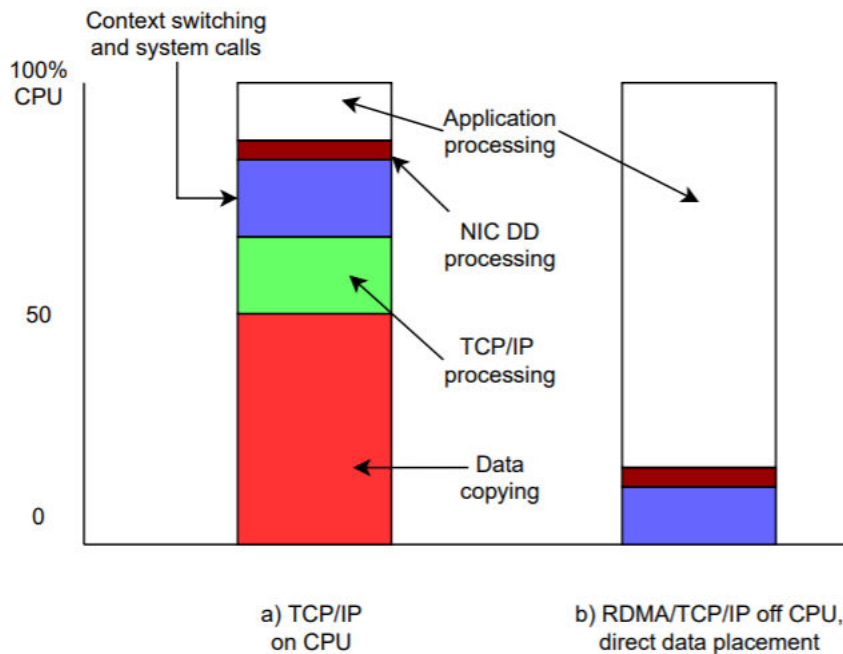


Figure 6. Comparison between TCP/IP on CPU and RDMA/TCP/IP off CPU

RDMA provides the following two benefits:

- Scales out bandwidth over multiple ports, which saves system resources.
- The higher network bandwidth results in more saving effects by using RDMA on the operating system.

IBM Storage Scale support matrix

IBM Storage Scale supports RDMA on Red Hat® Enterprise Linux and Ubuntu. IBM Storage Scale uses the VERBS programming interface to provide RDMA support. While IBM Storage Scale uses the VERBS programming interface for RDMA support, the underlying implementation of RDMA is vendor-specific.

IBM Storage Scale supports RDMA with RoCEv2 with in the following configurations:

- Supported on Red Hat Enterprise Linux and Ubuntu
- Supported on platform x86_64 and ppc64le
- Minimum Mellanox OFED Level 4.9 or later
- Minimum IBM Storage Scale System 7.0.1.x
- Minimum IBM Storage Scale System models 5000, 3000, 3200, and 3500.
- The minimum supported OS level is RHEL 9.4 for IBM Storage Scale System 7.0.1.x

RDMA requirements and limitations

The following restrictions apply for IBM Storage Scale RDMA:

- The protocols export over CES (NFS or SMB) does not use RDMA.
- Due to the TCP/IP implementation of current IBM Storage Scale releases, any additional network port apart from the `mmfsd` IP interface, needs to be on a different subnet. The `mmfsd` works itself for daemon-to-daemon communication.
- IBM Storage Scale does not support InfiniBand and Omni-Path partition keys.
- IPv6 must be enabled to use RoCE, if interfaces are selected by using the port name.
- IPv6 local link address needs to be conformed with `IPV6_ADDR_GEN_MODE=eui64`.

Mellanox requirements and limitations

- RDMA is not supported on a node when both Mellanox HCAs and Intel Omni-Path HFI are enabled for RDMA.
- You cannot use bonded ports across different adapters for use with RoCE.
- You need to have Connect X-6 adapters or later for RoCE.

RoCE configuration methods

You can deploy RoCE environment by using one of the following methods:

- **Single-port configuration**

Each node uses a single adapter port connected to the network. The same port handles both TCP/IP daemon communication and RoCE traffic concurrently.

- **Multi-port configuration**

Each node uses multiple ports connected to the network.

One port is used for both TCP/IP daemon communication and RDMA. The remaining ports are dedicated to RDMA and RoCE traffic. This approach allows you to scale out bandwidth to achieve the desired performance levels. Although it is more complex to configure, it offers greater flexibility and performance scalability.

- **Network bonding configuration**

IBM and Mellanox introduced a method to create a network bond by using two ports from the same adapter. This configuration

- Provides redundancy and fault tolerance, protecting against cable, port, or switch failures.
- Ensures high availability and reliability.

Network requirements

For running RoCE, you can get the best performance when the network is configured as a lossless network. Depending on the vendor, components, and the topology, the requirement for the lossless network can quickly become complex and is out of scope.

For more information, see [Network configuration examples for RoCE deployment](#).

Make sure that Mellanox OFED driver is installed on all the nodes in the cluster. The IBM Storage Scale System I/O server nodes are maintained by an IBM Storage Scale deployment. You can check the installed version by using the **ofed_info -s** command as shown in the following example:

```
ofed_info -s
MLNX_OFED_LINUX-x.x.x.x.x
```

Note: The minimum level of OFED version is documented in the release notes of IBM Storage Scale Systems.

Make sure that the IBM Storage Scale client nodes run the same MOFED level as the NSD and IBM Storage Scale System. However, in many projects and environments, it is a challenge to maintain all nodes with the same MOFED level.

It is possible that the client nodes are running the OFED software, which is distributed by the operating system. Such configurations are simple to operate and to maintain in the client clusters. However, in cases of trouble and network glitches, such configurations can cause unexpected failures.

Network topology

The acronym host is used for an endpoint in the network. It can be an IBM Storage Scale client system or an IBM Storage Scale System and NSD system.

A host configuration depends on the number of network ports that are used or needed. The number of ports a node connects with the network depends on the expected bandwidth, the number of subnets that are needed to get access to other environments and high availability requirements.

In TCP/IP environments, scaling bandwidth with multiple ports is commonly acquired with bonding network ports, which are known as link aggregation. But bonding can have some challenging complexities. It makes the deployment complex from the network perspective.

Bonding is a commonly used technology in data centers for balancing the traffic in a network. However, bonding does not help single-socket network traffic to use more than one cable. Further more, another major shortcoming of bonding is that the selection of network path is hard to predict or cannot be controlled from the application layer. Therefore, when you scale bandwidth over multiple ports, the use of RoCE is the most appropriate option.

To use a bond or not with an IBM Storage Scale system, the generic rule is to have at least one IP interface that is connected to the network for the daemon-to-daemon communication. A recommended network topology with bonding and RoCE is shown in the following figure.

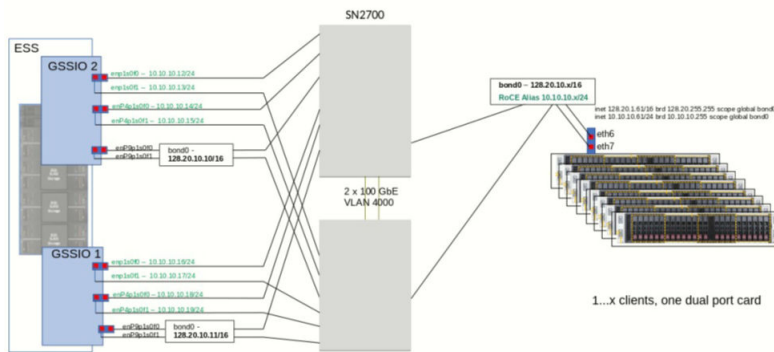


Figure 7. Displaying network topology with bonding

Depending on using a bond, it is considered that the LACP link aggregation needs to be set in the network. Alternatively, by using IBM Storage Scale and IBM Storage Scale System building blocks, you can also rely on higher HA layers in IBM Storage Scale such as NSD and recovery group server fail over and can consider skipping bonds in your topology to make the network set up less complex. For better performance and less complex setup, it is recommended to use a configuration without bond.

For better availability, use bonded configuration. An example for a configuration without bond is shown in the following figure.

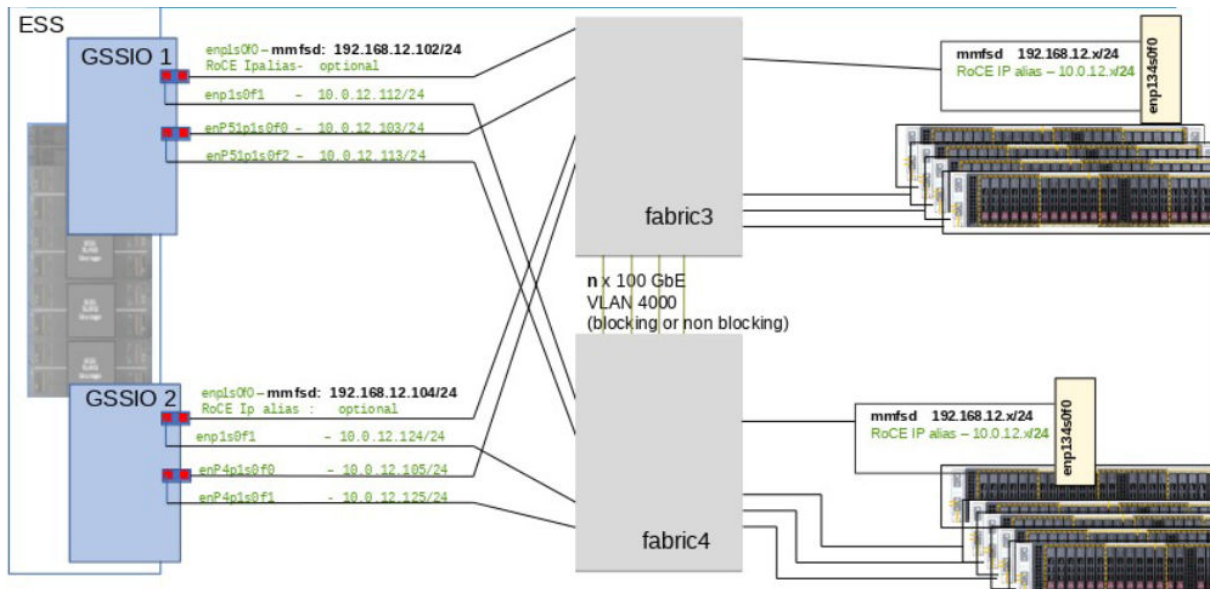


Figure 8. Showing configuration without bond

As shown in the figure, for network topology without bond, it is essential to have one IP address per network port. On the adapter that has the **mmfsd** IP address that is configured, additional RoCE IP address nor alias is needed until nodes in your cluster are able to communicate to this **mmfsd** IP address. You can configure as many aliases as the operating system version supports. You need one IP per adapter and RoCE can also use the existing IP, which is also used for TCP/IP traffic.

Running a configuration without bonds allows to later in the GPFS configuration, one can enhance the whole configuration by fabric numbers. By using such configuration, traffic on the ISL (inter-switch links) might be avoided.

Note: The adapters in a RoCE enabled environment can run TCP/IP traffic and RDMA traffic simultaneously.

MTU consideration

RDMA was introduced on InfiniBand networks. RDMA over InfiniBand supports maximum transmission unit (MTU) sizes from 256 up to 4096 Bytes.

It is recommended to adjust the MTU to jumbo frames, which is 9000 Bytes. Adjust this setting on all adapters and all nodes, communicating in your clusters. If you need to communicate with external nodes in remote networks and you cannot be sure that the path through the network supports MTU 9000 Bytes end-to-end, make sure that MTU Path Discovery is enabled. The MTU Path Discovery is enabled by default on Red Hat Enterprise Linux.

Tip: MTU can be set for the bond only when it is being getting created by using the `--create-bond` switch. Once the bond has been created, the MTU cannot be modified by using the **essgennetwork** command. The **essgennetwork** command does not support to reconfigure the MTU for an existing bond or an interface. A user must use the **nmcli** command or manually edit the `ifcfg` network configuration file and change the MTU value. Once MTU changes are applied, reload the new connection configuration by using the **nmcli config load** command and restart the bond or the interface by using the **ifdown <interface_name>** command followed by **ifup <interface_name>**.

Configuring a bond

For RoCEv2 to work properly, all interfaces need an IPv6 local link address and an IPv4 address.

The bond interfaces configuration is not needed to run RoCE. If you do not need a bonded TCP/IP interface, you can proceed with next section.

If you want avoid port failures for your IP interfaces, configure the bonds.

Note: If you do not need the daemon IP address to be protected by a bond, skip this step.

Note: To configure RoCE over bond, the bond must be created before you use the **essgennetwork** command. If a bond is created by using more than two physical network interfaces, then the existing bond must be broken and recreated by using same physical card. For example, if a bond is created by using four ports and two physical cards, then the bond must be broken and a new bond must be created by using two ports with the same physical card.

Complete the following to enable RoCE over bond.

1. Create a bond by using two interfaces.

```
essgennetworks -N <IOnode> --suffix=-ce --interface enp1s0f0,enp1s0f0f1 --create-bond --hash-policy layer3+4 --bond bond0 --mtu 9000 --miimon 1000
```

A sample output is as follows:

```
2021-10-25T03:30:24.978336 [INFO] Starting network generation...
2021-10-25T03:30:25.040319 [INFO] nodelist: <IOnode>
2021-10-25T03:30:25.040386 [INFO] suffix used for network hostname: -ce
2021-10-25T03:30:25.040430 [INFO] Bond will be created using --hash-policy layer3+4 which is
required to configure RoCE over Bond.
2021-10-25T03:30:25.040464 [INFO] Bond will be created using miimon 1000
2021-10-25T03:30:25.040495 [INFO] Bond will be created with --mtu 9000. Make sure switch is
configured to use 9000 MTU or set to auto negotiate.
2021-10-25T03:30:25.040526 [WARN] Make sure switch is configured to handle Jumbo frame
before creating bond.
2021-10-25T03:30:25.040556 [WARN] Without Jumbo frame configuration at switch may result to
network outage.
2021-10-25T03:30:25.040586 [WARN] Consider testing once configured. Example: ping -s 9000 -c
10 TARGET_NODE-hs
2021-10-25T03:30:50.356662 [INFO] Interface(s) available on node <IOnode>-ce
2021-10-25T03:30:50.356748 [INFO] Considered interface(s) of node <IOnode>-ce are
['enp1s0f0'] with RDMA Port ['mlx5_0'] for this operation
2021-10-25T03:30:50.360236 [INFO] Checking for IP address assignment on node(s)
2021-10-25T03:30:50.364861 [INFO] <IOnode>-ce: Current IP Address: IP not assigned
2021-10-25T03:30:50.367996 [INFO] Creating network bond bond0 on node <IOnode>-ce with IP
Address 192.168.2.51
2021-10-25T03:30:50.670300 [WARN] <IOnode>-ce: Bond created with one slave interface
2021-10-25T03:30:51.734465 [INFO] Network bond bond0 on node <IOnode>-ce with IP Address
192.168.2.51 has been created successfully.
2021-10-25T03:30:51.738006 [INFO] Network Bond with name bond0 is UP and connected at node
<IOnode>-ce
2021-10-25T03:30:51.738055 [INFO] Bond creation complete. Reloading connections.
```

Note: A bond is created by using the hash policy layer3+4.

Tip: MTU can only be set for the bond while it is getting created by using the **--create-bond** switch. After the bond has been created, the MTU cannot be modified by using the **essgennetwork** command. The **essgennetwork** command does not support to reconfigure the MTU for an existing bond or an interface. A user must use the **nmcli** command or manually edit the **ifcfg** network configuration file and change the MTU value. After MTU changes are applied, reload the new connection configuration by using the **nmcli config load** command and restart the bond or the interface by using the **ifdown <interface_name>** command followed by **ifup <interface_name>**.

2. Enable the RoCE by using the created bond.

```
essgennetworks -N <MSn> --suffix=-ce --bond bond0 --enableRoCE
```

A sample output is as follows:

```
2021-10-25T02:13:32.989606 [INFO] Starting network generation...
2021-10-25T02:13:33.053310 [INFO] nodelist: <MSn>
2021-10-25T02:13:33.053370 [INFO] suffix used for network hostname: -ce
2021-10-25T02:14:26.325569 [INFO] Interface(s) available on node <MSn>-ce
2021-10-25T02:14:26.325636 [INFO] Considered interface(s) of node <MSn>-ce are ['bond0']
with RDMA Port ['mlx5_bond_0'] for this operation
2021-10-25T02:14:26.693602 [INFO] Supported Mellanox RoCE card found at node <MSn>
2021-10-25T02:14:26.697197 [INFO] Supported version of Mellanox OFED found at node <MSn>-ce
2021-10-25T02:14:40.007004 [INFO] Bond validation passed and found bonds bond0 has been
created using same physical network adapter at node <MSn>-ce
2021-10-25T02:14:40.011375 [INFO] Bond MTU validation passed and found bonds MTU set to 9000
```

```

at node <MSn>-ce
2021-10-25T02:14:40.015525 [INFO] Interface bond0 have the IPv4 Address assigned at node
<MSn>-ce
2021-10-25T02:14:40.019593 [INFO] Interface bond0 have the IPv6 Address assigned at node
<MSn>-ce
2021-10-25T02:14:40.019648 [INFO] Enabling RDMA for Ports ['mlx5_bond_0']
2021-10-25T02:14:48.370397 [INFO] Enabled RDMA i.e. RoCE over Ethernet using bond bond0
2021-10-25T02:14:48.370446 [INFO] Please recycle the GPFS daemon on those nodes where RoCE
has been enabled.

```

Note: The GPFS daemon must be recycled to make the RoCE configuration working. After the daemon gets recycled, you can run the **mmdiag -network** command to check whether the RDMA is enabled over Ethernet.

Advantages and disadvantages of using bond with RoCE:

- A bonded interface protects against port and cable failures.
- For running RDMA, all ports of the bonded interface need to be on one, the same, physical PCI adapter. So, an adapter failure is not covered in such configurations.
- Creating bonds over multiple switches makes MLAG configuration mandatory in a network, which can cause unbalanced network use in the fabric.

Configuring regular IP interface

For RoCEv2 to work properly, all interfaces need an Ipv6 local link address and an IPv4 address. Configure all interfaces intended to use for RoCE communication, as shown in the following example.

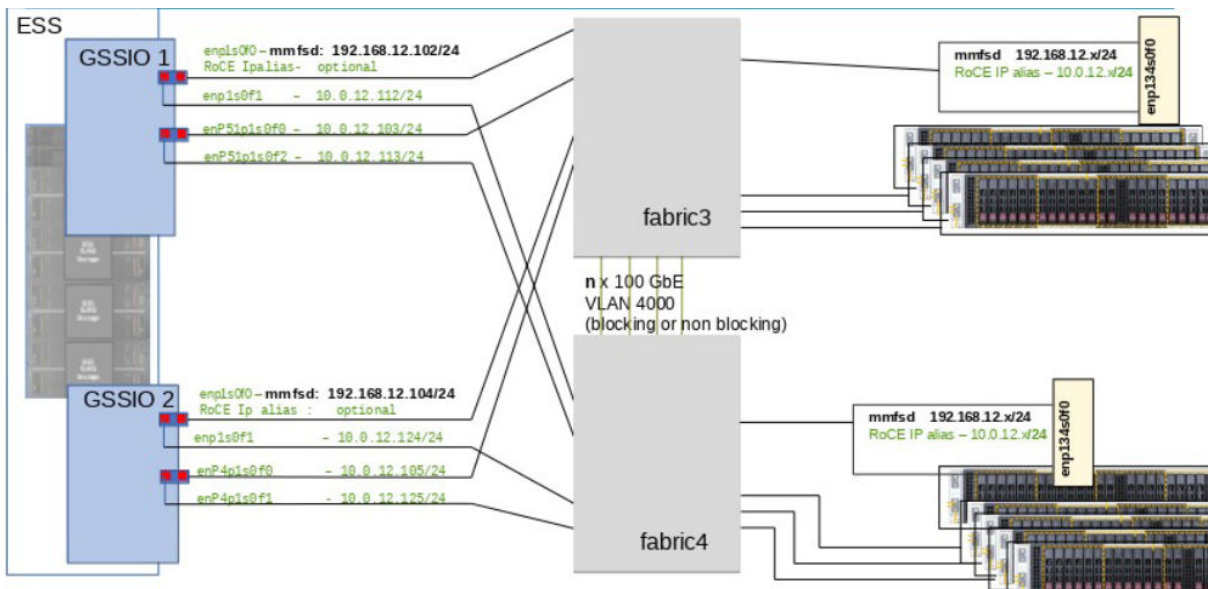


Figure 9. Configuring interfaces

As highlighted in black, make sure that only one IP interface has an IP address for the **mmfs** daemon communication. All other interfaces need to be on a different subnet.

1. Assign IPv4 to high-speed interface.

```

essgennetworks -N <IOnode> --suffix=-ce --interface enP48p1s0f0 --assignip 192.168.4.51

```

A sample output is as follows:

```

2021-10-25T04:40:58.779195 [INFO] Starting network generation...
2021-10-25T04:40:58.842575 [INFO] nodelist: <IOnode>
2021-10-25T04:40:58.842642 [INFO] suffix used for network hostname: -ce
2021-10-25T04:41:22.178448 [INFO] Interface(s) available on node <IOnode>-ce
2021-10-25T04:41:22.178533 [INFO] Considered interface(s) of node <IOnode>-ce are
['enP48p1s0f0'] with RDMA Port ['Inbuilt'] for this operation
2021-10-25T04:41:22.178591 [INFO] Checking for IP address assignment on node(s)

```

```

2021-10-25T04:41:22.182983 [INFO] <IOnode>-ce: Current IP Address: IP not assigned

2021-10-25T04:41:22.183057 [INFO] Interface name enP48p1s0f0 consider for IP assignment with
IP address 192.168.4.51 for node <IOnode>-ce
2021-10-25T04:41:22.308265 [INFO] Successfully assigned IP Address to interface enP48p1s0f0
at node <IOnode>-ce
2021-10-25T04:41:22.506151 [INFO] IP Assignment successful. Reloading connections.
[PASS] essgennetworks passed successfully

```

2. If you are using only one subnet, do not enable routing. For multiple networks in same subnet, must configure routing to route the RoCE traffic by using all subnets. To enable routing for a subnet, see the [“Configuring routing” on page 20](#) section. In the following example, only one subnet 192.168.4.0/24 is used. Therefore, you can enable RoCE by using the **-enableRoCE** command with the **-interface** switch. However, it is safe to enable routing even only one subnet is in use, by issuing the following command:

```

essgennetworks -N <IOnode> --suffix=-ce --interface enP48p1s0f0 --enableRoCE

```

A sample output is as follows:

```

2021-10-25T04:43:15.983035 [INFO] Starting network generation...
2021-10-25T04:43:16.045174 [INFO] nodelist: <IOnode>
2021-10-25T04:43:16.045243 [INFO] suffix used for network hostname: -ce
2021-10-25T04:43:39.040751 [INFO] Interface(s) available on node <IOnode>-ce
2021-10-25T04:43:39.040829 [INFO] Considered interface(s) of node <IOnode>-ce are
['enP48p1s0f0'] with RDMA Port ['mlx5_2'] for this operation
2021-10-25T04:43:39.348322 [WARN] MTU is NOT set to 9000 for interface enP48p1s0f0 at node
<IOnode>. It's advisable to set MTU as 9000 for better performance.
2021-10-25T04:43:39.348381 [WARN] Interface MTU NOT set to 9000 at node <IOnode>-ce
2021-10-25T04:43:39.348413 [WARN] Please re-configure the interface with --mtu 9000 and make
sure switch is also configured to use --mtu 9000 i.e. Jumbo frame for better performance at
node <IOnode>-ce
2021-10-25T04:43:39.352619 [INFO] Interface enP48p1s0f0 have the IPv4 Address assigned at
node <IOnode>-ce
2021-10-25T04:43:39.356763 [INFO] Interface enP48p1s0f0 have the IPv6 Address assigned at
node <IOnode>-ce
2021-10-25T04:43:39.356817 [INFO] Enabling RDMA for Ports ['mlx5_2']
2021-10-25T04:43:45.409405 [INFO] Enabled RDMA i.e. RoCE over Ethernet using Ethernet
interfaces enP48p1s0f0
2021-10-25T04:43:45.409464 [INFO] Please recycle the GPFS daemon on those nodes where RoCE
has been enabled.

```

Note: If you want to enable RoCE for bond and interfaces both, then use **-interface** and **-bond** switch with the **-enableRoCE** to enable it.

GPFS daemon must be recycled to make the RoCE configuration working. After the daemon is recycled you can run the **mmdiag -network** command to check whether the RDMA is enabled over Ethernet.

Tip: The multiple verbsPortsFabric configuration is not supported as of now by using the **essgennetwork** command. For example, if a user wants to configure fabric for bond as "/1/1" and for interfaces "/1/2", then the **essgennetwork** command does not have the option to configure it. The user needs to use the **mmchconfig verbsPorts="mlx5_bond_0/1/1 mlx5_2/1/2 mlx5_3/1/2"** command to enable the multiple fabrics for RoCE. GPFS daemon must be recycled after the **mmchconfig verbsPorts** parameter is used.

Configuring routing

In Figure 9 on page 19, the **mmfsd** communication runs in the 192.168.12.0/24 network. All other Mellanox cards interfaces are highlighted in green and configured to be in the subnet 10.10.10.x/24. All interfaces are intended to use for RDMA communication, while only the IP addresses in 192.168.12.0/24 network are used for TCP/IP communication.

With current IBM Storage Scale releases, only one IP address per node is supported for communication to the **mmfs** daemon's network.

To scale out over multiple ports, RoCE can be used. But according to the definition of OFED standards, each RDMA interface needs to have an IP address to maintain the connection.

To use multiple interfaces, you need to configure multiple subnets. It is complex for larger environments and technically not needed. It is possible to configure all interfaces that are intended to be used for RDMA into one separate subnet.

As a limitation of the Red Hat Enterprise Linux kernel, multiple interfaces to the same subnets are complex. When you have more than one interface per subnet on a node, answering arp requests, selecting rules for outgoing IP traffic, and other complexities need to be reconsidered. To resolve such issues, you must add some special routing tables, described in “2” on page 22. You need to specify routing entries for each interface and for all subnets, where you have more than one interface connected.

In example, each I/O server node has four interfaces that are connected to the same physical network. These interfaces need to get an IP address in a different IP-range(subnet) than the **mmfsd** but they are all in the same network. Refer to the following figure for topology overview.

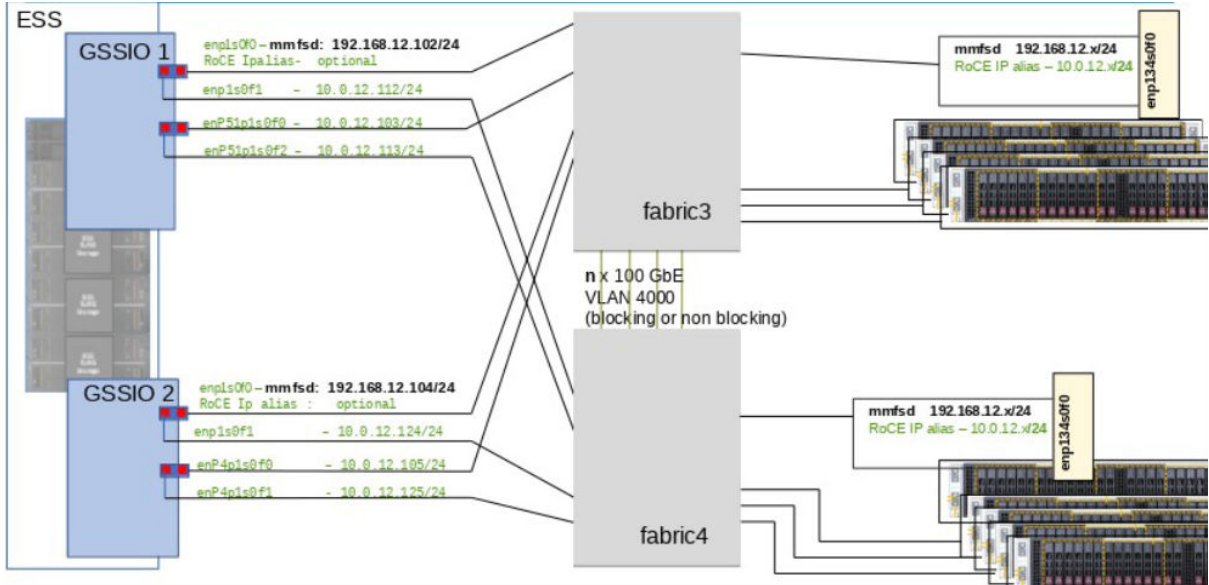


Figure 10. Displaying topology overview

For each green-marked interface, a routing entry is needed.

1. Enable routing for each RoCE interface.

```
essgennetworks -N <IOnode> --configureRouteForRoCE --roceRoutingTableId 101 --
roceRoutingTableName t1 --interface enP48pls0f0
```

A sample output is as follows:

```
2021-10-25T04:51:53.048486 [INFO] Starting network generation...
2021-10-25T04:51:53.111121 [INFO] nodelist: <IOnode>
2021-10-25T04:52:13.897913 [INFO] Interface(s) available on node <IOnode>
2021-10-25T04:52:13.897991 [INFO] Considered interface(s) of node <IOnode> are
['enP48pls0f0'] with RDMA Port ['mlx5_2'] for this operation
2021-10-25T04:52:14.202172 [WARN] MTU is NOT set to 9000 for interface enP48pls0f0 at node
<IOnode>. It's advisable to set MTU as 9000 for better performance.
2021-10-25T04:52:14.202235 [WARN] Interface MTU NOT set to 9000 at node <IOnode>
2021-10-25T04:52:14.202268 [WARN] Please re-configure the interface with --mtu 9000 and make
sure switch is also configured to use --mtu 9000 i.e. Jumbo frame for better performance at
node <IOnode>
2021-10-25T04:52:14.206459 [INFO] Interface enP48pls0f0 have the IPv4 Address assigned at
node <IOnode>
2021-10-25T04:52:14.210723 [INFO] Interface enP48pls0f0 have the IPv6 Address assigned at
node <IOnode>
2021-10-25T04:52:14.225432 [INFO] RoCE routing validation passes.
2021-10-25T04:52:14.225488 [INFO] Setting UP routing for interface RDMA for Interface
['enP48pls0f0'] at node <IOnode>
2021-10-25T04:52:14.228225 [INFO] Successfully added routing id and it's table name
node <IOnode>
2021-10-25T04:52:14.672155 [INFO] Successfully configured route for interface enP48pls0f0 at
node <IOnode>
```

```
2021-10-25T04:52:14.672219 [INF0] Routing configuration for subnet has configured successfully.
```

2. Check the routing table entry.

```
ip r
```

```
10.111.222.100/30 dev enP1p8s0f3 proto kernel scope link src 10.111.222.101 metric 101  
linkdown  
192.168.2.0/24 dev bond0 proto kernel scope link src 192.168.2.51 metric 300  
192.168.4.0/24 dev enP48p1s0f0 proto kernel scope link src 192.168.4.51 metric 104  
192.168.15.0/24 dev enP1p8s0f0 proto kernel scope link src 192.168.15.51 metric 100
```

Repeat these steps to configure multiple RoCE interfaces for all interfaces .

Configuring the sysctl file

In addition to the interface scripts, you need system-wide settings that are managed by `sysctl`. Depending on the IBM Storage Scale System version, a customized `sysctl` setting is available by a tuned profile, which is named as `scale`. You need to edit the `/etc/tuned/scale/tuned.conf` file. For any other client node in your cluster, you can deploy the same `sysctl` configuration file.

1. Ensure that the following `sysctl` settings are applied:

```
net.ipv6.conf.all.disable_ipv6=0  
net.ipv6.conf.default.disable_ipv6=0
```

Note: The IPv6 configuration can be enabled by the default IBM Storage Scale System deployment process.

2. Validate the `verbsPorts` configuration.

```
mmlsconfig  
[<MSn>-ce,<IOnode1>-ce,<IOnode2>-ce]  
verbsRdma enable  
verbsRdmaCm enable  
verbsRdmaSend yes  
[<MSn>-ce,<IOnode2>-ce]  
verbsPorts mlx5_bond_0  
[<IOnode1>-ce]  
verbsPorts mlx5_2
```

3. You can run the `mmdia` **--network** command and check for the `verbsRDMA` connection.

Configuring a switch VLAN

Configure an IBM Cumulus switch VLAN for high-performance, low-latency Ethernet, or InfiniBand network.

Manufacturing preconfigured the IBM Cumulus switch with the following VLAN:

- Service/FSP/BMC VLAN
 - Blue network - Bottom ports
 - VLAN 101
- Management/Provisioning VLAN
 - Yellow network - Top ports
 - VLAN 102
- Special ports
 - Ports 1 - 12
 - Trunk ports
 - Default routes traffic to management VLAN
 - Packets with VLAN tag 101 routed to service network

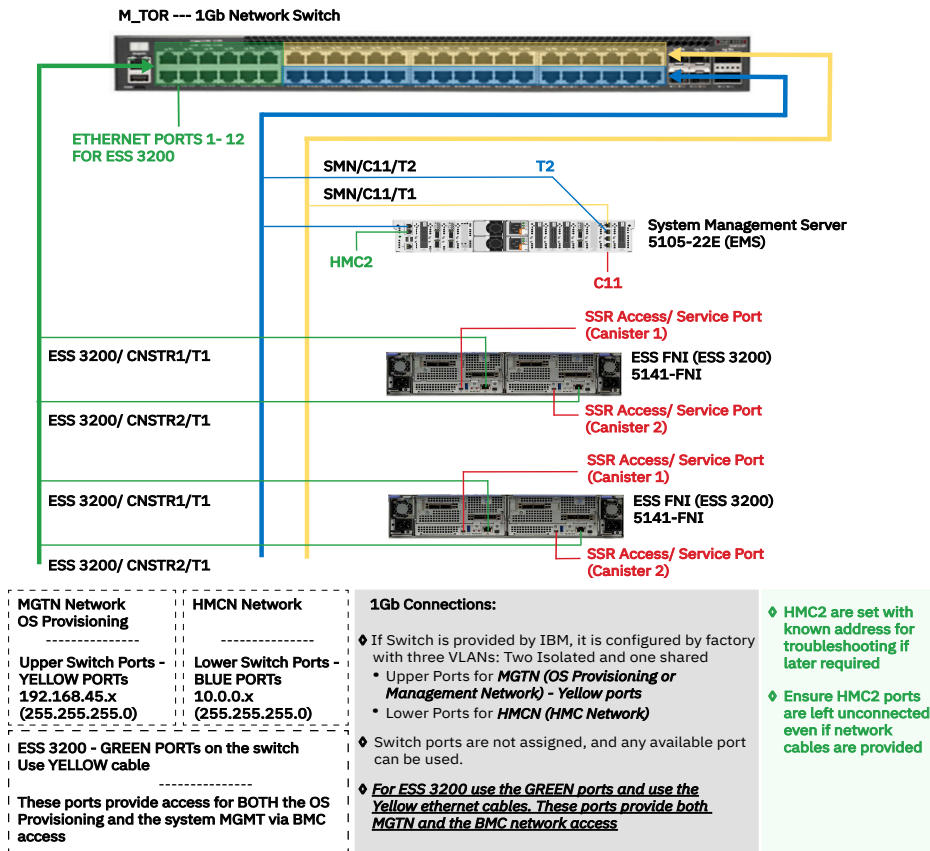


Figure 11. 1 Gb network switch

Adding management switches

If the customer is out of ports on a single management switch, then a second one can be used to extend VLANs.

Connect the existing switch (Switch 1) port 45 to the new switch (Switch 2) port 47, and Switch 1 port 46 to Switch 2 port 48. It extends both the management and service VLANs ports to the new switch.

Changing the switch default password

Change the default switch password when you set up the switch for the first time.

Before you begin

Verify the 11S label at the back of the switch.



Figure 12. 11S label

About this task

The required software update is cumulus-3.7.12a.

Procedure

1. Log in to the switch using the default credentials.

User ID

cumulus

Password

CumulusLinux!

If the CumulusLinux1 default password does not work, to reset the password see the [Password Recovery](#) section in the NVIDIA documentation.

2. Verify the serial number.

```
decode-syseeprom | grep Serial | awk '{print $5}' | cut --complement -c -3
```

The following 11S serial number is displayed:

```
01FT690YA50YD7BGABX
```

3. Change the default password to the 11S serial number.

```
passwd
current UNIX password: CumulusLinux!
Enter new UNIX password: <<<Paste the 11S serial number>>>
Retype new UNIX password: <<<Paste the 11S serial number>>>
The system displays:
```

The system output is as follows:

```
passwd: password updated successfully.
```

4. Log in through SSH or console by using the new 11S password to validate the changes.

Results

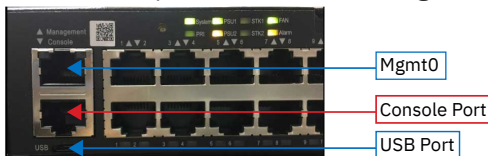
The default password must match the 11S serial number 01FT690YA50YD7BGABX. If not, the password must remain CumulusLinux!.

Connecting the PC to the switch console port

Describes how to connect the host PC to the switch console port for console-based access.

About this task

The console port and cable markings on the switch help identify the correct connection points.



Procedure

1. Connect the RJ45-to-serial cable to the switch console port.



Figure 13. RJ45 to serial cable and USB to serial cable

2. Attach the serial end of the cable to the USB-to-serial adapter.



Figure 14. USB cable

3. Connect the USB end of the adapter to the PC.

Configuring the host PC

Prepare the host PC to establish a console connection to the switch so that initial configuration can be performed.

Procedure

1. Ensure that the USB-to-serial cable driver is installed on the PC.
2. Open device manager and verify that the USB-to-serial cable is assigned a COM port.
3. Open `putty.exe` and configure the COM port connection.

Configure PuTTY with the following settings:

Baud rate	115200
Parity	None
Stop bits	1
Data bits	8
Flow control	None

4. Power on the switch and wait for the login prompt.
5. Log in with the default credentials.

User ID

`cumulus`

Password

`<11S serial number>`

If the switch uses Mellanox defaults, use:

User ID

`cumulus`

Password

`CumulusLinux!`

6. Download the `H48712_interfaces.rtf` VLAN configuration file from <https://ibm.box.com/v/H48712interfaces>.

If the configuration file link is unavailable, check the following output:

`H48712_interfaces.rtf`:

```
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).
source /etc/network/interfaces.d/*.intf
# The loopback network interface
auto lo
iface lo inet loopback
# The primary network interface
auto eth0
iface eth0
    address 192.168.45.60/24
    gateway 192.168.45.1
# EVEN Ports/Lower ports PVID 101 for FSP network
```

```

auto swp14
iface swp14
bridge-access 101
auto swp16
iface swp16
bridge-access 101
auto swp18
iface swp18
bridge-access 101
auto swp20
iface swp20
bridge-access 101
auto swp22
iface swp22
bridge-access 101
auto swp24
iface swp24
bridge-access 101
auto swp26
iface swp26
bridge-access 101
auto swp28
iface swp28
bridge-access 101
auto swp30
iface swp30
bridge-access 101
auto swp32
iface swp32
bridge-access 101
auto swp34
iface swp34
bridge-access 101
auto swp36
iface swp36
bridge-access 101
auto swp38
iface swp38
bridge-access 101
auto swp40
iface swp40
bridge-access 101
auto swp42
iface swp42
bridge-access 101
auto swp44
iface swp44
bridge-access 101
auto swp46
iface swp46
bridge-access 101
auto swp48
iface swp48
bridge-access 101

# ODD Ports/Upper ports PVID 102 for management network
auto swp13
iface swp13
bridge-access 102
auto swp15
iface swp15
bridge-access 102
auto swp17
iface swp17
bridge-access 102
auto swp19
iface swp19
bridge-access 102
auto swp21
iface swp21
bridge-access 102
auto swp23
iface swp23
bridge-access 102
auto swp25
iface swp25
bridge-access 102
auto swp27
iface swp27
bridge-access 102
auto swp29
iface swp29

```

```

bridge-access 102
auto swp31
iface swp31
bridge-access 102
auto swp33
iface swp33
bridge-access 102
auto swp35
iface swp35
bridge-access 102
auto swp37
iface swp37
bridge-access 102
auto swp39
iface swp39
bridge-access 102
auto swp41
iface swp41
bridge-access 102
auto swp43
iface swp43
bridge-access 102
auto swp45
iface swp45
bridge-access 102
auto swp47
iface swp47
bridge-access 102

# ESS 3200 ports (1 to 12) FSP + OS on single physical port
auto swp1
iface swp1
bridge-pvid 102
bridge-vids 101
auto swp2
iface swp2
bridge-pvid 102
bridge-vids 101
auto swp3
iface swp3
bridge-pvid 102
bridge-vids 101
auto swp4
iface swp4
bridge-pvid 102
bridge-vids 101
auto swp5
iface swp5
bridge-pvid 102
bridge-vids 101
auto swp6
iface swp6
bridge-pvid 102
bridge-vids 101
auto swp7
iface swp7
bridge-pvid 102
bridge-vids 101
auto swp8
iface swp8
bridge-pvid 102
bridge-vids 101
auto swp9
iface swp9
bridge-pvid 102
bridge-vids 101
auto swp10
iface swp10
bridge-pvid 102
bridge-vids 101
auto swp11
iface swp11
bridge-pvid 102
bridge-vids 101
auto swp12
iface swp12
bridge-pvid 102
bridge-vids 101

# Bridge setup
auto bridge

```

```
iface bridge
bridge-vlan-aware yes
bridge-ports glob swp1-48
bridge-pvid 101
bridge-pvid 102
bridge-stp off}
```

7. Gain sudo rights.

```
sudo su -
```

8. Copy the VLAN configuration file contents into /etc/network/interfaces.

Edit the file by using vi or any preferred editor.

9. Reload interfaces and verify status.

```
ifreload -a
ifquery -a
```

10. Check VLAN setup.

```
net show interface all
```

11. Configure switch network settings, if required.

```
net add interface eth0 IP address 192.168.44.20/24
net add interface eth0 IP gateway 192.168.44.1
net pending
net commit
```

12. Set the VLAN tag on each server canister.

Required VLAN tag: 101

```
/bin/ipmitool lan set 1 vlan id 101
```

13. Confirm the VLAN tag.

```
/bin/ipmitool lan print 1 | grep -i 'VLAN ID'
```

Modifying existing Cumulus switches

Before you begin

Consider the following points before you modify an existing Cumulus switch. For more information about ports, see

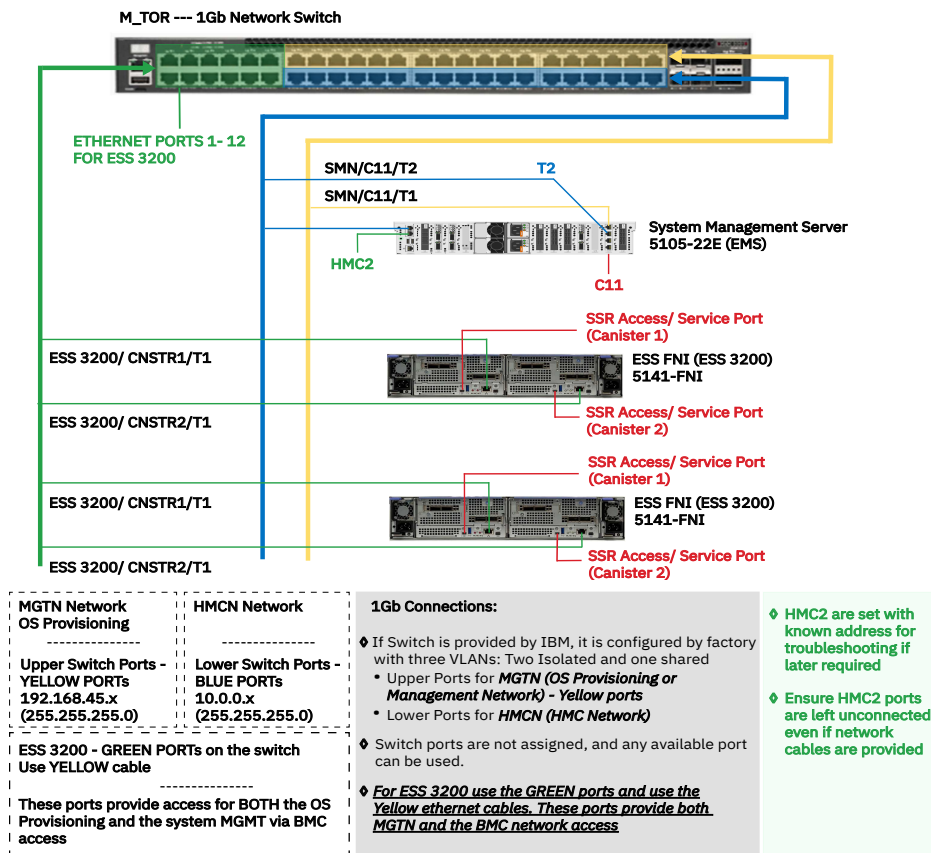


Figure 15. 1 Gb network switch

- If you are converting a switch that non-IBM Storage Scale System 3200 is using on any port in the range 1 - 12, you need to evacuate those ports one by one . If you are not using ports in the range 1 - 12, you need to apply the above process.
- If the switch you are converting is already in use in a non-IBM Storage Scale System 3200 environment and any cables are connected to ports 1–12, you must evacuate those ports one at a time. If no cables are connected to ports 1-12, apply the procedure directly.
- That means to move the cables on the upper ports in the range 1 - 12 to any free upper port that is not in the range ports 1 - 12. Equally, any lower cable plugged to any port in the range 1 - 12 needs to be moved to any lower port is not in the range of ports 1 - 12.
- You must move one cable at the time and wait until the link LED on the destination port comes up. After all ports in the range 1-12 are no longer cabled, you can apply the following procedure.

Note: If you have a non-IBM Cumulus switch, use VLAN switch configuration as a general reference on how to modify the switch. The key is to have a designated trunk ports that are apart of both VLANs.

About this task

If an existing Cumulus switch must be modified to support IBM Storage Scale Systems, complete the following steps:

Procedure

1. Free up minimum two ports (1 IBM Storage Scale System) on the existing switch. It is better if you can free up a block. Ideally, the current configuration is not scattered where it is easy to convert free ports for IBM Storage Scale Systems usage.
2. Take the existing interfaces file from the switch and modify it for the chosen IBM Storage Scale System ports.
3. Modify the interfaces file.

```

auto swp10
iface swp10
bridge-pvid 102
bridge-vids 101

```

Any ports that you designate as IBM Storage Scale System ports need to have this configuration. Consult the default IBM Storage Scale System interfaces file for more information.

4. Copy the new interfaces file to the switch.
5. Reload and verify the interfaces.
6. Set the VLAN tags on the IBM Storage Scale System canisters.

Configuring the MTM 5149-S54 switch

After the MTM 5149-S54 switch installation in the IBM Storage Scale System, set up this switch.

About this task

The next figure highlights the placement of the management and the serial console ports in an MTM 5149-S54 switch.

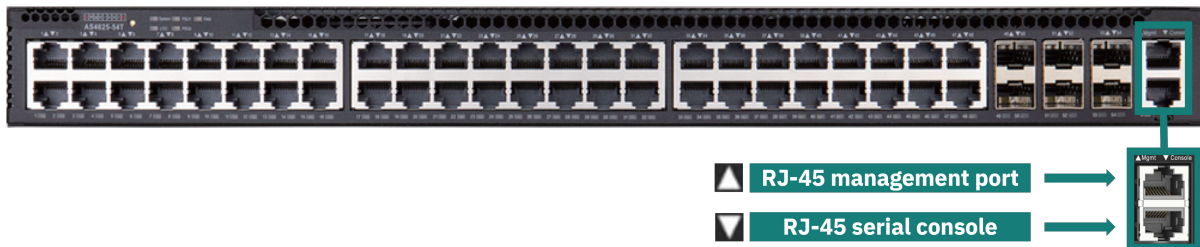


Figure 16. Locating the management and serial console ports of the MTM 5149-S54 switch

Procedure

1. Use the serial console port of the MTM 5149-S54 switch to connect it to your service computer.
2. In your service computer, use PuTTY or other SSH or serial application to apply the following settings:
 - a) Select the appropriate serial port (COM port 1 or COM port 2).
 - b) Set the baud rate to 115200 bps.
 - c) Set the data format to 8 data bits, 1 stop bit, and no parity.
 - d) Set flow control to none.
 - e) Set the emulation mode to VT100.
3. Power on the switch.

After the system completes the initialization cycle, the login screen is displayed.

4. In the **User Name** field, enter admin.
5. In the **Password** field, enter YourPaSsWoRd .

After a successful login, the following message is displayed:

```

You are on
  _____
 /  _  _  \
|  _ \| | | | | |
| |_) | | | |
|  _ \| | | |
|_| \_|_|_|_|

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Unauthorized access and/or use are prohibited.
All access and/or use are subject to monitoring.

```

6. Set a root password.

```
admin@sonic:~$ sudo passwd root
New password:
Retype new password:
```

Important: Make sure to securely save the password.

7. Configure the VLANs in the switch.

- a) Set up the factory configuration.

```
sudo config-setup factory keep-basic
```

- b) Reload the factory configuration.

```
sudo config reload -y
```

- c) Configure the BMC or FSP VLAN.

```
sudo config vlan add 101
```

- d) Configure the management VLAN.

```
sudo config vlan add 102
```

- e) Add ports or members to the management VLAN.

```
sudo for i in 0 2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46; do sudo
config vlan member add -u 102 Ethernet$i;done
```

- f) Add ports or members to the BMC VLAN.

```
sudo for i in 1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47; do sudo
config vlan member add -u 101 Ethernet$i;done
```

- g) Optional: Add a static management IP.

```
sudo config interface ip add eth0 192.168.1.10/24
```

- h) Save the configuration.

```
sudo config save -y
```

- i) Display the configuration on the screen and review it.

```
show vlan brief
```

- j) Reinitialize the switch.

```
reboot
```

- k) Verify that the configuration is persistent.

```
show vlan brief
```

Example output for show vlan brief

In the next snippet, it is corroborated that the configuration is persistent because the settings match what was set up in the previous steps.

```
show vlan brief
+-----+-----+-----+-----+-----+-----+
| VLAN ID | IP Address | Ports | Port | Proxy | DHCP Helper | DHCP |
| Relay Configuration | | Tagging | ARP | Address | |
| | | | | | | |
+=====+=====+=====+=====+=====+=====+
| 101 | | Ethernet0 | untagged | disabled | | |
Source Interface: |
```

Selection:		Ethernet2	untagged			Link
Server Vrf:		Ethernet4	untagged			
Server ID Override:		Ethernet6	untagged			
Primary Address:		Ethernet8	untagged			
Packet Size:		Ethernet10	untagged			Max
		Ethernet12	untagged			
		Ethernet14	untagged			
		Ethernet16	untagged			
		Ethernet18	untagged			
		Ethernet20	untagged			
		Ethernet22	untagged			
		Ethernet24	untagged			
		Ethernet26	untagged			
		Ethernet28	untagged			
		Ethernet30	untagged			
		Ethernet32	untagged			
		Ethernet38	untagged			
		Ethernet40	untagged			
		Ethernet42	untagged			
		Ethernet44	untagged			
		Ethernet46	untagged			

102		Ethernet1	untagged	disabled		
Source Interface:		Ethernet3	untagged			Link
Selection:		Ethernet5	untagged			
Server Vrf:		Ethernet7	untagged			
Server ID Override:		Ethernet9	untagged			
Primary Address:		Ethernet11	untagged			Max
Packet Size:		Ethernet13	untagged			
		Ethernet15	untagged			
		Ethernet17	untagged			
		Ethernet19	untagged			
		Ethernet21	untagged			
		Ethernet23	untagged			
		Ethernet25	untagged			
		Ethernet27	untagged			
		Ethernet29	untagged			
		Ethernet31	untagged			
		Ethernet33	untagged			
		Ethernet35	untagged			
		Ethernet37	untagged			

		Ethernet39	untagged		
		Ethernet41	untagged		
		Ethernet43	untagged		
		Ethernet45	untagged		
		Ethernet47	untagged		
+-----+-----+-----+-----+-----+					
+-----+-----+-----+-----+-----+					

8. Disconnect from the serial console port and try to connect to the switch by using the management port.

What to do next

If the configuration fails

Use the **show system status** command to investigate the possible cause of the failure.

Obtain the switch serial number

To obtain the switch serial number, from the command line, use **show platform syseeprom**.

Suppress console message

To suppress the console message when you are connected through the serial port, use **sudo dmesg -n 1**.

Add individual ports to a VLAN

To add individual ports to a VLAN, use **sudo config vlan member add -u <VLAN> <Ethernet_port>**.

If the system amber LED is on

Use the **sudo show system-health summary** command to investigate what component is failing.

For example, faulty fans are reported as "broken" in the next output.

```
admin@sonic:~$ sudo show system-health summary
System status summary

System status LED  STATUS_LED_COLOR_AMBER
Services:
  Status: OK
Hardware:
  Status: Not OK
  Reasons: FAN-3 is broken
           FAN-2 is broken
           FAN-1 is broken
admin@sonic:~$
```

After you know which is the faulty component, investigate further.

To continue with the same example of the faulty fans, the **fanshow** command is used to determine what is the cause of the issue. In the next output, the speed of the faulty fans is low: 21%.

```
admin@sonic:~$ fanshow
```

Drawer	LED	FAN	Speed	Direction	Presence	Status	Timestamp
FanTray-1	N/A	FAN-1	21%	intake	Present	Not OK	20230618 19:05:26
FanTray-2	N/A	FAN-2	21%	intake	Present	Not OK	20230618 19:05:26
FanTray-3	N/A	FAN-3	21%	intake	Present	Not OK	20230618 19:05:26
N/A	green	PSU-1	FAN-1	34%	N/A	Present	OK 20230618 19:05:27
N/A	green	PSU-2	FAN-1	33%	N/A	Present	OK 20230618 19:05:27

Extending VLAN across switches

To connect multiple switches, connect VLAN ports between switches with matching VLAN IDs.

1. Connect one cable from each VLAN on switch A to the same VLAN on switch B.

For example, connect Blue VLAN on switch A to Blue VLAN on switch B.

2. Select an available port in each VLAN on both switches and connect the VLAN ports directly.

This configuration extends management and BMC VLANs across switches.

The same method applies to all supported management switches, such as S54, S52, and 48-port switches.

New rules for VLAN tag usage in IBM Storage Scale System 3500

In earlier deployments of IBM Storage Scale System 3500, VLAN tags were required because the same BMC port supported management and interconnect traffic. This requirement no longer applies.

Do not configure VLAN tags on canisters by using IPMI commands. The S54 switch does not support VLAN-tagged traffic. When you deploy 3500 nodes or add S54 switches, use untagged traffic.

Configuring a dual 24-port (48 ports) MGMT switch

This topic describes how to configure a pair of ECS4100-28T to be the IBM Storage Scale System top of the rack (TOR) switches.

For more information about the port, see [ECS4100-28T](#).

This is intended for IBM manufacturing but can be also used by field engineers if needed.

When the management TOR is part of an order, IBM will deliver two of these switches as part of that order. The reason to deliver two instead of one is to keep similar number of ports available as with the 48 ports switch option.

Logical overview

From a logical perspective, the switches would look like the following figure:

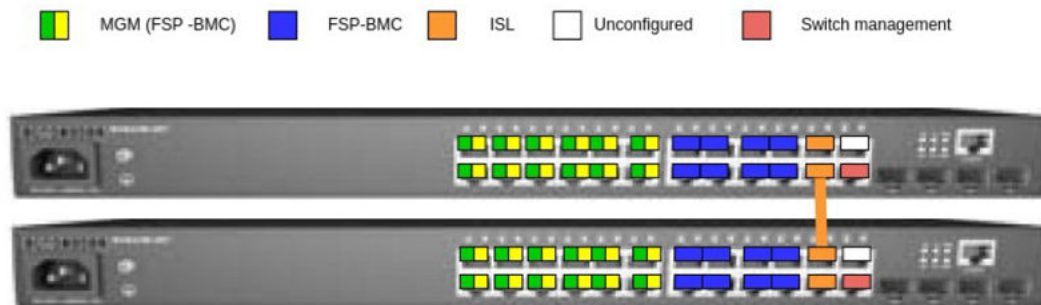


Figure 17. Logical view of two switches

The orange-colored cable shown in the figure must be connected between port 22 of the upper switch and 21 of the lower switch as a part of the configuration. That cable works as inter-switch link (ISL) between the two switches.

Ports definitions

The following are the ports definitions:

- Ports 1 through 12, named as MGM (FSP-BMC) in the green and yellow colors, are to be used for management connections to management server and I/O nodes. These ports are used for all IBM Storage Scale System models, regardless they have dual MAC ports (IBM Storage Scale System 3200 and IBM Storage Scale System 3500) or not (IBM Storage Scale System 5000). This is different from the previous IBM provided TOR rack where each logical color network had dedicated ports. Green and yellow ports share the same physical ports (1 through 12).
- Ports 13 through 20, named as FSP-BMC in the blue color, are to be used for systems that have dedicated FSPBMC connections. These are Power9 management server (both C11-T2 and HMC1 ports) and IBM Storage Scale System 5000 I/O nodes HMC1 ports.
- Ports 21 and 22, named as ISL in the orange color, are to be used for ISL between switches only. IBM allows to extend this setup to more than two switches on a line topology, meaning the first and last switch can only have one ISL connection to the following/previous switch. The switches that are not on the edge of that line topology have both ISL ports used to their previous and following switch on the line.
- Port 23, named as Unconfigured in the white color, is not used and is shut down. This port might be used in the future.
- Port 24, named as Switch management in the red color, is to be used to access the management functions of the switch. It is intended for customer switch management network and it is set up to get an IP address via the DHCP protocol.
 - The Switch management port is set as VLAN 1305 access port by default. It should work on any setup that provides an access port connected to it. If the field setup requires a different VLAN ID, change the following line:

```
VLAN 1305 name CUSTOMER media ethernet
```

- For the VLAN ID required. Match the VLAN ID with the ID in the following block:

```
interface vlan 1305
ip address dhcp
exit
```

- If you need to set a static IP address on the Switch management, replace the DHCP in the following block:

```
interface vlan 1305
ip address dhcp
exit
```

- For the IP and netmask required. In the following example, the IP address is set to 192.168.44.22 and the netmask is set to 255.255.255.0.

```
interface vlan 1305
ip address 192.168.44.22 255.255.255.0
exit
```

Customizing the switch

To each switch, the following customization must be done. In this section, only one switch is customized but this customization must be repeated to the second and any other subsequent switch from any order.

It is assumed that the “Switch management” port is not used and the serial connection will be used. For this, you need the RJ-45 to DB-9 cable that comes with the switch. You might need extra adapters and/or converters to connect to your computer if you do not have a DB-9 connection on it.

The serial port RJ-45 is located on the top right of the switch, the serial configuration settings are 115200 bps, 8 characters, no parity, one stop bit and no flow control.

Note: The factory user is “admin” and the factory password is “admin”, if the switch is personalized already at manufacturing.

For more information, see *Quick Start Guide* in the https://www.edge-core.com/_upload/images/ECS4100_series_models_QSG_R03_20180418.pdf.

Once logged in, apply the switch configuration information to both switches.

- Copy and paste the following configuration to the switch and press Enter.
- Or, copy the contents of the file in the [box](https://ibm.box.com/s/x3kzg5ykkfsu2t6536um5p0pmbg64xrl) (<https://ibm.box.com/s/x3kzg5ykkfsu2t6536um5p0pmbg64xrl>) to the switch and press Enter.
- Highlight all information and copy and paste to the switch command line after the login.

```
ip ssh crypto host-key generate
configure
ip ssh server
vlan database
VLAN 100 name ESS_MNG media ethernet
VLAN 101 name ESS_BMC media ethernet
VLAN 1305 name CUSTOMER media ethernet
exit
loopback-detection action none
no loopback-detection

interface ethernet 1/1-12
switchport mode hybrid
switchport native vlan 100
switchport allowed vlan add 100 untagged
switchport allowed vlan add 101 tagged
switchport allowed vlan remove 1
no spanning-tree loopback-detection
no shutdown
exit

interface ethernet 1/13-20
switchport mode access
switchport allowed vlan add 101 untagged
switchport native vlan 101
switchport allowed vlan remove 1
no shutdown
exit

interface ethernet 1/21-22
switchport mode hybrid
switchport native vlan 100
switchport allowed vlan add 100 untagged
switchport allowed vlan add 101 tagged
spanning-tree spanning-disabled
switchport allowed vlan remove 1
no shutdown
exit

interface ethernet 1/23
shutdown
no loopback-detection
spanning-tree spanning-disabled
exit

interface ethernet 1/24
switchport allowed vlan add 1305 untagged
switchport mode access
switchport native vlan 1305
switchport allowed vlan remove 1
no shutdown
exit

interface vlan 1305
ip address dhcp
exit

exit
copy running-config startup-config
```

Up to this point the configuration is always the same for every switch, the following lines are different for each switch. You need to know the serial number of the switch to continue. To get the serial, run show version command as shown in the following example:

```
Vty-1#show version
Unit 1
Serial Number : EC2028001435
Hardware Version : R02A
Number of Ports : 28
Main Power Status : Up
Role : Master
Loader Version : 1.0.1.9
Linux Kernel Version : 2.6.19-g496f2361-di
Operation Code Version : 1.2.71.203
```

In this example, the serial number EC2028001435 is used. You need to use the serial number of each switch on the commands.

```
configure
username guest password 0 EC2028001435
username admin password 0 EC2028001435
exit
copy running-config startup-config
exit
```

At this point, you are disconnected from the switch and the switch is personalized for IBM Storage Scale System usage.

Configuring ConnectX-5, ConnectX-6, or ConnectX-7 adapter

Configure ConnectX-5, ConnectX-6, or ConnectX-7 Virtual Protocol Interconnect (VPI) adapter to work as InfiniBand or Ethernet ports.

To configure the VPI card ports as suitable for your environment. You can choose one of the following options for each adapter (2 ports):

- Both InfiniBand ports
- InfiniBand port and Ethernet port
- Both Ethernet ports

Complete the following steps to check and enable adapter port configurations on a VPI adapter by using the **essgennetworks** command:

1. Check the port type of all attached devices.

```
essgennetworks -N localhost --query --devices all
```

The system output is as follows:

```
2020-09-23T04:24:03.397420 [INFO] Starting network generation
2020-09-23T04:24:03.579361 [INFO] nodelist: localhost
[ERROR] Mellanox Software Tools services are not running.
Make sure Mellanox Software Tools running configuring VPI adapters.
Make sure you must start Mellanox Software Tools using "/bin/mst start"
command before starting the configuration of the VPI adapters.
```

2. Start Mellanox Software Tools (MST).

```
/bin/mst start
```

The system output is as follows:

```
Starting MST (Mellanox Software Tools) driver set
Loading MST PCI module - Success
[warn] mst_pciconf is already loaded, skipping
Create devices
Unloading MST PCI module (unused) - Success
```

3. Check the port type of all attached devices again.

```
essgennetworks -N localhost --query --devices all
```

The system output is as follows:

```
2020-09-23T04:24:18.083995 [INFO] Starting network generation
2020-09-23T04:24:18.268935 [INFO] nodelist: localhost
[INFO] Device /dev/mst/mt4121_pciconf1 link type currently configured at system.
[INFO] Port 1 is set to InfiniBand
[INFO] Port 2 is set to InfiniBand
[INFO] Device /dev/mst/mt4121_pciconf0 link type currently configured at system.
[INFO] Port 1 is set to InfiniBand
[INFO] Port 2 is set to InfiniBand
```

4. Convert the *Port 1* of the device that is listed in the preceding command to Ethernet from InfiniBand.

```
essgennetworks -N localhost --change Ethernet --devices /dev/mst/mt4121_pciconf1 --port P1
```

The system output is as follows:

```
2020-09-23T03:45:52.322096 [INFO] Starting network generation
2020-09-23T03:45:52.510535 [INFO] nodelist: localhost
[INFO] Changing /dev/mst/mt4121_pciconf1 Port P1 link type to Ethernet
[INFO] Successfully changes the port type to Ethernet for Port 1
```

5. Reboot the node and check the port type of all attached devices again.

```
essgennetworks -N localhost --query --devices all
```

The system output is as follows:

```
2020-09-23T04:05:55.774019 [INFO] Starting network generation
2020-09-23T04:05:55.960088 [INFO] nodelist: localhost
[INFO] Device /dev/mst/mt4121_pciconf1 link type currently configured on the system.
[INFO] Port 1 is set to Ethernet
[INFO] Port 2 is set to InfiniBand
[INFO] Device /dev/mst/mt4121_pciconf0 link type currently configured on the system.
[INFO] Port 1 is set to InfiniBand
[INFO] Port 2 is set to InfiniBand
```

6. Verify that the port type of the *Port 1* is changed to Ethernet.

```
mlxconfig -d /dev/mst/mt4121_pciconf1 query | grep -i link_type
```

The system output is as follows:

LINK_TYPE_P1	ETH(2)
LINK_TYPE_P2	IB(1)

7. If any port is changed, the node must be rebooted to apply the changes.

Configuring Redfish service

Redfish manages IBM Storage Scale System 6000 and IBM Storage Scale System Utility Node operations.

To configure Redfish service, complete the following steps:

1. Configure a Redfish user.

By default, the Redfish user is **Administrator** and the password is **superuser**, which is disabled. Therefore, you need to set a new password.

Tip: You can set the system serial number as a new password, which is easy to remember.

- a. To connect the BMC/FSP network and make a `curl` patch call, run the following command from the management server:

```
curl -u Administrator:superuser -k -s -X PATCH https://<BMC IP>/redfish/v1/AccountService/Accounts/1 -d '{"Password":"<New Password>"}' -H "content-type: application/json" --header 'If-Match: *'
```

- b. Check whether the Redfish user is active by running the following command from the management server node:

```
curl -u Administrator:<New Password> -k -s https://<BMC IP>/redfish/v1/AccountService/Accounts/1 | python3 -mjson.tool
```

A sample output is as follows:

```
"Description": "Default Account",
"Enabled": true,
"Id": "1",
"Links": {
  "Role": {
    "@odata.id": "/redfish/v1/AccountService/Roles/Administrator"
  }
},
"Locked": false,
"Name": "Default Account",
"Password": null,
"PasswordChangeRequired": false,
"RoleId": "Administrator",
"UserName": "Administrator"
```

2. Use the Redfish crawler.

A sample script is created to retrieve all endpoints that Redfish supports. Run the sample script from a node that can connect to the BMC/FSP network such as the management server node. Complete the following steps to run the script:

a. Run the following command from the management server node:

```
/opt/ibm/ess/tools/samples/redfish_crawler -u Administrator -p <New Password> <BMC IP> |& tee /tmp/redfish_endpoints.txt
```

This command takes a few minutes to complete. Meanwhile, you can redirect to a file tee in this command so it can be closely inspected.

b. Check all discovered endpoints.

```
grep Discovered /tmp/redfish_endpoints.txt
```

This command displays many lines that are related to the system information, such as power status, voltages, temperatures, fan speed.

c. Gather all the BIOS settings.

```
curl -u Administrator:<New Password> -k -s https://<BMC IP>/redfish/v1/Systems/Self/Bios | python3 -mjson.tool
```

This command displays all current BIOS settings in the output. Based on the output, in the upcoming releases possible changes can be done if anything is not recommended.

Prerequisites

These prerequisites are applicable for Power9, IBM Storage Scale System Utility Node, and bring-your-own management server (BYOMS).

Manufacturing and SSR tasks

The following tasks must be complete before the installation:

1. Verify that all hardware is cleaned and all IP addresses are set and reachable over the appropriate networks by using the Code 20 operation.
2. Ensure that the `/etc/hosts` file is blank.
3. Place the correct IBM Storage Scale System **.tgz** installation file (appropriate edition) in the `/home/deploy` directory.
If an upgrade is needed, download the latest version from IBM Fix Central and replace the existing file.
4. Clear any existing network bridges.

- Remove all images and containers from the system.
- Clean and regenerate all SSH keys on the nodes.
- Confirm that all system components are updated to the latest supported versions before the system was shipped from manufacturing.

Components	3000	3200	3500	5000	6000	Utility Node	BYOMS
IBM Storage Scale	6.0.1.0 efix4	6.0.1.0 efix4	6.0.1.0 efix4	6.0.1.0 efix4	6.0.1.0 efix4	6.0.1.0 efix4	6.0.1.0 efix4
Operating system	RHEL 8.10	RHEL 8.10	RHEL 9.6	RHEL 8.10	RHEL 9.6	RHEL 9.6	RHEL 8.10
Kernel	4.18.0-553.124.1.el8_10	4.18.0-553.124.1.el8_10	5.14.0-570.113.1.el9_6	4.18.0-553.124.1.el8_10	5.14.0-570.113.1.el9_6	5.14.0-570.113.1.el9_6	4.18.0-553.124.1.el8_10
System d	239-82.el8_10.16	239-82.el8_10.16	252-51.el9_6.4	239-82.el8_10.16	252-51.el9_6.4	252-51.el9_6.4	239-82.el8_10.16
Network Manager	1.40.16-20.el8_10	1.40.16-20.el8_10	1.52.0-10.el9_6	1.40.16-20.el8_10	1.52.0-10.el9_6	1.52.0-10.el9_6	1.40.16-20.el8_10
Glibc	2.28-251.el8_10.34	2.28-251.el8_10.34	2.34-168.el9_6.24	2.28-251.el8_10.34	2.34-168.el9_6.24	2.34-168.el9_6.24	2.28-251.el8_10.34
OFED	25.10-2.4.1	25.10-2.4.1	25.10-2.4.1	25.10-2.4.1	25.10-2.4.1	25.10-2.4.1	25.10-2.4.1

Verifying code version

Ensure that you install correct code versions for the IBM Storage Scale System deployment.

- Download the unified container package, which supports both Data Management Edition and Data Access Edition from [IBM Fix Central](#).

```
Scale_System_UNIFIED-7.0.1.0-EMS.tgz
```

- Ensure deployment nodes are on a supported architecture:
 - x86_64: For IBM Storage Scale Utility Node or bring-your-own management server (BYOMS)
 - ppc64le: For POWER-based management servers. For details about functions supported by respective containers, see [Support Matrix](#).

Client nodes

- Ensure that client nodes are running MOFED 4.9.x or higher, and these nodes are converted to verbsRDMA core libraries.
- IBM Storage Scale System cluster node configuration is optimized to run IBM Storage Scale RAID functions. Protocols, other gateways, or any other non-IBM Storage Scale System services must not

be run on IBM Storage Scale System management server nodes or I/O server nodes. In a cluster with high I/O load, do not use IBM Storage Scale System nodes as a cluster manager or file system manager. For optimal performance the NSD client nodes that access IBM Storage Scale System nodes must be properly configured. IBM Storage Scale System ships with `gssClientConfig.sh` script located in `/usr/lpp/mmfs/samples/gss/` directory. This script can be used to configure the client as follows:

```
/usr/lpp/mmfs/samples/gss/gssClientConfig.sh <Comma Separated list of  
client nodes or nodeclass>
```

You can run the following to see configuration parameter settings without setting them:

```
/usr/lpp/mmfs/samples/gss/gssClientConfig.sh -D
```

After running this script, restart GPFS on the affected nodes for the optimized configuration settings to take effect.

Important: Do not run **gssClientConfig.sh** unless you fully understand the impact of each setting on the customer environment. Use the `-D` option to decide if all or some of the settings might be applied. Then, individually update each client node settings as required.

Best practices

General deployment and upgrade guidelines

- Use only the container image provided with IBM Storage Scale System.
Red Hat package installation outside the deployment or upgrade flow is not supported.
- Do not log in or run containers over the high-speed network.
Use a campus connection, dedicated management network, or virtual console instead.
- Run the container only from a direct SSH connection or virtual console.
Avoid starting it from an I/O node or any node that might reboot during the process.
- Stop all IBM Storage Scale tracing (**mmtrace**, **mmtracectl**) before starting the container or deploying any node.
The container blocks if active tracing is detected.
- Suspend all IBM Storage Scale and I/O operations before performing an upgrade. Wait for completion of the following commands:
 - **mmadddisk**, **mmdeldisk**, **mmrestripefile**, **mmrestripefs**, **mmrpldisk**
 - **mmapplypolicy**, **mmcheckquota**, **mmfsck**, **mmllsnapshot**, **mmrestorefs**
 - Snapshot operations: **mmcrsnapshot**, **mmdeletesnapshot**
- Place protocol nodes into file system maintenance mode before upgrades to prevent service disruption (recommended, not mandatory).
- Do not update the management server over the high-speed network.
Perform updates only via the management network, campus connection, or virtual console.
- Do not install any additional packages or services that the deployment container repository does not contain. If you want to include any package or service, open an IDEA/RFE for further action from IBM.
- The declustered arrays must contain minimum 20 drives for a 16+2p RAID configuration (18 code-width drives + 2 spare drives) and 21 drives for a 16+3p RAID configuration (19 code-width drives + 2 spare drives).

Configuration and compatibility

- Run the **essrun config load** command on all storage nodes, including management and protocol nodes before:
 - Enabling Admin Mode Central

- Deploying protocol nodes by using the installation toolkit
- Ensure all node hostnames and IPs are correctly defined, and that:
 - High-speed network names include a suffix of the management network name.
 - Management: *IOnode1.test.net*
 - High-speed: *IOnode1-hs.test.net*
- Do not install additional OS packages manually. If additional packages are required, contact IBM to request inclusion in future container releases.

System-level

- Enable TRIM support for IBM Storage Scale System 3000 where appropriate. By default, only 80% of space is allocated. Consult IBM Development if using beyond 80% to assess performance trade-offs.
- Configure a campus or secondary management connection before deploying the container. This enhances remote access and system manageability.
- Run **gnrhealthcheck** after adding an I/O node and before you create virtual disk sets to detect issues as follows:

Duplicate `hostid` values, which can cause environment instability.

Security and credentials

- Review cipher usage in your environment, especially for the campus connection.
 - The use of encryption ciphers versus **AUTHONLY** is under review.
 - Understand the security implications and consult IBM if needed.

Power9 management server

Power9 management server

- Only a single instance of all management services is supported (that is, a single management server per deployment).
- All nodes in the storage cluster must run the same IBM Storage Scale System release and IBM Storage Scale version to ensure compatibility and stability.
- It is recommended to upgrade to the latest available version before adding a new building block to the cluster.

Enabling remote management

To enable remote support in data centers with restrictive access, the following points must be considered:

- Connect HMC2 ports on all Power servers (IBM Storage Scale System 5000 or Power9 management servers) to the campus network and assign a campus IP address. This connection enables remote recovery and debugging during outages.
- Connect C11-T3 ports on Power9 nodes to the campus network to support remote access.
- Enable campus connectivity on IBM Storage Scale System 3200 by consulting IBM support about by using a USB-to-Ethernet dongle.
- Connect a free network port on each IBM Storage Scale System 3000 canister to the campus network. Also, install SMART PDUs in each 3000 frame to support remote power cycling.
- Connect Utility Nodes (x86-based) to the campus network. Although this is an optional task, this is recommended to improve remote support and management capabilities.

Preparing a stretch cluster

Consider A and B sites of the same cluster.

1. Ensure all management networks (FSP, BMC, MGMT) and the high-speed network (InfiniBand or Ethernet) are flat and operate within a single namespace.
2. Ensure each node has a unique *hostid*. If not, assign a unique IP address and hostname *hostid* to each node for both the management and high-speed networks.
 - Check *hostid* values manually.

```
mmdsh -N <NodeClass or NodeList> hostid
```

- If duplicates are found, generate a new *hostid*:

```
genhostid
```

3. Verify that all nodes are reachable from one another over all networks.
4. Define a common `/etc/hosts` file or configure DNS appropriately, and distribute it to all nodes by using IBM Storage Scale System deployment tools.
5. Run **gnrhealthcheck** before you create recovery groups in stretch clusters.

IBM Storage Scale System Utility Node (MTM 5149-23E)

The IBM Storage Scale System Utility Node (MTM 5149-23E), also called as Utility Node, is an x86 management server, which is fully sold and supported by IBM.

Support statements and matrix

Rack shipments are not supported. See the racking instructions included in the hardware guide for how to place the 5149-23E in a rack. This restriction may be lifted in the future.

Note: IBM Storage Scale System 3500 can be shipped in a rack.

This matrix shows what each management server can and cannot deploy or upgrade.

Node type	P9 management server	BYOMS host	BYOMS guest	Utility node host	Utility node guest	5000 I/O	3000	3200	3500/VM	P9 Protocol
P9 management server	Y	N	N	N	N	Y	Y	Y	Y	Y
BYOMS (guest)	N	N	Y	N	N	N	N	N	Y	N
Utility Node (guest)	Y	N	Y	*	Y	Y	Y	Y	Y	Y

* Deploy/upgrade of itself is possible under certain conditions.

This matrix shows what each management server can and cannot be mixed with the same cluster.

Management server	Power9 management server	BYOMS guest	Utility management server	5000 I/O	3000	3200	3500/VM	P9 Protocol
Power9 management server	Y	N	N	Y	Y	Y	Y	Y

Management server	Power9 management server	BYOMS guest	Utility management server	5000 I/O	3000	3200	3500/VM	P9 Protocol
BYOMS (guest)	N	N	N	N	N	N	Y	N
Utility node (guest)	Y	N	Y	Y	Y	Y	Y	Y
IBM Storage Scale System 5000 I/O	Y	N	N	Y	Y	Y	Y	Y
IBM Storage Scale System 3000	Y	N	N	Y	Y	Y	Y	Y
IBM Storage Scale System 3200	Y	N	N	Y	Y	Y	Y	Y
IBM Storage Scale System 3500/VM	Y	Y	Y	Y	Y	Y	Y	Y
P9 Protocol	Y	N	N	Y	Y	Y	I/O only	Y

Deployment requisites

- IBM Fix Central items

- QCOW (VM) image

```
EMSVM-23E.qcow2.xz
```

- 7.0.1.x x86 and ppc64le container (DAE or DME)

```
Scale_System_UNIFIED-6.2.3.0-EMS.tgz
```

- GitHub

- The EMSVM script (minimum version is 1.3). For more information, see [EMSVM script repository](#).

An example output of a host check by using the EMSVM script

```
./emsvm --check-host
2023-08-17 00:49:33,330 INFO: Welcome to emsvm version 1.3
2023-08-17 00:49:33,330 INFO: Please visit https://github.com/IBM/EMSVM for known issues
and ensure you are running latest version
2023-08-17 00:49:33,330 INFO: Log file with details for this run is saved on: /var/log/
emsvm/emsvm_check_host_2023_08_17_00_49_33.log
2023-08-17 00:49:33,340 INFO: This system is AMD EPYC based
```

```

2023-08-17 00:49:33,340 INFO: This host does not have AMD EPYC 1st Generation (Naples)
2023-08-17 00:49:33,401 INFO: This host CPU[s] are KVM capable and has extensions enabled
2023-08-17 00:49:33,401 INFO: Going to check if /dev/kvm exists
2023-08-17 00:49:33,401 INFO: The device /dev/kvm exists
2023-08-17 00:49:34,120 INFO: Red Hat Enterprise Linux 9.1 is a supported OS
2023-08-17 00:49:34,125 INFO: Checking RPM packages status
2023-08-17 00:49:34,247 INFO: All required RPM packages have the expected installation
status
2023-08-17 00:49:34,291 WARNING: Could not get current version from repository, please be
sure that you are running latest version from https://github.com/IBM/EMSVM
2023-08-17 00:49:34,291 INFO: Please manually check that you are running the latest
version. Any issue needs to be reproduced with the latest version
2023-08-17 00:49:39,296 INFO: Looking that we have at least one NIC for High Speed Network
(HSN)
2023-08-17 00:49:39,315 INFO: The PCI address 0000:01:00.0 has a negotiated link speed of
16 GT/s and 16 lines
2023-08-17 00:49:39,315 INFO: The NIC 'Infiniband controller: Mellanox Technologies MT28908
Family [ConnectX-6]' is OK as HSN NIC. Going to append PCI address '0000:01:00.0' as HSN NIC
2023-08-17 00:49:39,328 INFO: The PCI address 0000:01:00.1 has a negotiated link speed of
16 GT/s and 16 lines
2023-08-17 00:49:39,328 INFO: The NIC 'Infiniband controller: Mellanox Technologies MT28908
Family [ConnectX-6]' is OK as HSN NIC. Going to append PCI address '0000:01:00.1' as HSN NIC
2023-08-17 00:49:39,333 INFO: Found at least two HSN ports
2023-08-17 00:49:39,334 INFO: Looking that we have at least one quad port Ethernet NIC for
Low Speed Network (LSN)
2023-08-17 00:49:39,337 INFO: The NIC 'Ethernet controller: Intel Corporation Ethernet
Controller X710 for 10GBASE-T (rev 02)' is OK as LSN NIC. Going to append PCI address
'0000:25:00.' for ports 0, 1, 2 and 3
2023-08-17 00:49:39,337 INFO: Host has 32 core[s] which complies with 32 cores required
2023-08-17 00:49:39,337 INFO: Host memory of 125.24 GB is less than 247 GB required
2023-08-17 00:49:39,337 INFO: Host memory of 125.24 GB is equal or more than 247 GB
required for small setups of two or less EMS units
2023-08-17 00:49:39,664 INFO: Current tune profile is 'virtual-host'
2023-08-17 00:49:39,745 INFO: OS settings match the running profile
2023-08-17 00:49:39,812 INFO: KVM storage pool EMSVM already exists
2023-08-17 00:49:39,816 INFO: The path /emsvm has 683.15 GB free excluding reserved space
which is more than the required 575 GB
2023-08-17 00:49:39,819 INFO: KVM executable file /usr/libexec/qemu-kvm exists
2023-08-17 00:49:39,819 INFO:
2023-08-17 00:49:39,819 INFO: -----
2023-08-17 00:49:39,819 INFO: | Summary of this run |
2023-08-17 00:49:39,819 INFO: -----
2023-08-17 00:49:39,819 INFO:
2023-08-17 00:49:39,819 INFO: Log file with details for this run is saved on: /var/log/
emsvm/emsvm_check_host_2023_08_17_00_49_33.log
2023-08-17 00:49:39,819 INFO: Start of the checks on: 2023/08/17 00:49:33
2023-08-17 00:49:39,819 INFO: End of the checks on: 2023/08/17 00:49:39
2023-08-17 00:49:39,819 INFO: All tests were passed passed on 'utility1' host
2023-08-17 00:49:39,819 INFO: This EMSVM setup can only manage two or less ESS units. If
more units are to be managed this host is NOT supported
2023-08-17 00:49:39,819 INFO: The path /emsvm has 683.15 GB free excluding reserved space
which is more than the required 575 GB
2023-08-17 00:49:39,819 INFO: CPU model is AMD EPYC 7313 16-Core Processor
2023-08-17 00:49:39,819 INFO: This host has 2 socket(s)
2023-08-17 00:49:39,819 INFO: The VM CPU cores would be 30 core[s]
2023-08-17 00:49:39,819 INFO: The VM memory would be 114 GB
2023-08-17 00:49:39,819 INFO: The VM HSN PCI address[es] would be ['0000:01:00.0',
'0000:01:00.1']
2023-08-17 00:49:39,819 INFO: The VM LSN PCI addresseses would be ['0000:25:00.0',
'0000:25:00.1', '0000:25:00.2', '0000:25:00.3']
2023-08-17 00:49:39,819 INFO: OS is running and matching tune profile virtual-host

WARNING: All tests passed, you can run EMSVM on utility1 host, for small setups of two or less
ESS units only

```

Deployment information

- For process such as BYOMS (QCOW, EMSVM script, connecting to console, and so on), see [“Preparing bring-your-own management server”](#) on page 74.
- When the container is up and IPs are set. Copy the container to the VM, update the /etc/hosts file and follow the normal deployment flow (that is, Power9 management server). For more information, see the *Quick Deployment Guide* in the [IBM Storage Scale System documentation](#).

Bring-your-own management server

From IBM Storage Scale System 6.1.5.1 and higher, you can manage IBM Storage Scale System 3500 by using IBM Power9 hardware or you can bring your own hardware, and run a virtual machine (VM) on this hardware. This hardware is the management server for your cluster, which is BYOMS, and the VM is the management server virtual machine (MSVM).

Before you choose a BYOMS or an IBM Power9, you must know the difference between them. The following activities are supported on the management server that is validated, supported, and monitored by IBM. However, IBM does not support any of these activities on BYOMS. Customers must support the following activities on the BYOMS:

- Hardware monitoring on the host that runs the MSVM of IBM Storage Scale System.
- Call home related events on the hardware
- Firmware updates from IBM on the hardware that hosts the MSVM
- Operating system (OS) updates on the base OS that hosts the MSVM

Note: The OS of the MSVM is updated although it is a part of the BYOMS. IBM Storage Scale System, which has I/O nodes and IBM protocol nodes, is monitored and upgraded by the standard update flow that is run from the MSVM.

Requirements

These requirements are valid for IBM Storage Scale System 6.1.5.1 and higher. Check the latest documentation available in IBM Docs for the up-to-date requirements.

Host requirements

A non-IBM host must meet hardware requirements to run the MSVM. These requirements are applicable for small and standard setups. Small setups can be used to manage up to two IBM Storage Scale Systems. Standard setups can be used to manage the same number of IBM Storage Scale Systems as provided by IBM Power9 management server.

Note: These requirements might be a combination of BIOS, hardware wiring of the system board, and network cards, or specific to a vendor. Customer must make sure that these requirements are met. For any instructions on how to meet these requirements or combine them on a BYOMS hardware, the customer must contact the BYOMS vendor. IBM does not support these requirements on a BYOMS hardware because the BYOMS hardware is not an IBM hardware.

Customers must make sure that the BYOMS hardware fulfills the following requirements:

- AMD EPYC processor, single or dual socket (AMD EPYC Naples are not supported.)
- VT-x enabled
- KVM installed and enabled on OS
- A PCI pass-through capable system
- Minimum 600 GB free space on the /emsvm directory
- The /emsvm directory must be mounted on a file system of a local device (not on iSCSI, FC, NFS, or IBM Storage Scale)
- One quad-port Ethernet card for low-speed network (LSN), with all ports identical
- The LSN card must belong to its own IOMMU group, preferably each port on its own IOMMU group
- Red Hat Enterprise Linux 8.6 and 8.7 that are running in the host with a valid subscription
- SELinux must be disabled on the host.
- The host must be fully compliant with tuned virtual-host profile
- No other VMs must be running in the host
- No other workloads must be running on the host
- IBM Storage Scale software must not be installed on the host

- Minimum one ConnectX-6 VPI card for high-speed network (HSN)
- Each HSN card must belong to its own IOMMU group, for multiport preferable each port on its own IOMMU group
- No more than three HSN NIC
- Minimum two HSN ports
- No more than six HSN ports
- Setup requirements

Setup	Requirement
Small	Host with minimum 16 cores and 128 GB RAM
	Minimum PCIe 3.0 x16 lines for HSN slots
Standard	Host with minimum of 32 cores and 256 GB RAM
	Minimum PCIe 4.0 x16 lines for HSN slots

IBM created a tool that is called IBM EMSVM script to manage the hosts checks and the lifecycle of the MSVM. The tool is available on public GitHub. Customers can download and certify their hosts before they purchase anything from IBM. To install the tool, go to <https://github.com/IBM/EMSVM>.

Important: This tool helps to check requirements and compliance. However, it is customer responsibility to ensure that the setup is compliant with the requirements.

Recommended requirements on the host

Although a supported setup is not mandatory, the following configurations are recommended, and some requirements might become mandatory:

- A mirrored host OS drive
- Out-of-band management of the host
- Some Ethernet ports that are not part of the quad-port card or HSN cards. These cards are mapped to the MSVM for the remote access to the operating system of the host.
- Additional power supply

Network requirements

The MSVM has two types of networks. The low-speed network with a quad-port Ethernet card, where all four ports are identical. From one through three cards for the high-speed network (HSN) that provide between two to six ports.

All CX6 cards and the first valid four port NIC will be fully passed into the virtual machine. At this point, they are unavailable (or not seen) by the host.

Low-speed network

The LSN in the MSVM is used because it has for identical ports. The quad-port card is based on Power9 management server that connects to the IBM Storage Scale System management network, the IBM Storage Scale System FSP network, and the campus (the customer network). The lower port must not be cabled because it is reserved for IBM Power9 management server.

The following figure shows the ports of the quad-port card with their logical use:

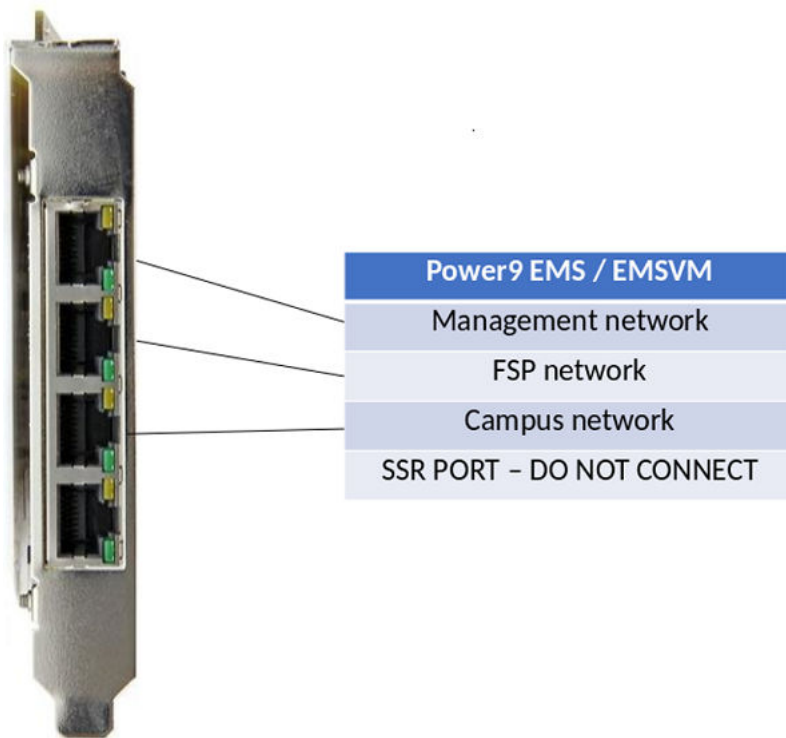


Figure 18. Ports for low-speed networks

For details about each network of IBM Storage Scale System, see the networking requirements section in the IBM Storage Scale System software deployment preparation chapter in the IBM Storage Scale System documentation that corresponds to your system.

High-speed network

The HSN used on the MSVM must be identical to the HSN on a Power9 management server node to connect to the IBM Storage Scale daemon and admin networks. For details, see the networking requirements section in the IBM Storage Scale System software deployment preparation chapter in the IBM Storage Scale System documentation that corresponds to your system.

Other requirements

To run the EMSVM script, which is an IBM management tool that is available in public GitHub, make sure that the following requirements are met for the host:

- Python 3.6 or higher version is installed.
- The following RPM packages are installed:
 - dmidecode
 - pciutils
 - numactl-libs
 - tuned
- The following coreutils are installed:
 - virt-manager
 - libvirt-daemon
 - libvirt-client
 - libvirt
 - libguestfs-tools
 - virt-install

- qemu-img
- python3-libvirt
- python3-requests
- python3-netifaces
- python3-ethtool
- xz
- xz-libs
- FIPS must be disabled on the host (needed for MD5 sums).

Managing more than one quad-port cards on the host

If the host has more than one quad-port Ethernet cards with identical ports, only the lowest PCI address is mapped to the MSVM. Other cards are not mapped to the MSVM. For example:

```
grep PCI_SLOT_NAME /sys/class/net/*/device/uevent
```

The Ethernet PCI address output is as follows:

```
/sys/class/net/eth0/device/uevent:PCI_SLOT_NAME=0000:45:00.0
/sys/class/net/eth1/device/uevent:PCI_SLOT_NAME=0000:45:00.1
/sys/class/net/eth2/device/uevent:PCI_SLOT_NAME=0000:45:00.2
/sys/class/net/eth3/device/uevent:PCI_SLOT_NAME=0000:45:00.3
/sys/class/net/eth4/device/uevent:PCI_SLOT_NAME=0000:c3:00.0
/sys/class/net/eth5/device/uevent:PCI_SLOT_NAME=0000:c3:00.1
/sys/class/net/eth6/device/uevent:PCI_SLOT_NAME=0000:c3:00.2
/sys/class/net/eth7/device/uevent:PCI_SLOT_NAME=0000:c3:00.3
/sys/class/net/ib0/device/uevent:PCI_SLOT_NAME=0000:27:00.0
/sys/class/net/ib1/device/uevent:PCI_SLOT_NAME=0000:27:00.1
```

The card on the 000:45:00 PCI address, at the lowest in hexadecimal, is mapped to the MSVM.

Note: Manage cards, only if the host has more than one quad-port Ethernet cards with identical ports. If you are using any other card for the host access, you need not to manage any cards.

EMSVM script deployment

For a successful deployment of the management server as a virtual machine on IBM Storage Scale System, validate the system on which you are planning to host the MSVM. The system must meet all requirements. For more information, see “Requirements” on page 46.

In addition to hardware and software requirements, the following additional information is required to complete the initial MSVM configuration:

- Access to the EMSVM script repository on public GitHub to download this IBM tool (<https://github.com/IBM/EMSVM>).
- An IBM account to access IBM Fix Central to download the EMSVM script disk image.
- IP address information for the initial MSVM configuration.

Network	IP address	Netmask	Gateway
Campus			
IBM Storage Scale System management			
OS FSP			

Note: It is also recommended to obtain an additional IP address. By using this IP address, the OS of the system that hosts the MSVM can be configured. This IP address cannot be configured on the 4-port Ethernet adapter that gets assigned to the MSVM.

Getting the IBM EMSVM script

The MSVM deployment is automated and managed by using the IBM management tool. This tool is used for the initial deployment and the management of the MSVM. The current version of the tool can be downloaded from the [GitHub repository](#).

This tool must be copied to the local file system of the host that is running the MSVM.

The IBM EMSVM script provides the following options:

-h

This tool provides several options. To get the options, the tool supports execution of the tool without additional options or with the -h option.

For example:

```
./emsvm -h
```

The IBM EMSVM script provides the following options:

```
usage: EMS VM [-h] (--check-host | --connect-EMS | --start-EMS | --check-EMS)
              [-d] [-v]

EMS VM host check and VM run

optional arguments:
  -h, --help            show this help message and exit
  --check-host          Assess the readiness of the host to run EMS VM
  --connect-EMS         Connect to the EMS VM console information
  --start-EMS           Start EMS VM in this host
  --check-EMS           Check if EMS VM is UP
  -d, --debug           Print verbose messages also to shell
  -v, --version         show program's version number and exit
```

--check-host

Performs checks on the host system to verify that all prerequisites to run the EMSVM script are validated. Check the output for any errors or warnings that are preventing the EMSVM script from starting.

This option can be used to validate the host system on which you are planning to run the EMSVM script. As this tool is hosted on public GitHub, these requirements can be checked before you decide whether you want to use the MSVM or the IBM Power9 as management server.

Note: The EMSVM script disk image from IBM Fix Central is not required to use this tool for validating the host.

--connect-EMS

Connects the current user session to the console of running MSVM on the host.

This option must be used on first start-up of a new MSVM to access the virtual machine and perform that **IBM SSR setup** task. The MSVM does not have any default IP addresses configured.

The **IBM SSR setup** task configures the following information on the MSVM according to the user-provided information.

- Password for user **root**
- SSH keys and passwordless SSH
- IP addresses for:
 - Campus network
 - IBM Storage Scale System management network
 - OS FSP network

Note: IBM SSR setup tool must be used for IP address configuration of the MSVM. Do not configure IP addresses manually by using OS provided tools.

--start-EMS

Starts the MSVM. The MSVM must be manually started first time when the virtual machine is being deployed, and if the virtual machine is manually shut down.

During the first start-up, the MSVM is set to auto start with the host. Do not run this step every time when the host boots.

--check-EMS

Validates the status of the MSVM. The IBM EMSVM script reports the MSVM domain status on the host.

This option returns the status of the MSVM domain in a readable format. In addition to the readable output, the command returns “0”, if the MSVM domain is found **active** and “1” for any other state.

This information can be used to validate the power state of the MSVM with an external monitoring tool that is running on the host.

For example:

```
./emsvm --check-EMS
```

A sample output is as follows:

```
2022-11-16 08:39:37,269 INFO: Welcome to EMS VM version 0.53
2022-11-16 08:39:37,269 INFO: Please visit https://github.com/IBM/EMSVM for issues and
new versions
2022-11-16 08:39:37,269 INFO: Log file with details for this run is saved on: /var/log/
emsvm/emsvm_check EMS_2022_11_16_08_39_37.log
2022-11-16 08:39:37,279 INFO: This system is AMD EPYC based
2022-11-16 08:39:37,279 INFO: This host does not have AMD EPYC 1st Generation (Naples)
2022-11-16 08:39:37,388 INFO: This host CPU[s] are KVM capable and has extensions enabled
2022-11-16 08:39:37,388 INFO: Going to check if /dev/kvm exists
2022-11-16 08:39:37,389 INFO: The device /dev/kvm exists
2022-11-16 08:39:38,417 INFO: Red Hat Enterprise Linux 8.6 is a supported OS
2022-11-16 08:39:38,428 INFO: Checking RPM packages status
2022-11-16 08:39:38,581 INFO: All required RPM packages have the expected installation
status
2022-11-16 08:39:38,665 INFO: Could not get current version from repository, please be
sure that you are running latest version from https://github.com/IBM/EMSVM
2022-11-16 08:39:43,712 INFO: EMSVM KVM domain exists
2022-11-16 08:39:43,712 INFO: KVM VM domain EMSVM is already active
```

Validating the host

The host system that runs the virtualization layer of the MSVM must fulfill hardware and software requirements. The IBM EMSVM script validates the host when the tool performs operations, such as deployment or starting the MSVM.

When you run the IBM EMSVM script, it writes a log file into the /var/log/emsvm host directory when the tool is run. The logs can be used for auditing the activity that is related to the MSVM and for debugging issues during deployments and start-ups of the MSVM.

Validate the host before you deploy the MSVM. Check the summary results of host validation at the end of the output.

For example:

```
./emsvm --check-host
```

A sample output of passed hosts validation is as follows:

```
2022-11-14 06:06:51,400 INFO: Welcome to EMS VM version 0.53
2022-11-14 06:06:51,400 INFO: Please visit https://github.com/IBM/EMSVM for issues and new
versions
2022-11-14 06:06:51,400 INFO: Log file with details for this run is saved on: /var/log/emsvm/
emsvm_check_host_2022_11_14_06_06_51.log
2022-11-14 06:06:51,409 INFO: This system is AMD EPYC based
2022-11-14 06:06:51,409 INFO: This host does not have AMD EPYC 1st Generation (Naples)
2022-11-14 06:06:51,516 INFO: This host CPU[s] are KVM capable and has extensions enabled
2022-11-14 06:06:51,516 INFO: Going to check if /dev/kvm exists
2022-11-14 06:06:51,516 INFO: The device /dev/kvm exists
2022-11-14 06:06:52,534 INFO: Red Hat Enterprise Linux 8.6 is a supported OS
2022-11-14 06:06:52,544 INFO: Checking RPM packages status
```

```

2022-11-14 06:06:52,697 INFO: All required RPM packages have the expected installation status
2022-11-14 06:06:52,779 INFO: Could not get current version from repository, please be sure
that you are running latest version from https://github.com/IBM/EMSVM
2022-11-14 06:06:57,785 INFO: Looking that we have at least one NIC for High Speed Network
(HSN)
2022-11-14 06:06:57,803 INFO: The NIC 'Infiniband controller: Mellanox Technologies MT28800
Family [ConnectX-6 Ex]' is OK as HSN NIC. Going to append PCI address '0000:27:00.0' as HSN NIC
2022-11-14 06:06:57,803 INFO: The NIC 'Infiniband controller: Mellanox Technologies MT28800
Family [ConnectX-6 Ex]' is OK as HSN NIC. Going to append PCI address '0000:27:00.1' as HSN NIC
2022-11-14 06:06:57,823 INFO: Found at least two HSN ports
2022-11-14 06:06:57,823 INFO: Looking that we have at least one quad port Ethernet NIC for
Low Speed Network (LSN)
2022-11-14 06:06:57,832 INFO: The NIC 'Ethernet controller: Intel Corporation I350 Gigabit
Network Connection (rev 01)' is OK as LSN NIC. Going to append PCI address '0000:45:00.' for
ports 0, 1, 2 and 3
2022-11-14 06:06:57,832 INFO: Host has 32 core[s] which complies with 32 cores required
2022-11-14 06:06:57,832 INFO: Total memory is 251.38 GB, which is more than the required 247
GB
2022-11-14 06:07:00,181 INFO: Current tune profile is 'virtual-host'
2022-11-14 06:07:00,358 INFO: OS settings match the running profile
2022-11-14 06:07:00,399 INFO: KVM storage pool EMSVM already exists
2022-11-14 06:07:00,449 INFO: The path /emsvm has 1181.29 GB free excluding reserved space
which is more than the required 600 GB
2022-11-14 06:07:00,455 INFO: KVM executable file /usr/libexec/qemu-kvm exists
2022-11-14 06:07:00,455 INFO:
-----
| Summary of this run |
-----
2022-11-14 06:07:00,455 INFO:
2022-11-14 06:07:00,455 INFO: Log file with details for this run is saved on: /var/log/emsvm/
emsvm_check_host_2022_11_14_06_06_51.log
2022-11-14 06:07:00,456 INFO: Start of the checks on: 2022/11/14 06:06:51
2022-11-14 06:07:00,456 INFO: End of the checks on: 2022/11/14 06:07:00
2022-11-14 06:07:00,456 INFO: All tests were passed passed on x86node
2022-11-14 06:07:00,456 INFO: The path /emsvm has 1181.29 GB free excluding reserved space
which is more than the required 600 GB
2022-11-14 06:07:00,456 INFO: CPU model is AMD EPYC 7302 16-Core Processor
2022-11-14 06:07:00,456 INFO: This host has 2 socket(s)
2022-11-14 06:07:00,456 INFO: The VM CPU cores would be 30 core[s]
2022-11-14 06:07:00,456 INFO: The VM memory would be 239 GB
2022-11-14 06:07:00,456 INFO: The VM HSN PCI address[es] would be ['0000:27:00.0',
'0000:27:00.1']
2022-11-14 06:07:00,456 INFO: The VM LSN PCI addresses would be ['0000:45:00.0',
'0000:45:00.1', '0000:45:00.2', '0000:45:00.3']
2022-11-14 06:07:00,456 INFO: OS is running and matching tune profile virtual-host

```

OK: All tests passed, you can run EMS VM on x86node host

Note: If the checks are passed, for small setups validations a “WARNING” message is displayed instead of “OK”.

Getting an MSVM disk image from IBM Fix Central

An IBM account with a valid subscription is required to download the MSVM disk image from [IBM Fix Central](https://www.ibm.com/support/fixcentral) (<https://www.ibm.com/support/fixcentral>).

1. Download the MSVM disk image from [IBM Fix Central](https://www.ibm.com/support/fixcentral) (<https://www.ibm.com/support/fixcentral>).
2. Transfer the downloaded disk image file to the /emsvm file system of the host that is running the MSVM.
EMSVM is delivered as a compressed qcow2 disk image file. Do not decompress or rename the file.
3. Store the EMSVM.qcow2.xz file in the /emsvm directory without making any changes.

```

pwd
/emsvm
ls -la
drwxr-xr-x. 2 root root          4096 Nov 14 06:42 .
dr-xr-xr-x. 23 root root          4096 Oct 20 06:15 ..
-rw-r--r-- 1 root root 1094970356 Nov 14 06:42 EMSVM.qcow2.xz

```

Starting the MSVM

1. Start the MSVM by using the IBM MSVM.

Start this tool manually if you are starting a new MSVM.

2. During the first manual start, set the EMSVM to auto start with the host system.

An error during start the MSVM without a virtual disk file is displayed in the following example:

```
./emsvm --start-EMS
...
2022-11-14 06:41:06,611 WARNING:      EMSVM KVM domain does not exist. Going to create it
2022-11-14 06:41:06,638 INFO:       KVM VM has been created
2022-11-14 06:41:06,638 INFO:       KVM VM domain EMSVM is set to autostart
2022-11-14 06:41:06,638 ERROR:      EMS VM compressed disk file /emsvm/EMSVM.qcow2.xz does not exist. Download the file from IBM and place it on the expected directory
```

An error when the MSVM is started with an IP address that is configured on the quad-port adapter is displayed in the following example:

```
./emsvm --start-EMS
...
2022-11-14 06:45:00,240 INFO:      OS device name: 'eth2' PCI address: '0000:45:00.0'
Description: 'Ethernet controller: Intel Corporation I350 Gigabit Network Connection (rev 01)'
2022-11-14 06:45:00,240 INFO:      OS device name: 'eth5' PCI address: '0000:45:00.1'
Description: 'Ethernet controller: Intel Corporation I350 Gigabit Network Connection (rev 01)'
2022-11-14 06:45:00,240 INFO:      OS device name: 'eth6' PCI address: '0000:45:00.2'
Description: 'Ethernet controller: Intel Corporation I350 Gigabit Network Connection (rev 01)'
2022-11-14 06:45:00,240 INFO:      OS device name: 'eth7' PCI address: '0000:45:00.3'
Description: 'Ethernet controller: Intel Corporation I350 Gigabit Network Connection (rev 01)'
2022-11-14 06:45:00,243 INFO:      Interface eth2 has IP 192.168.20.231
2022-11-14 06:45:00,244 ERROR:      We found IP address[es] on one or more interfaces that are going to be passthrough to the VM
```

If no errors are displayed during the prerequisites check that occurs after you start the MSVM, confirmation to start the MSVM on this host is requested.

For example:

```
./emsvm --start-EMS
```

Successful start of MSVM output is as follows:

```
2022-11-14 06:48:28,934 INFO:      Welcome to EMS VM version 0.53
2022-11-14 06:48:28,934 INFO:      Please visit https://github.com/IBM/EMSVM for issues and new versions
2022-11-14 06:48:28,934 INFO:      Log file with details for this run is saved on: /var/log/emsvm/emsvm_start EMS_2022_11_14_06_48_28.log
2022-11-14 06:48:28,941 INFO:      This system is AMD EPYC based
2022-11-14 06:48:28,942 INFO:      This host does not have AMD EPYC 1st Generation (Naples)
2022-11-14 06:48:29,052 INFO:      This host CPU[s] are KVM capable and has extensions enabled
2022-11-14 06:48:29,052 INFO:      Going to check if /dev/kvm exists
2022-11-14 06:48:29,052 INFO:      The device /dev/kvm exists
2022-11-14 06:48:30,068 INFO:      Red Hat Enterprise Linux 8.6 is a supported OS
2022-11-14 06:48:30,078 INFO:      Checking RPM packages status
2022-11-14 06:48:30,232 INFO:      All required RPM packages have the expected installation status
2022-11-14 06:48:30,316 INFO:      Could not get current version from repository, please be sure that you are running latest version from https://github.com/IBM/EMSVM
2022-11-14 06:48:35,321 INFO:      Looking that we have at least one NIC for High Speed Network (HSN)
2022-11-14 06:48:35,338 INFO:      The NIC 'Infiniband controller: Mellanox Technologies MT28800 Family [ConnectX-6 Ex]' is OK as HSN NIC. Going to append PCI address '0000:27:00.0' as HSN NIC
2022-11-14 06:48:35,338 INFO:      The NIC 'Infiniband controller: Mellanox Technologies MT28800 Family [ConnectX-6 Ex]' is OK as HSN NIC. Going to append PCI address '0000:27:00.1' as HSN NIC
2022-11-14 06:48:35,357 INFO:      Found at least two HSN ports
2022-11-14 06:48:35,357 INFO:      Looking that we have at least one quad port Ethernet NIC for Low Speed Network (LSN)
2022-11-14 06:48:35,365 INFO:      The NIC 'Ethernet controller: Intel Corporation I350 Gigabit Network Connection (rev 01)' is OK as LSN NIC. Going to append PCI address '0000:45:00.' for ports 0, 1, 2 and 3
2022-11-14 06:48:35,366 INFO:      Host has 32 core[s] which complies with 32 cores required
2022-11-14 06:48:35,366 INFO:      Total memory is 251.38 GB, which is more than the required 247 GB
2022-11-14 06:48:38,097 INFO:      Current tune profile is 'virtual-host'
2022-11-14 06:48:38,273 INFO:      OS settings match the running profile
2022-11-14 06:48:38,314 INFO:      KVM storage pool EMSVM already exists
2022-11-14 06:48:38,362 INFO:      The path /emsvm has 1181.29 GB free excluding reserved space which is more than the required 600 GB
2022-11-14 06:48:38,367 INFO:      KVM executable file /usr/libexec/qemu-kvm exists
2022-11-14 06:48:38,367 INFO:      All tests passed, starting EMS VM
```

```

2022-11-14 06:48:38,368 WARNING: EMSVM KVM domain does not exist. Going to create it
2022-11-14 06:48:38,394 INFO: KVM VM has been created
2022-11-14 06:48:38,394 INFO: KVM VM domain EMSVM is set to autostart
2022-11-14 06:48:38,395 INFO:
2022-11-14 06:48:38,395 INFO: You have requested to start the EMS VM in this host
2022-11-14 06:48:38,395 INFO: The following PCI devices are going to be assigned to the VM
and are not going to be accessible from the host
2022-11-14 06:48:38,395 INFO: If you are using any of them to connect to the host, you are
going to lose connectivity to this host
2022-11-14 06:48:38,395 INFO: Please be sure that you are not using any of the following
devices before continuing the start up of the EMS VM
2022-11-14 06:48:38,432 INFO: PCI address: '0000:27:00.0' Description: 'Infiniband
controller: Mellanox Technologies MT28800 Family [ConnectX-6 Ex]'
2022-11-14 06:48:38,433 INFO: PCI address: '0000:27:00.1' Description: 'Infiniband
controller: Mellanox Technologies MT28800 Family [ConnectX-6 Ex]'
2022-11-14 06:48:38,433 INFO: OS device name: 'eth2' PCI address: '0000:45:00.0'
Description: 'Ethernet controller: Intel Corporation I350 Gigabit Network Connection (rev 01)'
2022-11-14 06:48:38,433 INFO: OS device name: 'eth5' PCI address: '0000:45:00.1'
Description: 'Ethernet controller: Intel Corporation I350 Gigabit Network Connection (rev 01)'
2022-11-14 06:48:38,433 INFO: OS device name: 'eth6' PCI address: '0000:45:00.2'
Description: 'Ethernet controller: Intel Corporation I350 Gigabit Network Connection (rev 01)'
2022-11-14 06:48:38,433 INFO: OS device name: 'eth7' PCI address: '0000:45:00.3'
Description: 'Ethernet controller: Intel Corporation I350 Gigabit Network Connection (rev 01)'
2022-11-14 06:48:38,435 INFO: No IP addresses found on the interfaces to be passthrough to
the VM
2022-11-14 06:48:38,435 INFO: Do you want to continue with the start EMS VM? (y/n):
y
2022-11-14 06:49:47,807 INFO: It can take several minutes to start the VM, please wait ...
2022-11-14 06:50:32,206 INFO: EMS VM is started now ...

```

Configuring IP addresses

The downloaded MSVM disk image does not contain pre-configured IP addresses for any network interface. When you start the MSVM first time, the IP address must be configured to allow inbound and outbound communication of the EMSVM.

1. To configure IP addresses for the MSVM, use the `emsvm --connect-EMS` option to connect to the MSVM console from the host system.

```
./emsvm --connect-EMS
```

A sample output is as follows:

```

2022-11-14 06:54:50,900 INFO: Welcome to management server VM version 0.53
2022-11-14 06:54:50,900 INFO: Please visit https://github.com/IBM/EMSVM for issues and
new versions
2022-11-14 06:54:50,900 INFO: Log file with details for this run is saved on: /var/log/
emsvm/emsvm_connect EMS_2022_11_14_06_54_50.log
2022-11-14 06:54:50,909 INFO: This system is AMD EPYC based
2022-11-14 06:54:50,910 INFO: This host does not have AMD EPYC 1st Generation (Naples)
2022-11-14 06:54:51,018 INFO: This host CPU[s] are KVM capable and has extensions enabled
2022-11-14 06:54:51,018 INFO: Going to check if /dev/kvm exists
2022-11-14 06:54:51,018 INFO: The device /dev/kvm exists
2022-11-14 06:54:52,038 INFO: Red Hat Enterprise Linux 8.6 is a supported OS
2022-11-14 06:54:52,048 INFO: Checking RPM packages status
2022-11-14 06:54:52,201 INFO: All required RPM packages have the expected installation
status
2022-11-14 06:54:52,282 INFO: Could not get current version from repository, please be
sure that you are running latest
2022-11-14 06:54:57,331 INFO: version from https://github.com/IBM/EMSVM
2022-11-14 06:54:57,331 INFO: EMSVM KVM domain exists
2022-11-14 06:54:57,331 INFO: KVM VM domain EMSVM is already active

Connected to domain 'EMSVM'
Escape character is ^] (Ctrl + ])

Red Hat Enterprise Linux 8.6 (Ootpa)
Kernel 4.18.0-372.9.1.el8.x86_64 on an x86_64

localhost login: root
Password:
You are required to change your password immediately (administrator enforced)
Current password:
New password:
Retype new password:
Last login: Tue Nov 1 03:50:33 on ttyS0

```

2. When the Connected to domain 'EMSV' message is displayed press <Enter> to log in.

Note: The MSVM has a default password **ibmesscluster**, which is configured for the **root** user. The default password must be changed after the first login.

3. Configure IP address of the MSVM by using the `ess_ssr_setup` tool, which is provided with the EMSVM disk image.

A TDA table with necessary information is needed for the IP address configuration.

The `ess_ssr_setup` tool provides an option to configure the MSVM as a new deployment or adding management server to an existing IBM Storage Scale System. Regardless of selected option, the tool offers an option to test the management network connectivity to an existing device on that network. This test can be skipped, if new deployment option is selected.

An example of IP address configuration by using the `ess_ssr_setup` tool as follows:

```
ess_ssr_setup
```

A sample output is as follows:

```
2022-11-14 11:58:18,381 INFO: Welcome to IBM ESS 'code 20' helper tool
2022-11-14 11:58:18,381 INFO: You are going to need the TDA table information to
continue. If you do not have that information, you CANNOT continue.
2022-11-14 11:58:18,381 INFO: For debug output or issues, be sure to include the debug
file on /opt/ibm/ess/tools/bin/code20_debug.log

Do you want to continue? (y/n): y
2022-11-14 11:58:39,291 INFO: We strongly recommend setting a campus network in the EMS.

Do you want set up the campus network in this node? (y/n): y
Please type the campus IP of this node (i.e. 192.168.80.10): 192.168.64.231
Please type the campus netmask of this node (i.e. 255.255.255.0): 255.255.255.0
Please type the campus gateway IP (i.e. 192.168.80.1): 192.168.64.1

Please type the management IP of this node (i.e. 192.168.10.10): 192.168.21.231
Please type the management netmask of this node (i.e. 255.255.255.0): 255.255.255.0

Please type the OS FSP IP (C11-T2 for P9 EMS) of this EMS (i.e. 192.168.20.9): 192.168.20.232
Please type the OS FSP netmask of this node (i.e. 255.255.255.0): 255.255.255.0
2022-11-14 12:00:29,966 INFO: Going to ask for the node password. All the nodes must have
the same password for code 20
2022-11-14 12:00:29,966 INFO: Be aware that the password will not be prompted into the
screen and you will not see it on it
Password:
2022-11-14 12:00:34,965 INFO: Please type the same password again
Password:

Is this a new deployment or adding a block to a running cluster? (new/add): new
Is there already any node in this environment that we can test ping connectivity with?
(y/n): n
2022-11-14 12:01:26,158 WARNING: This is a new setup and we have no peers to check
with, no ping tests will happen

Please review the entered information before continuing.
No changes in the system had been performed at this point.

The entered IP in CIDR format for CAMPUS is 192.168.64.231/24
The entered IP in CIDR format for MANAGEMENT is 192.168.21.231/24
The entered IP in CIDR format for EMS OS FSP (C11-T2 for P9 EMS) is 192.168.20.232/24
No ping tests to be performed.
The entered root password to set is: cluster

Do you want to continue and perform changes and tests in this node? (y/n): y
2022-11-14 12:02:25,072 INFO: Going to set the root user password of this node to the
password typed before
2022-11-14 12:02:25,349 INFO: Run 'Root_password_set' completed successfully
2022-11-14 12:02:25,436 WARNING: Passwordless root SSH to localhost test is passed.
We are going to try to fix it. If a password is asked type same you where entered during
this run.
The authenticity of host 'localhost (:::1)' can't be established.
ECDSA key fingerprint is SHA256:eT/DjXNMyp61MxM8z4J+3eD9INi7u0cWcb/aEZpfq0A.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
2022-11-14 12:02:34,811 INFO: Self fix of root SSH to localhost worked, moving on
2022-11-14 12:02:34,814 INFO: Run 'Passwordless root SSH localhost' completed successfully
2022-11-14 12:02:34,814 INFO: Going to set Management Network IP on this node
2022-11-14 12:02:34,866 WARNING: The command sudo nmcli con show mgmt returned code
```

```

10
2022-11-14 12:02:34,867 INFO: The output of running the command sudo nmcli con show mgmt
was:
Error: mgmt - no such connection profile.

0 0 0 0
2022-11-14 12:02:35,051 INFO: Run 'Set MGMT IP' completed successfully
2022-11-14 12:02:35,051 INFO: Going to set Campus Network IP on this node
2022-11-14 12:02:35,102 WARNING: The command sudo nmcli con show campus returned
code 10
2022-11-14 12:02:35,102 INFO: The output of running the command sudo nmcli con show
campus was:
Error: campus - no such connection profile.

2022-11-14 12:02:35,276 INFO: Run 'Set Campus IP' completed successfully
2022-11-14 12:02:35,276 INFO: Going to set OS FSP Network IP on this node
2022-11-14 12:02:35,325 WARNING: The command sudo nmcli con show fsp returned code 10
2022-11-14 12:02:35,325 INFO: The output of running the command sudo nmcli con show fsp
was:
Error: fsp - no such connection profile.

2022-11-14 12:02:35,502 INFO: Run 'Set OS FSP IP' completed successfully
2022-11-14 12:02:35,502 WARNING: This is the first node of a new setup, no ping
tests are going to be performed
2022-11-14 12:02:35,506 INFO: Run 'Ping not tested' completed successfully
The following tasks were run in this node
TASK: DB_init was successfully run on 2022-11-14T11:58:18.141767
TASK: Root_password_set was successfully run on 2022-11-14T12:02:25.346899
TASK: Passwordless root SSH localhost was successfully run on 2022-11-14T12:02:34.812026
TASK: Set MGMT IP was successfully run on 2022-11-14T12:02:35.048829
TASK: Set Campus IP was successfully run on 2022-11-14T12:02:35.273364
TASK: Set OS FSP IP was successfully run on 2022-11-14T12:02:35.497673
TASK: Ping not tested was successfully run on 2022-11-14T12:02:35.502590
2022-11-14 12:02:35,508 INFO: All run tasks were successful
2022-11-14 12:02:35,509 INFO: All required tests were completed successfully
2022-11-14 12:02:35,509 INFO: CODE 20 completed.

```

A successful execution of the `ess_ssr_setup` tool displays the “CODE 20 completed.” message. The Install Complete app run (“Code 20”) must complete successfully to install the MSVM on an IBM Storage Scale cluster.

Note: To disconnect from the console session of the MSVM use ‘^]’ key combination on Linux systems. For windows systems, press **Ctrl - ^**.

4. After a successful completion of IP address setup by using the `ess_ssr_setup` tool, access the MSVM by using the SSH client.

Now, this MSVM is ready for usage.

5. To continue with the IBM Storage Scale System setup, refer the deployment process that is described in the *IBM Storage Scale System Deployment Guide*.

Other considerations

IBM Storage Scale Systems do not support the following host-related tasks:

Backing up and restoring of the MSVM

If the host goes into the irrecoverable mode, recover the host by reinstalling the OS or by using a remote copy:

- Recover the host by reinstalling the OS on the same host or different host that meets the requirements, and boot the MSVM.
 - Copy the `/emsvm/ESSVM.qcow2` file outside of the host.
 - Make sure that the copied file is compatible.

TIP: Copy the file while MSVM is shut down. Any freeze, other type of quiescence of the MSVM, or the underlying file system in the host might corrupt the MSVM OS or the quorum of the IBM Storage Scale cluster, which has the MSVM, might be lost.

- Make sure that the quorum of the IBM Storage Scale cluster is not lost, and shut down the MSVM for backup.

- You can recover the host by using a remote copy of the MSVM virtual disk.
 - Install the EMSVM script to recreate the MSVM for the host (the same or different host).
 - Copy the virtual disk into the /emsvm directory and start the MSVM with the IBM EMSVM script.

Shutting down the MSVM

The IBM EMSVM script does not shut down the MSVM. The MSVM must be stopped from the MSVM by using SSH or the IBM EMSVM script from the host.

- Make sure that shutting down the MSVM does not cause any problem in the IBM Storage Scale cluster, and then shut down the MSVM with standard OS commands.
- If the EMSVM OS does not respond, you can reboot the host to reboot the MSVM.

Note: The MSVM is set to auto start when the host boots.

- You can use standard **virsh** commands from the host to stop the MSVM. However, this practice is not recommended.

Monitoring the host hardware

IBM Power9 management server or MSVM does not monitor the host hardware. Customer must monitor the host hardware to avoid any risk. Customer must set up the host by following the vendor recommended best practices of hardware monitoring for the host of MSVM.

Accessing to the host

The access to the host is needed for the MSVM. It is recommended to set additional Ethernet ports into the host for the remote access and out-of-band management access to the host according to the best practices recommended by the vendor of the host.

Updating the host OS and the firmware

Updates of the OS and firmware currency and compatibility with MSVM in the host is entirely customer responsibility. IBM does not provide any support for the host OS and firmware updates.

- When upgrading an OS, make sure that the OS is supported by the MSVM.
- When upgrading the firmware, specially of the HSN cards, check with the vendor of cards whether the version you are installing is compatible with the Mellanox OFED version that MSVM uses.
- When you need to reboot the host, make sure that MSVM is down before the reboot and does not cause any issue in an IBM Storage Scale cluster.

Changing the port type of an NVIDIA ConnectX VPI adapter

If you need to change the VPI port type from Ethernet to InfiniBand or vice versa, change the port type from the host. Do not change it from the MSVM. Refer to the documentation of the system provider about how to change the VPI port. Usually, the **mst** command is used to change the port type.

Complete the following steps to change the VPI port type:

1. Shut down the MSVM, make sure that the quorum of the IBM Storage Scale cluster is not lost.
2. Change the needed ports to the required mode.
3. Reboot host, MSVM starts automatically.
4. Use the ports within the MSVM.

Alternatively, you can set the VPI type before you deploy the MSVM on your host.

Preparing for the deployment and the upgrade

Before IBM Storage Scale deployment or upgrade, download required packages and set up a management server.

Preparing a Power9 management server

Software package installation and the management server must be prepared before deployment or upgrade IBM Storage Scale System.

The following steps apply to any IBM Storage Scale System deployment or upgrade in a new cluster environment where no management server exists.

1. Log in to the Power9 management server by using the management IP, which the SSR set up by using the provided worksheet. The default password is `ibmesscluster`.
2. Optional: Set up a campus or a public connection (interface `enP1p8s0f2`). The connection might be named 'campus'. Connect an Ethernet cable to C11-T3 ports on the P9 management server and your lab network to access the GUI or call home to IBM. The container creates a bridge to the management network, thus a campus connection is highly advised.

Note: If you do not set up a campus or a public connection, you temporarily lose your connection when the container bridge is created in a later step.

This method is for configuring the campus network, not any other network in the management server node. Do not modify T1, T2, or T4 connections in the system after SSR set these connections, and use the SSR method only to configure T1 and T2 (if you must change after SSR is finished). That includes renaming the interface, setting IP, or any other interaction with those interfaces.

You can use the `nmtui` command to set the IP address of the campus interface. For more information, see [Configuring IP networking with nmtui tool](#).

3. Import the `/etc/hosts` file on the management server node. This file must contain the low-speed IP addresses for the management server and high-speed IP addresses for the cluster, FQDNs, and short names. The high-speed names must contain a suffix to the low-speed names (For example, `essio1-hs` (high-speed name) to `essio1` (low-speed name)).

```
127.0.0.1 localhost localhost.localdomain.local localhost4 localhost4.localdomain4

## Management IPs 192.168.45.0/24
192.168.45.20 <MS1>.localdomain.local <MS1>
192.168.45.21 <IOnode1>.localdomain.local <IOnode1>
192.168.45.22 <IOnode2>.localdomain.local <IOnode2>
192.168.45.23 <protnode1>.localdomain.local <protnode1>
192.168.45.24 <protnode2>.localdomain.local <protnode2>

## High-speed IPs 10.0.11.0/24
10.0.11.1 <MS1>-hs.localdomain.local <MS1>-hs
10.0.11.2 <IOnode1>-hs.localdomain.local <IOnode1>-hs
10.0.11.3 <IOnode2>-hs.localdomain.local <IOnode2>-hs
10.0.11.4 <protnode1>-hs.localdomain.local <protnode1>-hs
10.0.11.5 <protnode2>-hs.localdomain.local <protnode2>-hs

## Protocol CES IPs
10.0.11.100 <protnode>_ces1.localdomain.local <protnode>_ces1
10.0.11.101 <protnode>_ces1.localdomain.local <protnode>_ces1
10.0.11.102 <protnode>_ces2.localdomain.local <protnode>_ces2
10.0.11.103 <protnode>_ces2.localdomain.local <protnode>_ces2
```

Note:

- The `localdomain.local` domain name is an example that you cannot use for deployment. Replace the domain name with a valid fully qualified domain name (FQDN) during the `/etc/hosts` setup. The same domain name must be used across all defined network subnets. Set the domain on the management server.

```
hostnamectl set-hostname NAME
```

NAME must specify the fully qualified domain name (FQDN) of the management interface (T1) on the management server. If additional names are required for campus or other interfaces, configure names as aliases. Do not use these names as the primary hostname displayed by the `hostnamectl` command.

You can set up the management server FQDN manually or wait until prompted when the IBM Storage Scale System deployment binary is started. At that time, the scripts confirm the FQDN and you can make changes.

- If you want to set up a supported IBM Storage Scale System with the Power9 management server, add new IBM Storage Scale System hostnames to the `/etc/hosts` file by using the same structure. For example, low-speed (management) and high-speed (cluster) IP addresses, FQDNs, and short names.
- Do not use any special characters, underscores, or dashes in the hostnames other than the high-speed suffix; example, `-hs`.

Note: If you have protocol nodes, add the protocol nodes to the commands provided in these instructions. The default `/etc/hosts` file has hostnames `prt1` and `prt2` for protocol nodes. You might have more than two protocol nodes.

4. Extract the installation package.

Note: Check the installed version by manufacturing in SSR worksheet. If a new version is available in Fix Central, complete the following steps:

- a. Replace the existing image in the `/home/deploy` directory with a new image.
- b. Remove the old `.tgz` file before you extract the package.

```
cd /home/deploy
xz --decompress ess_7.0.1.x_0611-23.tar.xz
```

5. Complete the following steps, before you start the container:

- a. Create `/home/log` and `/home/backup`.
- b. Ensure that Chrony is running. For more information, see [“Setting a time zone” on page 106](#).

6. Accept the license and install the accepted image.

Important: Before you create a container, delete any preexisting container.

```
./ess_7.0.1.x_0611-23.tar.xz
```

Example of the contents of the extracted installation package:

```
├─ 70-persistent-net-ems.rules
├─ classes
│   ├── essmgr.py
│   ├── essmgr.yml.py
│   ├── _init_.py
│   └─ _pycache_
│   ├── essmgr.cpython-36.pyc
│   ├── essmgr.yml.cpython-36.pyc
│   └─ _init_.cpython-36.pyc
├─ ess_7.0.1.x_0611-23.tar.xz
├─ essmgr
├─ essmgr.yml
├─ essmkym1
├─ logs
│   ├── essmgr.yml_2022-05-16_18-16-28
│   └─ essmkym1_2022-05-16_18-16-28_log
```

```
|— podman_rh8_ppc64le.tgz
|— Release_note.ess_7.0.1.x_0611-23.tar.xz.txt
|— rhels-8.8-server-extra.iso
|— rhels-8.8-server-ppc64le.iso
|— rhels-8.8-server-x86_64.iso
```

In this step, you are prompted to provide these inputs:

- Container name, which must not be in the `/etc/hosts` file or resolvable by using DNS.
- Confirmation of the management server FQDN, which must match what is set for the management IP in the `/etc/hosts` file. If this value needs to be changed or set, you can change by using the **essmkym1** script. The **essmkym1** script is located in the extracted directory. For example,

```
/home/deploy/ess_7.0.1.x_0611-23_0611-23.tar.xz
```

7. Import the nodes configuration to the management sever. This command determines the node information based on virtual product data (VPD) and also exchanges the SSH keys.

```
essrun -N essio1,essio2,ems1 config load -p
```

Where:

essio1,essio2

Specifies all nodes of the same type in the building-blocks.

ems1

Specifies a management server.

-p

Specifies the root password. The default password is `ibmesscluster`. Change the root password after deployment is complete.

-N

Specifies all the nodes of the cluster to update the `hosts.yml` file for BMC. If this command is run on the selected nodes only, for example, only on I/O nodes, and management server node, the `hosts.yml` file is overwritten with the latest nodes.

After this command is run, you can use `-N NodeGroup` for future **essrun** steps (For example, `-N ess_ppc64le`). There are different node group names for IBM Storage Scale System 3000.

8. Check whether nodes have any issues before deployment or upgrade. Review the output, make needed changes.

```
essrun -N essio1,essio2,ems1 update --precheck
```

Adding IBM Storage Scale System Utility Node or IBM Storage Scale System 6000 to a Power9 management server

You can add an Utility Node or IBM Storage Scale System 6000 to a Power9 management server.

Prerequisites

Ensure that the following conditions are met before you start the conversion:

- Power9 management server environment is configured along with an IBM Storage Scale System 3500.
- A cluster is up and running on Power9 management server and IBM Storage Scale System 3500.
- The GUI is configured for the cluster:
 - Power9 management server is configured as a collector node.

- Other IBM Storage Scale System nodes, such as IBM Storage Scale System 3500, are configured as sensors.
- System health is in the optimal state. Run these commands to check:
 - **gnrhealthcheck**
 - **mmhealth**
- Utility Node passed correctly the Code 20 (installation).
- Existent cluster is up to date.
- Utility Node virtual machine is up and running, and up to date.
 - IBM Storage Scale System version must be the same as the existent cluster.
- The /etc/hosts file is updated and contains all new node entries.

Note: Include management and high-speed hostnames with the corresponding IP addresses.
- The deployment container must be up and running on the Power9 management server and the Utility Node management server.

Note: The same version must be running in both environments.

Adding an Utility Node

Consider the scenario where *ess35001a* is an existing node in the Power9 environment. *s6k1a* and *s6k1b* are IBM Storage Scale System 6000 canisters, and *utility-vm* is an Utility Node.

1. From the Power9 management server container, run the following command on Utility Node VM and all the existent IBM Storage Scale System nodes from the Power9 management server:

```
essrun -N utility-vm,s6k1a,s6k1b,<p9 EMS>,<existing io1>,<existing io2>,... config load -p
```

Where:

s6k1a* and *s6k1b

Are IBM Storage Scale System 6000 canisters

utility-vm

Is an Utility Node VM.

2. From the Power9 management server container, run the following command for a Power9 management server node:

```
essrun -N <p9 EMS> config check
```

3. From the container of the Utility Node VM, run the following command for a Utility Node VM node:

```
essrun -N utility-vm config check
```

4. From the Power9 management server container, run the following command to add an Utility Node, which uses an existing cluster node as a contact node, in this case *essio3*:

```
essrun -N ess35001a cluster --add-ems utility-vm --suffix=-hs
```

ess35001a is an existing node in the Power9 environment.

5. From the Power9 management server container, run the following command for the Utility management server:

```
essrun -N utility-vm gui --configure
```

6. From the Power9 management server *not* in the container, complete the following steps:
 - a. Run these commands to manually install the Utility Node and IBM Storage Scale System 6000 RPMS on the Power9 management server.

- `yum install gpfs.gnr.support-ess6000`
 - `yum install gpfs.gnr.support-essutility`
- b. After the RPMS installation, run the following commands:
- `mmaddcompspec default --replace`
 - `mmdiscovercomp -N all`
- c. In the **Edit Rack Components** wizard on the Power9 management server select the **Yes, discover new servers and enclosures first** option.
7. Log in to the GUI on the Power9 management server and the Utility Node VM.
- You should see all nodes in hardware monitoring view from both management server with some known issues.
 - One of the management servers is displayed as **Node Server** instead of **Management Server** and on the Utility Node VM GUI, you might see the state of the management server nodes as **Unknown**.
8. Configure the call home on Utility Node VM, and then on Power9 management server).
- a. If there is an IBM Storage Scale System 6000 in the system, add it to the cluster before you configure the call home.
 - b. Run the **esscallhomeconf** command on the Utility Node VM, and then on the Power9 management server. For more information, see [esscallhomeconf script](#).
9. For upcoming upgrades, make sure that you upgrade it from the associated management server:
- a. IBM Storage Scale System 6000 and Utility Node protocol from Utility management server VM.
 - b. IBM Storage Scale System 3500 is fluid, if originally associated with the Power9 management server, upgrade from Power9 management server. If originally associated with the Utility Node management server, upgrade from the Utility Node management server VM.

Adding IBM Storage Scale System 6000

In this example, `ess35001a` is an existing node in the Power9 environment, `s6k1a` and `s6k1b` are IBM Storage Scale System 6000 canisters, and `utility-vm` is a Utility management server node.

1. Must be done after Utility Node management server is added to the Power9 management server environment.
2. From the Utility Nodes management server VM container, run the following command to add the IBM Storage Scale System 6000 nodes into the cluster:

```
essrun -N utility-vm cluster --add-nodes s6k1a,s6k1b --suffix=-hs
```

3. Run the following command on each one for the nodes for the IBM Storage Scale System 6000 nodes:
 - **gnrhealthcheck**
 - **mmhealth node show -a**
4. The following example creates a separate file system for IBM Storage Scale System 6000. From the Utility Node management server VM container, run the following command for the file system creation:

```
essrun -N s6k1a,s6k1b filesystem --suffix=-hs
```

A simple 4M, 80% set size file system is created (8+2p).

If you want to create the vdisksets only, you can use the following command:

```
essrun -N s6k1a,s6k1b vdisk --suffix=-hs
```

VDisk sets that can be added to another existent file system are created.

5. From the Utility Node management server VM container, run the following command to add IBM Storage Scale System 6000 to the GUI or sensor configuration:

```
essrun -N s6k1a,s6k1b gui --add-to-hosts
```

6. From the Utility Node management server VM NOT in the container, run the following commands:

- `mmaddcompspec default --replace`
- `mmdiscovercomp -N all`

Run the **Edit Rack Components** wizard again on the Utility Node management server by using the **Yes, discover new servers and enclosures first** options.

Cleaning up the network bridges from a Power9 management server node

In earlier IBM Storage Scale System releases, the **mgmt_bridge** and **fsp_bridge** bridges are used on a Power9 management server node to serve the management and BMC network connectivity to the IBM Storage Scale System deployment container. Starting from IBM Storage Scale System 6.1.8.2, these bridges are not needed on a Power9 management server node.

The IBM Storage Scale System container uses Podman inbuilt CNI network to manage container IPs. Therefore, you need not to plan for any networking for the container. This simplifies the network design of the IBM Storage Scale System deployment container and reduce the networking overhead.

When the IBM Storage Scale System 6.1.8.2 container starts, an error appears if an IP address is not detected on the management interface and the **mgmt_bridge** exists. To perform the automatic bridging conversion process before starting the container, issue the following command:

```
essmkym1 --clean-network
```

A sample is as follows:

```
[root@essem1 deploy]# sh ess_7.0.1.x_0611-23.tar.xz --silent
-- [INFO] Supported version of the node 5105-22E found. Continuing... --
-- [INFO] Supported version of the OS found. Continuing... --
-- [INFO] Extracting License Acceptance Process Tool to ./ --
-- [INFO] Installing JRE --
-- [INFO] Invoking License Acceptance Process Tool --
-- [INFO] License Agreement Terms accepted --
-- [INFO] Extracting Product to ./ --
-- [INFO] Going to check SHA256 of ess_7.0.1.x_0611-23.tar.xz --
ess_7.0.1.x_0611-23.tar.xz: OK
-- [INFO] Checksum is valid, continuing --
-- [INFO] Checking if there is enough free space at ./ --
-- [INFO] Free space check PASSED --
-- [INFO] podman version 1.6.4 is already installed --
-- [INFO] Code extraction completed successfully --
-- [INFO] Python3 link found. Continuing --
2023-08-17 16:38:19,685 ERROR:      mgmt_bridge present at node. Starting 6182 "mgmt_bridge"
should be deleted and Management IP should be assigned to actual management interface like
"mgmt". Please go to the extracted directory (.dir) and run './essmkym1 --cleanup-network'
command to cleanup the network bridges. After that, rerun './essmkym1' command without any
switch/argument to start the container.
2023-08-17 16:38:19,685 INFO:      Please perform cleanup of network bridges before starting
container... Exiting...
-- [INFO] Removing License Acceptance Process Tool from ./ --
-- [INFO] Removing JRE from ./ --
[root@essem1 deploy]#
```

Notes:

- No change is noted in the BYOMS (EMSVM) and IBM Storage Scale System Utility Node (EMSVM-23E) because the above methodology was implemented for the original release. There is no need of the **mgmt_bridge** or **fsp_bridge** in these implementations.

Configuring dual management server

This appendix describes the basic requirements, use cases, and configuration steps for the dual management server support.

Until IBM Storage Scale System 6.1.6.0, the deployment did not formally support two management server in the same cluster. Many customers had one of three environments.

- Single P9 or BYOMS node, which controlled a single cluster and single site.
- Single P9 or BYOMS node, which controlled a single cluster and dual sites. Both sites networked via a stretch cluster (all networks stretch between locations).
- Dual P9 management nodes, two locations. Only the high-speed network stretched between sites. The management networks stayed local per site. Thus, each management node could only see its local hardware.

The dual management server provides a live backup management server to become available when the primary goes down. Also, dual management server allows for multiple GUI services and collectors to become active. The GUI, in particular, allows CSI applications to stay active when one management server goes down.

Requirements

Ensure that following requirements met for the dual management server support:

- The upgraded version needs to be V5.1.6.1 (IBM Storage Scale System 6.1.6.0) or higher.
- Two P9 management server
- Single cluster
- All networks are flat, single namespace (IBM Storage Scale System stretch cluster requirements)
- All nodes (including two management server) have unique FQDN/IPs
- The `/etc/hosts` file copied from a primary to a backup management server (the same as `hosts.yml`)

Support statement

Only P9 management server supported (no BYOMS, no Power8, or no mixing)

Restrictions

- This is not an HA IBM Storage Scale System but an HA GUI/ CSI.
- Concept of primary/secondary management server.
- If a primary management server goes down, a secondary management server can be used to install a container and takeover the primary management server role
- Consider one management server as primary but run all **essrun** commands on another management server in the list. You cannot move back and fourth between primary and secondary management server (for deployment or for GUI usage).

Configuring dual management server

1. Connect network in the backup management server as a primary management server (Mgmt/FSP/ Campus).
2. Add backup management server network hostnames to the `/etc/hosts` file in an existent cluster.
1. Connect network in the backup management server as primarily management server (Mgmt/FSP/ Campus).

```
bkpEMS.MGMT.Network.IP backupEms.domain backupEms
bkpEMS.HS.Network.IP backupEms-hs.domain backupEms-hs
```

2. Update the backup management server from the container that is running on the primary management server.

```
essrun -N backupEms config load -p
essrun -N backupEms update --precheck
essrun -N backupEms update --offline
```

3. Create network bonds.

```
essrun -N backupEms network --suffix=-hs
```

4. Add the backup management server to the existent cluster from the container that is running in the primary management server.

```
essrun -N primarilyEms cluster --add-ems backupEms --suffix=-hs
```

5. Check if successful checks.

```
essrun -N backupEms config check
```

6. Add the backup management server to the GUI configuration.

```
essrun -N backupEms gui --configure
```

Preparing a Utility Node

These instructions outline the steps to deploy systems with protocols from an x86 Utility Node management server.

Before you begin

The *hostnames* that are used in this procedure are only examples.

Table 3. Supported configurations	
Management server	Supported nodes
IBM Storage Scale System Utility Node	• IBM Storage Scale System 3000 nodes • IBM Storage Scale System 3200 nodes • IBM Storage Scale System 3500 nodes • IBM Storage Scale System 5000 nodes • IBM Storage Scale System 6000 nodes • IBM Storage Scale System Utility Node management server • IBM Power9 management server • IBM Power9 protocol node • IBM Storage Scale System 3500 protocol node (virtual machine) • IBM Storage Scale System Utility Node protocol node

Table 3. Supported configurations (continued)

Management server	Supported nodes
IBM Power9 management server	• IBM Storage Scale System 3000 nodes • IBM Storage Scale System 3200 nodes • IBM Storage Scale System 3500 nodes • IBM Storage Scale System 5000 nodes • IBM Power9 management server • IBM Power9 protocol node • IBM Storage Scale System 3500 protocol node (virtual machine)
Bring your own management server (BYOMS)	• IBM Storage Scale System 3500 nodes • IBM Storage Scale System 3500 protocol node (virtual machine) • BYOMS (virtual machine)

About this task

The following flow contains high-level steps to deploy the Utility Node management server, single IBM Storage Scale System 3500 building-block, and single Utility Node Protocol Node.



Attention: Check [Chapter 7, “Troubleshooting,”](#) on page 185 before you start the deployment.

Procedure

1. Corroborate that the SSR completed the installation (racking, cabling, software updates, hardware checkout).
2. To download the EMSVM-23E.qcow2.xz image:
 - a. Go to [IBM Fix Central](#), select the IBM Storage Scale System product and version, and click **Continue**.
 - b. To download the package, click the Scale_System_VM-<x.x.x.x>-x86_64-UTILITY-EMS.tgz file link.
 - c. Go to /serv and extract the Scale_System_VM-<x.x.x.x>-x86_64-UTILITY-EMS.tgz file. This .tgz file contains the /emsvm/EMSVM-23E.qcow2.xz file and the checksum file.
3. To download the deployment tool of the management server virtual machine:
 - a. Go to <https://github.com/IBM/EMSVM>.
 - b. Click **Code** (green button).
 - c. Click **Download ZIP**.
4. Download the x86 container image from [IBM Fix Central](#). Select the IBM Storage Scale System product and version, click **Continue** > **Unified License** > **Scale_System_Unified-6.2.3.0-EMS**.
5. Download the IBM Storage Scale installation toolkit (DAE or DME) from [IBM Storage Scale](#).
6. Copy all downloaded software to the /serv directory on the Utility Node host of the management server virtual machine.
7. Extract the EMSVM script package in the /serv directory.
8. Check the Utility Node.

```
cd /serv/EMSVM-main ; ./emsvm --check-host
```

9. Start the Utility Node.

```
./emsvm --start-EMS
```

10. Connect to the Utility Node.

```
./emsvm --connect-EMS
```

11. Press **Enter** and log in to the Utility Node VM (`root/ibmesscluster`) and change the password when prompted.
12. Configure the network, such as management, BMC, and campus IPs by the SSR tool.
 - a) Type `ess_ssr_setup` and press **Enter**.
 - b) Follow the instructions that are displayed in system output.
13. Exit the console and go back to the host. By using the SFTP, transfer the x86 container to the VM (`/home/deploy`) over the virtual bridge IP (10.23.16.2).

Note: The virtual bridge is not always the same; it varies from 10.23.16.2 to 10.23.16.6.

14. Complete the following substeps in the Utility Node VM.

- a) Set up `/etc/hosts`, and ensure that `/etc/hosts` contains the following items:
 - Utility Node VM (management and high speed)
 - I/O nodes (management and high speed)
 - Protocol node VM (management and high speed)
 - Protocol node host (management)
 - CES IPS
- b) Set the fully qualified domain name (FQDN).

```
hostnamectl hostname <FQDN>
```

FQDN must include complete domain name, for example, `server.test.net`.

c) Decompress the package.

```
cd /home/deploy ; xz --decompress <xz package>
```

d) Uncompress the compressed tarball.

```
tar xvf <uncompressed tarball>
```

15. To access the container and complete its deployment configuration, follow the next substeps.

a) Run the following command with the newly untarred binary file.

```
sh <untarred binary file>
```

This substep takes you to the container.

Note: When you start the container, delete the existing inventory file so that a new inventory file is generated. Because starting from the 7.0.1.x release, the VM and Utility Node are updated simultaneously. If an inventory file is not properly created, the VM and Utility Node upgrade does not complete successfully.

b) Load the configuration by running the following command, which determines the node information based on vital product data (VPD) and exchanges the SSH keys.

```
essrun -N utilityBareMetal config load
```

utilityBareMetal specifies an x86 Utility Node management server.

Check the configuration.

```
essrun -N MSVM,<IOnode1>,<IOnode2> config load -p
```

-p requests the password that is used for root login to the nodes. For more information, see [essrun](#) command in the *IBM Storage Scale Systems Command Reference*.

c) Check the nodes.

```
essrun -N utilityBareMetal,MSVM,<IOnode1>,<IOnode2> update --precheck
```

d) To exit the container, type `exit`.

Or, to detach from the container, press **Ctrl+p**, then press **Ctrl+q**.

16. Shut down the VM.

```
shutdown -h now
```

17. Reboot the Utility Node host.

```
systemctl reboot
```

18. Use ssh to get back to the management server host.

19. Start the management server.

```
cd /serv/EMSVM-main ; ./emsvm --start-EMS
```

20. Use ssh to access to the management server VM.

21. In the management server VM, go to the extracted deployment directory.

```
cd /home/deploy/<extracted .dir>
```

22. Restart the container.

```
./essmkym1 --restart
```

23. Check the nodes.

```
essrun -N utilityBareMetal,MSVM,<IOnode1>,<IOnode2> update --precheck
```

Converting a Power9 management server environment into a Utility Node unified management server

You can convert a Power9 management server environment to a unified management server Utility Node. After the conversion, the Power9 management server is re purposed or decommissioned. After this conversion, you can manage lifecycle from the single Utility Node for all supported node types (except BYOE).

The conversion limitations

- Initial support for the conversion is provided in IBM Storage Scale System 6.1.9.2.
- Initial support for co-existence was provided in IBM Storage Scale System 6.1.9.1.
- Co-existence is still a supported option for customers. See the [“Adding IBM Storage Scale System Utility Node or IBM Storage Scale System 6000 to a Power9 management server”](#) on page 60 for more information.
- The unified call home is supported. This support allows call home to work without ESA from the Utility Node on all node types (including Power).
- The conversion of dual Power9 management server nodes (HA GUI) is not supported.
- Dual management server Utility Node is not supported.
- Stretch cluster rules still apply.
 - All management networks (FSP/BMC/MGMT) and the high-speed network (InfiniBand/Ethernet) must be flat and single namespace.

- All nodes must have unique IP addresses and hostnames for the management networks and the high-speed network.
- All nodes must be reachable to each other over all networks.
- A common /etc/hosts file (or properly configured DNS) must be defined as per the best practices, and copied by using the deployment tools to all IBM Storage Scale System nodes in the cluster.
- .
- All nodes must be at IBM Storage Scale System 6.1.9.2 or higher before this conversion.
- After the conversion, Utility Node can deploy any supported node type (except BYOE).
- You might not mix Protocol nodes of different architectures (Power + x86).
- You cannot convert from BYOE environments.

Supported networks

- The network structure when you mix Power9 management server and IBM Storage Scale System 3500 is shown in the following figure:

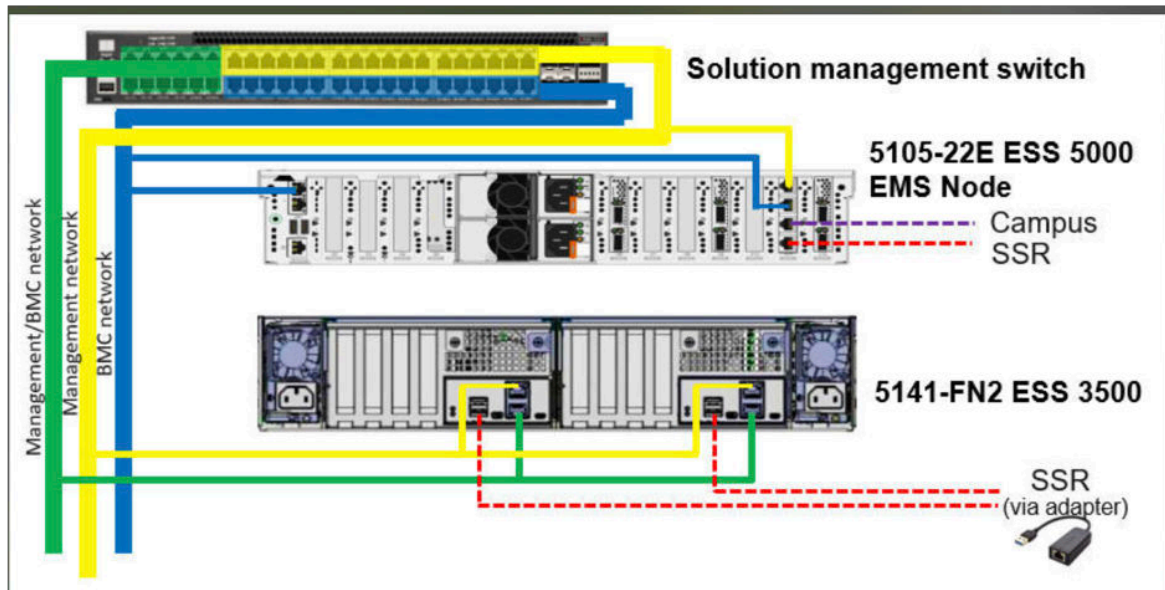


Figure 19. The network of Power9 management server and IBM Storage Scale System 3500

- The network structure when you mix a management server Utility Node, two utility protocol nodes, and an IBM Storage Scale System 3500 is shown in the following figure:

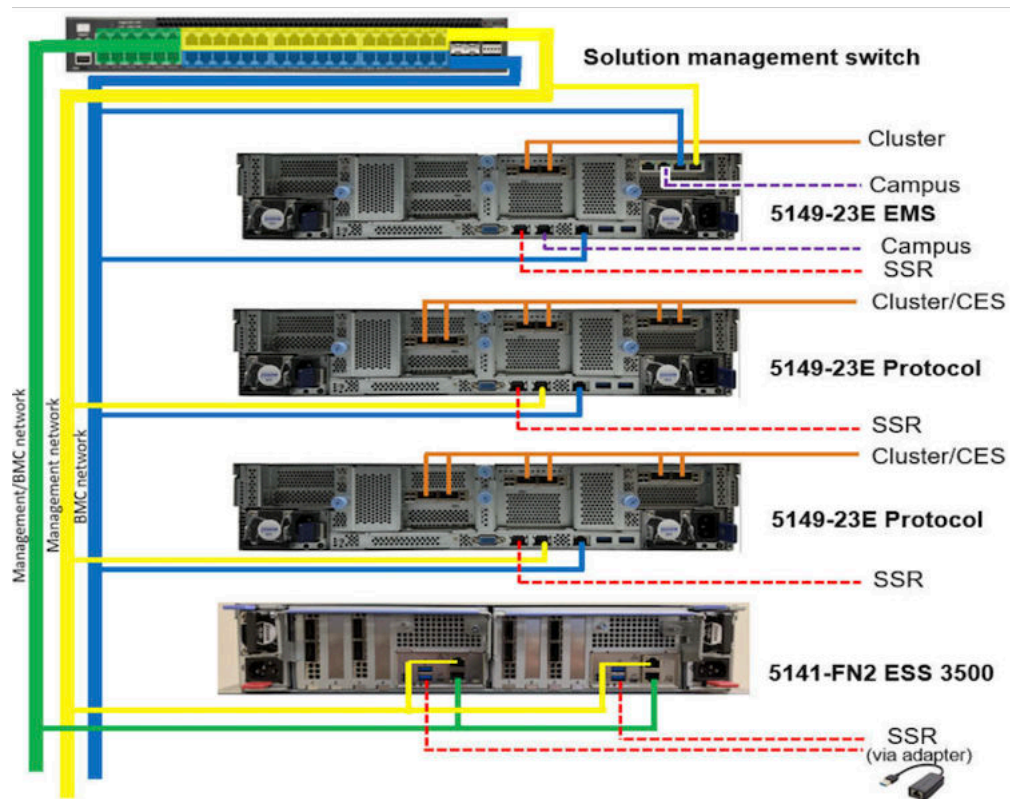


Figure 20. The network of a management server Utility Node, two utility protocol nodes, and an IBM Storage Scale System 3500

- The network structure when you mix a management server Utility Node and an IBM Storage Scale System 6000 is shown in the following figure:

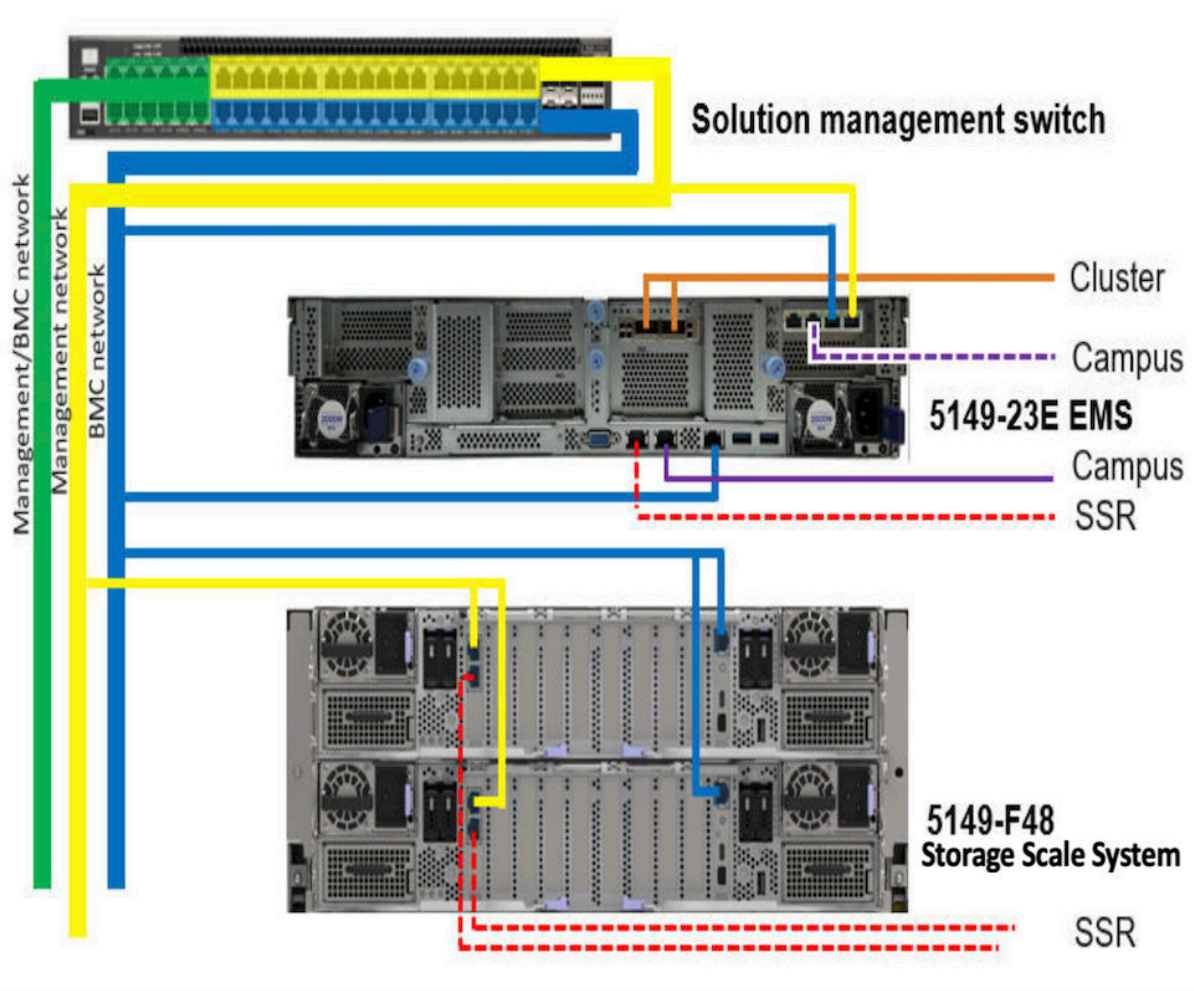


Figure 21. The network of a management server Utility Node and an IBM Storage Scale System 6000

Converting a management server Utility Node from a non co-existence environment

Complete the following steps to convert a production Power9 management server environment to a single unified management server Utility Node. These steps are high level because specific commands are not unique to conversion. Refer to other sections of the IBM Storage Scale System Deployment Guide, if required.

1. Ensure that all nodes are at 6.1.9.2 or higher before the conversion.
2. Ensure that existing production Power9 management server environment fully is configured and in production.
3. Set up management server Utility Node and IBM Storage Scale System 6000 setup via the SSR on same network fabric as the Power9 management server environment.
4. Upgrade the Power9 management server environment IBM Storage Scale System 6.1.9.2.
5. Deploy a VM and start the container on the management server Utility Node.
6. On the Utility Node container only, run the following command on add all nodes of the remote environment in the Utility Node container inventory:

```
essrun utility1NodeVM,remoteEMSp9,sssio6k2,sssio6k2 config load -p
```

- Passwordless SSH is resolved between all nodes.
- An inventory that contains all nodes is created.

7. On the Utility Node container only, run the following command to check the healthy of the nodes that are being added or converted.

```
essrun -N utility1NodeVm,remoteEMSp9,sssio6k2,sssio6k2 update precheck
```

This command verifies whether the entire environment is healthy to perform a conversion.

8. Upgrade the management server Utility Node and IBM Storage Scale System 6000 environment fully to IBM Storage Scale System 6.1.9.2.
9. Create the high-speed network bonds (if applicable). The new nodes can talk on the same cluster fabric as the Power9 environment.
10. Add the management server Utility Node and IBM Storage Scale System 6000 to the existing Power9 management server environment. Perform this operation from the management server Utility Node container.

```
essrun -N sssio6k1 cluster --add-ems utilityNodeVm --suffix=-hs
```

11. Create VDisk sets on the Utility Node, add new pool or policy to an existing file system (or create a VDisk set).
12. Remove the Power9 management server, which is a collector node (from the Utility Node VM, out of the container).

```
/usr/lpp/mmfs/bin/mmperfmon config update --collectors utilityNodeVm-hs
```

13. Remove the Power9 management server from the cluster and shut down the node.
 - a. Remove Power9 management server from cluster.
 - b. Shut down IBM Storage Scale.
 - c. Shut down the Power9 management server.
 - d. Disconnect power and network.
 - e. Repurpose or decommission the Power9 management server.
14. Clean up the existing configuration on the cluster.
 - a. Change the call home node the Utility Node management server to delete the old regular EMS P9.

15. Configure the GUI for the management server Utility Node on the Utility Node container.

```
essrun -N UnGDL1vmgmt,ess3500rw1a,ess3500rw1b gui --configure
```

16. Add Utility Node nodes to the GUI or sensor configuration (from the Utility Node management server container).

```
essrun -N sssio6k1,sssio6k2 gui --add-to-hosts
```

17. Start the GUI.

- a. Run the following command on the Utility Node VM:

```
mmaddcompspec default --replace
```

- b. Run **edit rack components on GUI**.
 - c. Select **discover new components**.

18. Re-configure the unified call home from management server Utility Node on all cluster nodes.
19. Perform final health checks.

- Run the following command on each node:

```
essinstallcheck
```

- Run the following command only on the management server:

```
gnrhealthcheck
```

- Run the following command on each node:

```
mmhealth node show -a
```

Converting a management server Utility Node from a co-existence environment

To convert a management server Utility Node, complete the following high-level steps:

1. Ensure that all nodes are at 6.1.9.2 or higher before the conversion.
2. Upgrade all nodes to IBM Storage Scale System 6.1.9.2.

Upgrade the management server' first, then the attached nodes. That is, update the IBM Storage Scale System 6000 and any management server Utility protocol nodes from the EMSVM and rest of the hardware from the Power9 management server.

3. From the management server Utility Node, run the following command on all nodes in the cluster:

```
essrun config load
```

4. Remove the Power9 management server, which is a collector node (from the Utility Node VM, out of the container).

```
/usr/lpp/mmfs/bin/mmperfmon config update --collectors utilityNodeVm-hs
```

5. Remove the Power9 management server from the cluster and shut down the node.

- a. Remove the Power9 management server from the cluster.
- b. Shut down IBM Storage Scale.
- c. Shut down the Power9 management server.
- d. Disconnect power and networks.
- e. Re purpose or decommission Power9 management server.

6. Start the GUI.

- a. Run **edit rack components**.
- b. Select **discover new components**.

7. Re-configure the unified call home from management server Utility Node on all cluster nodes.

8. Perform final health checks.

- Run the following command on each node:

```
essinstallcheck
```

- Run the following command only on the management server:

```
gnrhealthcheck
```

- Run the following command on each node:

```
mmhealth node show -a
```

Preparing bring-your-own management server

These instructions outline the steps to deploy systems with protocols from an x86 Utility Node management server.

Before you begin

For a successful deployment of the management server as a virtual machine on IBM Storage Scale System, validate the system on which you are planning to host the MSVM. The system must meet all requirements. For more information, see [“Bring-your-own management server” on page 46](#).

In addition to hardware and software requirements, the following additional information is required to complete the initial MSVM configuration:

- Access to the EMSVM script repository on public GitHub to download the IBM EMSVM script (<https://github.com/IBM/EMSVM>).
- An IBM account to access IBM Fix Central to download the MSVM disk image.
- IP address information for the initial MSVM configuration.

Network	IP address	Netmask	Gateway
Campus			
IBM Storage Scale System management			
OS FSP			

About this task

Procedure

1. Start the MSVM by using the IBM EMSVM script.
If you are starting a new MSVM, start this tool manually.
2. During the first manual start, set the MSVM to auto start with the host system.

Example

An error during start the MSVM without a virtual disk file is displayed in the following example:

```
./emsvm --start-EMS
...
2022-11-14 06:41:06,611 WARNING:      EMSVM KVM domain does not exist. Going to create it
2022-11-14 06:41:06,638 INFO:       KVM VM has been created
2022-11-14 06:41:06,638 INFO:       KVM VM domain EMSVM is set to autostart
2022-11-14 06:41:06,638 ERROR:      EMS VM compressed disk file /emsvm/EMSVM.qcow2.xz does not
exist. Download the file from IBM and place it on the expected directory
```

The following example shows that an error occurred when the MSVM is started with an IP address that is configured on the quad-port adapter:

```
./emsvm --start-EMS
...
2022-11-14 06:45:00,240 INFO:       OS device name: 'eth2' PCI address: '0000:45:00.0'
Description: 'Ethernet controller: Intel Corporation I350 Gigabit Network Connection (rev 01)'
2022-11-14 06:45:00,240 INFO:       OS device name: 'eth5' PCI address: '0000:45:00.1'
Description: 'Ethernet controller: Intel Corporation I350 Gigabit Network Connection (rev 01)'
2022-11-14 06:45:00,240 INFO:       OS device name: 'eth6' PCI address: '0000:45:00.2'
Description: 'Ethernet controller: Intel Corporation I350 Gigabit Network Connection (rev 01)'
2022-11-14 06:45:00,240 INFO:       OS device name: 'eth7' PCI address: '0000:45:00.3'
Description: 'Ethernet controller: Intel Corporation I350 Gigabit Network Connection (rev 01)'
2022-11-14 06:45:00,243 INFO:       Interface eth2 has IP 192.168.20.231
2022-11-14 06:45:00,244 ERROR:      We found IP address[es] on one or more interfaces that are
going to be passthrough to the VM
```

If no errors are displayed during the prerequisites check that occurs after you start the MSVM, confirmation to start the MSVM on this host is requested.

For example:

```
./emsvm --start-EMS
```

Successful start of MSVM output is as follows:

```
2022-11-14 06:48:28,934 INFO: Welcome to EMS VM version 0.53
2022-11-14 06:48:28,934 INFO: Please visit https://github.com/IBM/EMSVM for issues and new
versions
2022-11-14 06:48:28,934 INFO: Log file with details for this run is saved on: /var/log/emsvm/
emsvm_start EMS_2022_11_14_06_48_28.log
2022-11-14 06:48:28,941 INFO: This system is AMD EPYC based
2022-11-14 06:48:28,942 INFO: This host does not have AMD EPYC 1st Generation (Naples)
2022-11-14 06:48:29,052 INFO: This host CPU[s] are KVM capable and has extensions enabled
2022-11-14 06:48:29,052 INFO: Going to check if /dev/kvm exists
2022-11-14 06:48:29,052 INFO: The device /dev/kvm exists
2022-11-14 06:48:30,068 INFO: Red Hat Enterprise Linux 8.6 is a supported OS
2022-11-14 06:48:30,078 INFO: Checking RPM packages status
2022-11-14 06:48:30,232 INFO: All required RPM packages have the expected installation status
2022-11-14 06:48:30,316 INFO: Could not get current version from repository, please be sure
that you are running latest version from https://github.com/IBM/EMSVM
2022-11-14 06:48:35,321 INFO: Looking that we have at least one NIC for High Speed Network
(HSN)
2022-11-14 06:48:35,338 INFO: The NIC 'Infiniband controller: Mellanox Technologies MT28800
Family [ConnectX-6 Ex]' is OK as HSN NIC. Going to append PCI address '0000:27:00.0' as HSN NIC
2022-11-14 06:48:35,338 INFO: The NIC 'Infiniband controller: Mellanox Technologies MT28800
Family [ConnectX-6 Ex]' is OK as HSN NIC. Going to append PCI address '0000:27:00.1' as HSN NIC
2022-11-14 06:48:35,357 INFO: Found at least two HSN ports
2022-11-14 06:48:35,357 INFO: Looking that we have at least one quad port Ethernet NIC for
Low Speed Network (LSN)
2022-11-14 06:48:35,365 INFO: The NIC 'Ethernet controller: Intel Corporation I350 Gigabit
Network Connection (rev 01)' is OK as LSN NIC. Going to append PCI address '0000:45:00.' for
ports 0, 1, 2 and 3
2022-11-14 06:48:35,366 INFO: Host has 32 core[s] which complies with 32 cores required
2022-11-14 06:48:35,366 INFO: Total memory is 251.38 GB, which is more than the required 247
GB
2022-11-14 06:48:38,097 INFO: Current tune profile is 'virtual-host'
2022-11-14 06:48:38,273 INFO: OS settings match the running profile
2022-11-14 06:48:38,314 INFO: KVM storage pool EMSVM already exists
2022-11-14 06:48:38,362 INFO: The path /emsvm has 1181.29 GB free excluding reserved space
which is more than the required 600 GB
2022-11-14 06:48:38,367 INFO: KVM executable file /usr/libexec/qemu-kvm exists
2022-11-14 06:48:38,367 INFO: All tests passed, starting EMS VM
2022-11-14 06:48:38,368 WARNING: EMSVM KVM domain does not exist. Going to create it
2022-11-14 06:48:38,394 INFO: KVM VM has been created
2022-11-14 06:48:38,394 INFO: KVM VM domain EMSVM is set to autostart
2022-11-14 06:48:38,395 INFO:
2022-11-14 06:48:38,395 INFO: You have requested to start the EMS VM in this host
2022-11-14 06:48:38,395 INFO: The following PCI devices are going to be assigned to the VM
and are not going to be accessible from the host
2022-11-14 06:48:38,395 INFO: If you are using any of them to connect to the host, you are
going to lose connectivity to this host
2022-11-14 06:48:38,395 INFO: Please be sure that you are not using any of the following
devices before continuing the start up of the EMS VM
2022-11-14 06:48:38,432 INFO: PCI address: '0000:27:00.0' Description: 'Infiniband
controller: Mellanox Technologies MT28800 Family [ConnectX-6 Ex]'
2022-11-14 06:48:38,433 INFO: PCI address: '0000:27:00.1' Description: 'Infiniband
controller: Mellanox Technologies MT28800 Family [ConnectX-6 Ex]'
2022-11-14 06:48:38,433 INFO: OS device name: 'eth2' PCI address: '0000:45:00.0'
Description: 'Ethernet controller: Intel Corporation I350 Gigabit Network Connection (rev 01)'
2022-11-14 06:48:38,433 INFO: OS device name: 'eth5' PCI address: '0000:45:00.1'
Description: 'Ethernet controller: Intel Corporation I350 Gigabit Network Connection (rev 01)'
2022-11-14 06:48:38,433 INFO: OS device name: 'eth6' PCI address: '0000:45:00.2'
Description: 'Ethernet controller: Intel Corporation I350 Gigabit Network Connection (rev 01)'
2022-11-14 06:48:38,433 INFO: OS device name: 'eth7' PCI address: '0000:45:00.3'
Description: 'Ethernet controller: Intel Corporation I350 Gigabit Network Connection (rev 01)'
2022-11-14 06:48:38,435 INFO: No IP addresses found on the interfaces to be passthrough to
the VM
2022-11-14 06:48:38,435 INFO: Do you want to continue with the start EMS VM? (y/n):
y
2022-11-14 06:49:47,807 INFO: It can take several minutes to start the VM, please wait ...
2022-11-14 06:50:32,206 INFO: EMS VM is started now ...
```

What to do next

The downloaded MSVM disk image does not contain preconfigured IP addresses for any network interface. When you start the MSVM for the first time, the IP address must be configured to allow inbound and outbound communication of the MSVM.

1. To configure IP addresses for the MSVM, use the `emsvm --connect-EMS` option to connect to the MSVM console from the host system.

```
./emsvm --connect-EMS
```

A sample output is as follows:

```
2022-11-14 06:54:50,900 INFO: Welcome to EMS VM version 0.53
2022-11-14 06:54:50,900 INFO: Please visit https://github.com/IBM/EMSVM for issues and
new versions
2022-11-14 06:54:50,900 INFO: Log file with details for this run is saved on: /var/log/
emsvm/emsvm_connect EMS_2022_11_14_06_54_50.log
2022-11-14 06:54:50,909 INFO: This system is AMD EPYC based
2022-11-14 06:54:50,910 INFO: This host does not have AMD EPYC 1st Generation (Naples)
2022-11-14 06:54:51,018 INFO: This host CPU[s] are KVM capable and has extensions enabled
2022-11-14 06:54:51,018 INFO: Going to check if /dev/kvm exists
2022-11-14 06:54:51,018 INFO: The device /dev/kvm exists
2022-11-14 06:54:52,038 INFO: Red Hat Enterprise Linux 8.6 is a supported OS
2022-11-14 06:54:52,048 INFO: Checking RPM packages status
2022-11-14 06:54:52,201 INFO: All required RPM packages have the expected installation
status
2022-11-14 06:54:52,282 INFO: Could not get current version from repository, please be
sure that you are running latest version from https://github.com/IBM/EMSVM
2022-11-14 06:54:57,331 INFO: EMSVM KVM domain exists
2022-11-14 06:54:57,331 INFO: KVM VM domain EMSVM is already active

Connected to domain 'EMSVM'
Escape character is ^] (Ctrl + ])

Red Hat Enterprise Linux 8.6 (Ootpa)
Kernel 4.18.0-372.9.1.el8.x86_64 on an x86_64

localhost login: root
Password:
You are required to change your password immediately (administrator enforced)
Current password:
New password:
Retype new password:
Last login: Tue Nov 1 03:50:33 on ttyS0
```

2. When the Connected to domain 'EMSVM' message is displayed press **Enter** to log in.

Note: The MSVM has a default password **ibmesscluster**, which is configured for the **root** user. The default password must be changed after the first login.

3. Configure IP address of the MSVM by using the `ess_ssr_setup` tool, which is provided with the MSVM disk image.

A TDA table with necessary information is needed for the IP address configuration.

The `ess_ssr_setup` tool provides an option to configure the MSVM as a new deployment or adding management server to an existing IBM Storage Scale System. Regardless of selected option, the tool offers an option to test the management network connectivity to an existing device on that network. This test can be skipped, if new deployment option is selected.

An example of IP address configuration by using the `ess_ssr_setup` tool as follows:

```
ess_ssr_setup
```

A sample output is as follows:

```
2022-11-14 11:58:18,381 INFO: Welcome to IBM ESS 'code 20' helper tool
2022-11-14 11:58:18,381 INFO: You are going to need the TDA table information to
continue. If you do not have that information, you CANNOT continue.
2022-11-14 11:58:18,381 INFO: For debug output or issues, be sure to include the debug
file on /opt/ibm/ess/tools/bin/code20_debug.log

Do you want to continue? (y/n): y
```

```

2022-11-14 11:58:39,291 INFO: We strongly recommend setting a campus network in the EMS.

Do you want set up the campus network in this node? (y/n): y
Please type the campus IP of this node (i.e. 192.168.80.10): 192.168.64.231
Please type the campus netmask of this node (i.e. 255.255.255.0): 255.255.255.0
Please type the campus gateway IP (i.e. 192.168.80.1): 192.168.64.1

Please type the management IP of this node (i.e. 192.168.10.10): 192.168.21.231
Please type the management netmask of this node (i.e. 255.255.255.0): 255.255.255.0

Please type the OS FSP IP (C11-T2 for P9 EMS) of this EMS (i.e. 192.168.20.9): 192.168.20.232
Please type the OS FSP netmask of this node (i.e. 255.255.255.0): 255.255.255.0
2022-11-14 12:00:29,966 INFO: Going to ask for the node password. All the nodes must have
the same password for code 20
2022-11-14 12:00:29,966 INFO: Be aware that the password will not be prompted into the
screen and you will not see it on it
Password:
2022-11-14 12:00:34,965 INFO: Please type the same password again
Password:

Is this a new deployment or adding a block to a running cluster? (new/add): new
Is there already any node in this environment that we can test ping connectivity with?
(y/n): n
2022-11-14 12:01:26,158 WARNING: This is a new setup and we have no peers to check
with, no ping tests will happen

Please review the entered information before continuing.
No changes in the system had been performed at this point.

The entered IP in CIDR format for CAMPUS is 192.168.64.231/24
The entered IP in CIDR format for MANAGEMENT is 192.168.21.231/24
The entered IP in CIDR format for EMS OS FSP (C11-T2 for P9 EMS) is 192.168.20.232/24
No ping tests to be performed.
The entered root password to set is: cluster

Do you want to continue and perform changes and tests in this node? (y/n): y
2022-11-14 12:02:25,072 INFO: Going to set the root user password of this node to the
password typed before
2022-11-14 12:02:25,349 INFO: Run 'Root_password_set' completed successfully
2022-11-14 12:02:25,436 WARNING: Passwordless root SSH to localhost test is passed.
We are going to try to fix it. If a password is asked type same you where entered during
this run.
The authenticity of host 'localhost (::1)' can't be established.
ECDSA key fingerprint is SHA256:eT/DjXNMyp61MxM8z4J+3eD9INi7u0cWcb/aEZpfqOA.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
2022-11-14 12:02:34,811 INFO: Self fix of root SSH to localhost worked, moving on
2022-11-14 12:02:34,814 INFO: Run 'Passwordless root SSH localhost' completed successfully
2022-11-14 12:02:34,814 INFO: Going to set Management Network IP on this node
2022-11-14 12:02:34,866 WARNING: The command sudo nmcli con show mgmt returned code
10
2022-11-14 12:02:34,867 INFO: The output of running the command sudo nmcli con show mgmt
was:
Error: mgmt - no such connection profile.

0 0 0 0
2022-11-14 12:02:35,051 INFO: Run 'Set MGMT IP' completed successfully
2022-11-14 12:02:35,051 INFO: Going to set Campus Network IP on this node
2022-11-14 12:02:35,102 WARNING: The command sudo nmcli con show campus returned
code 10
2022-11-14 12:02:35,102 INFO: The output of running the command sudo nmcli con show
campus was:
Error: campus - no such connection profile.

2022-11-14 12:02:35,276 INFO: Run 'Set Campus IP' completed successfully
2022-11-14 12:02:35,276 INFO: Going to set OS FSP Network IP on this node
2022-11-14 12:02:35,325 WARNING: The command sudo nmcli con show fsp returned code 10
2022-11-14 12:02:35,325 INFO: The output of running the command sudo nmcli con show fsp
was:
Error: fsp - no such connection profile.

2022-11-14 12:02:35,502 INFO: Run 'Set OS FSP IP' completed successfully
2022-11-14 12:02:35,502 WARNING: This is the first node of a new setup, no ping
tests are going to be performed
2022-11-14 12:02:35,506 INFO: Run 'Ping not tested' completed successfully
The following tasks were run in this node
TASK: DB_init was successfully run on 2022-11-14T11:58:18.141767
TASK: Root_password_set was successfully run on 2022-11-14T12:02:25.346899
TASK: Passwordless root SSH localhost was successfully run on 2022-11-14T12:02:34.812026
TASK: Set MGMT IP was successfully run on 2022-11-14T12:02:35.048829
TASK: Set Campus IP was successfully run on 2022-11-14T12:02:35.273364
TASK: Set OS FSP IP was successfully run on 2022-11-14T12:02:35.497673
TASK: Ping not tested was successfully run on 2022-11-14T12:02:35.502590

```

```
2022-11-14 12:02:35,508 INFO: All run tasks were successful
2022-11-14 12:02:35,509 INFO: All required tests were completed successfully
2022-11-14 12:02:35,509 INFO: CODE 20 completed.
```

A successful execution of the `ess_ssr_setup` tool displays the “CODE 20 completed.” message. The Install Complete app run (“Code 20”) must complete successfully to install the MSVM on an IBM Storage Scale cluster.

Note: To disconnect from the console session of the MSVM use ‘^J’ key combination on Linux systems. For windows systems, press **Ctrl - ^**.

4. After a successful completion of IP address setup by using the `ess_ssr_setup` tool, access the MSVM by using the SSH client.

Now, this MSVM is ready for usage.

5. To continue with the IBM Storage Scale System setup, refer the deployment process that is described in the *IBM Storage Scale System Deployment Guide* in IBM Documentation.

Configuring dual management server

This appendix describes the basic requirements, use cases, and configuration steps for the dual management server support.

Until IBM Storage Scale System 6.1.6.0, the deployment did not formally support two management server in the same cluster. Many customers had one of three environments.

- Single P9 or BYOMS node, which controlled a single cluster and single site.
- Single P9 or BYOMS node, which controlled a single cluster and dual sites. Both sites networked via a stretch cluster (all networks stretch between locations).
- Dual P9 management nodes, two locations. Only the high-speed network stretched between sites. The management networks stayed local per site. Thus, each management node could only see its local hardware.

The dual management server provides a live backup management server to become available when the primary goes down. Also, dual management server allows for multiple GUI services and collectors to become active. The GUI, in particular, allows CSI applications to stay active when one management server goes down.

Requirements

Ensure that following requirements met for the dual management server support:

- The upgraded version needs to be V5.1.6.1 (IBM Storage Scale System 6.1.6.0) or higher.
- Two P9 management server
- Single cluster
- All networks are flat, single namespace (IBM Storage Scale System stretch cluster requirements)
- All nodes (including two management server) have unique FQDN/IPs
- The `/etc/hosts` file copied from a primary to a backup management server (the same as `hosts.yml`)

Support statement

Only P9 management server supported (no BYOMS, no Power8, or no mixing)

Restrictions

- This is not an HA IBM Storage Scale System but an HA GUI/ CSI.
- Concept of primary/secondary management server.
- If a primary management server goes down, a secondary management server can be used to install a container and takeover the primary management server role

- Consider one management server as primary but run all **essrun** commands on another management server in the list. You cannot move back and fourth between primary and secondary management server (for deployment or for GUI usage).

Configuring dual management server

1. Connect network in the backup management server as a primary management server (Mgmt/FSP/Campus).
2. Add backup management server network hostnames to the /etc/hosts file in an existent cluster.

1. Connect network in the backup management server as primarily management server (Mgmt/FSP/Campus).

```
bkpEMS.MGMT.Network.IP backupEms.domain backupEms
bkpEMS.HS.Network.IP backupEms-hs.domain backupEms-hs
```

2. Update the backup management server from the container that is running on the primary management server.

```
essrun -N backupEms config load -p
essrun -N backupEms update --precheck
essrun -N backupEms update --offline
```

3. Create network bonds.

```
essrun -N backupEms network --suffix=-hs
```

4. Add the backup management server to the existent cluster from the container that is running in the primary management server.

```
essrun -N primarilyEms cluster --add-ems backupEms --suffix=-hs
```

5. Check if successful checks.

```
essrun -N backupEms config check
```

6. Add the backup management server to the GUI configuration.

```
essrun -N backupEms gui --configure
```

Updating nodes

Update nodes before an IBM Storage Scale System deployment or upgrade because the entire cluster depends on consistent OS, firmware, kernel modules, and networking behavior. If the underlying nodes are not healthy or aligned, deployment or upgrade cannot be successful.

Before you begin

Before you start deployment, check whether management servers, protocols, and I/O nodes on an IBM Storage Scale System need any updates, and update these nodes if required.

```
essrun -n emsvm,IOnodes update --precheck
```

About this task

Procedure

1. Update a management server.
 - To update the Power9 management server, issue the following command:

```
essrun -N <MS1> update --offline
```

- To update the Utility Node management server virtual machine (Utility Node VM), issue the following command:

```
essrun -N MSVM update --offline
```

Note: The Utility Node, a bare-metal host, does not need to be updated separately. Because this host is updated automatically when you update the Utility Node VM node on the host.

Use ssh to get back to the Utility Node VM when is up and running through the virtual bridge.

Note: The virtual bridge is not always the same; it varies from 10.23.16.2 to 10.23.16.6.

Alternatively, use ssh through the campus or use the management address that you assigned to the Utility Node VM.

If the kernel is changed, run the following commands:

- Exit the container.

```
exit
```

- Reboot the Power9 management server or Utility Node VM node.

```
systemctl reboot
```

- Navigate to the extracted IBM Storage Scale System 7.0.1.x directory, and then restart the container.

```
/home/deploy/ess_7.0.1.x_0611-23.tar.dir
```

```
./essmkym1 --restart
```

- Update management server or Utility Node VM nodes again.

```
essrun -N <MS1> update --offline
```

- After a successful updates, reboot the Power9 management server or Utility Node VM node.

```
systemctl reboot
```

- Update the I/O nodes from Power9 management server or Utility Node VM.

```
essrun -N <nodeclasses> update --offline
```

Where *nodeclasses* are as follows:

- IBM Storage Scale System 5000: ess_ppc64le
- IBM Storage Scale System 3000: ess_x86_64
- IBM Storage Scale System 3200: ess3200_x86_64
- IBM Storage Scale System 3500: ess3500_x86_64
- Protocol Power9: ces_ppc64le
- IBM Storage Scale System 6000: s6k_x86_64

Update all nodes of all building blocks except the Power9 management server or Utility Node VM.

- Update I/O and protocol nodes at once from Power9 management sever.

```
essrun -N prt01,prt02 update --offline
```

- If the protocol node is part of a GPFS cluster, run the **mmbuildgpl** command after the protocol node update is completed from the Power9 management server.

For more information, see [mmbuildgpl command](#) in IBM Storage Scale documentation.

Creating network bonds

Network bonding is needed to give IBM Storage Scale Systems nodes reliable and high-performance networking. Bonding combines multiple network interface cards (NICs) so that the cluster does not stop if a cable or port fails. You must create high-speed network bonds on I/O nodes and management server nodes before the cluster creation.

About this task

Note: When you create network bonds, use only the original hostname, `hostname -hs`, in the **essrun** commands.

Complete the following steps to create network bonds:

Procedure

1. Create network bonds I/O and management nodes.

- Create network bonds on I/O nodes.

```
essrun -N <IOnode1>,<IOnode2>,<IOnodeN>... network --suffix=-hs
```

- Create network bonds on management nodes.

```
essrun -N <MS1> network --suffix=-hs
```

- Create bonds on I/O and management server nodes at the same time.

```
essrun -N <IOnode1>,<IOnode2>,<MS1> network --suffix=-hs
```

2. Ensure that the network adapters are configured for the correct protocol, that is, InfiniBand or Ethernet. For more information about adapters protocols, see [“Configuring ConnectX-5, ConnectX-6, or ConnectX-7 adapter”](#) on page 37.
3. Optional: Run the network test.

Use the `nsdperf` tool to verify whether the newly created network bonds are healthy.

- a) Connect the container to an I/O node or a management server node.

```
ssh <IOnode1>  
ESSENV=TEST essnettest -N <IOnode1>,<IOnode2> --suffix=-hs
```

Note: After the container is connected to an I/O node, exit the container. When the test completes, type `exit` to return to the container.

4. Review test output.
 - Check for errors or warnings indicating dropped packets or threshold violations.
 - If InfiniBand is configured, the tool also runs an RDMA test to verify end-to-end communication.
 - Ensure that all tests complete successfully before cluster creation. For more information, see [Creating a cluster](#).

Creating a cluster

Create a cluster and add multiple nodes to provide shared storage.

Procedure

1. Create a cluster.

```
essrun -N <IOnodeclasses> cluster --suffix=-hs
```

Where the I/O node classes can be similar to the ones in the following examples:

- IBM Storage Scale System 3000: ess30001a and ess30001b
- IBM Storage Scale System 3500: ess35001a and ess35001b
- IBM Storage Scale System 6000: ess60001a and ess60001b

Tip: Use the optional parameters of the **essrun** command to customize your clusters.

For more information, see [essrun command](#) in the *IBM Storage Scale Systems Command Reference*.

2. Add a management server node to the cluster.

```
essrun -N <IOnode1> cluster --add-ems <MS1> --suffix=-hs
```

3. Create a file system.

```
essrun -N ess3500_x86_64 filesystem --name fs1 --suffix=-hs
```

Tip: Use the optional parameters of the **essrun** command to customize your file systems.

For more information, see [essrun command](#) in the *IBM Storage Scale Systems Command Reference*

Note:

- By default, this command attempts to use all the available space. If you need to create multiple file systems or a CES shared root file system for protocol nodes, consider by using less space. For example:

```
essrun -N <IOnodes> filesystem --suffix=-hs --size 80%
```

For more options such as blocksize, filesystem size, or RAID code, see the **essrun command** in the *IBM Storage Scale System Command Reference*.

- This step creates combined metadata + data vdisk sets by using a default RAID code and block size. You can use additional flags to customize or use the **mmvdisk** command directly for advanced configurations.
 - If you are updating IBM Storage Scale System 3000, IBM Storage Scale System 3200, and IBM Storage Scale System 3500 the default set-size is 80% and it must not be increased. If you are updating IBM Storage Scale System 5000, the default set-size is 100%. For additional options, see *essrun command*. The default block size for PPC64LE is 16M whereas for IBM Storage Scale System 3000 it is 4M.
 - If you are deploying protocol nodes, make sure that you leave space for CES shared root file system. Adjust the set-size slightly lower when you are creating this required file system for protocol nodes.
 - For IBM Storage Scale System 3500, you must keep 1.5 TB or more space free if future capacity MES is planned (performance to hybrid). Thus, it is recommended to not use all available space when you create a file system for the performance model. The default allocation is 80% of available space when you use the **essrun filesystem** command (for x86 nodes).
 - Before you create a recovery group or a file system, check whether a self-encrypted drive (SED) is required. You cannot enroll recovery groups for the SED support after they are created.
4. Add the protocol node to the cluster.
- For more information, see [“Creating a shared root file system for CES”](#) on page 92.
 5. Create a cesSharedRoot file system (if you are deploying protocols).

```
essrun -N <IOnode1>,<IOnode2> filesystem --ces
```

For more information, see [“Adding and configuring protocols in an exiting cluster by using IBM Storage Scale installation toolkit”](#) on page 96.

Adding nodes or building blocks

Additional nodes or building blocks in to an existing environment increase storage capacity of a cluster during the deployment or upgrade. You can add building blocks or nodes by using the **essrun** command or GUI.

Before you add nodes or building blocks, ensure that the following prerequisites are met:

- Power9 management server and one or more nodes are deployed in the cluster, which includes three or more quorum nodes.
- A server with two canisters is added without code upgrade in the existing management server
- The file system includes 4M block size, 8+2p RAID code, 80% default set-size.

Complete the following steps to add nodes to an existing cluster:

1. Edit the `/etc/hosts` file to add a building block or nodes and I/O node names to an existing management server including management and high-speed network.
2. Reinstall the deployed code image if it is not in management server, and run the container clean to bring the new `/etc/hosts` in the container.

```
essrun -N ems,essio1,essio2,essio3,essio4 config load -p
```

3. Check whether nodes need any updates..

```
essrun -N ems,essio1,essio2,essio3,essio4 update --precheck
```

4. Create network bonds for the added nodes.

```
essrun -N essio3,essio4 network --suffix=-hs
```

5. Outside the container, issue the **essnettest** command against the new nodes.

Run this command from one of the nodes in the cluster and use this command to test the health of the high-speed network connections. For more information, see the [essnettest](#) command.

6. Add new nodes to the existing cluster by issuing the following command from the container:

```
essrun -N ems cluster --add-nodes essio3,essio4 --suffix=-hs
```

Note: Add building blocks with the same node type at a time. For example:

```
essrun -N ems1 cluster --suffix=-hs --add-nodes ess3k1a,ess3k1b
```

Do not add building blocks with node type as follows:

```
essrun -N ems1 cluster --suffix=-hs --add-nodes ess3k1a,ess3k1b,ess5k1a,ess5k1b
```

7. Create node class and recovery groups by issuing the following command from the container:

```
essrun -N essio3,essio4 filesystem --rg-only --suffix=-hs
```

8. Get the list of recovery groups by issuing the following command from the system host:

```
mmvdisk rg list
```

9. Define the VDisk set.

Note: For IBM Storage Scale System 3500, you must keep 1.5 TB or more space free if future capacity MES is planned (performance to hybrid). Thus, it is recommended to not use all available space when you create a file system for the performance model. The default allocation is 80% of available space when you use the **essrun filesystem** command (for x86 nodes).

- a. Use the **ssh** command to connect to one of the new nodes.
- b. Define the VDisk set.

```
mmvdisk vs define --vs vs_fs1_essio3_hs_essio4_hs --rg ess3500_essio3_hs_essio4_hs
--code 8+2p --bs 4M --ss 80% --nsd-usage dataAndMetadata --sp system
```

Note: Assume that the existing file system has the attributes that are mentioned in the command. For VDisk set name, use a unique name similar to the existing VDisk set.

For more information, see [mmvdisk online command reference](#).

10. Add the VDisk set to an existing file system.

```
mmvdisk fs add --file-system fs1 --vdisk-set vs_fs1_essio3_hs_essio4_hs
```

Note: Assume that the original file system name is *fs1*.

11. Update the component database on the management server by issuing the following command from system host:

```
mmaddcompspec default --replace
```

12. Restripe the file system.

```
mmrestripefs fs1 -b
```

13. Add new nodes to the GUI for monitoring.

```
mmchnode --perfmon -N essio5-hs,essio6-hs
```

To monitor new nodes from the GUI, complete following steps:

Adding nodes to the GUI for monitoring

1. From the container, issue the following command to configure and start the performance monitoring sensors.

```
cd /home/deploy/ess_7.0.1.x_0604-21_dme_ppc64le.dir
./essmkym1
essrun -N essio3,essio4 gui --add-to-hosts
```

Note: Ensure that the passwordless SSH is configured from new nodes to the management server.

2. Check whether GUI web page shows new IBM Storage Scale System nodes.
3. Add the new nodes as sensors, reconfigure call home, and rerun the GUI wizard to identify the new nodes.

Adding AFM gateway nodes to a cluster by using the installation toolkit

You can add Active File Management (AFM) gateway nodes to an existing IBM Storage Scale cluster. To add an AFM gateway node, you need to install the necessary software on the installer node, designate gateway nodes, populate configuration, validate setup, run pre-installation checks, and finalize license and gateway node assignments.

Before you begin

Ensure that all prerequisites are met.

-
- Planning for AFM gateway node, caching, and AFM to cloud object storage in an IBM Storage Scale System. For more information, see [IBM Storage Scale documentation](#).
 - The *General guidelines and recommendations for AFM-DR* section in the *IBM Storage Scale: Concepts, Planning, and Installation Guide*
 - The *General recommendations for AFM gateway node configuration* section in the *IBM Storage Scale: Concepts, Planning, and Installation Guide*

- Planning for adding an AFM gateway node. For more information, see [IBM Storage Scale documentation](#).

Install IBM Storage Scale by using the installation toolkit and set up AFM gateway node. Use the IBM Storage Scale documentation for installing IBM Storage Scale by using the installation toolkit and for enabling the required services for the customer environment.

- The *Using the installation toolkit to perform installation tasks: Explanations and examples* section in the *IBM Storage Scale: Concepts, Planning, and Installation Guide*
- The *Installing IBM Storage Scale on Linux nodes with the installation toolkit* section in the *IBM Storage Scale: Concepts, Planning, and Installation Guide*
- The *ESS awareness with the installation toolkit* section in the *IBM Storage Scale: Concepts, Planning, and Installation Guide*

Procedure

1. Extract and install the IBM Storage Scale installation package on the installer node.

```
tar -xvzf ./Storage_Scale_DME_522_x86_64_LNX.tar.gz
```

2. Designate an AFM gateway node as an installer node.

- a) Specify an added protocol node as an installer node in the directory.

```
/usr/lpp/mmfs/5.2.2.0/ansible-toolkit
```

- b) Specify the setup.

```
./spectrumscale setup -s InstallerNodeIP -st ess
```

- c) Populate the existing IBM Storage Scale System configuration into the toolkit configuration file. In IBM Storage Scale System cluster, you must specify a management server node.

```
./spectrumscale config populate --node EMSNode
```

- d) Define AFM gateway nodes in the cluster.

```
./spectrumscale node add Node1
```

3. Add nodes to an existing cluster and check the configuration.

```
./spectrumscale node list
```

4. If the configuration correct, check the pre-installation environment.

```
./spectrumscale install --precheck
```

5. Add nodes to the cluster.

```
./spectrumscale install
```

6. After successful cluster installation, verify the cluster.

```
/usr/lpp/mmfs/bin/mmlscluster
```

7. Assign gateways nodes as server licenses.

```
mmchlicense server --accept -N Node1,Node2,...
```

8. Designate AFM gateways to the list of newly added gateway nodes.

```
mmchnode --gateway -N Node1,Node2,...
```

9. Configure the AFM cache.

For more information, see [IBM Storage Scale documentation](#).

- The *Creating an AFM relationship by using the NFS protocol* section in the *IBM Storage Scale: Administration Guide*
 - The *Creating an AFM relationship by using GPFS protocol* section in the *IBM Storage Scale: Administration Guide*
 - The *Configuring Active File Management* section in the *IBM Storage Scale: Administration Guide*
10. Configure the AFM to cloud object storage.
For more information, see [IBM Storage Scale documentation](#).
- The *Configuring AFM to cloud object storage* section in the *IBM Storage Scale: Administration Guide*
 - The *Administering AFM to cloud object storage* section in the *IBM Storage Scale: Administration Guide*

Adding an AFM gateway node to clusters by using the `essrun` command

Add Active File Management (AFM) gateway nodes to move data between the local cluster and remote sites. Configure these nodes to provide caching, replication, and data transfer in IBM Storage Scale System clusters.

About this task

From IBM Storage Scale System 7.0.0.2, use the **essrun** command. This a preferred method that automates, validates, and standardizes the AFM gateway node deployment.

Procedure

1. Ensure that a Cluster Export Services virtual machine (CES VM) is created. If a CES VM is not created, create it. For more information, see [Creating a CES VM](#).
2. Add the CES VM as an AFM gateway node to the IBM Storage Scale System cluster.

```
essrun -N ems_vm cluster --add-afmgateway-nodes node_name
```

Where:

ems_vm

Specifies a management server virtual machine (MSVM), which runs on the Utility Node, of the IBM Storage Scale System.

node_name

Specifies an AFM gateway node.

Example:

```
essrun -N emsgd12 cluster --add-afmgateway-nodes ess3500rw2b-essvm
```

Configuring a mixed cluster and file system

Use these instructions for setting up IBM Storage Scale System 3000 and IBM Storage Scale System 5000 mixed cluster and file system.

The following high-level tasks need to be done for setting up IBM Storage Scale System 3000 and IBM Storage Scale System 5000 mixed cluster:

- Deploy an IBM Storage Scale System 3000 system (including cluster, file system, GUI).
- Deploy an IBM Storage Scale System 5000 system (adding to cluster, create recovery groups, and so on).
- Create the IBM Storage Scale System 5000 vdisks and add to the existing file system.
- Create a policy file.
- Adjust sensors.
- Add IBM Storage Scale System 5000 nodes to the GUI.

Note: These instructions contain summarized steps and references to documents that cover the items in more detail. The goal is to give an example scenario to help clients understand aspects of this procedure. At the end of this procedure, if you have Power9 protocol nodes, for guidance in implementing them into your environment, see [“Creating a shared root file system for CES” on page 92.](#)

Prerequisites

- SSR has completed Code 20 on both the IBM Storage Scale System 3000 and IBM Storage Scale System 5000 nodes (including management server)

SSR works on Power® nodes and the management server node first, then the IBM Storage Scale System 3000 system.

- Public connection setup on C11-T3 (f3 connection on management server)
- IBM Storage Scale System 3000 and IBM Storage Scale System 5000 nodes are added to the `/etc/hosts` file.
 - Low-speed names FQDNs , short names, and IP addresses
 - High-speed names FQDNs, short names, and IP addresses (add suffix of low-speed names)
- Host name and domain set on management server.
- Latest code for IBM Storage Scale System 3000 and IBM Storage Scale System 5000 stored in the `/home/deploy` on the management server.
- For information about how to deploy the IBM Storage Scale System system, see [IBM Storage Scale System: Quick Deployment Guide](#).
- For information about how to use the **mmvdisk** command, see [mmvdisk in IBM Storage Scale System documentation](#).

Adding IBM Storage Scale System 3000 building blocks

1. Extract the IBM Storage Scale System 3000 installation package.

```
tar zxvf ESS3000InstallationPackage
```

2. Accept the license and deploy the container.

```
./ess_7.0.1.x_0604-21_dme_ppc64le
```

After logging in to the container, complete the following steps:

1. Run the config load command.

```
essrun -N ESS3000Node1,ESS3000Node2,EMSNode config load -p
```

Note: Use the low-speed names.

2. If required, update the management server node.

```
essrun -N EMSNode update --offline
```

3. Update the IBM Storage Scale System 3000 nodes.

```
essrun -N ESS3000Node1,ESS3000Node2 update --offline
```

4. Create network bonds.

```
essrun -N ESS3000Node1,ESS3000Node2,EMSNode network --suffix=Suffix
```

5. Create a cluster.

```
essrun -N ESS3000NodeGroup cluster --suffix=Suffix
```

Note: To obtain the group name, use **-t group**.

6. Add the management server node to the cluster.

```
essrun -N ESS3000Node1 cluster --suffix=Suffix --add-ems EMSNode
```

7. Create a file system.

```
essrun -N ESS3000NodeGroup filesystem --suffix=Suffix
```

Note: This command creates a combined data and metadata vdisk in the system pool. The file system name must be `fs3k`.

Type `exit` and press `Enter` to exit the container. Proceed with the instructions on how to setup the collector, sensors, and run the GUI wizard.

The status of the current IBM Storage Scale System 3000 container should be exited. To confirm, use the **podman ps -a** command. For example:

```
# podman container attach cems0
ESS 3000 CONTAINER root@cems0:/ # exit
exit
# podman ps -a
CONTAINER ID   IMAGE                                COMMAND                                CREATED
STATUS        PORTS   NAMES
bfc2980bdfc9   localhost/ess_7.0.1.x_dme_xcat:0604-21 /myStartupScript... 7 days ago Exited
(0) 2 seconds ago                cems0
```

If the IBM Storage Scale System 3000 container is not in the stopped state, use the **podman stop ContainerName** command.

Adding IBM Storage Scale System 5000

1. Extract the IBM Storage Scale System 5000 installation package.

```
tar xvf ESS5000InstallationPackage
```

2. Verify the installation package.

```
sha256sum -c Extractedsha256sumFile
```

3. Accept the license and deploy the container

```
./ess_7.0.1.x_0604-21_dme_ppc64le7.0.1.x
```

After you have logged into the container, complete the following steps:

1. Run the config load command.

```
essrun -N ESS5000Node1,ESS5000Node2,ESS3000Node1,ESS3000Node2,EMSNode config load -p
```

Note: If you plan to add protocol nodes in the cluster, include them in the list of nodes that you are specifying in this command.

2. Update the nodes.

```
essrun -N ESS5000Node1,ESS5000Node2 update --offline
```

3. Create network bonds.

```
essrun -N ESS5000Node1,ESS5000Node2 network --suffix=Suffix
```

4. Add the IBM Storage Scale System 5000 nodes to an existing cluster.

- a. Use SSH to connect to one of the IBM Storage Scale System 5000 I/O server nodes. For example:

```
ssh ESS5000Node1
```

- b. Accept the license. Use high-speed names.

```
essaddnode -N ESS5000Node1-hs,ESS5000Node2-hs --cluster-node ESS3000Node-hs --nodetype
ess5k --accept-license
```

Where:

ESS5000Node1-hs,ESS5000Node2-hs, ESS3000Node-hs

Are high-speed names.

--nodetype

You can add the following nodes to any environment:

- For management server node, use *ems*.
- For IBM Storage Scale System 3000, use *ess3k*.
- For IBM Storage Scale System 5000, use *ess5k*.
- For IBM Storage Scale System 3200, use *ess3200*.
- Default is *ems*.

If there is an error, you might need to log in to each IBM Storage Scale System 5000 node and start GPFS.

```
mmbuildgpl mmstartup
```

Type `exit` and press Enter to exit the container. Running these commands, takes you to the IBM Storage Scale System 5000 node.

5. Create **mmvdisk** artifacts.

a. Create a node class.

```
mmvdisk nc create --node-class ess5k_ppc64le_mmvdisk -N
ListOfESS5000Nodes_highspeedsuffix
```

b. Configure the node class.

```
mmvdisk server configure --nc ess5k_ppc64le_mmvdisk --recycle one
```

c. Create recovery groups.

```
mmvdisk rg create --rg ess5k_rg1,ess5k_rg2 --nc ess5k_ppc64le_mmvdisk
```

d. Define vdiskset.

```
mmvdisk vs define --vs vs_fs5k_1 --rg ess5k_rg1,ess5k_rg2 code 8+2p --bs 16M --ss 80%
--nsd-usage dataOnly --sp data
```

e. Create a vdiskset.

```
mmvdisk vs create --vs vs_fs5k_1
```

f. Add the vdiskset to the file system.

```
mmvdisk fs add --file-system fs3k --vdisk-set vs_fs5k_1
```

g. Add the policy file.

Define your policy file. This can be used to move data from the system pool to the data pool when thresholds reached. For more information, see [Overview of policies](#).

You can also use the GUI to define policies. For more information, see [Creating and applying ILM policy by using GUI](#).

The following example rule ingests the writes on the IBM Storage Scale System 3000 and moves the data to IBM Storage Scale System 5000 when it reaches 75% capacity on the IBM Storage Scale System 3000:

- Add callback for automatic movement of data between pools:

```
mmaddcallback MIGRATION --command /usr/lpp/mmfs/bin/mmstartpolicy --event  
lowDiskSpace,noDiskSpace --parms "%eventName %fsName"
```

- Write the policy into a file with the following content:

```
RULE 'clean_system' MIGRATE FROM POOL 'system' THRESHOLD(75,25) WEIGHT(KB_ALLOCATED)  
TO POOL 'data'
```

Note: You need to understand the implications of this rule before applying it in your system. When capacity on IBM Storage Scale System 3000 reaches 75%, it migrates files (larger ones first) out of the system pool to the data pool until the capacity reaches 25%.

- h. On the management server node, run the following command.

```
mmaddcompspec default --replace
```

- i. Add the IBM Storage Scale System 5000 nodes to the pmsensors list and use the **Edit rack components** option in the GUI to slot the new nodes into the frame.
- j. If you want to add protocol nodes, see [“Creating a shared root file system for CES” on page 92](#).

Configuring IBM Storage Scale native REST API on IBM Storage Scale Systems cluster nodes

The `gpfs.scaleapi` installation enables secure certificate-based communication through the native REST API. You need to prepare all nodes for REST-based cluster management.

Procedure

1. **Note:** You must run the **essrun** command from the IBM Storage Scale Systems container.

Install the `gpfs.scaleapi` RPM on all IBM Storage Scale Systems nodes.

- a) Configure the repositories for `gpfs.scaleapi` on all IBM Storage Scale Systems nodes.

```
essrun -N ess_nodes config repos
```

Replace `ess_nodes` with a comma-separated list of all IBM Storage Scale Systems node names.

- b) Install the `gpfs.scaleapi` RPM on all IBM Storage Scale Systems nodes.

```
yum install gpfs.scaleapi
```

Note: Ensure that the version of the IBM Storage Scale System container that you want to install matches with the deployed version on the existing nodes. The `scaleadmd` daemon is installed on the cluster for the native REST API calls. Verify whether the daemon runs on the cluster.

```
systemctl status scaleadmd
```

For more information about the RPM installation, see [Manually installing IBM Storage Scale native REST API with the RPM](#).

2. Install on non-IBM Storage Scale System nodes (client or protocol nodes).

If IBM Storage Scale System nodes are attached with client or protocol nodes, install `gpfs.scaleapi` on non-IBM Storage Scale System nodes by using the manual installation or the **essrun** command:

Manual installation

Install `gpfs.scaleapi` manually on all non-IBM Storage Scale System nodes. For more information about the RPM installation, see [Manually installing IBM Storage Scale native REST API with the RPM](#).

The **essrun** command

Install `gpfs.scaleapi` by using the **essrun** command.

```
ALLOW_SCALEADM_PACKAGE_INSTALL=true; essrun -N emsnode cluster --add-cesvm-nodes  
listofcesvmnode
```

Where:

emsnode

Specifies a management node.

listofcesvmnode

Comma-separated list of CES VM hostnames to add to the cluster.

For example,

```
export ALLOW_SCALEADM_PACKAGE_INSTALL=true; essrun -N emsnode cluster --add-cesvm-  
nodes cesvm01,cesvm02
```

3. Generate or import a node identity certificate. For more information about generating a certificate, see [Node identities](#).

To import the generated node identity, run the following command on each node (IBM Storage Scale System and non-IBM Storage Scale System):

```
scalectl node config set --cert path to certificate --key path to private key --chain path  
to ca chain
```

For example,

```
scalectl node config set --cert /root/tls/server.pem --key /root/tls/server.key --chain /  
root/tls/ca.crt
```

4. Migrate an existing IBM Storage Scale System cluster from the mm-commands CLI to the use of native REST API. For more information about the migration of an existing cluster, see [Migration](#).
 - a) Validate each node and the cluster requirements before you make any updates during the migration from one of the quorum node.

```
scalectl cluster migrate --precheck
```

- b) Migrate from one of the quorum node.

```
scalectl cluster migrate
```

Note: Restart the I/O daemons on each node in the cluster managed that is managed by the native REST API to ensure proper communication with the newly migrated administration daemons.

5. Replace SSH and SCP with the native REST API. For more information, see [SSH and SCP replacements with IBM Storage Scale native REST API](#).
6. Validate and view the SSH/SCP wrapper.

```
scalectl cluster list
```

For more information on the native Rest API, see [IBM Storage Scale native REST API](#).

Setting up Cluster Export Services

By using the IBM Storage Scale installation toolkit, or the **essrun** command, you can set up Cluster Export Services (CES) on an IBM Storage Scale System to add and configure protocols on cluster nodes.

Note: Starting with IBM Storage Scale System 7.0.0.2, the recommended approach for the protocol deployment is to use the **essrun** command.

Creating a CES VM by using the **essrun** command

From IBM Storage Scale System 7.0.0.2, use the **essrun** command to create a Cluster Export Services virtual machine (CES VM) in an IBM Storage Scale System.

About this task

Note: The protocol VM creation is not supported on IBM Storage Scale System 6000 nodes.

Complete the following steps from the IBM Storage Scale System container:

Procedure

1. Run configuration load against the protocol node host.

```
essrun -N <protnode_host> config load -p
```

2. Check the protocol node host.

```
essrun -N <protnode_host> update --precheck
```

3. Update the protocol node host.

```
essrun -N <protnode_host> update
```

4. Create a protocol node VM.

```
essrun -N <protnode_host> cesvm --create --vm-name prtvm1
```

5. Log in to the protocol node host, and issue the following commands:

```
virsh list --all  
virsh console <protocol VM domain>
```

Domain is the output of the **virsh list** command:

6. Log in to the protocol VM (root/cluster) and change the password when prompted.
7. Run the **ess_ssr_setup** script to set up a management IP.
8. Log in to the management server virtual machine (MSVM), which runs on the Utility Node, again.
9. Run configuration load against the protocol node guest VM.

```
essrun -N <protnode_protnodeVM1> config load -p
```

10. Run precheck against the protocol node guest VM.

```
essrun -N <protnode_protnodeVM1> update --precheck
```

11. Run the following command to update protocol nodes in the offline mode.

```
essrun -N <protnode_protnodeVM1> update --offline
```

12. Create a high-speed network for the protocol node guest VM.

```
essrun -N <protnode_protnodeVM1> network --suffix =-hs
```

Creating a shared root file system for CES

A shared root file system must be created for the CES by using the IBM Storage Scale installation toolkit or the **essrun** command. This file system is needed for the CES because all protocol nodes must access the same configuration, state, and protocol-related data to provide consistent services.

Note: The following instructions for protocol node deployment by using the installation toolkit is an example. For detailed and latest information, see the following topics.

- [Installing IBM Storage Scale on Linux nodes with the installation toolkit](#)

- [Configuring the CES and protocols](#)

Prerequisites

During file system creation, adequate space is available for CES shared root file system

In a default IBM Storage Scale System setup, you can use the Ansible® based file system task to create the recovery groups, vdisk sets, and file system. By default, during this task, 100% of the available space is attempted to be consumed. If you plan to include protocol nodes in your setup, you must leave enough free space for the required CES shared root file system. Use the **--size** flag to adjust the space consumed accordingly.

For example: **essrun -N ess_ppc64le filesystem --suffix=-hs --size 80%**

Running this command leaves approximately 20% space available for the CES shared root file system or additional vdisks. If you are in a mixed storage environment, you might not use the **essrun filesystem** task due to more complex storage pool requirements. In that case, when using **mmvdisk**, make sure that you leave adequate space for the CES shared root file system. The CES shared root file system requires around 20 GB of space for operation.

Note: When you are updating IBM Storage Scale CES on an IBM Storage Scale System protocol node, use the toolkit code to update IBM Storage Scale and then update the node from the deployment container (such as MOFED, Kernel). This practice ensures whether IBM Storage Scale is compatible with the underlying operating system.

IBM Storage Scale System container has the protocol node management IP addresses defined

Before running the IBM Storage Scale System container, add the protocol node management IP addresses to the /etc/hosts file. These IP addresses are given to the SSR through the TDA process and they are already set. The customer needs to define host names and add the IP addresses to the management server node /etc/hosts file before running the container.

You also need to define the high-speed IP address and host names. The IP addresses get set when running the Ansible network bonding task but the host names and IP addresses must be defined in the /etc/hosts file before the container starts. The high-speed hostnames must add a suffix of the management names. The IP addresses are user definable. Consult the network administrator for guidance.

For example:

```
Protocol management IPs
```

A sample output is as follows:

```
192.168.45.23 prt1.localdomain prt1
192.168.45.24 prt2.localdomain prt2
```

```
Protocol high-speed IPs
```

A sample output is as follows:

```
11.0.0.4 pr1-hs.localdomain prt1-hs
11.0.0.5 pr2-hs.localdomain prt2-hs
```

where, localdomain is an example domain. The domain must be changed and also match that of the other nodes.

IBM Storage Scale System container has the CES IP addresses defined

The final item that must be defined before starting the IBM Storage Scale System container are the CES IP addresses. The following example shows the usage of two IP addresses per node over the high-speed network. Consult the IBM Storage Scale documentation for best practices.

```
11.0.0.100 prt_ces1.localdomain prt_ces1
11.0.0.101 prt_ces2.localdomain prt_ces2
11.0.0.102 prt_ces3.localdomain prt_ces3
11.0.0.103 prt_ces4.localdomain prt_ces4
```

Starting state in the example scenario

- IBM Storage Scale System storage is deployed and configured.
- Adequate space (approximately 20 GB) is available for CES shared root file system.
- Protocol node required host names and IP addresses is defined on the management server prior to starting the container.
- You are logged in from the IBM Storage Scale System container.

Adding protocol nodes in an IBM Storage Scale System

Complete the following steps from the IBM Storage Scale System container.

1. Ping the management IP addresses of the protocol nodes.

```
ping IPAdress1,...IPAdressN
```

Where, *IPAdress1,...IPAdressN* specifies IP addresses of the protocol nodes.

Each protocol node must respond to the ping test indicating they have an IP address set and it is on the same subnet as the container.

2. Run the config load task.

```
essrun -N MS1,essio1,essio2, prt1,prt2 config load -p
```

Where:

MS1

Specifies a management server virtual machine (MSVM), which runs on the Utility Node.

essio1,essio2

Specifies I/O nodes.

prt1,prt2

Specifies protocol nodes.

If you have more than one node, you can specify them in a comma-separated list. Make sure that you add all IBM Storage Scale System nodes in this **config load** command before continuing.

3. Create network bonds.

Note: Make sure that the nodes are connected to the high-speed switch before doing this step.

```
essrun -N prt1,prt2 network --suffix=-hs
```

4. Install the CES shared root file system.

```
essrun -N ess_ppc64le filesystem --suffix=-hs --ces
```

5. Install IBM Storage Scale by using the installation toolkit and set up CES.

Use the IBM Storage Scale documentation for installing IBM Storage Scale by using the installation toolkit and for enabling the required services for the customer environment.

You can obtain the IBM Storage Scale installation toolkit from IBM Fix central. For example,

```
Storage_Scale_Data_Management-6.0.0.1-x86_64-Linux-install
```

(data management edition example, you must download data access if you have that edition)

For more information, see:

- [Using the installation toolkit to perform installation tasks: Explanations and examples.](#)
- [Adding CES HDFS nodes into the centralized file system.](#)
- [ESS awareness with the installation toolkit.](#)
- [Using the installation toolkit to perform installation tasks: Explanations and examples](#)
- [Installing IBM Storage Scale on Linux nodes and deploying protocols](#)
- [Deploying protocols](#)
- [Upgrading IBM Storage Scale protocol nodes](#)

Adding and configuring protocols in an existing cluster by using the **essrun** command

Create a Cluster Export Service virtual machine (CES VM) and configure CES by using the **essrun** command from IBM Storage Scale System 7.0.0.2. When you use the **essrun** command, you do not need the IBM Storage Scale installation toolkit to deploy protocols, such as NFS, SMB, or S3.

Before you begin

Ensure that you have set up CES on an IBM Storage Scale System by using the **essrun** command. For more information, see [“Creating a CES VM by using the essrun command”](#) on page 92.

About this task

To add protocol nodes, complete the following steps:

Procedure

1. Add a CES virtual machine (CES VM) nodes to an IBM Storage Scale System cluster.

```
essrun -N emsnode cluster --add-cesvm-nodes listofcesvmnode
```

where:

emsnode

Specifies a management server virtual machine (MSVM), which runs on the Utility Node.

listofcesvmnode

Specifies comma-separated list of CES VM hostnames that need to be added to the cluster.

```
essrun -N emsnode cluster --add-cesvm-nodes cesvm01,cesvm02
```

Where, **cesvm01,cesvm02** specify the CES VMs hostnames.

You can add a CES VM node as an AFM gateway to a cluster.

```
essrun -N emsnode cluster --add-afmgateway-nodes listofcesvmnode
```

2. Check all the **essrun** command options to install and configure protocols on CES VM.

```
essrun protocol --help
```

For more information, see https://ibmdocs-test.dcs.ibm.com/docs/en/ESS_test?topic=commands-essrun-command.

3. Install and configure protocols services on CES VMs.

```
essrun -N cesvm01,cesvm02 protocol configure --enable s3 nfs smb --filesystem cesSharedRoot
--export-ip-pool 172.16.45.66
```

where:

cesvm01,cesvm02

Specify CES VMs hostnames.

s3 NFS smb

Specifies protocols that need to be configured.

cesSharedRoot

Specifies a shared root file system for CES.

Note: If an existing Utility Node VM is configured as a protocol node and is running the protocol service, do not specify a shared root file system and export IP address when you add another Utility Node VM as a protocol node.

```
essrun -N cesvm03 protocol configure --enable nfs smb s3
```

Adding and configuring protocols in an exiting cluster by using IBM Storage Scale installation toolkit

Use the IBM Storage Scale installation toolkit to set up the Cluster Export Services (CES) in an IBM Storage Scale System.

About this task

Complete the following steps from the management server virtual machine (MSVM), which runs on the Utility Node. This procedure configures CES and enables NFS, SMB, and S3 protocols on protocol nodes in an existing cluster.

Important: Run all commands from the MSVM, not from protocol or container nodes.

Procedure

1. Access a protocol VM by using SSH, and set a single CES IP.

```
nmcli connection add type ethernet con-name eth5 ifname eth5 ipv4.addr 192.168.145.206/16
ipv4.method manual connection.autoconnect yes
```

Replace the IP address and subnet mask with values appropriate for your environment.

2. Copy IBM Storage Scale installation toolkit (with a correct edition) to the MSVM (the management server virtual machine, not the Protocol or container) and extract the package.

```
./Storage_Scale_Data_Management-7.0.1.0-x86_64-Linux-install --text-only
```

If the OS-level Python version is earlier than 3.8, upgrade Python to 3.8 and later version, and then run the installation toolkit.

3. Specify an installer node by issuing the following command:

```
./spectrumscale setup -s 192.168.1.40 -st ess (EMSVM high speed IP)
```

where:

-s

Specifies the EMSVM high-speed IP address.

-st

Specifies the system type (ess for IBM Storage Scale Systems).

4. Populate the cluster definition file with the current cluster state by issuing the following command:

```
./spectrumscale config populate -N 192.168.1.40 (EMSVM high speed IP)
```

5. Add protocol nodes in your environment to use for the deployment by issuing the following command:

```
./spectrumscale node add 192.168.1.41 -p -n (protocol high speed IP)
```

where:

-p

Designates the node as a protocol node.

-n

Specifies the node designation.

Replace *192.168.1.41* with your protocol node high-speed IP address.

6. Configure protocols to point to a file system, which will be used as a shared root by issuing the following command:

```
./spectrumscale config protocols -f cesSharedRoot -m /gpfs/cesSharedRoot
```

where:

-f

Specifies the file system name to be used as the CES shared root.

-m

Specifies the mount path for the shared root file system.

7. Enable NFS on all protocol nodes by issuing the following command:

```
./spectrumscale enable nfs
```

8. Enable SMB on all protocol nodes by issuing the following command:

```
./spectrumscale enable smb
```

9. Enable S3 on all protocol nodes by issuing the following command:

```
./spectrumscale enable S3
```

Note: S3 protocol support is available on IBM Storage Scale 5.2.1.0 or later.

10. Configure a pool of export IPs by issuing the following command:

```
./spectrumscale config protocols -e 192.168.20.204/24,192.168.20.205/24 -i eth2,eth3.  
(amount of IPS and interfaces are up to user)
```

where:

-e

Specifies a comma-separated list of export IP addresses with subnet masks.

-i

Specifies a comma-separated list of network interfaces.

The number of IP addresses and interfaces depends on your environment requirements.

11. Review the configuration before the deployment by issuing the following command. Verify that all protocol nodes are listed correctly and that the enabled protocols are displayed.

```
./spectrumscale node list
```

12. Install IBM Storage Scale on the new protocol VM nodes by issuing the following command:

a) Run the pre-installation check.

```
./spectrumscale install --pr
```

b) Install the software.

```
./spectrumscale install
```

13. Deploy protocols on your defined environment by issuing the following command:

a) Run the pre-deployment check.

```
./spectrumscale deploy --precheck
```

b) Deploy the protocols.

```
./spectrumscale deploy
```

What to do next

After deployment completes, verify the CES configuration by using the following command:

```
/usr/lpp/mmfs/bin/mmlscluster --ces
```

You can rerun the **./spectrumscale deploy** command to add more protocol nodes or enable additional protocols as needed.

Deploying a protocol virtual machine on IBM Storage Scale System 3500 I/O nodes

Until IBM Storage Scale System 6.1.2, the deployment of protocol nodes was supported through Power 64 LE nodes. However, from IBM Storage Scale System 6.1.3.1, the protocol nodes deployment is moved on IBM Storage Scale System 3500 nodes to reduce the initial deployment cost and support the protocol deployment in the IBM Storage Scale System 3500 environment.

From IBM Storage Scale System 6.1.3.1, you can run a virtual machine (VM) on an IBM Storage Scale System 3500 I/O node canister to support protocol services such as NFS, S3, and SMB, which are enabled for an IBM Storage Scale System cluster.

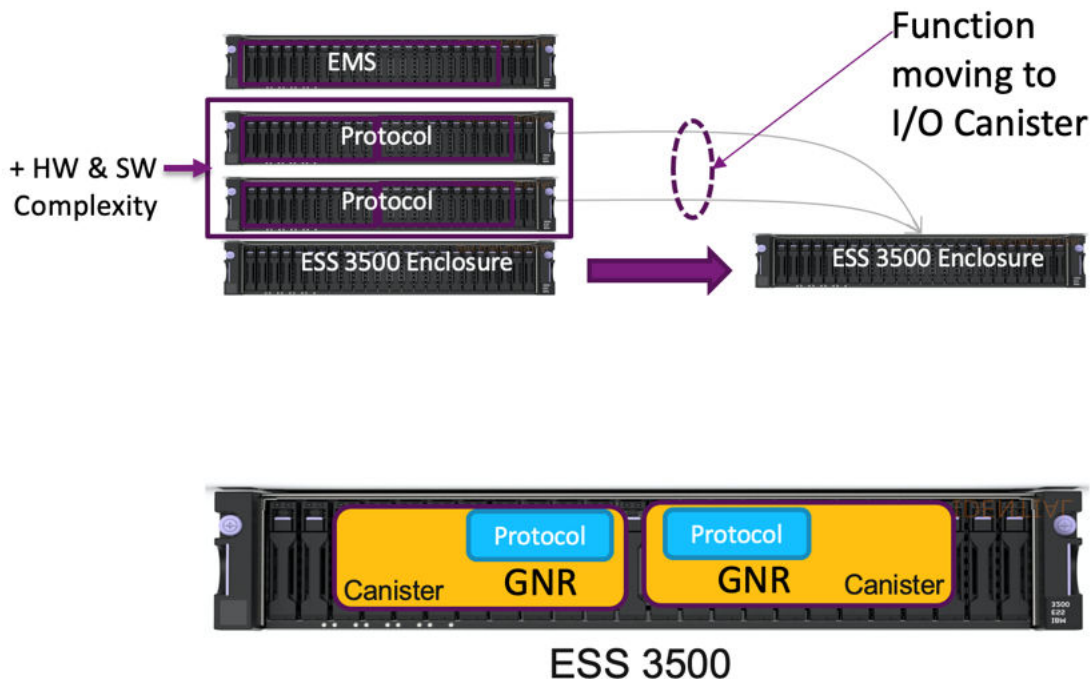


Figure 22. Protocol VM deployment on IBM Storage Scale System 3500 I/O nodes

Hardware specifications and requirements

The hardware requirements to run protocol nodes as a VM on an I/O node are as follows:

- Features
 - Optimized for entry configuration
 - Hardware simplification and cost reduction
 - Eliminates the need for a dedicated protocol node
 - Supports only NFS, S3, and SMB
- Management
 - Through a bridged network from the host management network
 - No BMC network
 - Fixed VM configuration
 - VM creation and destruction handled through script or Ansible
- Canister
 - One VM per canister
 - 8 CPU cores allocated to the VM (from a total of 48 cores: 1 socket, 8 threads)
 - 64 GB memory allocated to the VM (from a total of 512 GB)
- Supports up to 8 building blocks
- Client connections
 - NFS: up to 1000 connections
 - SMB: up to 512 connections
- 195710 MiB boot drive mapped to a dedicated mirrored partition on the host.
- Network
 - One or two network adapters passed through via PCIe Passthrough
 - Adapters are not shared with the host (each has its own IOMMU group)

Protocol VM implementation

The deployment of protocol binaries and services remains the same by using the IBM Storage Scale install toolkit.

Ensure that the following prerequisites are met before you deploy and use a protocol VM in the IBM Storage Scale System 3500 environment:

- IBM Storage Scale System 3500 nodes are completely deployed.
- A CES shared root file system is created.

See the *IBM Storage Scale System Quick Deployment Guide* to deploy and create a CES shared root file system in the IBM Storage Scale System 3500 or any IBM Storage Scale System environment.

In the **essrun** command, an argument **cesvm** is introduced. The **cesvm** argument accepts the node name as a requirement for the **--create** and **--delete** options with several physical Mellanox cards, which the protocol VM uses.

Note: Only one protocol VM can be run on a single I/O node canister. Therefore, for one building block of IBM Storage Scale System 3500, you can run two VMs per canister node.

The protocol VM has limited CPU and memory resources and does not support highly scalable NFS, S3, or SMB workloads. Use this implementation for small-scale or limited workload scenarios.

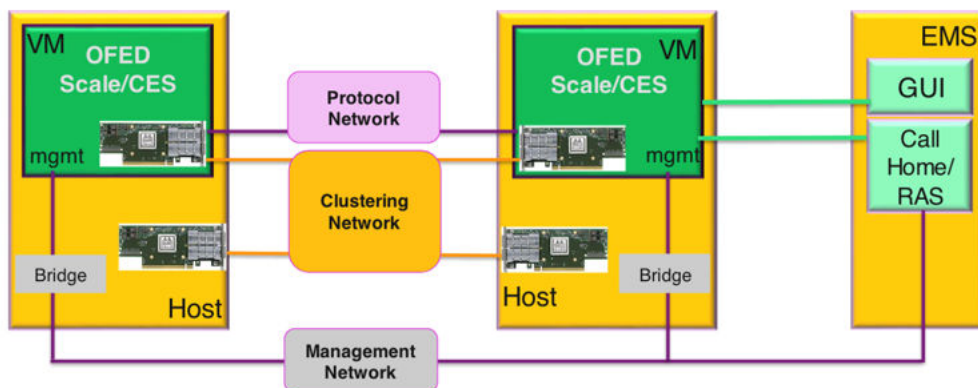
You cannot change any settings on the KVM definition and IBM Storage Scale settings that affect the protocol nodes VM. For example, you cannot change pagepool, files to cache, CES NFS settings, KVM XML, unless IBM support directs.

Tip: If you want to run a massive workload by using an NFS protocol and/or an SMB protocol, you must use dedicated protocol nodes instead a protocol VM.

Network requirements

Before you set up the protocol VM, you must decide on its network aspect. Usually, IBM Storage Scale System 3500 nodes come with two or three number of, two ports per adapter, Mellanox Ethernet or Mellanox InfiniBand cards.

Figure 23. Protocol in a VM - logical network



The following figure contains slot mapping details:

Slot Number	Usage	VM Usage when Configured
Slot 1 – Network	Network: CX6 or CX7 (Dual port)	CX6 – host IOMMU group
Slot 2 – Network or Aux Card or SAS	Network: CX6 or SAS: 9500-16e	CX6 – host or guest IOMMU group SAS – host IOMMU group
Slot 3 – Network	Network: CX6 or CX7 (Dual port)	CX6 – guest IOMMU group
Slot 4 – Network (Aux card or CX6) or SAS (Note 1)	Network: CX6 or SAS: 9500-16e	SAS – host IOMMU group CX6 – host or guest IOMMU group

Network: CX6 or CX7
 Network or SAS: CX6 or SAS
 Network: CX6 or CX7
 Network or SAS: CX6 or SAS

Figure 24. IBM Storage Scale System 3500 slot placement summary

To run a protocol VM, you must assign one of the Mellanox network adapters (2 ports) to the VM that provides the following connections:

- CES (first port of the network adapter) connection
- GPFS admin or daemon (second port of the network adapter) connection

In the following figures, two types of network connections are shown:

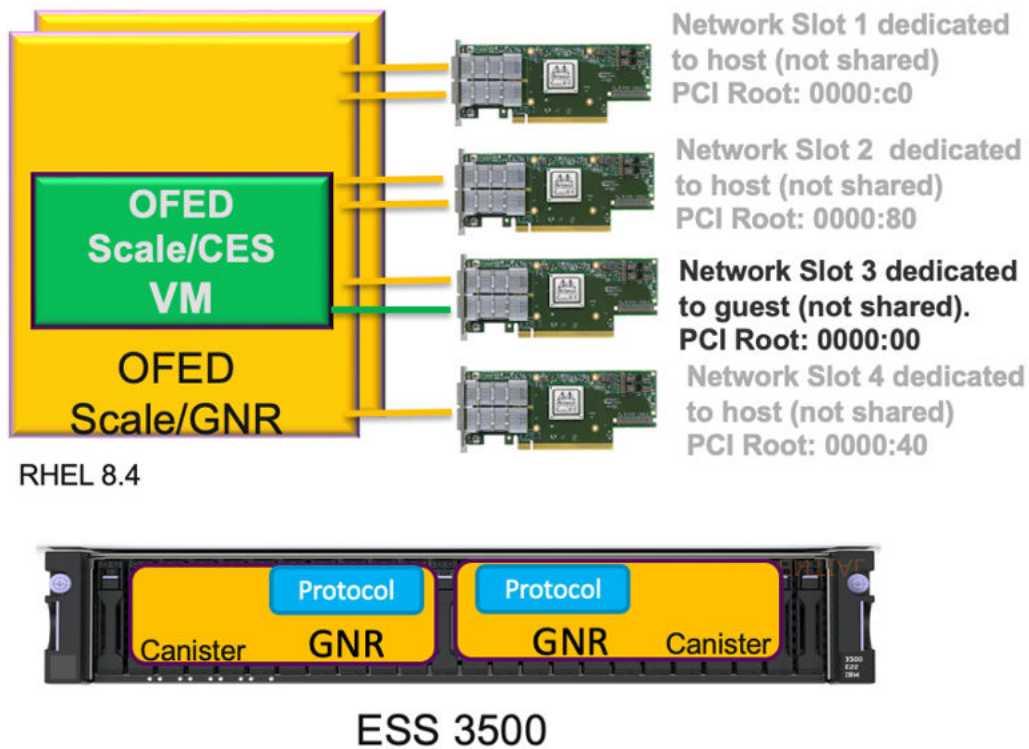


Figure 25. One dedicated card for VM

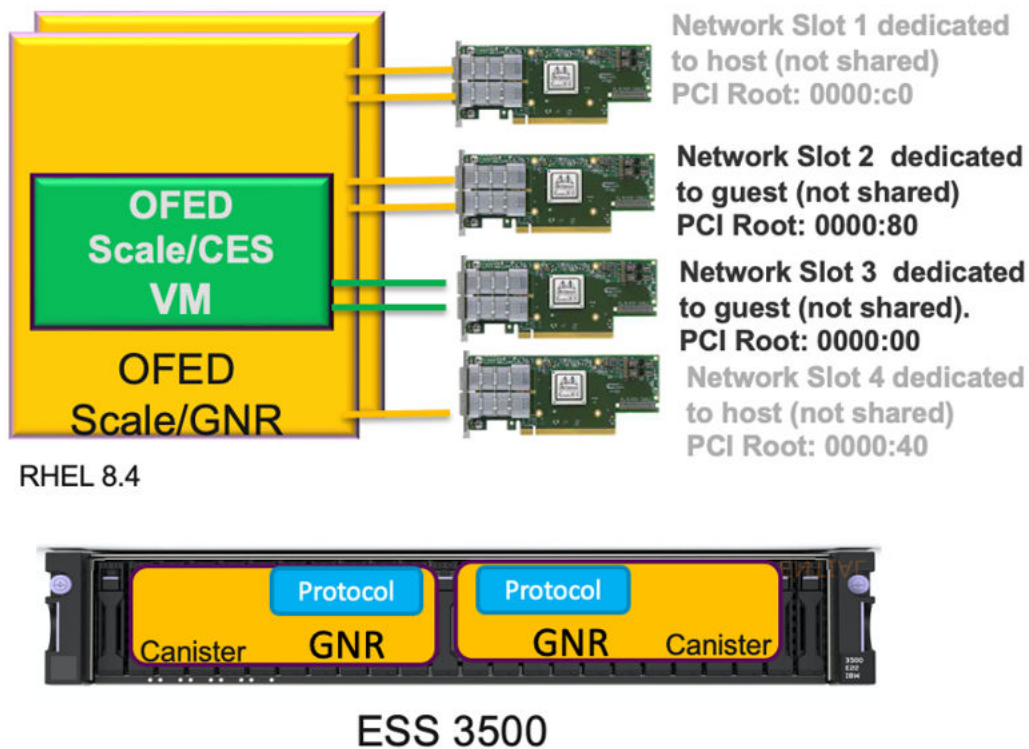


Figure 26. Two dedicated cards for VM

- Yellow is for GPFS admin or daemon network.
- Green is for CES network.

Apart from the new CES networks, the other network requirement of the IBM Storage Scale System remains the same as the management and interlink network.

Important: When a network card is assigned to the VM, the network interfaces disappear from the host. You can see them in the VM.

The VM also needs Mellanox OFED and other requirements, such as physical I/O nodes to deploy a protocol VM. Complete the following steps before a VM protocol implementation:

1. Initialize the management network in the VM by using the **ess_instnic.essvm** command, to get the management network connectivity in the VM.
2. Reboot the I/O nodes after the VM operating system is updated.
3. Update the VM by using the **yum update** command.
4. Install OFED manually by using the **ess_ofed.essvm** command.

Creating or deleting a VM

- Create a VM.

Before you create a VM, ensure that the */etc/hosts* have the entries for VM management network IP address and hostname mapping along with VM high-speed GPFS admin or daemon. All possible CES IP address must be in */etc/hosts*.

```
essrun -N essio3,essio4 cesvm --create --number-of-mellanox-device-passthru 1
[INFO] Will continue with default order.
[INFO] Will continue with default order.

PLAY [Setup VM on IO node canister for CES Deployment.]
*****
*****
*****
Monday 11 April 2022  08:27:38 +0000 (0:00:00.051)    0:00:00.051 *****

TASK [Gathering Facts]
*****
*****
*****
ok: [essio4]
ok: [essio3]
Monday 11 April 2022  08:27:40 +0000 (0:00:01.494)    0:00:01.546 *****
Monday 11 April 2022  08:27:40 +0000 (0:00:00.267)    0:00:01.813 *****

TASK [/opt/ibm/ess/deploy/ansible/roles/kvmCreation : Sync ESS facts | create dir]
*****
*****
*****
ok: [essio4]
ok: [essio3]
Monday 11 April 2022  08:27:40 +0000 (0:00:00.372)    0:00:02.186 *****
```

After a VM is created, complete the following steps to deploy a protocol node:

1. For the VM management, initialize the management network by attaching it to the VM console. Issue the following command to initialize the management network:

```
virsh console essio3-essvm
```

```
ess_ssr_setup
```

Wait until **ess_ssr_setup** finishes before proceeding to the next step.

2. Update the VM by issuing the following commands on the container:

```
essrun -N essio3-essvm config load -p
essrun -N essio3-essvm update --precheck
essrun -N essio3-essvm update --offline
```

For more information, see [Deploying protocols](#).

If you want to use the CES or GPFS admin or daemon network bond in the VM, use the **essgennetwork** command to create the CES or GPFS admin or daemon network bond.

You can deploy the protocol binaries by using the IBM Storage Scale Install Toolkit. Download and use the **x86_64** version of install-toolkit. Designate a protocol VM as an installer node though the setup type of install toolkit must be IBM Storage Scale System. For more information about several types of installation, see the *IBM Storage Scale Install Toolkit*.

- Delete the VM.

```
essrun -N essio3,essio4 cesvm -delete --number-of-mellanox-device-passthru 1
[INFO] The node essio3 is not registered in the Inventory
[INFO] Will continue with default order.
[INFO] The node essio4 is not registered in the Inventory
[INFO] Will continue with default order.

PLAY [Setup VM on IO node canister for CES Deployment.]
*****
*****
Monday 11 April 2022 08:27:38 +0000 (0:00:00.051)    0:00:00.051 *****

TASK [Gathering Facts]
*****
*****
ok: [essio4]
ok: [essio3]
Monday 11 April 2022 08:27:40 +0000 (0:00:01.494)    0:00:01.546 *****
Monday 11 April 2022 08:27:40 +0000 (0:00:00.267)    0:00:01.813 *****

TASK [/opt/ibm/ess/deploy/ansible/roles/kvmCreation : Sync ESS facts | create dir]
*****
*****
ok: [essio4]
ok: [essio3]
Monday 11 April 2022 08:27:40 +0000 (0:00:00.372)    0:00:02.186 *****
```

When a VM is deleted by using **-delete** command, the network interfaces are returned to the host.

Note: After a VM is created or deleted, you can run the **virsh** command to manage the VM.

For more information about the **essrun** command, see this command in the IBM Storage Scale System Command Reference.

Deleting unwanted network manager connections from the host

After passing adapter from host to VM, there might be some stale network manager connections on the host. Delete these stale connections by completing the following steps:

1. Check Mellanox adapters assigned to the host and VM.

```
lspci | grep Mell
```

A sample output is as follows:

```
08:00.0 Ethernet controller: Mellanox Technologies MT28908 Family [ConnectX-6]
08:00.1 Infiniband controller: Mellanox Technologies MT28908 Family [ConnectX-6]
41:00.0 Infiniband controller: Mellanox Technologies MT28908 Family [ConnectX-6]
41:00.1 Infiniband controller: Mellanox Technologies MT28908 Family [ConnectX-6]
87:00.0 Ethernet controller: Mellanox Technologies MT28908 Family [ConnectX-6]
87:00.1 Infiniband controller: Mellanox Technologies MT28908 Family [ConnectX-6]
c1:00.0 Infiniband controller: Mellanox Technologies MT28908 Family [ConnectX-6]
c1:00.1 Infiniband controller: Mellanox Technologies MT28908 Family [ConnectX-6]
```

```
lspci | grep Mellanox
```

A sample output is as follows:

```
05:00.0 Ethernet controller: Mellanox Technologies MT28908 Family [ConnectX-6]
06:00.0 Infiniband controller: Mellanox Technologies MT28908 Family [ConnectX-6]
07:00.0 Infiniband controller: Mellanox Technologies MT28908 Family [ConnectX-6]
08:00.0 Ethernet controller: Mellanox Technologies MT28908 Family [ConnectX-6]
```

2. Check active IB devices on host (such as, ib0, ib1, ib3 & ib4 are active).

```
echo "dev bus-info"; for dev in $(nmcli --terse --fields DEVICE device status | grep -v '^lo$'); do echo -n "$dev "; ethtool -i "$dev" | awk -F': ' '/^bus-info:/ {print $2}'; done
```

A sample output is as follows:

```
dev bus-info
bond0
ess_mgmt_br0 N/A
ib0 0000:c1:00.0
ib1 0000:c1:00.1
ib3 0000:41:00.0
ib4 0000:41:00.1
interlink 0000:c2:00.0
mgmt 0000:c2:00.1
ssr 3-2:1.0
virbr0 N/A
vnet0 tap
```

3. Check **nmcli** connections on the host and delete inactive IB connections (take backup of this output).

```
nmcli connection show
```

A sample output is as follows:

NAME	UUID	TYPE	DEVICE
bond-bond0	f6b80506-41b3-4faf-9b96-d19656bd9d9c	bond	bond0
ess_mgmt_br0	0db44caf-c041-4d68-af87-66a0617cfe97	bridge	ess_mgmt_br0
SSR	36fad259-833c-41de-a15f-79e54f1b8b8c	ethernet	ssr
interlink	4b4b9c1c-770a-4de4-a18d-621800d2441c	ethernet	interlink
virbr0	19c544ef-8ad0-4c81-96f2-85cd7c1d2fe0	bridge	virbr0
bond-slave-ib0	29db6a1b-dd3f-4daa-a0fa-35d7471688e5	infiniband	ib0
bond-slave-ib1	39c839b0-bd2f-4243-b59d-be960f25575b	infiniband	ib1
bond-slave-ib3	49d5e200-8261-44a8-b33a-229a00481b63	infiniband	ib3
bond-slave-ib4	f99b4949-9063-4492-9031-11892d28950c	infiniband	ib4
bridge-slave-mgmt	673508d4-f3e9-40cd-a08b-92febec601e8	ethernet	mgmt
vnet0	ed62df5c-bf8b-4aa9-815d-6dfab2510c2e	tun	vnet0
ib0	7cd87e7f-6244-4bc6-862d-17dcd9cc2bbe	infiniband	--
ib1	a3cb14d5-b5ab-4da7-9f35-99e2db2f6e74	infiniband	--

ib2 inactive	c277bca3-2dd4-47cc-8338-228e198db95c	infiniband	--	##ib2 is
ib3	9b0c0538-cb79-4101-93ac-7d4b222c5162	infiniband	--	
ib4	13a26d75-3644-47e1-9114-79d4e1a97edd	infiniband	--	
ib5 inactive	0c39a335-63da-419a-8b15-7c2a659c9081	infiniband	--	##ib5 is
ib6 inactive	00b00a04-debb-47e5-a454-df95ea466473	infiniband	--	##ib6 is
ib7 inactive	703e2f57-6ae3-49bc-b1fc-4e19d4c739f0	infiniband	--	##ib7 is
interlink	40eb9938-f462-4759-bb03-70ab091ea4fa	ethernet	--	
mgmt	597221e0-9473-4368-8883-289399cdf24e	ethernet	--	

4. Delete ib2, ib5, ib6, and ib7 from host as those are passed to the VM.

```
nmcli con delete ib2
nmcli con delete ib5
nmcli con delete ib6
nmcli con delete ib7
```

Verifying installed nodes and health

After the software deployment is complete, check all nodes.

About this task

Procedure

1. Log in to each node to check that all software and cluster versions are up to date.

```
essinstallcheck -N localhost
```

Note: For IBM Storage Scale System 5000, a check is added that flags whether the WCE bit is enabled on any drive. If the WCE bit is enabled, refer to the published flash for the recommended action.

2. From the management server node, out of the container, run the following final health check commands to verify your system health.

To check health status of IBM Storage Scale System 3200, issue the following command:

```
gnrhealthcheck
```

To check the status of all nodes, issue the following command:

```
mmhealth node show -a
```

Configuring the GUI hardware monitoring

Configure the GUI to monitor hardware components such as nodes and enclosures for system health and performance. The GUI provides a hardware monitoring view where you can monitor all nodes in a cluster.

Procedure

1. From the container, issue the following command to configure and start the performance monitoring sensors.

```
essrun -N MSVM,<IOnode1>,<IOnode2> gui --configure
```

2. Start the GUI and configure it by using the configuration assistant

Setting a time zone

Enabling Chrony timeset in an IBM Storage Scale System environment is done as follows. It can be enabled for management server, I/O server nodes, and protocol nodes by using the **timeset** sub-command of the **essrun** command.

About this task

By default, any node in an IBM Storage Scale System cluster has the Chrony timeset disabled. This command can be run after the deployment of the management server node or I/O server nodes is complete.

Note: The Chrony timeset works in a client-server model. You need to edit the client's `chrony.conf` file with the management IP address of the Chrony server. You also need to edit the server's `chrony.conf` with the comma-separated IP addresses of clients. The file location is `/etc/chrony.conf`.

Procedure

1. Enable Chrony on client nodes:

```
essrun -N nodeList timeset client --server-ip your.ntp.server.ip
```

Note: If you need to set up a node as a server issue the following command:

```
essrun -N ems1 timeset server --server-network your.ntp.server.network/mask
```

This command requires management server to have firewall enabled. Because NTP requests come from certain ports that are blocked for management server nodes by default.

2. To apply the changes, restart Chrony:

```
systemctl restart chronyd  
chronyc makestep  
chronyc ntpdata  
timedatectl
```

Setting up chronyd in non-IBM Storage Scale System nodes

Before you begin

Note: The following time server setup documentation is for general reference. You can configure the time server as suitable for your environment. In the simplest example, the management server host is used as the time server and the I/O nodes (or protocol nodes) are used as clients. Customers might want to have all nodes point to an external time server. Use online references for more detailed instructions for setting up Chrony.

Chrony is a preferred method of setting up a time server. NTP is considered deprecated. Chrony uses the NTP protocol.

For the following example steps, it is assumed that the management server node is the `chronyd` time server and there is no public internet synchronization.

Procedure

- complete the following steps on the management server node, outside of the container.
 - a) Set the time zone and the date locally.

b) Edit the contents of the `/etc/chrony.conf` file.

Note: Replace the server and the allow range with the network settings specific to your setup.

```
# Use public servers from the pool.ntp.org project.
# Please consider joining the pool (http://www.pool.ntp.org/join.html).
#server 0.rhel.pool.ntp.org iburst
#server 1.rhel.pool.ntp.org iburst
#server 2.rhel.pool.ntp.org iburst
#server 3.rhel.pool.ntp.org iburst
server 192.168.7.1 prefer iburst

# Record the rate at which the system clock gains/losses time.
driftfile /var/lib/chrony/drift
local stratum 8
manual

# Allow the system clock to be stepped in the first three updates
# if its offset is larger than 1 second.
makestep 1.0 3

# Enable kernel synchronization of the real-time clock (RTC).
rtcsync

# Enable hardware timestamping on all interfaces that support it.
#hwtimestamp *

# Increase the minimum number of selectable sources required to adjust
# the system clock.
#minsources 2

# Allow NTP client access from local network.
#allow 192.168.0.0/16
allow 192.168.7.0/24

# Serve time even if not synchronized to a time source.
#local stratum 10

# Specify file containing keys for NTP authentication.
#keyfile /etc/chrony.keys

# Specify directory for log files.
logdir /var/log/chrony

# Select which information is logged.
#log measurements statistics tracking
```

c) Save the changes in `/etc/chrony.conf` file.

d) Restart chronyd.

```
systemctl restart chronyd

chronyc makestep
chronyc ntpdata
timedatectl
```

- Do the following steps on the client nodes (canister nodes or IBM Storage Scale System nodes).

a) Edit the contents of the `/etc/chrony.conf` file.

Note: Replace the server and the allow range with the network settings specific to your setup.

```
# Use public servers from the pool.ntp.org project.
# Please consider joining the pool (http://www.pool.ntp.org/join.html).
#server 0.rhel.pool.ntp.org iburst
#server 1.rhel.pool.ntp.org iburst
#server 2.rhel.pool.ntp.org iburst
#server 3.rhel.pool.ntp.org iburst
server 192.168.7.1 prefer iburst

# Record the rate at which the system clock gains/losses time.
server master iburst
driftfile /var/lib/chrony/drift
logdir /var/log/chrony
log measurements statistics tracking

# Allow the system clock to be stepped in the first three updates
# if its offset is larger than 1 second.
```

```

makestep 1.0 3

# Enable kernel synchronization of the real-time clock (RTC).
rtcsync

# Enable hardware timestamping on all interfaces that support it.
#hwtimestamp *

# Increase the minimum number of selectable sources required to adjust
# the system clock.
#minsources 2

# Allow NTP client access from local network.
#allow 192.168.0.0/16
#allow 192.168.7.0/24

# Serve time even if not synchronized to a time source.
#local stratum 10

# Specify file containing keys for NTP authentication.
#keyfile /etc/chrony.keys

# Specify directory for log files.
logdir /var/log/chrony

# Select which information is logged.
#log measurements statistics tracking

```

- b) Save the changes in the `/etc/chrony.conf` file.
- c) Restart chronyd.

```

systemctl restart chronyd

chronyc makestep
chronyc ntpdata
timedatectl

```

Configuring call home

In IBM Storage Scale System 3000, IBM Storage Scale System 3200, IBM Storage Scale System 5000, and IBM Storage Scale System 6000, IBM Storage Scale System version 6.1.1.x can generate call home events when a drive in an attached enclosure needs to be replaced. In IBM Storage Scale System 3000, IBM Storage Scale System 3200, IBM Storage Scale System 3500, IBM Storage Scale System 5000, and IBM Storage Scale System 6000, can also generate call home events for other hardware-related events in the I/O server nodes, protocol nodes, and client nodes that need service.

IBM Storage Scale System 5000 hardware events rely on POWER® system OPAL logs. IBM Storage Scale System 3000, IBM Storage Scale System 3200, IBM Storage Scale System 3500, and IBM Storage Scale System 6000 hardware events rely on **mmcallhome** and **mmhealth** commands.

Starting from IBM Storage Scale System 6.1.9.2, the call home is no longer using Electronic Service Agent (ESA). ESA was completely removed (RPM is no longer available) from IBM Storage Scale System 6.2.1.0 onward.

To ensure a smooth transition, re run the **esscallhomeconf** command after you complete a code update from an earlier code level of IBM Storage Scale System 6.1.9.2 to IBM Storage Scale System 6.1.9.2 and higher.

Examples

Use the **esscallhomeconf** command by following the syntax that is shown in the following example, where the variables are enclosed in `<>`.

Remember: Replace the variables with the values that correspond to your deployment. The management server is not the only node you can set up for call home. You can add all the nodes in the cluster.

```
esscallhomeconf --suffix=-hs -N <EMS,IoNode1,IoNode2...> --icn '<ID_NUMBER>' -m '<COMPANY>' -n  
'<USER_NAME>' -e '<EMAIL>' -u '<COUNTRY>' -t '<PHONE_NUMBER>' -g '<COUNTRY>' -a '<STATE>' -z  
'<ZIP_CODE>' -y '<CITY>' -r '<ADDRESS>'
```

When you run the **mmcallhome info list** command, the output is similar to the next example.

```
# mmcallhome info list  
group      customerName  customerID  email          countryCode  type  primary-  
contact    machine-country  machine-state  primary-contact-phone  machine-city  machine-zip  machine-  
address  
-----  
-----  
global     IBM              <ID_NUMBER>  <EMAIL>        <COUNTRY>    production  
<USER_NAME> <PHONE_NUMBER>  <USER_NAME> <PHONE_NUMBER> <CITY>  
<ZIP_CODE>  <ADDRESS>      <COUNTRY>   <STATE>
```

Disabling the Call Home auto expansion

The call home auto expand feature was introduced in IBM Storage Scale System 6.2.3, and this feature is enabled by default.

This feature automatically expands call-home groups to include all IBM Scale System nodes in a cluster. This expansion ensures that Call Home, including Unified Call Home, is enabled newly added I/O nodes. The Call Home is required for automatic service ticket creation when hardware components fail.

The feature starts within 2 hours after the `gpfs.base` package is updated to IBM Storage Scale 5.2.3.0 on a node that serves as the call home group server. During this interval or later if needed, you can disable the feature by running the following command:

```
mmcallhome capability disable --feature autoexpand
```

You can also run this command before the call home configuration for new setups.

Important: If more than one call home groups exist, the autoexpand feature is automatically disabled. This behavior assumes that the customer has a customized call home configuration and does not require automatic group management.

Disk call home

The IBM Storage Scale RAID pdisk is an abstraction of a physical disk. A pdisk corresponds to exactly one physical disk, and belongs to exactly one de-clustered array within exactly one recovery group.

Overall flow for call home

The management server, either Power9 or Utility Node VM, is the central point for call home in the entire cluster. The management server is responsible of collecting call home information from all the nodes that it manages and sending it to IBM. The data that is sent belongs to one of these categories:

- Daily or weekly heartbeat data: General cluster information that is used for proactive alerts. It can also be used as data points when resolving support tickets.
- Support tickets: An internal issue is detected by **mmhealth** (typically a hardware issue) that requires IBM intervention (and possible part replacement).

In the following systems, the I/O nodes collect and store call home data locally, and the management server retrieves local data later.

- IBM Storage Scale System 3000
- IBM Storage Scale System 3200
- IBM Storage Scale System 3500

- IBM Storage Scale System 5000

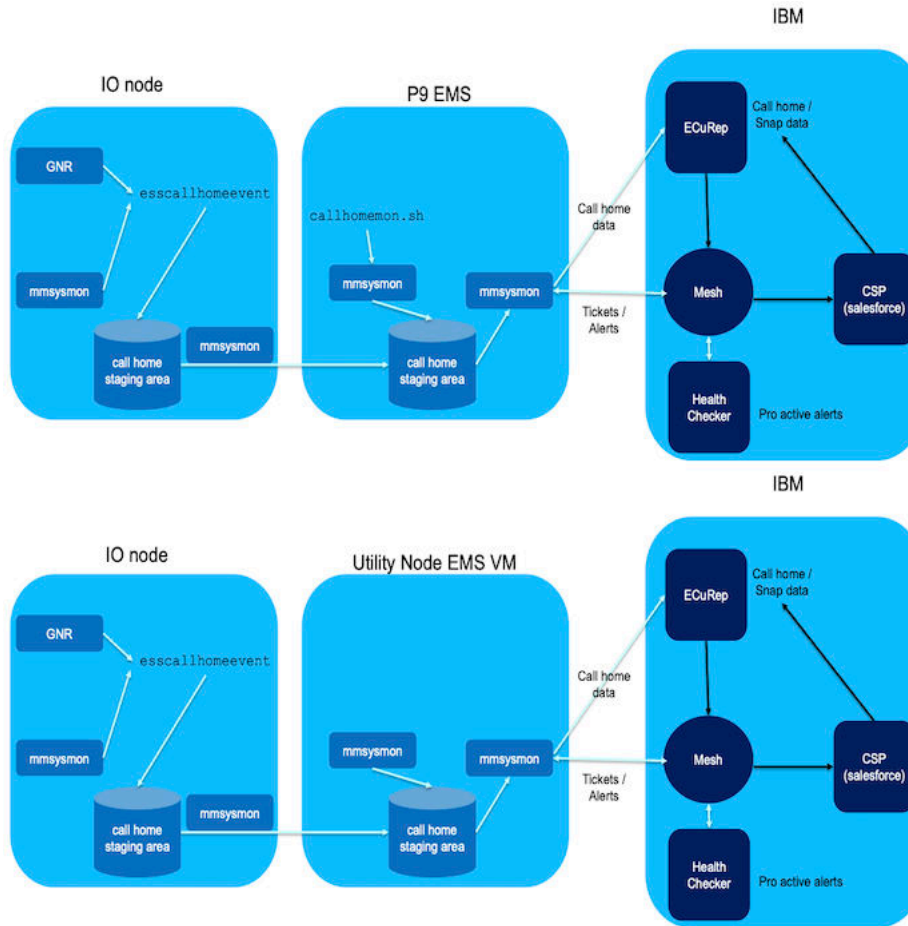


Figure 27. Call home flow

Proactive alerts: Internal IBM tools, which are improved continuously, generate these alerts. Therefore, these alert are not the same although IBM Storage Scale System on the cluster is not changed.

Internal issues: These issues are detected by the IBM Storage Scale System code on the cluster – their detection mechanism is specific to the cluster code level.

From IBM Storage Scale System 6.2.1.0 onward, IBM Electronic Service Agent (ESA) is removed (the RPM is no longer available); the unified call home is used to create service requests. Although ESA is still installed on Power9 management server for backward compatibility, ESA is deactivated after the node is updated.

The IBM Storage Scale RAID pdisk is an abstraction of a physical disk. A pdisk maps to a physical disk and belongs a declustered array in a recovery group.

The disk hospital is a key feature of the IBM Storage Scale RAID that asynchronously diagnoses errors and faults in the storage subsystem.

A pdisk includes the following attributes:

- State

ok

The pdisk is healthy and operational.

diagnosing

The IBM Storage Scale RAID disk hospital is performing a diagnosis when an error occurred.

missing

The IBM Storage Scale RAID is cannot communicate with a disk. If a missing disk is reconnected and operational, the disk state displayed as ok.

For a complete list of pdisk states and further information on pdisk configuration and administration, see [IBM Storage Scale RAID Administration](#).

Any pdisk with the dead, missing, failing, or slow state is a non-functioning pdisk.

- Unique worldwide name (WWN)
- Field replaceable unit (FRU) code
- Physical location code

The disk hospital applies the replace flag to replace disks immediately, regardless of the replacement threshold of their declustered array. A call home notification is generated when the total number of non-functioning pdisks exceeds the replacement threshold, but only for disks that receive the replace flag.

Note: Missing disks must be examined and replaced if no other problem is found.

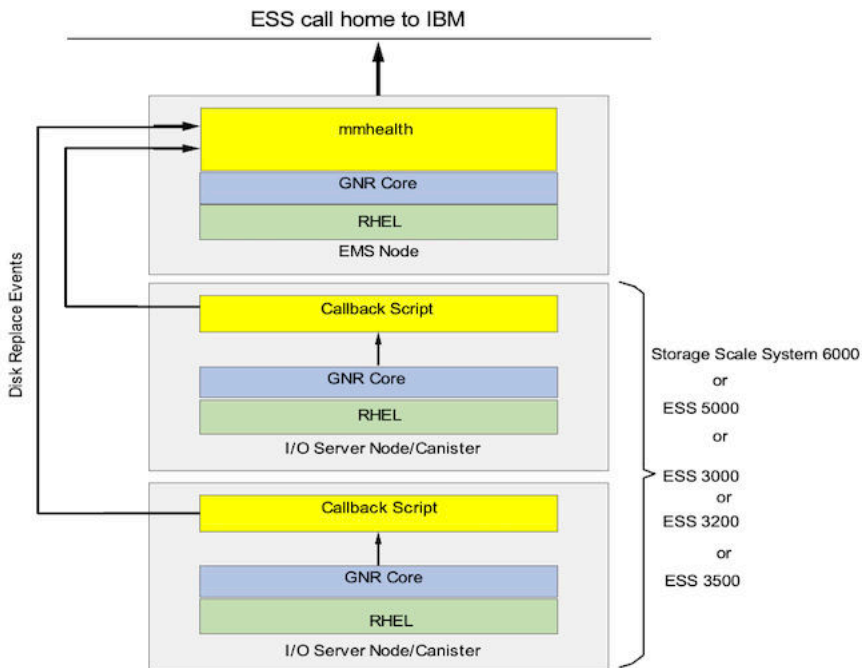


Figure 28. IBM Storage Scale System call home block

The callback script communicates with **mmhealth** and captures a call home ticket information in the *call home staging area* of the node. The call home node (the management server, MS) uploads this data and submit a support ticket to IBM through **mmcallhome**.

Configuring proxy for call home

You can configure proxy for call home events by providing the proxy related parameters during or after configuration of call home.

The following switches are required for setting up proxy for call home.

```
--proxy-ip PROXY-IP-HOSTNAME
    Provide hostname or IP for proxy configuration
    configuration.
--proxy-port PROXY-PORT
    Provide port number for proxy
    configuration.
--proxy-userid PROXY-USERNAME
    Provide userid for proxy configuration.
```

```
--proxy-password PROXY-PASSWORD
Provide password for proxy
```

Proxy configuration by itself fails if it is used without a call home configuration.

This is an example of configuring proxy along with the configuration for call home events.

```
# /opt/ibm/ess/tools/bin/esscallhomeconf -N essem1 --proxy-ip host.example.com
--proxy-port 5028 --proxy-userid johndoe --proxy-password secret -m IBM -n "Jane Doe" -e
janedoe@example.com -t 5121234567 -g "United States" -u "United States" -r "11400 Burnet Rd"

-y Austin -a TX -z 78758 --icn 123456789
[WARNING] Only one node should be used here. Use daemon hostname for EMS or IO node name here.
Preferably EMS daemon node name.
[INFO] Picking first nodename from the provided list to configure callhome.
Started configuring proxy
Setting proxy customer details
Successfully set the customer details for proxy.
Successfully set the DAILY schedule for proxy.
Successfully set the WEEKLY schedule for proxy.
Direct connection will be used for proxy.
Successfully set the direct connection for proxy.
Successfully set the group configuration for proxy.
Proxy configuration completed.
```

```
# mmcallhome info list
group      customerName  customerID  email                      countryCode  type  primary-
contact    primary-contact-phone  machine-city  machine-zip  machine-address  machine-
country    machine-state
-----
global     IBM             123456789   janedoe@example.com      US           production Jane
Doe        5121234567      Austin      78758        11400 Burnet Rd  United
States     TX
```

Call home logs and location

The **esscallhomeconf** command logs the progress and error messages in the `/var/log/messages` file, the `/var/log/ess/` folder, and `/var/adm/ras/mmsysmonitor.log` file.

There is a **--verbose** option that provides more details of the progress and error messages. The following example displays the type of information sent to the `/var/log/messages` file on the management server node by the **esscallhomeconf** command.

```
# grep emsdev1 /var/log/messages | grep esscallhomeconf
Dec 11 12:33:45 emsdev1 /esscallhomeconf[1187703]: [I] Unified call home configuration
completed.
```



Attention: The **esscallhomeconf** command also configures the IBM Storage Scale call home setup. The IBM Storage Scale call home feature collects files, logs, traces, and details of certain system health events from the I/O and management server nodes and services running on those nodes. These details are shared with the IBM support center for monitoring and problem determination. For more information on IBM Storage Scale call home, see IBM Storage Scale documentation in IBM Documentation.

Overview of a problem report

For Power9 hardware failure can be an IBM Storage Scale System 5000 I/O node or an management server node or a protocol node. Hardware failure event can be DIMM failure, power supply failure, or fan failure in Power9 nodes. Any of the failure events reported by the Power nodes are recorded in OPAL log. IBM Storage Scale System hardware call home reads the OPAL events, which need service and it is a call home event.

IBM Storage Scale System 5000 and Power management server hardware failure event raised in OPAL log reported by event type

- disk_call_home, disk_call_home2 – GNR disk needs replacement.
- nodeEvent - Any Power9 node hardware failure which is reported in the OPAL log is a call home event that is reported with this event.
- raid_root_disk_bad – Power boot disk failure.
- raid_sas_adapter_bad, sas_adapter_down_trained, sas_adapter_failed, sas_cable_failed - Power SAS adapter failures.
- sas_adapter_missing – Expected SAS adapter is not detected by the Power system.
- power_supply_failed, power_supply_fan_critical, power_supply_iout_critical, power_supply_vout_critical, ps_fan_failed – Expansion enclosure PSU failures.
- temp_bus_failed, temp_sensor_failed, voltage_bus_failed, voltage_sensor_failed, temp_bus_failed, temp_sensor_failed – Expansion enclosure sensor failures.
- power_high_current, power_high_voltage – Expansion enclosure sensor reporting input power outside expected range.

Similarly, IBM Storage Scale System 3000, IBM Storage Scale System 3200, IBM Storage Scale System 3500, IBM Storage Scale System Utility Node, IBM Storage Scale System 6000 x86 nodes can also report any of the hardware failure events with **mmhealth**.

IBM Storage Scale System 3000 supported hardware events

- disk_call_home, disk_call_home2 – GNR disk needs replacement.
- bootDrvFail - When boot drive failed at canister.
- canFailed - When canister failed.
- bootDrvMissing - When boot drive missing.
- bootDrvSmtFailed - When boot drive smart failed.
- canFanFailed - When canister FAN stopped working.
- fanFailed - When FAN failed.
- psFailed - Power supply failure.
- psFanFailed - Power supply FAN failure.

IBM Storage Scale System 3200, IBM Storage Scale System 3500, IBM Storage Scale System Utility Node, and IBM Storage Scale System 6000 supported hardware events

- disk_call_home, disk_call_home2 – GNR disk needs replacement.
- nvmeof_raw_disk_smart_failed, nvmeof_raw_disk_smart_unknown – non GNR disk needs replacement.
- bootdrive_mirror_degraded, bootdrive_mirror_failed – Boot drive mirror no longer redundant.
- bootdrive_missing – Expected boot drive not detected by the system.
- bootdrive_power_mode_degraded – Boot drive power mode is not at the expected value.
- bootdrive_smart_failed – Pro active failure reported by the boot drive.
- can_fan_missing – Utility Node internal fan not detected by the system.
- canister_failed – Node hardware failure.
- pair_canister_failed – Partner node is not reachable.
- can_power_supply_failed, can_power_supply_missing - Utility Node PSU failures.
- can_temp_bus_failed, can_temp_sensor_failed, temp_bus_failed, temp_sensor_failed, voltage_bus_failed, voltage_sensor_failed – Enclosure sensor failures.
- coin_battery_low, coin_battery_missing – CMOS battery failures.
- cpu_unit_in_wrong_slot, cpu_unit_missing, cpu_unit_speed_wrong – CPU health check errors.

- `dimm_config_mismatch`, `dimm_module_in_wrong_slot`, `dimm_module_missing`, `dimm_module_size_wrong`, `dimm_module_speed_wrong` – DIMM health check errors.
- `enc_temp_critical` – Enclosure reporting elevated temperature, thermal shutdown might follow.
- `esm_failed`, `expander_failed` – Expansion IO module failures.
- `fan_absent` – Fan not detected in fan bay.
- `can_fan_failed` – Utility node internal fan failure.
- `fan_failed`, `fan_fault_indicated`, `fan_speed_high`, `fan_speed_low` – enclosure fan failures.
- `power_high_current`, `power_high_voltage` – Enclosure sensor reporting input power outside expected range.
- `power_supply_config_mismatch` – Incompatible of PSU types in the same enclosure.
- `power_supply_failed`, `power_supply_fan_critical`, `power_supply_iout_critical`, `power_supply_vout_critical`, `ps_fan_failed` – PSU failures.
- `sas_adapter_down_trained`, `sas_adapter_failed`, `sas_adapter_missing`, `sas_cable_failed` – Connectivity to expansion enclosure failures.

Problem and IBM tickets details

After the system successfully open a ticket, the ticket is displayed by issuing the **mmcallhome ticket list** command. The problem descriptions of the events can be seen by selecting the endpoint. You can select an endpoint by clicking the red **X**.

You can use the **mmcallhome ticket list/delete** command to display unified call home information.

```
mmcallhome ticket list
```

A sample output is as follows:

Tickets Opened	Ticket Number	Creation Date	Status	Description
TS011927298	2023-12-09 22:24:49	New Case	Opened	78P000C:N/A:N/A:N/A:Test symptom generated by Electronic Service Agent
TS011927322	2023-12-10 21:48:07	New Case	Opened	I22C001:N/A:N/A:N/A:Test symptom generated by Electronic Service Agent
TS011927385	2023-12-11 14:13:20	New Case	Opened	I22C001:canister:I22C001::The boot drive mirroring is degraded. TS011927420
TS011927420	2023-12-11 18:00:53	New Case	Opened	78L0001:enclosure:78L0001/27:psu4_fan_id27:Fan psu4_fan_id27 is absent. TS011927421
TS011927421	2023-12-11 18:04:29	New Case	Opened	78L0001:enclosure:78L0001/75:psu4_vout_id75:Voltage sensor psu4_vout_id75 is FAILED. TS011927556
TS011927556	2023-12-12 15:23:18	New Case	Opened	78L0001:enclosure:78C100F/1:right_esm_id1:ESM right_esm_id1 is FAILED. TS011927590
TS011927590	2023-12-12 21:31:18	New Case	Opened	78L0001:canister::78L0001A:The coin battery has low voltage. TS011927932
TS011927932	2023-12-15 19:32:27	New Case	Opened	78L0001:enclosure:78L0001/27:psu4_fan_id27:Fan psu4_fan_id27 is absent. TS011927933
TS011927933	2023-12-15 20:45:55	New Case	Opened	78L0001:enclosure:78L0001/9:fan_module3_id9:Fan fan_module3_id9 has a fault. TS011927988
TS011927988	2023-12-18 10:53:34	New Case	Opened	I22C001:canister:Slot-3:Slot-3:The fan for location Slot-3 is missing. TS011928011
TS011928011	2023-12-18 20:27:37	New Case	Opened	78L0001:N/A:N/A:N/A:Test symptom generated by Electronic Service Agent
TS011928038	2023-12-19 10:22:51	New Case	Opened	78L0001:enclosure:78L0001/27:psu4_fan_id27:Fan psu4_fan_id27 is absent. TS011928039
TS011928039	2023-12-19 10:26:00	New Case	Opened	78L0001:enclosure:78L0001/75:psu4_vout_id75:Voltage sensor psu4_vout_id75 is FAILED. TS015207286
TS015207286	2024-03-08 15:43:31	New Case	Opened	78P000C:canister::78P000C:The coin battery has low voltage. TS015207299
TS015207299	2024-03-08 16:46:49	New Case	Opened	00E5006:canister::00E5006A:The coin battery has low voltage. TS015207300
TS015207300	2024-03-08 16:50:38	New Case	Opened	78P000C:canister:Slot-1:Slot-1:Fan Slot-1 has failed. TS015207301
TS015207301	2024-03-08 16:52:50	New Case	Opened	78L0001:canister::78L0001A:The coin battery has low voltage.

[root@emsdev3 ~]#

Note: This command displays the tickets opened for the whole cluster. It can be run on any node with the same output.

Call home monitoring of Power management server and IBM Storage Scale System 5000 systems

To report the OPAL errors correctly to the unified call home, the `callhomemon.sh` script located in the `/opt/ibm/ess/tools/samples` must be run on a regular interval on the node.

You can set up as a cron job. On each of the Power node, create a cronjob as follows:

1. Open crontab editor by using the following command:

```
# crontab -e
```

2. Set up a periodic cronjob by adding the following line:

```
*/30 * * * * /opt/ibm/ess/tools/samples/callhomemon.sh
```

3. View the cronjob by using the following command:

```
# crontab -l
*/30 * * * * /opt/ibm/ess/tools/samples/callhomemon.sh
```

Upload data

The following support data is uploaded when the IBM Storage Scale System disks in enclosures display a drive replace notification on an IBM Storage Scale System 3000, 3200, 3500, 5000, or .

- The output of **mm lspdisk** command for the pdisk that is in replace state.
- Additional support data is provided only when the event is initiated as a response to a callback. The following information is supplied in a .tgz file as additional support data:
 - Last 10000 lines of `mmfs.log.latest` from the node which generates the event.
 - Last 24 hours of the kernel messages (from journal) from the node which generates the event.
 - For IBM Flash Core Module (FCM) drives, a diagnostic report (drive debug log) per drive. This report only contains internal counters or debug statistics for failure analysis; it does not contain any data stored on the drive. Drive debug logs are in plain text format, they can be reviewed, and they are stored on the I/O node at `/var/log/ess/hal/drivedumps`.

The following support data is uploaded when the system displays any hardware issue in an IBM Storage Scale System 5000 system.

- The output of the **opal_elog_parse** command for the serviceable event that caused failure.
- Additional support data is provided only when the event is initiated as a response to a callback. The following information is supplied in a .tgz file as additional support data:
 - Last 10000 lines of `mmfs.log.latest` from the node which generates the event.
 - Last 24 hours of the kernel messages (from journal) from the node which generates the event.

The following support data is uploaded when the system displays any hardware issue in an IBM Storage Scale System 3000 or an IBM Storage Scale System 3200 system.

- The output of the **mmhealth** command and the actual component that caused failure.
- Additional support data is provided only when the event is initiated as a response to a callback. The following information is supplied in a .tgz file as additional support data:
 - Last 10000 lines of `mmfs.log.latest` from the node which generates the event.
 - Last 24 hours of the kernel messages (from journal) from the node which generates the event.

Call home pre-installation worksheets

Fill in the following call home worksheets before the installation.

General Information	
Primary Contact for Providing Site Access: (Required)	
Contact Information for Primary Contact (Phone & Email): (Required) Note: • If the phone number is provided, indicate if it is a mobile number or office number. • Do not use a personal email address. Use a group or task email address, like itsupport@mycompanyname.com.	
System Location Settings	
Country or Region: (Required) Note: You need to provide the Alpha-2 Code for this entry. You can obtain this information by searching for the country code on ISO Online Browsing Platform .	
State or Province: (Required)	
Postal Code: (Required)	
City: (Required)	
Street Address: (Required)	
Building, Floor, Office: (Required)	
Telephone Number: (Required)	
Service Contact Settings - Company Information	
Company Name: (Required)	

Service Contact Settings - Company Information	
Street Address: (Optional)	
City: (Optional)	
State or Province: (Optional)	
Country or Region: (Required) Note: You need to provide the Alpha-2 Code for this entry. You can obtain this information by searching for the country code on ISO Online Browsing Platform .	
Postal Code: (Optional)	
Service Contact Settings - Primary Contact	
Contact Name: (Required)	
Telephone Number: (Required)	
Alternate Telephone Number: (Optional)	
E-mail Address: (Required)	
Alternate E-mail Address: (Optional)	
Pager Number: (Optional)	
Notifications Settings - Email Notification	
Enable E-mail Notification (Y or N): (Required)	
SMTP Server Name: (Required)	
Port: (Required)	

Service Contact Settings - Company Information	
User ID: (Optional)	
Password: (Optional)	
Email Address(s) to Receive Notifications: (Required)	
Notifications Settings - SNMP Traps	
Enable SNMP Trap (Y or N): (Required)	
Target Network Host: (Required)	
Community: (Required)	
Port: (Required)	
IBM ID Settings IBM provides customized web tools and functions that use information collected by IBM Electronic Service Agent. The access to these functions is managed by an association between your IBM ID(s) and the Electronic Service Agent information from your systems. The association is made using this page. To obtain an IBM ID, which is used by many IBM web sites, go to http://www.ibm.com/registration	
IBM ID(s) to Add: (Required)	

Chapter 3. Deployment from an x86 Utility Node by using the GUI

Starting with IBM Storage Scale Systems 6.2.3.0, you can use a graphical user interface (GUI) to perform new deployments from an x86 Utility Node. The GUI is an alternative to CLI and it simplifies the user experience during the initial deployment.

The new deployment of IBM Storage Scale System by using the GUI includes the following steps:

1. Verify the supported systems for deployment through the GUI. For more information, see [“Supported systems”](#) on page 119.
2. Make sure that the network preparation is completed. For more information, see [“Planning and configuring network”](#) on page 7.
3. Make sure that the preparation and prerequisites for the new deployment are completed. For more information, see [“Prerequisites”](#) on page 39.
4. Copy the required files to the Utility Node management server host (/serv). For more information, see [“Copy the required files to the Utility Node management server host”](#) on page 119.
5. Set up the IBM Storage Scale System deployment GUI container. For more information, see [“Setting up the IBM Storage Scale System deployment GUI container”](#) on page 129.
6. Set up the IBM Storage Scale System API server container. For more information, see [“Setting up the IBM Storage Scale System API server container”](#) on page 120.
7. Log in to the GUI and complete the deployment steps. For more information, see [“Using the deployment GUI”](#) on page 135.

Supported systems

Deployment by using the GUI supports only new deployments and available only for the following systems:

- IBM Storage Scale System Utility Node (MTM 5149-23E)
- IBM Storage Scale System 6000 (MTM 5149-F48)
- IBM Storage Scale System 3500 (MTM 5141-FN2)

The deployment by using the GUI has following limitations:

- Only one EMSVM can be deployed per cluster. Any number of I/O Building Blocks, Protocol VMs, AFM Gateway VMs, and Client VMs are supported.
- Adding nodes to an existing cluster is not supported.

Note: I/O nodes are categorized into building blocks based on their serial numbers. For example, nodes with serial numbers 00x0000A and 00x0000B are grouped into one building block.

Copy the required files to the Utility Node management server host

Copy the following items to the Utility Node management server host (/serv):

- EMSVM-23E.qcow2.xz. To download the EMSVM-23E.qcow2.xz image, complete the following steps:
 1. Go to [IBM Fix Central](#).
 2. Select Scale_System_VM-<x.x.x.x>-x86_64-UTILITY-EMS.
 3. Download the Scale_System_VM-<x.x.x.x>-x86_64-UTILITY-EMS.tgz file.
 4. Go to /serv and extract the Scale_System_VM-<x.x.x.x>-x86_64-UTILITY-EMS.tgz file there. This .tgz file contains the EMSVM-23E.qcow2.xz file and the checksum file.
 5. Rename the QCOW image to /emsvm/EMSVM-23E.qcow2.xz.

- Management server virtual machine (EMSVM) script package. To download the EMSVM script package:
 1. Go to <https://github.com/IBM/EMSVM>.
 2. Click **Code** (green button).
 3. Select **Download ZIP**.
 4. If the downloaded file name is EMSVM-main.zip, rename it to EMSVM.zip.
 5. Extract the EMSVM script package in /serv.
- x86 container (DAE or DME): [IBM Fix Central](#)

Note: The root (/) partition of the Utility Bare Metal server must have a minimum of 60 GB of available space.

Setting up the IBM Storage Scale System API server container

The IBM Storage Scale System API server container is a dedicated container that runs on the IBM Storage Scale System Utility Node and serves REST endpoints for the deployment artifacts of IBM Storage Scale System. This API server container enables IBM Storage Scale System deployment over the REST interface.

Ensure that the prerequisites are completed on the system before setting up the container. For more information, see [“Prerequisites” on page 129](#).

To set up the API server container, complete the following steps:

1. [Create user and user group](#).
2. [Pull the image of the API server container and **startAPIContainer** utility](#).
3. [Start the API server container on the Utility Node and verify that it is active](#).
4. [If required, replace the autogenerated self-signed certificate with your organization-owned, CA-signed certificates](#).
5. [Set the password for **installgui** user](#).
6. [Verify that the API server container is configured correctly on the Utility Node](#).

Create user and user group

Create the **apiadmin** user and user group to host the API server container. The **apiadmin** user is a superuser that can change any authorization. This user must be familiar with the IBM Storage Scale native REST API feature. For more information about the IBM Storage Scale native REST API, see [IBM Storage Scale native REST API](#).

If required, you can also add the **apiuser** to the API server container. This user has non-privileged access and restricted to change the authorization.

To create the user and user group and set the user password, complete the following steps:

1. Log in as the root user on the Utility Node.
2. Create the **apiadmin** user group.

```
groupadd apiadmin
```

3. Create the **apiadmin** user and assign it to the **apiadmin** user group.

```
useradd -m -g apiadmin apiadmin
```

4. Set the password for the **apiadmin** user.

```
passwd apiadmin
```

When the **New password:** prompt appears, enter the new password. When the **Retype new password:** prompt appears, re-enter the password to confirm it.

5. Verify that the **apiadmin** user is created.

```
cat /etc/passwd
```

6. Log out as the root user and log in as the **apiadmin** user.

Pull the API sever container image and the startAPIContainer utility

Complete the next steps to pull the image of the API server container from the IBM Cloud container repository and the **startAPIContainer** utility package from the [GitHub](#). Internet access is required on the node to complete this step. If the Utility Node does not have internet access, pull the container image from a node that is connected to internet and move the downloaded files to the Utility Node. Alternatively, you can download these files from the [IBM Fix Central](#).

To pull the image of the API server container and **startAPIContainer** utility package, complete one of the following methods:

For a Utility Node with internet access

1. In a browser, log in to [MyIBM Container Software Library](#) by using the IBMid and password that are associated with your IBM Storage Scale System.
2. In the **Entitlement keys** section, click **Copy** to get the entitlement key in the clipboard. If no key exists, click **Add new key**, then copy the new key to the clipboard.

Note: To verify that your entitlement key is valid, go to the left pane and click **Container software library**. You can view the list of products that you are entitled to.

3. On the Utility Node, log in as the **apiadmin** user and issue the following command to log in to the IBM Cloud container Repository:

```
podman login cp.icr.io --username cp --password <APIKEY>
```

Where,

<APIKEY>

Specifies the entitlement key that is obtained from the **Container software library**.

4. Clone the **SSSAPI** repository from [GitHub](#) on the Utility Node at /home/apiadmin. Alternatively, use the **Download ZIP** option on [GitHub](#) to manually download the **SSSAPI** repository on the Utility Node at /home/apiadmin. This repository includes the **startAPIContainer** utility file, which starts the API server container.
5. If downloaded manually, extract the downloaded SSSAPI-main.zip file at /home/apiadmin.

```
unzip <SSSAPI-main.zip>
```

6. In the CLI, change the path to the extracted directory.

```
cd </home/apiadmin/SSSAPI-main>
```

7. Start the **startAPIContainer** utility on the Utility Node and enter the image version.

```
./startAPIContainer
```

A sample output is as follows:

```
Enter the campus interface name (default: campus):
Enter the image version (default: 7.0.1.x):
Enter the API port (default: 46443):
Enter the IBM Storage Scale System Utility Node password to configure passwordless SSH
(default: cluster):
Specify whether EMSVM-23E exists (default: no):
Enter the EMSVM-23E password to configure passwordless SSH (default: cluster):
```

Detailed logs are located in the ./logs directory.

```
2025-11-24 07:38:52,708 INFO: All configurable variables were validated successfully.
2025-11-24 07:38:52,827 INFO: No images related to api-official were found for deletion.
```

```

2025-11-24 07:38:52,829 INFO: The API server container image installation is about to
begin. No changes are applied if the image is already installed.
-- [INFO] The container image is about to be pulled from the IBM repository. --
-- [WARNING] The existing container image cp.stg.icr.io/cp/scalesystem/sss_api:7.0.1.x
is being deleted forcefully. --
Untagged: cp.stg.icr.io/cp/scalesystem/sss_api:7.0.1.x
Deleted: 3794e1364fe4021cb09c9cf0cbd1b082e32269ec25bcf862ed281c908be5e69a
Pulling the container image: cp.stg.icr.io/cp/scalesystem/sss_api:7.0.1.x...
Getting image source signatures
Copying blob 6513ff0e52ba done |
Copying blob d3cc7dd07291 done |
Copying blob 1df576680603 done |
Copying blob f07779ff8072 done |
Copying blob 53408b222912 done |
Copying blob 714ba5fda8ed done |
Copying blob 1042a1990615 done |
Copying blob 72a5323fcdbe done |
Copying blob f61ac7034cc9 done |
Copying blob 792808c78253 done |
Copying blob 73d00ef8b89b done |
Copying config 3794e1364f done |
Writing manifest to image destination
3794e1364fe4021cb09c9cf0cbd1b082e32269ec25bcf862ed281c908be5e69a
-- [INFO] The container image was pulled successfully. --
2025-11-24 07:39:14,222 INFO: The container image installation completed
successfully.
2025-11-24 07:39:14,223 INFO: The container is about to start. On later runs, use
the 'startAPIContainer' command with optional arguments to manage the container.
-- [INFO] Running the container image - cp.stg.icr.io/cp/scalesystem/sss_api:7.0.1.x --
-- [WARNING] The '-x' or '--force' option removes containers that are in the EXIT
state. --
-- [INFO] Automatic initialization of the container will begin shortly. --
-- [INFO] Startup can take several minutes. --
e48b3ea490cc1ecfc205f28dd855cf08a78deaebc8bfdade05a33348b9fffbce
-- [WARNING] The '/home/apiadmin/SSSAPI-main/credentials.properties' file was deleted. --
-- [INFO] The API server container is being configured to start as a systemd service. --
-- [INFO] The API server container is set to autostart. --

Deprecated command:
It is recommended to use quadlets for running containers and pods under systemd.

Refer to podman-systemd.unit(5) for details.
/home/apiadmin/container-utilityBareMetal-api-official.service
-- [INFO] The container was created successfully in the background. To start
this container, run the 'systemctl --user start container-utilityBareMetal-api-
official.service' command. --
-- [INFO] To log in to the container, run the 'podman exec -it utilityBareMetal-api-
official /bin/bash' command. --
2025-11-24 07:39:19,768 INFO: To start a new container, run the "/home/apiadmin/SSSAPI-
main/startAPIContainer" command.

```

After this command runs successfully, the Utility Node hosts the API server container, but the container is not started.

For a Utility Node without internet access

1. Download the API server container image `sss_api_7.0.1.x.tar` and the utility package `SSSAPI-main.zip` from IBM Fix central or MyIBM Container software library.

Using IBM Fix central

- a. From a node that is connected to the internet, go to [IBM Fix Central](#).
- b. Select `Scale_System_GUI-7.0.1.x`.
- c. Download the `Scale_System_GUI-7.0.1.x.tgz` file. This file contains the API server container image `sss_api_7.0.1.x.tar` and the utility package `SSSAPI-main.zip` file.
- d. Transfer the `sss_api_7.0.1.x.tar` and `SSSAPI-main.zip` files to `/home/apiadmin` on the Utility Node.
- e. On the Utility Node, extract the `SSSAPI-main.zip` file at `/home/apiadmin`.

```
unzip <ZipFileName>
```

Using MyIBM Container software library

- a. From a node that is connected to the internet, log in to [MyIBM Container Software Library](#) by using the IBMid and password that are associated with your IBM Storage Scale System.
- b. In the **Entitlement keys** section, click **Copy** to get the entitlement key in the clipboard. If no key exists, click **Add new key**, then copy the new key to the clipboard.

Note: To verify that your entitlement key is valid, go to the left pane and click **Container software library**. You can view the list of products that you are entitled to.

- c. Log in to the IBM Cloud container Repository.

```
podman login cp.icr.io --username cp --password <APIKEY>
```

Where,

<APIKEY>

Specifies the entitlement key that is obtained from the **Container software library**.

- d. Pull the image of the API server container.

```
podman pull cp.icr.io/cp/scalesystem/sss_api:7.0.1.x
```

- e. Export the pulled image as a .tar file on the node that is connected to the internet.

```
podman save cp.icr.io/cp/scalesystem/sss_api:7.0.1.x -o sss_api_7.0.1.x.tar
```

- f. Download the **SSSAPI-main** repository by using the **Download ZIP** option on [GitHub](#) on the same node. This repository includes the **startAPIContainer** utility file, which starts the API server container.
- g. Move the generated sss_api_7.0.1.x.tar file and the downloaded ZIP **SSSAPI-main** file to the /home/apiadmin on the Utility Node.
- h. On the Utility Node, extract the SSSAPI-main files at /home/apiadmin.

```
unzip <ZipFileName>
```

2. In the CLI, change the path to the extracted directory.

```
cd </home/apiadmin/SSSAPI-main>
```

3. Restore the .tar file by using the **-f** option through the **startAPIContainer** utility program.

```
./startAPIContainer -f ../sss_api_7.0.1.x.tar
```

Note: To get help information for the **startAPIContainer** utility, issue the following command:

```
./startAPIContainer --help
```

A sample output is as follows:

```
usage: startAPIContainer [-h] [-v] [-f FILENAME] [-p PORT] [-i CAMPUS_INTERFACE]
[-vn IMAGE_VERSION] [-up UTILITYBAREMETAL_PASSWORD] [-e IS_EMSVM23E_EXIST] [-ep
EMSVM23E_PASSWORD]

optional arguments:
  -h, --help            Display the help message and exit.
  -v, --verbose          Display verbose messages in the console.
  -f FILENAME, --file-name FILENAME
                        Restore the container from a local tar file. The default
value is SSSDEPLOYGUI.tar.
  -p PORT, --port PORT  Port on which the API server listens. The default value is
46443.
  -i CAMPUS_INTERFACE, --campus-interface CAMPUS_INTERFACE
                        Name of the interface connected to the campus network. The
default value is campus.
  -vn IMAGE_VERSION, --image-version IMAGE_VERSION
                        API server version to deploy. The default value is 7.0.1.x.
  -up UTILITYBAREMETAL_PASSWORD, --utilitybaremetal-password UTILITYBAREMETAL_PASSWORD
```

```

        Password for the root user on the 'utilityBareMetal' host.
The default value is cluster.
-e IS_EMSVM23E_EXIST, --is-emsvm23e-exist IS_EMSVM23E_EXIST
        Specify whether EMSVM-23E exists. The default value is no.
-ep EMSVM23E_PASSWORD, --emsvm23e-password EMSVM23E_PASSWORD
        Password for the root user on EMSVM-23E. The default value is
cluster.

```

4. Start the **startAPIContainer** utility on the Utility Node and enter the image version.

```
./startAPIContainer
```

A sample output is as follows:

```

Enter the campus interface name (default: campus):
Enter the image version (default: 7.0.1.x):
Enter the API port (default: 46443):
Enter the IBM Storage Scale Utility Node password to configure passwordless SSH
(default: cluster):
Specify whether EMSVM-23E exists (default: no): yes
Enter the EMSVM-23E password to configure passwordless SSH (default: cluster):

Detailed logs are located on ./logs directory

2025-12-02 23:46:09,966 INFO:      All configurable variables were validated successfully.
2025-12-02 23:46:10,082 INFO:      No images related to api-official were found for
deletion.
2025-12-02 23:46:10,084 INFO:      The API server container image installation is about
to begin. No changes are applied if the image is already installed.
-- [INFO] The API container image is about to be pulled from the IBM repository. --
-- [WARNING] The existing container image cp.stg.icr.io/cp/scalesystem/sss_api:7.0.1.xis
being deleted forcibly. --
Untagged: cp.stg.icr.io/cp/scalesystem/sss_api:7.0.1.x
Deleted: bc77eb54e68150279d1c1e7afd72109ff33e0b5899bb6c2f0cf4a62cdc69ace3
Trying to pull cp.stg.icr.io/cp/scalesystem/sss_api:7.0.1.x...
Getting image source signatures
Copying blob 8d6cf32b1d96 done |
Copying blob 35d6a5cce6a1 done |
Copying blob b5033592c3e0 done |
Copying blob 29d4b2abb1d4 done |
Copying blob 354a1969e4a5 done |
Copying blob 0382955305cb done |
Copying blob c8cad13252da done |
Copying blob fbb2a441be54 done |
Copying blob fa7bd7163b6b done |
Copying blob 394966977691 done |
Copying blob f3e63b1012d6 done |
Copying config cf62860064 done |
Writing manifest to image destination
cf628600645cfe7d95fc528504b5406ef5d43f1e55d47717b1f487717784e7c3
-- [INFO] The container image was pulled successfully. --
2025-12-02 23:46:38,901 INFO:      The container image installation completed
successfully.
2025-12-02 23:46:38,904 INFO:      The container is about to start. On later runs, use
the 'startAPIContainer' command with optional arguments to manage the container.
-- [INFO] Running the container image - cp.stg.icr.io/cp/scalesystem/sss_api:7.0.1.x
-- [WARNING] The '-x' or '--force' option removes containers that are in the EXIT state.
--
utilityBareMetal-api-official
-- [INFO] Automatic initialization of the container will begin shortly. --
-- Startup can take several minutes. --
456cbc4056b214186b2d0ed9a48784857cb8782f30b6c8671ec75576ee1c9b61
-- [WARNING] The /home/apiadmin/SSSAPI-main_7000/credentials.properties file was deleted.
-- [INFO] The API server container is being configured to start as a systemd service. --
-- [INFO] The API server container is set to autostart. --

DEPRECATED command:
It is recommended to use Quadlets for running containers and pods under systemd.

Please refer to podman-systemd.unit(5) for details.
/home/apiadmin/container-utilityBareMetal-api-official.service
-- [INFO] The container was created successfully in the background. To start
the container, run the "systemctl --user start container-utilityBareMetal-api-
official.service" to start using container. --
-- [INFO] To log in to the container, run the "podman exec -it utilityBareMetal-api-
official /bin/bash" command. --
2025-12-02 23:46:45,233 INFO:      To start a new container again run the /home/apiadmin/
SSSAPI-main_7000/./startAPIContainer command.

```

After this command runs successfully, the Utility Node hosts the API server container, but the container is not started.

Start the API server container

To start the API server container, complete the following steps:

1. If you are not already logged in, log in as the **apiadmin** user on the Utility Node.
2. Start the API server container by using the **systemd** service of this container.

```
systemctl --user start container-utilityBareMetal-api-official.service
```

Note: When the API server container is started, a self-signed certificate is autogenerated for the authentication. If required, replace this certificate with your organization-owned, CA-signed certificate. For more information, see [“Replace the autogenerated self-signed certificate with your organization-owned, CA-signed certificate”](#) on page 125.

3. Verify that the API server container is active.

```
systemctl --user status container-utilityBareMetal-api-official.service
```

A sample output is as follows:

```
container-utilityBareMetal-api-official.service - Podman container-utilityBareMetal-api-official.service
   Loaded: loaded (/home/apiadmin/.config/systemd/user/container-utilityBareMetal-api-official.service; enabled; preset: disabled)
   Active: active (running) since Tue 2025-05-06 04:30:36 EDT; 6s ago
     Docs: man:podman-generate-systemd(1)
   Process: 156908 ExecStart=/usr/bin/podman start utilityBareMetal-api-official (code=exited, status=0/SUCCESS)
    Main PID: 156952 (common)
      Tasks: 18 (limit: 820520)
     Memory: 17.0M
        CPU: 140ms
   CGroup: /user.slice/user-1002.slice/user@1002.service/app.slice/container-utilityBareMetal-api-official.service
           └─156928 /usr/bin/slrp4netns --disable-host-loopback --mtu=65520 --enable-sandbox --enable-seccomp --enable-ipv6 -c -r 3 -e 4 --netns-type=path /run/user/1002/netns>
               └─156930 rootlessport
                   └─156936 rootlessport-child
                       └─156952 /usr/bin/common --api-version 1
   -c 9a8d34d2ea684840873a06f7f5d76e26afd1c95c3818625f8cdb877ee46ccab2 -u 9a8d34d2ea684840873a06f7f5d76e26afd1c95c3818625f8cdb8>
```

Replace the autogenerated self-signed certificate with your organization-owned, CA-signed certificate

When the API server container is started, a TLS or SSL self-signed certificate is autogenerated and placed at `/home/apiadmin/keys`. By default, this self-signed certificate is imported to the API server container to start the API server. Replace this self-signed certificate of the API server container with your organization-owned, CA-signed certificates. Alternatively, generate a new self-signed certificate for your organization and import it to the API server container.

To replace the certificate, complete the following steps:

1. If you are not already logged in, log in as the **apiadmin** user on the Utility Node.
2. Get the CA certificate and its keys that are owned by your organization and save it in the `/home/apiadmin/backup` folder on the Utility Node.
3. Log in to the API server container by issuing the following command:

```
podman exec -it utilityBareMetal-api-official /bin/bash
```

4. Go to the /home/backup folder and verify that your certificates are listed by issuing the following commands:

```
cd /home/backup
```

```
ls -la
```

A sample output is as follows:

```
total 8
drwxr-xr-x 2 root root 4096 Jun  2 07:18 .
drwxr-xr-x 1 root root 4096 Jun  4 06:57 ..
-rw----- 1 root root 1704 Feb 12 07:56 ca.key
-rw-r--r-- 1 root root 1440 Feb 12 07:56 ca.pem
-rw-r--r-- 1 root root  41 Feb 12 07:56 ca.srl
-rw-r--r-- 1 root root  90 Feb 12 07:56 san.conf
-rw-r--r-- 1 root root 1054 Feb 12 07:56 server.cr
-rw----- 1 root root 1708 Feb 12 07:56 server.key
-rw-r--r-- 1 root root 1509 Feb 12 07:56 server.pem
```

5. Delete the existing TLS or SSL configuration:

```
/usr/lpp/mmfs/bin/essctl node config delete
```

A sample output is as follows:

```
Node configuration deleted successfully
```

6. Import the new TLS or SSL configuration by issuing the following command:

```
/usr/lpp/mmfs/bin/essctl node config set --cert /home/apiadmin/server.pem --key /home/apiadminserver.key --chain /home/backup
```

A sample output is as follows:

```
Node configuration set successfully
```

7. Verify the status of the **deployadmd** service by issuing the following command:

```
systemctl status deployadmd
```

A sample output is as follows:

```
• deployadmd.service - Deployment API Admin Daemon Service
   Loaded: loaded (/usr/lib/systemd/system/deployadmd.service; enabled; preset: disabled)
   Active: active (running) since Wed 2025-02-12 07:22:51 UTC; 39min ago
     Process: 52 ExecStartPre=/bin/sh -c targetLink=$(/bin/readlink -f /var/mmfs/run) ;
    [[ -n "$targetLink" && "$targetLink" == */mmfs ]]>
   Main PID: 55 (scaleadmd)
     Tasks: 18 (limit: 1638)
    Memory: 17.2M
       CPU: 284ms
    CGroup: /system.slice/deployadmd.service
            └─55 /usr/lpp/mmfs/bin/scaleadmd -deployment -serverType ess -hostname
utilityBareMetal-api-official.gpfs.local

Feb 12 07:22:51 utilityBareMetal-api-official.gpfs.local deployadmd.sh[55]:
%home%:70_MMFSCFG::9:sdrNotifyAuthEnabled:yes:.....>
Feb 12 07:22:51 utilityBareMetal-api-official.gpfs.local deployadmd.sh[55]:
-----
Feb 12 07:22:51 utilityBareMetal-api-official.gpfs.local deployadmd.sh[55]: Starting GRPC
server on unix:///var/mmfs/run/scaleadmd.grpc.sock,>
Feb 12 07:22:51 utilityBareMetal-api-official.gpfs.local deployadmd.sh[55]: REST server
stopped with error: unable to start admin daemon rest>
Feb 12 07:22:51 utilityBareMetal-api-official.gpfs.local deployadmd.sh[55]: TCP gRPC server
stopped with error: unable to start admin daemon >
Feb 12 08:00:39 utilityBareMetal-api-official.gpfs.local deployadmd.sh[55]: REST server
stopped with error: http: Server closed
Feb 12 08:01:38 utilityBareMetal-api-official.gpfs.local deployadmd.sh[55]: Starting GRPC
server on https://[:]:50052, boot mode 2
Feb 12 08:01:38 utilityBareMetal-api-official.gpfs.local deployadmd.sh[55]: Starting REST
API server on https://0.0.0.0:46443, boot mode 2
```

Verify that the last two lines of output state that the GRPC and REST API server are started with boot mode 2. This mode indicates that the TLS or SSL certificate is imported successfully into the API server container.

Set the password for **installgui** user

By default, when the API server container starts, the **installgui** user is automatically created with a password requirement to securely access API requests during GUI deployment. You must set this password before using the deployment GUI. To set the password for **installgui** user, complete the following steps:

1. If you are not already logged in, log in as the **apiadmin** user on the Utility Node.
2. Log in to the API server container.

```
podman exec -it utilityBareMetal-api-official /bin/bash
```

3. Set the password for the **installgui** user.

```
passwd installgui
```

When the **New password:** prompt appears, enter the new password. When the **Retype new password:** prompt appears, re-enter the password to confirm it.

Verifying that the API server container is configured on the Utility Node

If the API server is configured correctly on the Utility Node, you can log in to the OpenAPI specification page and access the API endpoints. To verify that the API server container is configured correctly on the Utility Node, complete the following steps:

1. In a browser, navigate to the following URL:

Replace **<utility_node_IP>** with your system IP address.

```
https://<utility_node_IP>:46443/openapi/#
```

This URL opens the OpenAPI specification page for the IBM Storage Scale native REST API endpoints.

2. If the **This Connection Is Not Private** warning message appears, click the **Visit this website** link in the message. This message appears if the autogenerated self-signed certificate is not replaced with the organization owned CA certificates. For more information, see [“Replace the autogenerated self-signed certificate with your organization-owned, CA-signed certificate” on page 125.](#)
3. After the successful authentication, verify that the IBM Storage Scale native REST API endpoints are listed on the page. If the endpoints are listed, then the API server container is configured correctly on the Utility Node. You can now access the endpoints.

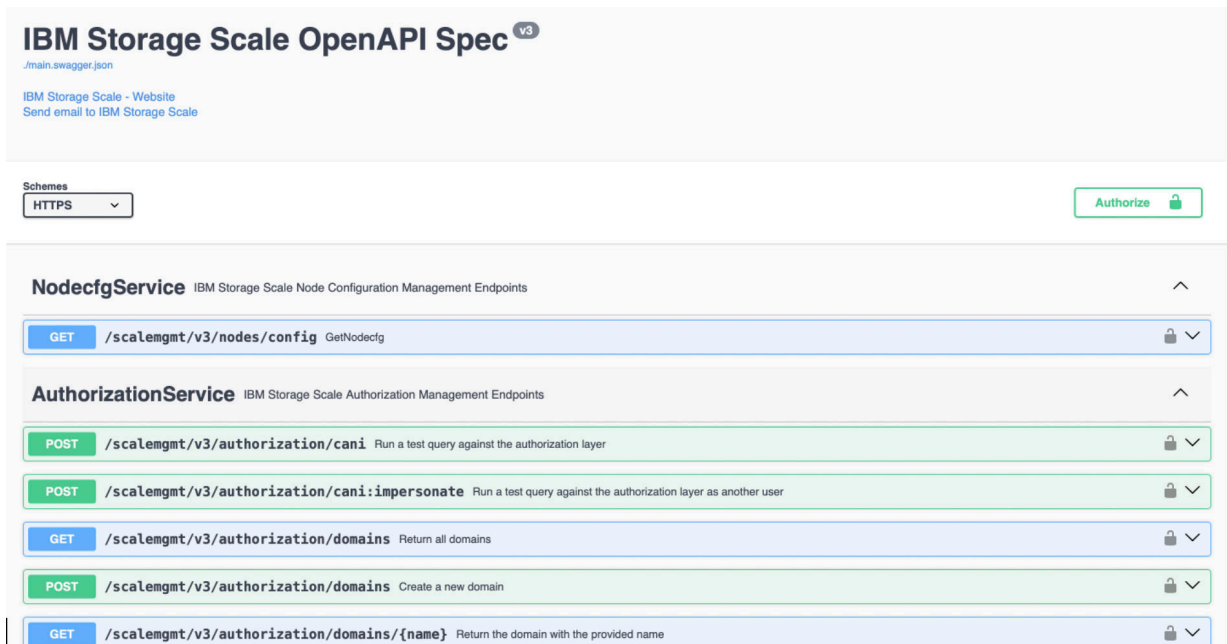


Figure 29. IBM Storage Scale Native REST API endpoints

4. In the OpenAPI specification page, click **Authorize** button.
5. On the **Available authorizations** dialog, enter **Username** and **Password**. Then, click **Authorize**. You can use the **apiadmin** or **apiuser** credentials as required.

Note: All authentication in the API server container use the pluggable authentication modules (PAM). To add a new user, create the user in the API server container by using the **useradd** command. Then, set the required authorization by using the **AuthorizationService** endpoints. Any changes to authorization require authentication as the *apiadmin* user. For more information about configuring PAM authentication, see [Authentication for IBM Storage Scale native REST API](#).

6. After a successful authentication, access the endpoints. For example, the following procedure initiates the **VersionService** endpoint to retrieve the version of the API server container:
 - a. In the OpenAPI specification page, navigate to the **VersionService** endpoint.
 - b. Click **GET /scalemgmt/v3/version** to expand it.
 - c. Click **Try it out** > **Execute**. The version of **scaleadmd** is displayed in the **Responses** section.

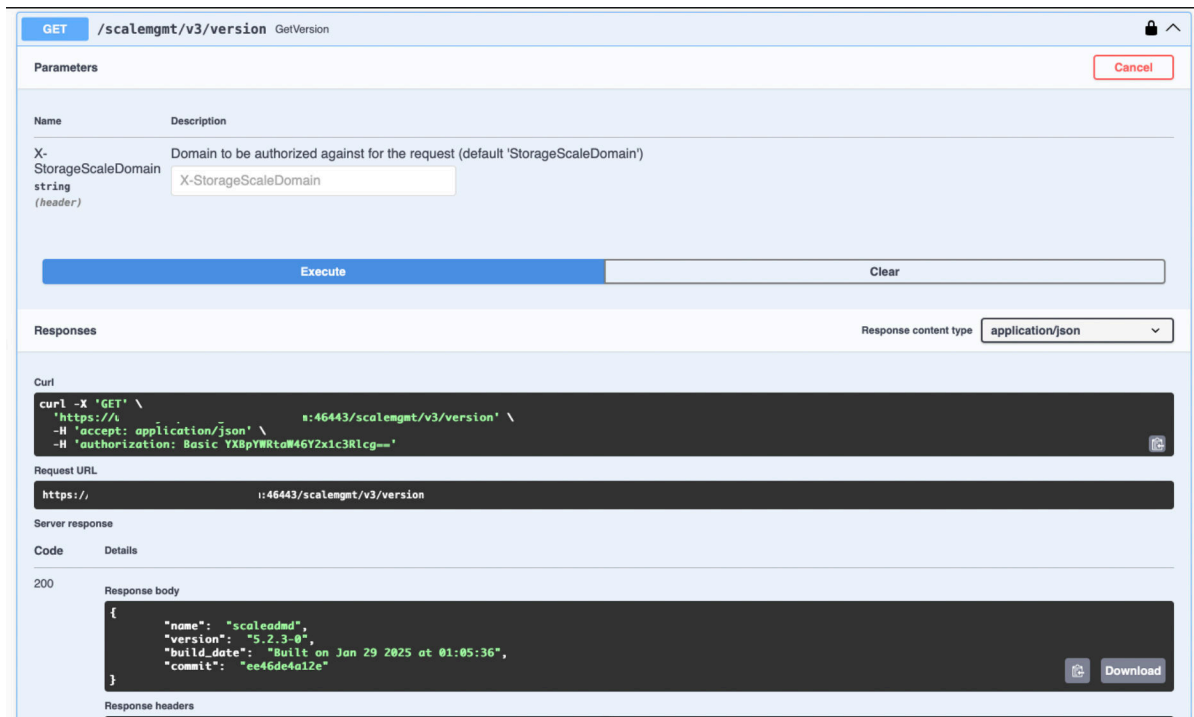


Figure 30. VersionService

Setting up the IBM Storage Scale System deployment GUI container

The IBM Storage Scale System deployment GUI container enables the deployment of the IBM Storage Scale System through the graphical user interface (GUI).

Prerequisites

Ensure that the following prerequisites are completed on the system before setting up the container.

- Ensure that the Utility Node can access the public network through the campus network interface. The campus interface must have a single IPv4 addresses configured. If direct access to the public network is not available, use a proxy server to establish the internet connectivity for the Utility Node.
- Ensure that the system administrator has root access to create other users, which are required to host containers. To verify that a user has root access, run the `whoami` command. If the output is `root`, the user has root privileges.
- Before you set up the containers on the Utility Node, make sure that the Podman and guestfs-tools packages are installed.
- Before you start any containers, make sure that the host file is saved at `/etc/hosts` on all IBM Utility Nodes. Verify that the IP address specified in the hosts file matches the IP address of your system. A sample host file includes the following details:

```
127.0.0.1 localhost localhost.localdomain localhost4 localhost4.localdomain4
::1      localhost localhost.localdomain localhost6 localhost6.localdomain6

# *****
# IBM Utility Node (virbr1) (where EMSVM-23E/CESVM-23E running)
# *****
10.23.16.1 utilityBareMetal

# *****
# EMSVM-23E/CESVM-23E (ras interface IP on EMSVM-23E/CESVM-23E)
# *****
10.23.16.2 emsvm cesvm
10.23.16.3 emsvm cesvm
```

```
10.23.16.4 emsvm cesvm
10.23.16.5 emsvm cesvm
10.23.16.6 emsvm cesvm
```

Important:

- Add only the required host entries to the `/etc/hosts` file to minimize security risks and prevent exposure of information about other nodes in the environment.

Procedure

To set up a deployment GUI container on the IBM Storage Scale System Utility Node, complete the following steps:

1. Create user and user group.
2. Pull the deployment GUI container image and **startDeployGUIContainer** utility.
3. Start the deployment GUI container on the Utility Node and verify that it is active.

Create user and user group

Create the **deployguiadmin** user and user group to host the deployment GUI container. To create the user and user group, and set the user password, complete the following steps:

1. Log in as the root user on the Utility Node.
2. Create the **deployguiadmin** user group.

```
groupadd deployguiadmin
```

3. Create the **deployguiadmin** user and assign it to the **deployguiadmin** user group.

```
useradd -m -g deployguiadmin deployguiadmin
```

4. Set the password for the **deployguiadmin** user.

```
passwd deployguiadmin
```

When the **New password:** prompt appears, enter the new password. When the **Retype new password:** prompt appears, re-enter the password to confirm it.

5. Verify that the **deployguiadmin** user is created.

```
cat /etc/passwd
```

6. Log out as the root user and log in as the **deployguiadmin** user.

Pull the Deployment GUI container image and the startDeployGUIContainer utility

Complete the next steps to pull the image of the Deployment GUI container from the IBM Cloud container repository and the **startDeployGUIContainer** utility package from the [GitHub](#). Internet access is required on the node to complete this step. If the Utility Node does not have internet access, pull the container image from a node that is connected to internet and move the downloaded files to the Utility Node. Alternatively, you can download these files from the [IBM Fix Central](#).

To pull the image of the Deployment GUI container and **startDeployGUIContainer** utility package, complete one of the following methods:

For a Utility Node with internet access

1. In a browser, log in to [MyIBM Container Software Library](#) by using the IBMid and password that are associated with your IBM Storage Scale System.
2. In the **Entitlement keys** section, click **Copy** to get the entitlement key in the clipboard. If no key exists, click **Add new key**, then copy the new key to the clipboard.

Note: To verify that your entitlement key is valid, go to the left pane and click **Container software library**. You can view the list of products that you are entitled to.

3. On the Utility Node, log in as the **deployguiadmin** user and issue the following command to log in to the IBM Cloud container Repository:

```
podman login cp.icr.io --username cp --password <APIKEY>
```

Where,

<APIKEY>

Specifies the entitlement key that is obtained from the **Container software library**.

4. Clone the SSSDEPLOYGUI repository from GitHub on the Utility Node at /home/deployguiadmin. Alternatively, use the **Download ZIP** option on GitHub to manually download the SSSDEPLOYGUI-main repository on the Utility Node at the /home/deployguiadmin. This repository includes the **startDeployGUIContainer** utility file, which starts the deployment GUI container.
5. If downloaded manually, extract the downloaded SSSDEPLOYGUI-main.zip file at /home/deployguiadmin.

```
unzip <SSSDEPLOYGUI-main.zip>
```

6. In the CLI, change the path to the extracted directory.

```
cd </home/deployguiadmin/SSSDEPLOYGUI-main>
```

7. Start the **startDeployGUIContainer** utility and enter the image version.

```
./startDeployGUIContainer
```

A sample output is as follows:

```
Enter the campus interface name (default: campus):
Enter the image version (default: 7.0.1.x):
Enter the API port (default: 46443):
Enter the deployment GUI port (default: 30443):

Detailed logs are located in the ./logs directory.

2025-11-24 07:41:27,791 INFO:      All configurable variables were validated successfully.
2025-11-24 07:41:27,919 INFO:      No images are installed. No cleanup is required.
2025-11-24 07:41:27,921 INFO:      The deployment GUI container image installation is
about to begin. No changes are applied if the image is already installed.
-- [INFO] The container image is about to be pulled from the IBM repository. --
-- [WARNING] The existing container image cp.stg.icr.io/cp/scalesystem/sss_api:7.0.1.x
is being deleted forcefully. --
Pulling the container image: cp.stg.icr.io/cp/scalesystem/sss_deploygui:7.0.1.x...
Getting image source signatures
Copying blob a2ffa89d44b0 done |
Copying blob 1df576680603 skipped: already exists
Copying blob a1b9ed56766e done |
Copying blob c5bb15b88a5f done |
Copying blob e5d1718abf95 done |
Copying blob d3cc7dd07291 skipped: already exists
Copying blob 144cf415b4b9 done |
Copying blob 993fbc37c02f done |
Copying blob 656649aae3da done |
Copying blob f5129bf44a94 done |
Copying config 95574030d2 done |
Writing manifest to image destination
95574030d206d398f8b07df2c17e741a441cfca35ec0795f25ac42e97a87c81f
-- [INFO] The container image was pulled successfully. --
2025-11-24 07:41:40,695 INFO:      The container image installation completed
successfully.
2025-11-24 07:41:40,697 INFO:      The container is about to start. On later runs, use
the 'startDeployGUIContainer' command to manage the container.
-- [INFO] Running the container image - cp.stg.icr.io/cp/scalesystem/
sss_deploygui:7.0.1.x
-- [WARNING] The '-x' or '--force' option removes containers that are in the EXIT state.
--
-- [INFO] Automatic initialization of the container will begin shortly. --
-- [INFO] Startup can take several minutes. --
```

```
e3c6a47ad4ef893a8918b714ae9f2b13da450fcadff137586847ee889f9e3566
-- [INFO] The deployment GUI container is being configured to start as a systemd
service. --
-- [INFO] The deployment GUI container is set to autostart. --

DEPRECATED command:
It is recommended to use quadlets for running containers and pods under systemd.

Refer to podman-systemd.unit(5) for details.
/home/deployguiadmin/container-utilityBareMetal-deploygui-official.service
-- [INFO] The container was created successfully in the background. To start
this container, run the 'systemctl --user start container-utilityBareMetal-deploygui-
official.service' command. --
-- [INFO] To log in to the container, run the 'podman exec -it utilityBareMetal-
deploygui-official /bin/bash' command. --
2025-11-24 07:41:44,140 INFO: To start a new container, run the '/home/deployguiadmin/
SSSDEPLOYGUI-main/./startDeployGUIContainer' command.
```

After this command runs successfully, the Utility Node hosts the deployment GUI container, but the container is not started.

For a Utility Node without internet access

1. Download the Deployment GUI container image `sss_deploygui_7.0.1.x.tar` and the utility package `SSSDEPLOYGUI-main.zip` from IBM Fix central or MyIBM Container software library.

Using IBM Fix central

- a. From a node that is connected to the internet, go to [IBM Fix Central](#).
- b. Select `Scale_System_GUI-7.0.1.x`.
- c. Download the `Scale_System_GUI-7.0.1.x.tgz` file. This file contains the Deployment GUI container image `sss_deploygui_7.0.1.x.tar` and the utility package `SSSDEPLOYGUI-main.zip` file.
- d. Transfer the `sss_deploygui_7.0.1.x.tar` and `SSSDEPLOYGUI-main.zip` files to `/home/deployguiadmin` on the Utility Node.
- e. On the Utility Node, extract the `SSSDEPLOYGUI-main` file at `/home/deployguiadmin`.

```
unzip <ZipFileName>
```

Using MyIBM Container software library

- a. From a node that is connected to the internet, log in to [MyIBM Container Software Library](#) by using the IBMid and password that are associated with your IBM Storage Scale System.
- b. In the **Entitlement keys** section, click **Copy** to get the entitlement key in the clipboard. If no key exists, click **Add new key**, then copy the new key to the clipboard.

Note: To verify that your entitlement key is valid, go to the left pane and click **Container software library**. You can view the list of products that you are entitled to.

- c. Log in to the IBM Cloud container Repository.

```
podman login cp.icr.io --username cp --password <APIKEY>
```

Where,

<APIKEY>

Specifies the entitlement key that is obtained from the **Container software library**.

- d. Pull the image of the Deployment GUI container.

```
podman pull cp.icr.io/cp/scalesystem/sss_deploygui:7.0.1.x
```

- e. Export the pulled image as a .tar file on the node that is connected to the internet.

```
podman save cp.stg.icr.io/cp/scalesystem/sss_deploygui:7.0.1.x -o SSSDEPLOYGUI.tar
```

- f. Download the **SSSDEPLOYGUI-main** repository by using the **Download ZIP** option on [GitHub](#) on the same node. This repository includes the **startDeployGUIContainer** utility file, which starts the Deployment GUI container.
- g. Move the generated `sss_deploygui_7.0.1.x.tar` file and the downloaded ZIP **SSSDEPLOYGUI-main** file to the `/home/deployguiadmin` on the Utility Node.
- h. On the Utility Node, extract the SSSDEPLOYGUI-main file at `/home/deployguiadmin`.

```
unzip <ZipFileName>
```

2. In the CLI, change the path to the extracted directory.

```
cd </home/deployguiadmin/SSSDEPLOYGUI-main>
```

3. Restore the .tar file by using the **-f** option through the **startDeployGUIContainer** utility program.

```
./startDeployGUIContainer -f ../SSSDEPLOYGUI.tar
```

Note: To get help information for the **startDeployGUIContainer** utility, issue the following command:

```
./startDeployGUIContainer --help
```

A sample output is as follows:

```
usage: startDeployGUIContainer [-h] [-v] [-f FILENAME] [-p PORT] [-ap API_PORT] [-i
CAMPUS_INTERFACE] [-vn IMAGE_VERSION]

optional arguments:
  -h or --help            Display the help message and exit.
  -v or --verbose         Display verbose messages in the console.
  -f FILENAME or --file-name FILENAME
                          Restore the container from a local tar file. The default value
is SSSDEPLOYGUI.tar.
  -p PORT or --port PORT  Port on which the GUI server listens. The default value is
30443.
  -ap API_PORT, --api-port API_PORT
                          Port on which the API server listens. The default value is
46443.
  -i CAMPUS_INTERFACE or --campus-interface CAMPUS_INTERFACE
                          Name of the interface connected to the campus network. The
default value is campus.
  -vn IMAGE_VERSION or --image-version IMAGE_VERSION
                          API Server version to be deployed. The default value is
7.0.1.x.
```

4. Start the **startDeployGUIContainer** utility and enter the image version.

```
./startDeployGUIContainer
```

A sample output is as follows:

```
Enter the campus interface name (default: campus):
Enter the image version (default: 7.0.1.x):
Enter the API port (default: 46443):
Enter the deployment GUI port (default: 30443):

Detailed logs are located in the ./logs directory.

2025-11-24 07:41:27,791 INFO:      All configurable variables were validated successfully.
2025-11-24 07:41:27,919 INFO:      No images are installed. No cleanup is required.
2025-11-24 07:41:27,921 INFO:      The deployment GUI container image installation is
about to begin. No changes are applied if the image is already installed.
-- [INFO] The container image is about to be pulled from the IBM repository. --
-- [WARNING] The existing container image cp.stg.icr.io/cp/scalesystem/sss_api:7.0.1.x
is being deleted forcefully. --
Pulling the container image: cp.stg.icr.io/cp/scalesystem/sss_deploygui:7.0.1.x...
Getting image source signatures
Copying blob a2ffa89d44b0 done |
Copying blob 1df576680603 skipped: already exists
Copying blob a1b9ed56766e done |
```

```

Copying blob c5bb15b88a5f done |
Copying blob e5d1718abf95 done |
Copying blob d3cc7dd07291 skipped: already exists
Copying blob 144cf415b4b9 done |
Copying blob 993fbc37c02f done |
Copying blob 656649aae3da done |
Copying blob f5129bf44a94 done |
Copying config 95574030d2 done |
Writing manifest to image destination
95574030d206d398f8b07df2c17e741a441cfca35ec0795f25ac42e97a87c81f
-- [INFO] The container image was pulled successfully. --
2025-11-24 07:41:40,695 INFO: The container image installation completed
successfully.
2025-11-24 07:41:40,697 INFO: The container is about to start. On later runs, use
the 'startDeployGUIContainer' command to manage the container.
-- [INFO] Running the container image - cp.stg.icr.io/cp/scalesystem/
sss_deploygui:7.0.1.x
-- [WARNING] The '-x' or '--force' option removes containers that are in the EXIT state.
--
-- [INFO] Automatic initialization of the container will begin shortly. --
-- [INFO] Startup can take several minutes. --
e3c6a47ad4ef893a8918b714ae9f2b13da450fcadff137586847ee889f9e3566
-- [INFO] The deployment GUI container is being configured to start as a systemd
service. --
-- [INFO] The deployment GUI container is set to autostart. --

DEPRECATED command:
It is recommended to use quadlets for running containers and pods under systemd.

Refer to podman-systemd.unit(5) for details.
/home/deployguiadmin/container-utilityBareMetal-deploygui-official.service
-- [INFO] The container was created successfully in the background. To start
this container, run the 'systemctl --user start container-utilityBareMetal-deploygui-
official.service' command. --
-- [INFO] To log in to the container, run the 'podman exec -it utilityBareMetal-
deploygui-official /bin/bash' command. --
2025-11-24 07:41:44,140 INFO: To start a new container, run the '/home/deployguiadmin/
SSSDEPLOYGUI-main/./startDeployGUIContainer' command.

```

After this command runs successfully, the Utility Node hosts the deployment GUI container, but the container is not started.

Start the deployment GUI container

To start the deployment GUI container, complete the following steps:

1. If you are not already logged in, log in as the **deployguiadmin** user on the Utility Node.
2. Start the deployment GUI container by using the systemd service of this container.

```
systemctl --user start container-utilityBareMetal-deploygui-official.service
```

A sample output is as follows:

```

container-utilityBareMetal-deploygui-official.service - Podman container-utilityBareMetal-
deploygui-official.service
  Loaded: loaded (/home/deployguiadmin/.config/systemd/user/container-utilityBareMetal-
deploygui-official.service; enabled; preset: disabled)
  Active: active (running) since Wed 2025-06-04 02:49:47 EDT; 1min 19s ago
    Docs: man:podman-generate-systemd(1)
  Process: 362703 ExecStart=/usr/bin/podman start utilityBareMetal-deploygui-official
(code=exited, status=0/SUCCESS)
   Main PID: 362744 (common)
     Tasks: 16 (limit: 820520)
    Memory: 12.8M
       CPU: 169ms
   CGroup: /user.slice/user-1001.slice/user@1001.service/app.slice/container-
utilityBareMetal-deploygui-official.service
           └─362721 /usr/bin/slrp4netns --disable-host-loopback --mtu=65520 --enable-
sandbox --enable-seccomp --enable-ipv6 -c -r 3 -e 4 --netns-type=path /run/user/1001/netns/
netn>
               └─362723 rootlessport
                   └─362729 rootlessport-child
                       └─362744 /usr/bin/common --api-version 1
-c 2e2b0db831b07c013ab2d2c1e402d1ed2b26660ef0b719e80e5705775aacb7dc -u
2e2b0db831b07c013ab2d2c1e402d1ed2b26660ef0b719e80e5705775a>
lines 1-14/14 (END)

```

3. Verify whether the deployment GUI container is active.

```
systemctl --user status container-utilityBareMetal-deploygui-official.service
```

4. Start the deployment GUI service. By default, the deployment GUI service is disabled for security. To start the service, complete the following steps:

- a. Log in to the **deployguiadmin** container.

```
podman exec -it utilityBareMetal-deploygui-official /bin/bash
```

- b. If the service is not running, start the GUI.

```
systemctl start gpfsgui
```

- c. Verify that the service is active.

```
systemctl status gpfsgui
```

If the service is not yet started, start the service again.

```
systemctl start gpfsgui
```

After the deployment GUI service starts, access the deployment GUI through the IP address of the Utility Node. For example:

```
https://<utilityBareMetalIP>:30443/
```

Next, set up the API server container if it is not already configured.

Using the deployment GUI

After successful configuration of the IBM Storage Scale System deployment GUI container and IBM Storage Scale System API server container, you can access the deployment GUI to complete the deployment of IBM Storage Scale Systems.

Before you begin

Ensure that the prerequisites are completed on the system. For more information, see [“Prerequisites” on page 129](#).

Procedure

To deploy IBM Storage Scale System by using the GUI, complete the following steps:

1. In a browser, go to the following URL to access the deployment GUI. Replace the <utilityBareMetalIP> with the IP address of the Utility Node.

```
https://<utilityBareMetalIP>:30443/
```

The IBM Storage Scale System deployment GUI wizard is displayed only if the deployment GUI server container is configured correctly. Also, ensure that the machine running the browser can access the GUI port of the Utility Bare-metal server.

2. In the **Welcome** section, select the **I have placed the Management Server VM QCOW2, Script and the Storage Scale System Build file in the / serv directory on the Utility Bare Metal** checkbox.
3. Select the **I accept the License Agreement** checkbox to review and accept the license agreement.
4. After you review the license agreement, click **Close** on **License Agreement**. Then, click **Next**.
5. In the **Network Settings** section, follow these steps:
 - a) Enter the **Management Mask** and **FSP OS Mask** values.
 - b) If required, turn on the **Use High-speed Network** option.
The system enables additional options.

- c) Turn on **Skip Automatic High-speed Network Setup (Requires an existing high-speed network)**.
When this option is enabled, the system does not create network bonds on the I/O nodes. Because the system creates only the EMS VM, it creates the required network bonds only on the EMS VM by using the provided configuration.
- d) Select a **High-speed Network Interface**, and enter the **High-speed Suffix**.
- e) Click **Next**.
6. In the **Enterprise Management** section, follow these steps:
 - a) Enter the **Management Server FQDN**.
 - b) Choose a **License Type** from the drop-down list.
 - c) Enter the **Bare Metal Server Root Password**.
 - d) Turn on the following options based on your requirements:
 - **Re-use existing VM EMSVM and skip EMSVM creation**
 - **Re-use existing VM CESVM and skip CESVM creation**
 - e) Click **Next**.
7. In the **Node Configuration** section, follow these steps:
 - a) Follow the on-screen instructions to download the CSV template, enter the node details, and upload the file to add the configuration details for multiple nodes.
 - b) To add nodes individually, click **Add**, and enter the details manually.

Note:

 - Use short hostnames (for example, node01), not fully qualified domain names (such as node01.example.com).
 - For the management server virtual machine, use the BMC user and BMC password that are used for the underlying IBM Utility Bare-metal server.
 - For serial numbers, add the IBM serial number provided by the IBM Service Support Representative (SSR) during installation. You can also retrieve the serial number of the nodes by running the `essgetconfig` command from the CLI.
 - The **Utility Hostname** and **Utility Host IP** fields are applicable only for the **Protocol VM**, **AFM Gateway**, and **Client VM** node types.
 - c) Turn on **Reuse Existing EMS VM**.
 - With the **Skip Automatic High-speed Network Setup** enabled, the system uses the existing EMS VM and does not create any network bonds. The system expects all required bonds on both the IO nodes and the EMS VM to already exist. You must manually configure all network bonds before deployment.
 - With the **Skip Automatic High-speed Network Setup** disabled, the system uses the existing EMS VM but automatically creates the required network bonds on both the IO nodes and the EMS VM based on the configuration that you provide. No manual bond configuration is required unless otherwise specified.
 - d) Click **Next**.
8. In the **Cluster Settings** section, follow these steps:
 - a) Enter the **Cluster Name** and the **Storage Scale System Build file name**.
 - b) Click **Next**.
9. (Optional) In the **Campus Settings** section, update the following values, and then click **Next**:
 - Campus Gateway
 - Campus IP
 - Campus Mask
10. In the **Create File Systems** section, update the following details:

Note: If you have enabled the **Protocol VM** option, you must create a file system and then configure protocols. Otherwise, you can skip this section.

- a) Enter a **Filesystem Name**.
 - b) Select the nodes from the drop-down list
 - c) Turn on **Create CES Shared Root**, and then update the **Select Nodes** details.
 - d) Click **Next**.
11. In the **Configure Protocols** section, update the required details based on your previous selections.
 12. (Optional) In the **Configure GUI** section, create the GUI credentials by entering a username and password, and then click **Next**.
 13. In the **Review Configuration** section, verify the details, and then click **Start Deployment**.
The system displays a confirmation pop-up.
 14. If you want the deployment to stop when an existing EMS VM or Deployment Container is detected, turn on the required option in the **Clean-up** section.

Note: If you have already enabled the Reuse existing EMS VM option in the **Node Configuration** section, the system doesn't display the **Clean-up** options.

15. Click **Start Deployment**.
The deployment begins, and the system displays a message after the deployment is complete.

Note:

- The deployment fails if any existing management server VM or deployment container is detected.
 - Deployment runs independently of the browser. You can refresh or close the browser and log in again to view the progress, as long as the deployment continues without errors.
 - If the deployment stops due to an error, download the log file to review the output and troubleshoot the issue. After resolving the error, you must restart the deployment from the beginning.
 - In the **Deployment has been completed successfully** dialog, the management GUI for your IBM Storage Scale System is generated automatically based on the IP address provided in the node configuration.
 - If the deployment fails or is partially completed, clean up the incomplete deployment before you retry or restart the deployment. Otherwise, the system might detect a partially deployed cluster and redirect you to the Upgrades page instead of starting a new deployment.
16. Click **Finish** to complete the deployment.

Chapter 4. Software upgrade and firmware update

To improve stability, performance, and compatibility, upgrade an existing cluster and a file system.

Important: Before you upgrade, download the latest precheck tool from [GitHub](#) or obtain a static copy from IBM Fix Central unified container packaging. Extract the tool and run it on the ESS nodes in your cluster to validate system health. For instructions, see the README file in the GitHub repository.

Upgrade methods

You can upgrade IBM Storage Scale System online or offline based on system availability and maintenance planning.

Upgrade can be done by using the following methods.

Offline upgrade

This method is faster than an online upgrade. The nodes are upgraded in parallel but the system is typically taken down for a period. You can update all nodes at the same time or a subset of them. Preferred when no cluster exists in the environment, or during a scheduled outage for the system.

Online upgrade

This method requires a cluster and file systems, or multiple clusters and file systems. The cluster stays fully available, but requires nodes to be upgraded in a specific order (usually one per building block) to keep recovery groups accessible all the time.

Protocol nodes

- When upgrading protocol nodes, the **essrun** command does not automatically upgrade IBM Storage Scale. On protocol nodes, use the installation toolkit (recommended) or use the **xpm** command to upgrade IBM Storage Scale. For more information, see [Upgrading IBM Storage Scale protocol nodes](#) and [Upgrading IBM Storage Scale components with installation kit](#).
- If protocol nodes are part of the upgrade procedure, see [IBM Storage Scale supported upgrade paths](#) for more information.

Firmware updates

After upgrading, use the **essinstallcheck** command to determine whether a firmware update is required.

- During an upgrade, only the firmware for SAS adapters of applicable I/O nodes is updated along with the IBM Storage Scale System.
- All other firmware must be updated separately.
- Power9 system firmware must be updated after the IBM Storage Scale System servers are upgraded. For information about how to update the Power9 system firmware, see [“Upgrading the Power9 firmware” on page 158](#).

Support matrix

Table 4. IBM Storage Scale System supported online upgrade paths													
To →	7.0.1 .0	7.0.0 .2	6.2.3 .6	6.2.3 .5	6.2.3 .2	6.2.3 .1	6.2.2 .0	6.2.1 .1	6.2.1 .0	6.1.9 .8	6.1.9 .4	6.1.5 .1	6.1.2 .9
From ↓													
7.0.1 .0	--	--	--	--	--	--	--	--	--	--	--	--	--
7.0.0 .2	☑	--	--	--	--	--	--	--	--	--	--	--	--

Table 4. IBM Storage Scale System supported online upgrade paths (continued)													
To →	7.0.1 .0	7.0.0 .2	6.2.3 .6	6.2.3 .5	6.2.3 .2	6.2.3 .1	6.2.2 .0	6.2.1 .1	6.2.1 .0	6.1.9 .8	6.1.9 .4	6.1.5 .1	6.1.2 .9
From ↓													
7.0.0 .1	☑	☑	--	--	--	--	--	--	--	--	--	--	--
7.0.0 .0	☑	☑	--	--	--	--	--	--	--	--	--	--	--
6.2.3 .6	☑	⊗	--	--	--	--	--	--	--	--	--	--	--
6.2.3 .5	☑	☑	☑	--	--	--	--	--	--	--	--	--	--
6.2.3 .4	☑	☑	☑	☑	--	--	--	--	--	--	--	--	--
6.2.3 .3	☑	☑	☑	☑	--	--	--	--	--	--	--	--	--
6.2.3 .2	☑	☑	☑	☑	--	--	--	--	--	--	--	--	--
6.2.3 .1	☑	☑	☑	☑	☑	--	--	--	--	--	--	--	--
6.2.3 .0	☑	☑	☑	☑	☑	☑	--	--	--	--	--	--	--
6.2.2 .1	☑	☑	☑	☑	☑	☑	--	--	--	--	--	--	--
6.2.2 .0	☑	☑	☑	☑	☑	☑	--	--	--	--	--	--	--
6.2.1 .1	⊗	☑	☑	☑	☑	☑	☑	--	--	--	--	--	--
6.2.1 .0	⊗	☑	☑	☑	☑	☑	☑	☑	--	--	--	--	--
6.2.0 .1	⊗	⊗	☑	☑	☑	☑	☑	☑	☑	--	--	--	--
6.2.0 .0	⊗	⊗	☑	☑	☑	☑	☑	☑	☑	--	--	--	--
6.1.9 .8	⊗	⊗	☑	☑	⊗	⊗	⊗	⊗	⊗	--	--	--	--
6.1.9 .7	⊗	⊗	☑	☑	☑	☑	⊗	⊗	⊗	☑	--	--	--
6.1.9 .6	⊗	⊗	☑	☑	☑	☑	⊗	⊗	⊗	☑	--	--	--
6.1.9 .5	⊗	⊗	☑	☑	☑	☑	☑	⊗	⊗	☑	--	--	--

Table 4. IBM Storage Scale System supported online upgrade paths (continued)													
To →	7.0.1 .0	7.0.0 .2	6.2.3 .6	6.2.3 .5	6.2.3 .2	6.2.3 .1	6.2.2 .0	6.2.1 .1	6.2.1 .0	6.1.9 .8	6.1.9 .4	6.1.5 .1	6.1.2 .9
From ↓													
6.1.9 .4	✗	✗	☑	☑	☑	☑	☑	☑	✗	☑	--	--	--
6.1.9 .3	✗	✗	✗	✗	✗	✗	☑	☑	☑	☑	☑	--	--
6.1.9 .2	✗	✗	✗	✗	✗	✗	☑	☑	☑	☑	☑	--	--
6.1.9 .1	✗	✗	✗	✗	✗	✗	☑	☑	☑	☑	☑	--	--
6.1.9 .0	✗	✗	✗	✗	✗	✗	☑	☑	☑	☑	☑	--	--
6.1.8 .3	✗	✗	✗	✗	✗	✗	✗	☑	☑	☑	☑	--	--
6.1.8 .2	✗	✗	✗	✗	✗	✗	✗	☑	☑	☑	☑	--	--
6.1.8 .1	✗	✗	✗	✗	✗	✗	✗	☑	☑	☑	☑	--	--
6.1.8 .0	✗	✗	✗	✗	✗	✗	✗	☑	☑	☑	☑	--	--
6.1.6 .1	✗	✗	✗	✗	✗	✗	✗	✗	✗	☑	☑	--	--
6.1.6 .0	✗	✗	✗	✗	✗	✗	✗	✗	✗	☑	☑	--	--
6.1.5 .1	✗	✗	✗	✗	✗	✗	✗	✗	✗	☑	☑	--	--
6.1.5 .0	✗	✗	✗	✗	✗	✗	✗	✗	✗	☑	☑	☑	--
6.1.4 .1	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	☑	--
6.1.4 .0	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	☑	--
6.1.3 .1	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	☑	--
6.1.3 .0	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	☑	--
6.1.2 .9	✗	✗	✗	✗	✗	✗	✗	✗	✗	☑	☑	✗	--
6.1.2 .8	✗	✗	✗	✗	✗	✗	✗	✗	✗	☑	☑	✗	☑

Table 4. IBM Storage Scale System supported online upgrade paths (continued)													
To →	7.0.1 .0	7.0.0 .2	6.2.3 .6	6.2.3 .5	6.2.3 .2	6.2.3 .1	6.2.2 .0	6.2.1 .1	6.2.1 .0	6.1.9 .8	6.1.9 .4	6.1.5 .1	6.1.2 .9
From ↓													
6.1.2 .7	✗	✗	✗	✗	✗	✗	✗	✗	✗	☑	☑	✗	☑
6.1.2 .6	✗	✗	✗	✗	✗	✗	✗	✗	✗	☑	☑	✗	☑
6.1.2 .5	✗	✗	✗	✗	✗	✗	✗	✗	✗	☑	☑	☑	☑
6.1.2 .4	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	☑	☑
6.1.2 .3	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	☑	☑
6.1.2 .2	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	☑	☑
6.1.2 .1	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	☑	☑
6.1.2 .0	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	☑	☑

Legend

- --: Not applicable
- ☑: Supported
- ✗: Not supported

General upgrade requirements

- Contact IBM Support when you want to upgrade from versions earlier than 6.1.2.0.
- Downgrades are not supported.

LTS-to-LTS upgrade rules

- Upgrade to the latest level in the current LTS stream before you move to the next LTS level.
- Do not skip LTS levels.

For example:

- 6.1.2.x → 6.1.2.9 → 6.1.9.8 → 6.2.3.x
- Refer to the upgrade support matrix table for supported paths.
- Obtain required intermediate versions from IBM Support if these versions are not available on Fix Central.

Note:

- Do not upgrade systems that are already on an LTS level to match incoming orders.
- Follow the latest LTS path to reach the target version.

Building-block integration

- Upgrade to the current LTS stream before adding a new building block.

- Do not integrate a building block that runs a newer LTS level than the existing system.

Specific upgrade paths

- Upgrade from 6.1.9.4 to 6.2.3.4 by using 6.1.9.8 as an intermediate level.
- Direct upgrade from 6.1.2.9 to 6.2.3.x is not supported.

N-3 upgrade support

Direct upgrades are supported within the N-3 range. For example:

- 6.2.0.x , 6.2.1.x, or 6.2.2.x to 6.2.3.x
- N-3 coexistence is supported only during the upgrade.

Platform-specific restrictions

- Upgrade from 6.2.0.0 or 6.2.0.1 to 6.2.3.x is not supported on Storage Scale System 3500 nodes.
- Upgrade from:
 - 6.1.9.4 , 6.2.0.x, or 6.2.1.x (RHEL 9.2) to 6.2.3.5 (RHEL 9.6) is not supported.
 - Perform an intermediate upgrade as defined in the upgrade support matrix.

For more information about this restriction, see [Nodes may kernel panic during OS upgrade from RHEL 9.2 to RHEL 9.6](#).

Supported configurations

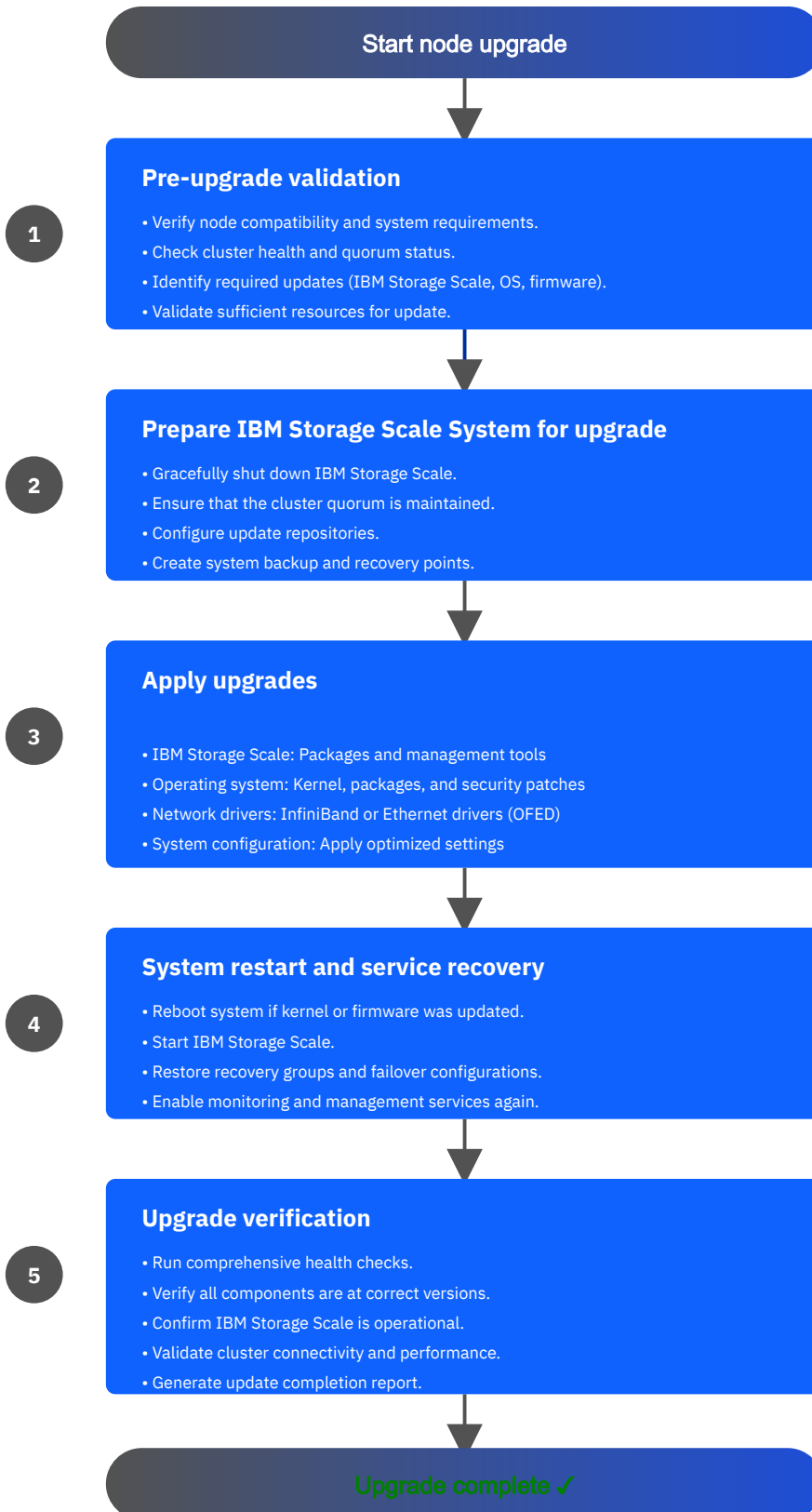
<i>Table 5. Supported configurations</i>	
Management server	Supported nodes
IBM Storage Scale System Utility Node	• IBM Storage Scale System 3000 nodes • IBM Storage Scale System 3200 nodes • IBM Storage Scale System 3500 nodes • IBM Storage Scale System 5000 nodes • IBM Storage Scale System 6000 nodes • IBM Storage Scale System Utility Node management server • IBM Power9 management server • IBM Power9 protocol node • IBM Storage Scale System 3500 protocol node (virtual machine) • IBM Storage Scale System Utility Node protocol node
IBM Power9 management server	• IBM Storage Scale System 3000 nodes • IBM Storage Scale System 3200 nodes • IBM Storage Scale System 3500 nodes • IBM Storage Scale System 5000 nodes • IBM Power9 management server • IBM Power9 protocol node • IBM Storage Scale System 3500 protocol node (virtual machine)

<i>Table 5. Supported configurations (continued)</i>	
Management server	Supported nodes
Bring your own management server (BYOMS)	• IBM Storage Scale System 3500 nodes • IBM Storage Scale System 3500 protocol node (virtual machine) • BYOMS (virtual machine)

Upgrade workflow

During the IBM Storage Scale System node upgrade, high-level operations are run to upgrade storage software, the operating system, and network drivers without any impact on cluster availability.

The following upgrade flowchart contains the upgrade tasks that are completed on IBM Storage Scale System nodes.



Important:

A timing report is generated for each upgrade and stored in the `/var/log/ess/7.0.1.0/` directory on the deployment container. The report records the start time, end time, and duration of high-level operations during the upgrade.

These details provide visibility into the overall upgrade timeline and the duration of individual operations. Use the report for performance analysis, monitoring, troubleshooting, and identification of operations that take longer.

Preparation and prerequisites

1. Before the online upgrade, make sure that all IBM Storage Scale System nodes are active by running the following command from one of the cluster nodes:

```
/usr/lpp/mmfs/bin/mmgetstate -N $(/usr/lpp/mmfs/bin/mmisnodeclass -Y |
/usr/lpp/mmfs/bin/mmfields nodeClass allMembers |
grep -iE "ess3500_x86_64|ess3200_x86_64|ess3500_x86_64|ess5k_ppc64le|s6k_x86_64" |
awk '{ if (NF > 1) print $(NF-1)}' | paste -d',' -s)
```

2. Download the target IBM Storage Scale System release from [IBM Fix Central](#).
3. If IBM Storage Scale native REST API is enabled on the IBM Storage Scale System cluster, disable the SSH/SCP wrapper during the upgrade. To return to SSH and SCP, you can issue the following command:

```
mmchcluster -r /usr/bin/ssh -R /usr/bin/scp
```

4. Copy the IBM Storage Scale System release image, which is downloaded from IBM Fix Central, to the `/home/deploy` directory on the management server.

Remember:

- The GUI is stopped during the management server upgrade.
- Firmware is not automatically updated. Consider this while you plan the maintenance window in the environment to be upgraded.

Important:

- An update must be done over the management network. Use the correct hostname.
- High-speed network might become inaccessible during the upgrade process.

Note: Offline upgrade assumptions (management server or protocol nodes only)

- You assume the risks of potential quorum loss.
- The IBM Storage Scale GUI and collector must be down.

Setting up a container

Set up the target release container on the Utility Node MSVM, Power9 management server, or any other management server, decompress installation packages, and start the container for system updates and prechecks.

Set up the target release container by completing these steps:

1. Decompress the package.

```
cd /home/deploy ; xz --decompress <xz package>
```

2. Decompress the generated tarball.

```
tar -xvf <uncompressed tarball>
```

3. Start the container.

```
sh <untarred binary file>
```

Note:

- If this is not the first time that you run the container or if there is no hardware change on the system, it is not necessary to cleanup the inventory. In a later step, the inventory file is filled with the latest hardware information of the system by running the **essrun config load** command.
- It is highly recommended to run the **essrun config load** command in a later step, even if the hardware inventory from the container is not deleted.
- Manage the deployment container by using the **essmkym1** script, not by using direct **podman** commands.

4. Import nodes configuration in to the management node.

```
essrun -N <management hostnames> config load -p
```

Use applicable *managment hostnames* to a management server. The following command is an example of which other nodes configuration can be imported.

```
essrun -N emsvm,ems,ems2,essio1,essio2,prt01,prt02 config load -p
```

Where:

emsvm

Specifies a management sever virtual machine.

ems

Specifies a management server.

essio1,essio2

Specifies I/O nodes.

prt01,prt02

Specifies protocol nodes.

5. Check nodes for issues.

```
essrun -N ems,ems2,essio1,essio2,prt01,prt02 update --precheck
```

Use [IBM Storage Scale Systems Precheck tool](#) to check all nodes before the upgrade.

If you run the precheck during the upgrade, the upgrade takes longer time. Therefore, it is recommended to run the precheck before you start the upgrade. For more information, see [“Checking the system health by using the decoupled precheck tool” on page 147](#).

Checking the system health by using the decoupled precheck tool

The upgrade precheck tool is a stand-alone utility that verifies the health of IBM Storage Scale System clusters. Use this tool at any time to assess system health and validate critical components, configurations, and hardware status across all cluster nodes. Use the tool before upgrades to confirm system readiness. You can also use the tool for ongoing health checks and routine cluster maintenance.

Before you begin

Make sure that the following requirements are met:

- Python 3.10 or later is installed on IBM Storage Scale System Utility Node.

Important: For EMS P9 or BYOE with RHEL 8.8, use Python 3.11.

- The **sshpas** package is installed on IBM Storage Scale System Utility Node.

```
yum install -y sshpas
```

- You have SSH access to Utility Node.

About this task

The precheck tool validates system health. This tool checks the following components:

GNR health check (**gnrhealthcheck**)

- Validates GNR health status.
- Checks for enclosure and component issues.
- Verifies that hardware components are functioning properly.

System HAL check (**system_check**)

- Verifies system hardware abstraction layer (HAL) health by running the following command:

```
/opt/ibm/ess/hal/bin/system_check -c all
```

- Validates system-level hardware components.

Node type and OS version validation

- Validates node types across the cluster.
- Checks OS version compatibility (for IBM Storage Scale System 6000 nodes).
- Make sure that IBM Storage Scale System 6000 nodes are running minimum Red Hat Enterprise Linux 9.4.

Storage quick check (**essstoragequickcheck**)

- Runs comprehensive storage validation on I/O nodes.
- Checks storage adapters, enclosures, and drives.
- Validates storage configuration and health.

Storage firmware check

- Verifies firmware versions on I/O nodes.
- Checks enclosure firmware level.
- Uses the **mmfsmfirmware** command for validation.

Firewall status and port check

- Validates firewall configuration on all cluster nodes.
- Checks that required ports are open for cluster communication.
- Verifies ports for SSH, GPFS, GUI, monitoring, and other services.

Network verification (**mmnetverify**)

- Validates network configuration and connectivity.
- Checks communication between all cluster nodes.
- Runs the **mmnetverify -N all** command to verify network health.

Node health check (**mmhealth**)

- Checks overall health of all cluster nodes.
- Identifies unhealthy components (DEGRADED, FAILED, CHECKING states).
- Runs the **mmhealth node show --unhealthy -a** command.

The tool displays one of the following statuses for each check:

✓ HEALTHY

The component is functioning correctly. No action is required.

⚠ WARNING

Minor issues are detected. Address these issues, but these issues do not block the upgrade.

x CRITICAL

Serious issues are found. Resolve these issues before upgrade.

⚠ ERROR

The system check failed.

Procedure

1. Download the precheck tool from [GitHub](#).
2. Extract the tool on Utility Node.
 - a) Navigate to the download directory.

```
cd /path/to/download
```

- b) Extract the archive.

```
tar -xzf Scale_System_PRECHECK_TOOL-main.tar.gz
```

- c) Open the extracted folder.

```
cd Scale_System_PRECHECK_TOOL-main
```

3. Make the `upgrade_precheck.py` script executable.

```
chmod +x upgrade_precheck.py
```

4. Run the tool.

- For standard configurations:

```
./upgrade_precheck.py
```

- For EMS P9 or BYOE with RHEL 8.8:

```
python3.11 upgrade_precheck.py
```

5. Enter the connection information.

- EMS SSH hostname (the management server hostname or IP address)
- SSH username (default: root)
- SSH password (no password needed if you use key-based authentication)

Example:

```
Enter the EMS SSH hostname: ems_vm1
Enter the SSH username (default root):
Enter the SSH password (leave blank for key-based auth):
```

Note: The tool uses SSH to connect to the EMS node and then runs health checks across all nodes in the cluster.

6. Review the health check progress.

The tool displays real-time progress:

```
Starting comprehensive system health check...
=====
[1/8] Checking ESS GNR Health (gnrhealthcheck)... ✓
[2/8] Checking System HAL... ✓
[3/8] Checking Node Types and OS Versions... ✓
[4/8] Checking ESS Storage Quick Check... ✓
[5/8] Checking Storage Firmware... ✓
[6/8] Checking Firewall and Ports... ✓
[7/8] Checking ESS Network (mmnetverify)... ✓
[8/8] Checking ESS Node Health (mmhealth)... ✓
```

7. Review the health report.

The tool generates a report that includes the following information:

- Summary of all checks with pass or fail status.
- Detailed resolution guidance for any found issues.
- Time estimates to resolve problems.
- Whether the system can proceed with upgrade.

Example summary:

```
SUMMARY:
  ✓ Healthy: 7    □ Warning: 1    ✗ Critical: 0    □ Error: 0

COMPONENT HEALTH STATUS
=====
COMPONENT                STATUS          CAN UPGRADE
-----
ESS GNR Health            ✓ HEALTHY      YES
System HAL                ✓ HEALTHY      YES
Node Types and OS         ✓ HEALTHY      YES
ESS Storage Quick Check   ✓ HEALTHY      YES
Storage Firmware          □ WARNING      YES
Firewall and Ports        ✓ HEALTHY      YES
ESS Network               ✓ HEALTHY      YES
ESS Node Health           ✓ HEALTHY      YES
```

8. Resolve any critical issues.

Important: Resolve all CRITICAL issues before upgrade. The report provides specific steps to address each issue. If critical issues are not resolved, do not proceed with the upgrade.

Tip: If timeout errors occur (on systems with many storage enclosures), you can adjust timeout values in the script. See the *Timeout Configuration* section in the [README.md](#) file for instructions to increase timeout values.

Results

The tool saves the health check report to a log file in the current directory:

```
Report saved to: health_report_<hostname>_<timestamp>.log
```

You can review this log file for reference or troubleshooting.

What to do next

Complete the following tasks based on the health check results:

- If all checks pass or show only warnings, you can proceed with the upgrade. Warnings indicate minor issues that must be addressed but do not block the upgrade.
- If critical issues are found, resolve the issues before upgrade. Follow the detailed resolution guide in the report, which provides specific steps for each issue.
- After you resolve issues, run the precheck tool again to verify that issues are resolved successful and no new issues appear.
- Keep the health report log file for documentation, audit purposes, and future reference.

For complete documentation, troubleshooting guidance, and timeout configuration, see the [README.md](#) file in the [GitHub repository](https://github.com/IBM/Scale_System_PRECHECK_TOOL) (https://github.com/IBM/Scale_System_PRECHECK_TOOL).

Upgrading GUI high availability nodes

In a GUI high availability (HA) environment, upgrade all GUI nodes before you upgrade the cluster nodes by using the **essrun** CLI. Upgrade the GUI nodes one at a time so that at least one GUI node is available during the upgrade process.

Before you begin

Make sure that the upgrade packages are available on each GUI node and identify the master and other two GUI nodes.

About this task

A cluster with GUI HA can have three IBM Storage Scale GUI nodes. One of these nodes is called master GUI node and the other two are called primary and secondary nodes.

Master GUI node

This node is automatically elected to complete cluster-wide tasks that must run only once in the cluster. These tasks include snapshot schedules, email notifications, and SNMP notifications. If the current master GUI node becomes unavailable, another GUI node is automatically elected as the master GUI node.

Primary and secondary nodes

One of these nodes can become a master node when the current master GUI node fails.

Procedure

1. Find the GUI nodes.

```
/usr/lpp/mmfs/gui/cli/lsnode
```

A sample output is as follows:

Hostname	IP	Description	Role	Product version	Connection status
GPFS status Last updated					
gpfsgui-11.novalocal	10.0.100.12	Master GUI Node	management,storage	7.0.1.0	HEALTHY
HEALTHY	7/10/17 10:19 AM				
gpfsgui-12.novalocal	10.0.100.13		storage,ces	7.0.1.0	HEALTHY
HEALTHY	7/10/17 10:19 AM				
gpfsgui-13.novalocal	10.0.100.14		storage,ces	7.0.1.0	HEALTHY
HEALTHY	7/10/17 10:19 AM				

where:

gpfsgui-11.novalocal

Is a master GUI node that fails over automatically.

gpfsgui-12.novalocal

Is the primary GUI node in other nodes.

gpfsgui-13.novalocal

Is the secondary GUI node in the other nodes.

2. Stop the GUI service on the secondary GUI node.

```
systemctl stop gpfsgui-13.novalocal
```

Verify that the GUI service is stopped.

```
systemctl status gpfsgui-13.novalocal
```

3. Stop the GUI service on the primary GUI node.

```
systemctl stop gpfsgui-12.novalocal
```

Verify that the GUI service is stopped.

```
systemctl status gpfsgui-12.novalocal
```

4. Upgrade the primary GUI node.

Install the required IBM Storage Scale GUI packages and complete any release-specific postinstallation steps.

```
rpm -Uvh gpfsgui-12.novalocal*.rpm
```

5. Start the GUI service on the primary GUI node.

```
systemctl start gpfsgui-12.novalocal
```

Verify that the GUI service is running and that all GUI nodes are healthy.

```
systemctl status gpfsgui-12.novalocal  
/usr/lpp/mmfs/gui/cli/lsnode
```

6. Upgrade the secondary GUI node.

Install the required IBM Storage Scale GUI packages and complete any release-specific postinstallation steps.

```
rpm -Uvh gpfsgui-13.novalocal*.rpm
```

7. Start the GUI service on the secondary GUI node.

```
systemctl start gpfsgui-13.novalocal
```

Verify that the GUI service is running and the node is healthy.

```
systemctl status gpfsgui-13.novalocal  
/usr/lpp/mmfs/gui/cli/lsnode
```

8. After all GUI nodes are upgraded and verified, start the cluster upgrade by using the **essrun** command.

9. After an IBM Storage Scale System upgrade is complete, start the GUI.

Related information

Updating management servers

Update and reboot the management server, such as Power9 or IBM Storage Scale System Utility Node management server virtual machine (MSVM), restart the management server VM, rerun container processes, and complete the final **essrun** update for IBM Storage Scale.

Note: Upgrade only Power9 firmware. The IBM Utility Node, a bare-metal host, does not need to be updated separately. Because this host is updated automatically when you update the management server VM node on the host.

You can upgrade a single management server or two management servers by completing the following steps:

1. Run the initial update.

```
essrun -N ems update --offline
```

2. If kernel or OFED is changed, exit the container, reboot, and restart the container.

```
exit
reboot
```

Note: You need not reboot the secondary management server because this server automatically reboots, and then reactivates both IIBM Storage Scale and the [I/O nodes](#).

3. If kernel or OFED is not changed, restart the performance monitoring sensor and update I/O nodes.

```
systemctl start pmcollector
systemctl start pmsensors
```

4. Restart the container by using the `essmkym1` script.

```
./essmkym1 --restart
```

The `essmkym1` script is in the extracted directory of the IBM Storage Scale System image that was used when you started the container. Usually, the path is `/home/deploy/ess_6.2.x.x_...dir/`.

5. Update the management node again.

```
essrun -N ems update --offline
```

6. Reboot the management server.

```
exit
reboot
```

7. Reload the management server by using the `reloadEms.sh` script.

```
/opt/ibm/ess/tools/samples/reloadEms.sh
```

8. Restart the performance monitoring services and the IBM Storage Scale GUI.

```
systemctl restart pmsensors
systemctl restart gpfsGUI
```

9. Restart the container again.

```
./essmkym1 --restart
```

10. Update the Power9 firmware. The management server firmware update applicable for Utility Node and bring-your-own management server (BYOMS).

```
cd /install/ess/otherpkgs/rhels8/ppc64le/firmware/
```

Note: The system reboots automatically and the firmware is upgraded. This upgrade can take up to 1.5 hours per node. Consider this for planning the maintenance.



Warning: Do not revert the system firmware level to an earlier version because it can cause a system breakage. Instead, upgrade the firmware to any latest firmware level available.

- a. Verify the package.

```
update_flash -v -f 01VL950_131_045.img
```

- b. Flash the update to set the new image as temporary.

```
update_flash -f 01VL950_131_045.img
```

- c. Commit the new firmware image to permanent.

```
update_flash -c
```

11. Restart the container in the primary management server.

The `essmkym1` script is in the extracted directory of the IBM Storage Scale System image that was used when you started the container. Usually, the path is `/home/deploy/ess_6.2.x.x_...dir/`.

```
cd /home/deploy/ess_6.2.x.x_...dir/ ; ./essmkym1 --restart
```

Updating the system firmware

Use this procedure to update POWER9 firmware on ppc64le nodes.

Procedure

1. Update the management server firmware.

```
cd /install/ess/otherpkgs/rhels8/ppc64le/firmware/
```



Warning: Do not revert the system firmware level to an earlier version because it can cause a system breakage. Instead, upgrade the firmware to any latest firmware level available.

Remember: The system reboots automatically and the firmware is upgraded. This process can take up to 1.5 hours per node. Consider this for planning the maintenance window.

2. Verify the package.

```
update_flash -v -f 01VL950_131_045.img
```

3. Flash the update to set the new image as temporary.

```
update_flash -f 01VL950_131_045.img
```

4. Commit the new firmware image to permanent.

```
update_flash -c
```

Updating the I/O nodes

You can upgrade I/O nodes to keep the nodes up to date, fix issues, improve performance, and make the nodes compatible with the system.

1. Update I/O nodes online or offline.

- To update all I/O nodes.

Online upgrade

```
essrun -N essio1,essio2,essioN... update
```

Offline upgrade

```
essrun -N essio1,essio2,essioN... update --offline
```

- To update a subset of the I/O nodes.

Online upgrade

```
essrun -N essio1,essio2,essioN... update --serial N
```

Offline upgrade

```
essrun -N essio1,essio2,essioN... update --serial N --offline
```

- To update a subset of I/O nodes offline, issue the following command:

For more information about the `--serial` option, see the [essrun](#) command.

Ensure that daemon (IBM Storage Scale), recovery groups, and file system are active on the I/O nodes.

2. Restart the performance monitoring service on each upgraded I/O node.

```
systemctl restart pmsensors
```

3. Update the firmwares of the updated I/O nodes.

Note: When a GPFS cluster exists in the environment and IBM Storage Scale is down, update the enclosures and drives offline. Verify whether IBM Scale is down by issuing the `mmgetstate -a` command, and shut down active IBM storage Scale by using the `mmshutdown` command to update enclosures offline.

- a. Access the node through SSH.
- b. Update the enclosures firmware.

Online update

```
mmchfirmware --type storage-enclosure
```

Offline update

```
mmchfirmware --type storage-enclosure --fast-offline
```

Local update without a GPFS cluster

```
mmchfirmware --type storage-enclosure --local-only
```

Multiple enclosure updates

```
mmchfirmware --type storage-enclosure --update-id PREVIOUS --serial-number  
serialnumber
```

For x86 systems, this command upgrades the BIOS/BMC/FPGA of the canisters.

- c. Update the drives firmware.

Online update

```
mmchfirmware --type drive
```

Offline update

```
mmchfirmware --type drive --fast-offline
```

Local update without a GPFS cluster

```
mmchfirmware --type drive --local-only
```

- d. Update the system firmware. For detailed procedure, see [“Updating the system firmware”](#) on page 154.



Warning: Do not revert the system firmware level to an earlier version because it can cause a system breakage. Instead, upgrade the firmware to any latest firmware level available.

Remember: The system reboots automatically and the firmware is upgraded. This process can take up to 1.5 hours per node. Consider this for planning the maintenance window.

- e. Check health of the I/O nodes that are upgraded.

```
essinstallcheck
```

If an error is displayed, see [Chapter 7, “Troubleshooting,”](#) on page 185 for more information.

Note:

Upgrading the protocol nodes

Upgrade the protocol nodes (virtual machines and Power protocol nodes) by using IBM Storage Scale installation toolkit or the **essrun** command.

About this task

You can upgrade the protocol nodes after the target IBM Storage Scale System release image is available on the management server and, if applicable, on the protocol nodes.

Run the **essrun** commands from the management server or from the active IBM Storage Scale System container, depending on your upgrade workflow.

Note: On IBM Storage Scale System 7.0.1, only online upgrades are supported.

Procedure

1. Upgrade IBM Storage Scale on the protocol nodes.

To upgrade the protocol nodes by using the **essrun** command, complete the following upgrade steps:

- a) Populate the protocol node inventory.

```
essrun -N <ems_node> protocol upgrade configpopulate
```

For example:

```
essrun -N utility1-vm1-hs protocol upgrade configpopulate
```

- b) Check the configuration of populated nodes.

```
essrun -N <ems_node> protocol upgrade nodelist
```

For example:

```
essrun -N utility1-vm1-hs protocol upgrade nodelist
```

- c) Check all protocol nodes before the upgrade.

```
essrun -N <ems_node> protocol upgrade precheck
```

For example:

```
essrun -N utility1-vm1-hs protocol upgrade precheck
```

- d) Upgrade the protocol nodes.

```
essrun -N <ems_node> protocol upgrade run
```

For example:

```
essrun -N utility1-vm1-hs protocol upgrade run
```

For protocol nodes upgrade you can use the installation toolkit. For more information, see [Performing online upgrade by using the installation toolkit](#) in the IBM Storage Scale documentation.

Important: A LEAPP upgrade of an operating system can remove IBM Storage Scale and protocol packages from one or more nodes. If this issue occurs, you can reinstall packages. To install the packages, see [“GPFS and protocol packages removed after LEAPP operating system upgrade”](#) on page 187.

2. Upgrade the operating system and other components from the IBM Storage Scale System container.

- a) Shut down the protocol nodes.

```
mmshutdown -N <protocol_node>
```

where, *protocol_node* that is upgraded for the operating system.

For more information, see the [mmshutdown](#) command.

- b) Start IBM Storage Scale daemon on protocol nodes.

```
mmstartup -N <protocol_node>
```

For more information, see the [mmstartup](#) command.

- c) Upgrade the operation system and other components.

```
essrun -N prt1,prt2 update --offline
```

- d) After the operating system upgrade (kernel changes), create an IBM Storage Scale portability layer.

```
/usr/lpp/mmfs/bin/mmbuildgpl
```

For more information, see [building the GPL on Linux nodes](#).

3. Update the system firmware of the Power protocol nodes.



Warning: Do not revert the system firmware level to an earlier version because it can cause a system breakage. Instead, upgrade the firmware to any latest firmware level available.

Remember: The system reboots automatically and the firmware is upgraded. This process can take up to 1 hour and 30 minutes per node. Consider this for planning the maintenance window.

- a) Go to the firmware directory and look for the 1VL950_131_045.img file.

```
cd /install/ess/otherpkgs/rhels8/ppc64le/firmware/
```

- b) Verify the package.

```
update_flash -v -f 01VL950_131_045.img
```

If the result of the verification does not flag any issues, proceed with the update.

- c) Start the update.

```
update_flash -f 01VL950_131_045.img
```

Flashing the firmware sets the new image as temporary. After you are comfortable with the new firmware, you must commit it to permanent:

```
update_flash -c
```

Restarting nodes and services

After the online or the offline update is complete, complete the following final steps:

Procedure

1. Run this command to start the performance monitoring collector on the EMS node:

```
systemctl start pmcollector
```

2. Run this command to start the performance monitoring sensors on each node:

```
mmdsh -N NodeList | NodeGroup "systemctl restart pmsensors"
```

3. Run this command to enable **gpfsgui** start on reboot:

```
systemctl enable gpfsGUI
```

4. Run this command to start the GUI on the EMS node:

```
systemctl start gpfsGUI
```

5. Run these commands to manually check the health:

```
gnrhealthcheck  
mmhealth node show -a  
mmnetverify -N all
```

Upgrading the Power9 firmware

The Power9 firmware must be upgraded manually. If the firmware is not at the latest level, complete the following steps.

Procedure

1. Run these commands to copy the firmware `img` file from the container to the Power9 management server or the Power9 protocol node:

```
cd /install/ess/otherpkgs/rhels8/ppc64le/firmware/  
sftp EMSNode  
mput 01VL950_131_045.img
```

2. Run this command to shut down the container:

```
podman stop ContainerName
```

3. Upgrade the firmware.



Warning: Do not revert the system firmware level to an earlier version because it might cause a system breakage. Instead, upgrade the firmware to any latest firmware level available.

- a) Run the following command:

```
update_flash -v -f 01VL950_131_045.img
```

- b) If there are no issues, run this update:

```
update_flash -f 01VL950_131_045.img
```

The system restarts and the firmware is upgraded. This process might take up to 1 hour and 30 minutes for each node.

What to do next

Flashing the firmware sets the new image as temporary. After you are comfortable with the new firmware, you *must* make it permanent. To do this, run the following command:

```
update_flash -c
```

Manually updating a boot drive for IBM Storage Scale System 5000

About this task

Complete the following steps to update a boot drive:

Procedure

1. Get the HDD10KV9-4K-9F26-LINUX.rpm file that is available in the /install/ess/otherpkgs/rhels8/ppc64le/firmware container and transfer it to the I/O nodes.

```
scp <complete file with path> ionode:/root
```

2. Install the rpm on an I/O node.

```
# rpm -ivh /root/HDD10KV9-4K-9F26-LINUX.rpm
```

3. Check the installed package.

```
# rpm -ql ibm.sas.HDD10KV9-4K-9F26-1.noarch
/lib/firmware/IBM-ST1200M.A1800017.39463236
/lib/firmware/IBM-ST1800M.A1800017.39463236
/lib/firmware/IBM-ST600MM.A1800017.39463236
```

4. Check whether any boot drives need to be updated.

```
# lsscsi -g | grep -i storage
```

A sample output is as follows:

[1:0:0:0]	storage IBM	ST1800MM0139	9F14	-	/dev/sg1
[1:0:1:0]	storage IBM	ST1800MM0139	9F14	-	/dev/sg2

Here,

ST1800MM0139

This is the product ID.

/dev/sg1 and /dev/sg2

These are the storage devices.

9F14

This is the device version.

5. Check whether the firmware file matches to the product ID.
6. If the firmware file and the product ID are matching, upgrade the firmware.

```
/lib/firmware/IBM-ST1800M.A1800017.39463233
```

7. Upgrade the code on each device.

```
# iprconfig -c update-ucode /dev/sg1 /lib/firmware/IBM-ST1800M.A1800017.39463236
```

A sample output is as follows:

```
# iprconfig[147710]: 0:0:0:0: Updating device microcode using /lib/firmware/IBM-ST1800M.A1800017.39463236 from 39463233 (9F23) to 39463236 (9F26)
```

8. Upgrade the same code on the /dev/sg2 device.

```
# iprconfig -c update-ucode /dev/sg2 /lib/firmware/IBM-ST1800M.A1800017.39463233
```

A sample output is as follows:

```
# iprconfig[147710]: 0:0:0:0: Updating device microcode using /lib/firmware/IBM-ST1800M.A1800017.39463236 from 39463233 (9F23) to 39463236 (9F26)
```

9. After both devices are successfully updated, reboot the I/O node.
10. Check whether the firmware version is changed.

```
# lsscsi -g | grep -i storage
```

A sample output is as follows:

[0:0:0:0]	storage IBM	ST1800MM0139	9F26	-	/dev/sg1
[0:0:1:0]	storage IBM	ST1800MM0139	9F26	-	/dev/sg2

The firmware version is changed from 9F14 to 9F26.

Upgrade flow in IBM Storage Scale System 5000

For customers who want to add storage to an existing building block, this option is now supported. The goal of a capacity upgrade is to expand a customer's available storage, either by expanding an existing file system or adding new file systems, without the need to buy an entirely new building block. Capacity upgrade is designed to be an online operation and to not interrupt customer workloads.

Supported paths

SL models (5U92)

- SL1 -> SL2
- SL2 -> SL3
- SL2 -> SL4
- SL3 -> SL4
- SL3 -> SL5
- SL4 -> SL5
- SL4 -> SL6
- SL5 -> SL6
- SL5 -> SL7
- SL6 -> SL7

SC models (4U106)

- SC1 -> SC2
- SC2 -> SC3
- SC2 -> SC4
- SC3 -> SC4
- SC3 -> SC5
- SC4 -> SC5
- SC4 -> SC6
- SC5 -> SC6
- SC5 -> SC7
- SC6 -> SC7
- SC6 -> SC8
- SC7 -> SC8
- SC7 -> SC9
- SC8 -> SC9

Prerequisites

1. All new or existing building blocks must be at IBM Storage Scale System 6.1.0.0 or later. If there are protocol nodes in the setup, they must also be upgraded to the matching IBM Storage Scale System version.

2. If space needs to be made, for example for moving of the management server, this has to be planned for accordingly.
3. LBS must wear an ESD wrist band when physically working on the hardware (like plugging in SAS cables).

Capacity upgrade considerations

- Do not try to configure call home before the capacity upgrade is complete, that is until resizing is done.
- You can perform additional capacity upgrades while the DA's are rebalancing.
- You can restripe the file system while the DA's are rebalancing.
- Although it is recommended, you do not have to rebalance the file system if NSDs are added during the capacity upgrade.
- For customers who have call home enabled, remember to re-configure call home after the capacity upgrade is complete. This is the last step in the following flow or you can use the instructions in the call home configuration appendix for guidance.

SAS cable plug-in tips

- Unlatch the cable arm from the I/O server node into which you will be plugging in the SAS cable.
- Remove the blue cap.
- Make sure the location code label from the cable matches the port location code and the port number.
- Remove the cap from the SAS cable connector and plug it into the port.
- You should hear a click when the cable is inserted correctly.

SSR tasks

SSR is responsible for the following tasks.

1. Code 20 of the new enclosures - replacing parts as needed.
2. Running or labeling the new SAS cable connections.
3. Potentially making space in the frame - Moving the management server.

SSR is not responsible for checking system health using **essutils** like in a rackless or a rackful solution.

LBS tasks

LBS is responsible for the following tasks.

1. Upgrade of IBM Storage Scale System 6.1.0.0 - prior to the capacity upgrade engagement.
2. Post capacity upgrade health checks.
3. Plugging the SAS cables into the adapters and enclosures.
4. Performing capacity upgrade software functions such as conversion and resizing.
5. New storage management functions such as adding new space to existing file system and creating a new file system.
6. Restripping the file system.
7. Replacing any bad parts such as disks or cables.
8. Pre and post engagement operations

Flow

TDA process ensures that the customer is prepared for the capacity upgrade. Considerations such as if there is enough room in the rack or usage of the file system space are planned out.

LBS

1. LBS performs normal IBM Storage Scale System software upgrade. Customer must be at IBM Storage Scale System 6.1.0.0 for the capacity upgrade. This upgrade is treated as a separate engagement than the future capacity upgrade operation.

=== Capacity upgrade begins ===

SSR

1. The SSR arrives at the customer site. If the management server needs to be moved, the SSR shuts down GPFS and powers down the server to move. For more information, see *Shutting down and powering up IBM Storage Scale System* in *IBM Storage Scale System Quick Deployment Guide*.
2. The SSR places the new enclosures in the rack and establishes power connection. Based on the lights, the SSR performs a code 20 operation. If lights indicate any problem, they might need to take a service action.
3. The SSR runs the new SAS cable connections and labels in a bundle and hooks them to the frame. Later when LBS comes, they simply plug in the connections when required in the flow.
4. The SSR places the management server (if required) back into the existing frame or a new frame. Network connections and power are restored. Once the server is powered on, the SSR (or customer) can start GPFS to return the management server back into the cluster.

LBS

1. Power on the new enclosure(s).
 - For SLx or SCx, the power cord should be connected. Press the switch to turn on the enclosures.
2. Verify that the system is converted to mmvdisk.
 - a. **mmvdisk nodeclass list** - This command shows if the **mmvdisk** node class exists.
This command can be run on any node in the cluster.
3. Upon arrival LBS should first perform the normal upgrade-related system verification steps. Run the following from the management server:
 - a. **gnrhealthcheck** - This command determines if there are any issues in various areas of IBM Storage Scale System. Any problems that show up must be addressed before capacity upgrade starts. Run this command from any node in the cluster. This command only needs to be run once.
 - b. **essinstallcheck -N localhost** - This command checks the system to ensure all components match IBM Storage Scale System 6.1.x.x levels. Run from each node in the cluster including protocol nodes.
Note: For IBM Storage Scale System 5000, a check is added that flags whether the WCE bit is enabled on any drive. If the WCE bit is enabled, refer to the published flash for the recommended action.
 - c. **mmhealth node show -N all --verbose** - This command shows any system health related issues to address. Run this command from any node in the cluster.
 - d. Look for any events or tips that are open in the GUI. These also show up when you issue **mmhealth** but it is good to check in the GUI as well. The GUI is run from the management server node.
 - e. **/usr/sbin/opal-elog-parse -s** - Run this command from each node in the cluster to check for any serviceable events.
4. Verification steps:
 - a. **mmgetstate -a** - Issue this command to ensure that all daemons are active.
 - b. **mmismount all -L** - Issue this command to ensure that all mount points are still up. The file system must only be mounted on the management server and protocol nodes (if applicable).After these issues are resolved, capacity upgrade can begin.

5. Start by moving both recovery groups to essio2-hs.

Note: The following recovery group names are examples.

Move the recovery group in the current I/O server node to the peer I/O server node in the same building block.

- a. To list the recovery groups and the current master server, run:

```
mmvdisk recoverygroup list
```

- b. To move the recovery group from the current active I/O server node (rg_essio1-hs) to the peer I/O server node (essio2-hs) in the same building block, run the following commands in the shown order:

```
mmvdisk recoverygroup change --recovery-group rg_essio1-hs --active essio2-hs
```

Running **mmvdisk recoverygroup list** should show both RGs actively managed by essio2-hs.

6. Plug in the SAS cables for essio1 on the server and enclosure ends. Shut down GPFS, only on the server just modified, and then reboot the I/O node. Wait for 5 minutes for the node to reboot and paths to be rediscovered. Run the following commands to ensure that essio1 has discovered the new enclosures.

Note: Before shutting down GPFS, make sure that autoloading is turned off (**mmchconfig autoloading=no**).

- a. **essstoragequickcheck -N localhost**
- b. **essfindmissingdisks -N localhost**

Both commands should return with no issues and recognize the new enclosure and disk counts. The paths should also be without error. After this is complete, start IBM Storage Scale on the node in question by using **mmstartup**. After determining that IBM Storage Scale is active by using **mmgetstate** proceed to the next step.

7. Move the recovery group ownership to essio1-hs. Use the same commands as used in [this step](#) but make sure to use the correct node name (essio1-hs).

After the preceding steps are complete, new enclosures have been successfully cabled to both servers, proceed with the following final steps.

8. Rebalance both recovery groups by running from any node in the storage cluster.

- a. **mmvdisk rg list**
- b. **mmvdisk recoverygroup change --recovery-group rg1 --active essio1-hs**
- c. **mmvdisk recoverygroup change --recovery-group rg2 --active essio2-hs**
- d. Check that the ownership has changed using the **mmvdisk recoverygroup list** command.

9. Perform the [system verification steps](#) again before proceeding.
10. Update enclosure and drive firmware. If there are any issues, you should stop and replace any disks or enclosures that could not be updated for some reason.

CurrentIoServer implies running the command from either of I/O server nodes in the building block.

Note: It might take up to an hour for the firmware upgrade to complete. You might notice that the fan starts to run at high speed. This is a known issue.

- a. **CurrentIoServer\$ mmchfirmware --type storage-enclosure**
- b. **CurrentIoServer\$ mmchfirmware --type drive**
- c. **mmhealth node show -N all --verbose**- This command shows any system health related issues to address. (Run from any node in the storage cluster.)

- d. **gnrhealthcheck** - This command determines if there are any issues in various areas of IBM Storage Scale System. Any problems that show up must be addressed before capacity upgrade starts.

11. Add new storage into recovery groups.

```
mmvdisk rg resize --rg rg_essio1-hs,rg_essio2-hs -v no
```

12. Verify that the new storage is available and the DA is rebalancing.

```
mmvdisk recoverygroup list --recovery-group rg1 --all  
mmvdisk recoverygroup list --recovery-group rg2 --all
```

Run for both recovery groups. Note that the additional free space available in the DA and the background task for each DA is showing as 'rebalance'.

On the management server node, update the component database.

```
mmaddcompspec default --replace
```

13. Start up the GUI and use **Edit rack components** to have the GUI discover the new topologies and make changes to the frame accordingly. Changes such as modify IBM Storage Scale System model to consume more U space, move management server, and so on.

14. Reconfigure call home.

```
esscallhomeconf -E ems1 -N EMSNode,IONode1,IONode2 --suffix=-hs --register=all
```

At this point, discussions with the customers need to occur on what to do with the free space.

1. Add to the existing file system?

- See the add building block flow in *IBM Storage Scale System Quick Deployment Guide* for tips on creating new NSDs and adding to an existing file system.
- See the add building block flow ([“Adding nodes or building blocks” on page 83](#)) for tips on creating new NSDs and adding to an existing file system.
- Consider file system restripe at the end which might take time. (**mmrestripefs** *FileSystem* -b)

2. Create a new file system.

- See the installation section on how to use **essrun** on creating a new file system from inside the container. You may also use **mmvdisk** commands directly to perform this operation.

Chapter 5. Upgrade from an x86 Utility Node by using the GUI

Starting with IBM Storage Scale Systems 6.2.3.0, you can use a graphical user interface (GUI) to perform upgrades from an x86 Utility Node. The GUI is an alternative to CLI and it simplifies the user experience during the upgrade.

The upgrade of IBM Storage Scale Systems by using the GUI includes the following steps:

1. Verify the supported systems for upgrade through the GUI. For more information, see [“Supported systems”](#) on page 165.
2. Make sure that the network preparation is completed. For more information, see [“Planning and configuring network”](#) on page 7.
3. Make sure that the preparation and prerequisites are completed. For more information, see [“Prerequisites”](#) on page 39.
4. Copy the required files to the Utility Node management server host (/serv). For more information, see [“Copy the required files to the Utility Node management server host”](#) on page 165.
5. Set up the IBM Storage Scale Systems deployment GUI container. For more information, see [“Setting up the IBM Storage Scale System deployment GUI container”](#) on page 129.
6. Set up the IBM Storage Scale Systems API server container. For more information, see [“Setting up the IBM Storage Scale System API server container”](#) on page 120.
7. Log in to the GUI and complete the deployment steps. For more information, see [“Perform the upgrade by using the GUI”](#) on page 166.

Supported systems

The GUI-based upgrade is supported only on the following systems:

- IBM Storage Scale System Utility Node (MTM 5149-23E)
- IBM Storage Scale System 6000 (MTM 5149-F48)
- IBM Storage Scale System 3500 (MTM 5141-FN2)

These are the limitations:

- GUI upgrade is not supported if the cluster includes building blocks other than the supported systems.
- Only one EMSVM can be deployed per cluster, and multiple I/O building blocks are supported.
- Protocol, AFM, and client nodes in the cluster are not supported.
- BYOEMS and Power9 systems are not supported.
- Protocol VMs on IBM Storage Scale System 3500 and Utility Nodes are not supported.
- Adding nodes to an existing cluster is not supported.

Note: I/O nodes are categorized into building blocks based on their serial numbers. For example, nodes with serial numbers 00x0000A and 00x0000B are grouped into one building block.

Copy the required files to the Utility Node management server host

Copy the x86 container (DAE or DME) to the Utility Node management server host (/serv). For more information, see [IBM Fix Central](#).

Note: The root (/) partition of the Utility Bare Metal server must have a minimum of 60 GB of available space.

Video: Upgrade the system using the GUI

The following video provides the complete step-by-step procedure for upgrading the system using the GUI:

Perform the upgrade by using the GUI

Use a graphical user interface (GUI) to perform new upgrades from an x86 Utility Node.

Before you begin

Remember: Upgrading using the GUI is not recommended if you are using the Remote Code Load (RCL) service.

The following prerequisites must be completed before you proceed:

- “Setting up the IBM Storage Scale System API server container” on page 120
- “Setting up the IBM Storage Scale System deployment GUI container” on page 129

Procedure

To upgrade IBM Storage Scale Systems using the GUI, complete the following steps:

1. In a browser, go to the following URL to access the GUI. Replace the <utilityBareMetalIP> with the IP address of the Utility Node.

```
https://<utilityBareMetalIP>:30443/
```

2. Log in using the credentials set for the **installgui** user on the API container.
3. Select an upgrade type:
 - **Software upgrade:** This includes the IBM Storage Scale upgrade
 - **Firmware upgrade:** This includes the Storage Enclosure firmware upgrade
 - **Full upgrade:** This includes both IBM Storage Scale and Storage Enclosure firmware upgrades
4. Click **Next**.
5. In the **Tarball Name** field, enter the name of the IBM Storage Scale Systems x86 container build placed on the Utility Bare Metal.
6. Enter **Utility password** and **Emsvm password**.
7. Select a serial number of the node by clicking the checkbox in the first column of the table.
8. Click **Next**.
9. Select an **Upgrade mode** from the settings page:
 - **Online:** The file system remains online, but individual blocks go offline temporarily during the upgrade.
 - **Offline:** The file system goes offline to upgrade selected node blocks and returns online after completion.
10. Select an **Upgrade order** based on your previous selection:
 - **In parallel:** All the top canisters will be updated parallelly, then the bottom canisters.
Note: The **Offline** upgrade mode only supports **In parallel** upgrade order.
 - **Sequentially:** All the nodes will be updated one by one.
11. Click **Next** and select **Run Pre-check**.

The system validates the essential requirements and displays a success message if the pre-check completes without errors.
12. Click **Upgrade** and select **Start Upgrade**.

You can monitor the upgrade progress until it completes.

13. Click **Finish upgrade** to complete.

Chapter 6. Configuring remote code load

Remote code load (RCL) is a service that enables IBM service personnel to securely connect and complete code updates on IBM Storage Scale Systems from a remote location. The RCL service is the preferred method for delivering code updates, which does not require an onsite visit from IBM service personnel. It provides a secure and efficient alternative to on-premises microcode upgrades. You can use the RCL to update firmware for host adapters, enclosures, and drives. If a firmware update fails during a standard deployment, use the RCL to retry the update instead of restarting the entire upgrade procedure.

Important: RCL is available only to customers with Expert Care offerings.

To update the code through the RCL service, complete the following tasks on the IBM Storage Scale System Utility Node:

1. Verify the [supported versions and system requirements](#).
2. Verify that the [prerequisites](#) are met for the RCL service.
3. Set up the IBM Storage Scale System RCL service container.
4. Set up the IBM Storage Scale System API server container.
5. Contact IBM service personnel to perform the code update through the RCL service. For more information, see [IBM Remote Code Load](#) in the IBM Support page.
6. Optionally, clean up the RCL service container after the code update is complete. For more information, see [“Cleaning up the RCL service container”](#) on page 170.

Supported versions and system requirements

The minimum supported version from which you can run the RCL is the N-3 version of 6.2.3.0.

The RCL service uses Utility Node to perform the code update for other systems and it is supported only for the following systems:

- IBM Storage Scale System 6000 (MTM 5149-F48)
- IBM Storage Scale System Utility Node (MTM 5149-23E)
- IBM Storage Scale System management server (EMSVM-23E)

Note: RCL supports mixed environments that include only supported systems.

Prerequisites

The following prerequisites must be met on the system before setting up containers:

- Ensure that the Utility Node can access the public network through the campus network interface. This access is required for IBM service personnel to connect to the system remotely and perform the code update by using the RCL service.

Note:

- The campus interface must have a single IPv4 addresses configured.
- If direct access to the public network is not available, use a proxy server to establish the internet connectivity for the Utility Node.
- Ensure that the system administrator has root access to create other users, which are required to host containers. To verify that a user has root access, run the **whoami** command. If the output is *root*, the user has root privileges.
- Ensure that the root (/) file system has at least 60 GB of available space to support the RCL and API server operations.
- Before you set up the containers on the Utility Node, make sure that the Podman and guestfs-tools packages are installed.

- Before you start any containers, make sure that the host file is saved at /etc/hosts on the Utility Node. Verify that the IP address specified in the hosts file matches the IP address of your system. A sample host file includes the following details:

```
127.0.0.1 localhost localhost.localdomain localhost4 localhost4.localdomain4
::1      localhost localhost.localdomain localhost6 localhost6.localdomain6

# *****
# IBM Utility Node (virbr1) (where EMSVM-23E running)
# *****
10.23.16.1 utilityBareMetal

# *****
# EMSVM-23E (ras interface IP on EMSVM-23E)
# *****
10.23.16.2 emsvm
```

Important:

- Add only the required host entries to the /etc/hosts file to minimize security risks and prevent exposure of information about other nodes in the environment.
- Get familiar with the IBM Storage Scale native REST API feature, which is used to perform the RCL service through the API server container. For more information about the IBM Storage Scale native REST API, see [IBM Storage Scale native REST API](#).

Note: The API server container is a streamlined version of the IBM Storage Scale native REST API server and does not include core IBM Storage Scale functions. The API server container is intended for deployment in a single-host environments and does not support communication or state exchange with other API server containers.

Cleaning up the RCL service container

After the code update is complete, you can optionally remove the RCL service container.

1. Log in to the IBM Storage Scale System Utility Node as the `rcladmin` user.
2. Stop the RCL service.

```
systemctl --user stop container-utilityBareMetal-rcl-official.service
```

3. List the installed container images.

```
podman images
```

4. List the installed containers.

```
podman ps -a
```

5. Stop the RCL service container.

```
podman stop <ContainerName>
```

6. Remove the container.

```
podman rm <ContainerName>
```

7. Remove the container image.

```
podman image rm <ImageName> -f
```

Setting up the IBM Storage Scale System RCL service container

The IBM Storage Scale System RCL service container enables the code update through the RCL service by using the command-line interface (CLI).

Before you set up the container, make sure that the prerequisites are completed on the system. For more information, see [“Prerequisites” on page 169](#).

To set up the RCL service container on the Utility Node, complete the following steps:

1. [Create user and user group](#).
2. Pull the RCL service container image and **startRCLContainer** utility.
3. [Start the RCL service container on the Utility Node and verify that it is active](#).

Create user and user group

Create the **rcladmin** user and user group to host the RCL service container. To create the user and the user group, and set the user password, complete the following steps:

1. Log in as the root user on the Utility Node.
2. Create the **rcladmin** user group.

```
groupadd rcladmin
```

3. Create the **rcladmin** user and assign it to the **rcladmin** user group.

```
useradd -m -g rcladmin rcladmin
```

4. Set the password for the **rcladmin** user.

```
passwd rcladmin
```

When the **New password:** prompt appears, enter the new password. When the **Retype new password:** prompt appears, reenter the password to confirm it.

5. Verify that the **rcladmin** user is created.

```
cat /etc/passwd
```

6. Add the **rcladmin** user to the `sudoers.d` folder to provide restricted administrative access without giving the root password. To add the user, complete the following steps.

- a. In the `/etc/sudoers.d/` folder, create a file that is named **rcladmin_sudoers**.
- b. Paste the following content into the **rcladmin_sudoers** file:

```
rcladmin    ALL=(ALL)        NOPASSWD: /usr/sbin/dmidecode, /bin/cat, /sys/devices/
virtual/dmi/id/product_serial
```

7. Log out as the root user and log in as the **rcladmin** user.

Pull the RCL service container image and the startRCLContainer utility

Complete the next steps to pull the image of the RCL service container from the IBM Cloud container repository and the **startRCLContainer** utility package from the [GitHub](#). Internet access is required on the node to complete this step. If the Utility Node does not have internet access, pull the container image from a node that is connected to internet and move the downloaded files to the Utility Node. Alternatively, you can download these files from the [IBM Fix Central](#).

To pull the image of the RCL service container and **startRCLContainer** utility package, complete one of the following methods:

For a Utility Node with internet access

1. In a browser, log in to [MyIBM Container Software Library](#) by using the IBMid and password that are associated with your IBM Storage Scale System.
2. In the **Entitlement keys** section, click **Copy** to get the entitlement key in the clipboard. If no key exists, click **Add new key**, then copy the new key to the clipboard.

Note: To verify that your entitlement key is valid, go to the left pane and click **Container software library**. You can view the list of products that you are entitled to.

3. On the Utility Node, log in as the **rcldadmin** user and issue the following command to log in to the IBM Cloud container Repository:

```
podman login cp.icr.io --username cp --password <APIKEY>
```

Where,

<APIKEY>

Specifies the entitlement key that is obtained from the **Container software library**.

4. Clone the SSSRCL repository from [GitHub](#) on the Utility Node at /home/rcldadmin. Alternatively, use the **Download ZIP** option on [GitHub](#) to manually download this SSSRCL-main.zip file on the Utility Node at /home/rcldadmin. This repository includes the **startRCLContainer** utility file, which starts the RCL service container.
5. If downloaded manually, extract the downloaded SSSRCL-main.zip file at /home/rcldadmin.

```
unzip <SSSRCL-main.zip>
```

6. In the CLI, change the path to the extracted directory.

```
cd </home/rcldadmin/SSSRCL-main>
```

7. Start the **startRCLContainer** utility on the Utility Node and enter the image version.

```
./startRCLContainer
```

A sample output is as follows:

```
Enter the campus interface name (default: campus):
Enter the image version (default: 7.0.1.x):

Detailed logs are located on ./logs directory

2025-12-02 23:50:36,021 INFO:      All 4 endpoints are reachable on port 22
2025-12-02 23:50:36,025 INFO:      All configurable variables were validated successfully.
2025-12-02 23:50:36,129 INFO:      No images are installed. No cleanup is required.
2025-12-02 23:50:36,131 INFO:      The RCL container image installation is about to
begin. No changes are applied if the image is already installed.
-- [INFO] The container image is about to be pulled from the IBM repository. --
-- [WARNING] The existing container image cp.stg.icr.io/cp/scalesystem/sss_rcl:7.0.1.x
is being deleted forcefully --
Trying to pull cp.stg.icr.io/cp/scalesystem/sss_rcl:7.0.1.x...
Getting image source signatures
Copying blob ee7e8b316b25 done |
Copying blob 35d6a5cce6a1 done |
Copying blob f988781302f6 done |
Copying blob 1ecaa84e2377 done |
Copying blob 2c3055ea4e9b done |
Copying blob b5033592c3e0 done |
Copying blob ba279a2362fa done |
Copying blob c4ee0281ec09 done |
Copying blob 03a3b7808ecb done |
Copying blob 9e19a643aa24 done |
Copying config f4615fa598 done |
Writing manifest to image destination
f4615fa5988fef226ece1661d7e766c9a58901108fe3e0ce01cca42108a798d8
-- [INFO] The container image was pulled successfully. --
2025-12-02 23:50:46,116 INFO:      The container image installation completed
successfully.
```

```

2025-12-02 23:50:46,118 INFO:      The container is about to start. On later runs, use
the 'startRCLContainer' command to manage this container.
-- [INFO] Running the container image: cp.stg.icr.io/cp/scalesystem/sss_rcl:7.0.1.x
-- [WARNING] The '-x' or '--force' option removes containers that are in the EXIT state
--
-- [INFO] Automatic initialization of the container begin shortly --
-- [INFO] Startup can take several minutes. --
ae702692d757acfb9877200e6f1e92275f49893a6f8c8268f72698855ba79167
-- [INFO] The RCL service container is being configured to start as a systemd service. --
-- [INFO] The RCL service container is set to autostart --

DEPRECATED command:
It is recommended to use Quadlets for running containers and pods under systemd.

Please refer to podman-systemd.unit(5) for details.
/home/rcladmin/container-utilityBareMetal-rcl-official.service
-- [INFO] The container was created successfully in the background. To
start this container, run the "systemctl --user start container-utilityBareMetal-rcl-
official.service" command. --
-- [INFO] To log in to the container, run the "podman exec -it utilityBareMetal-rcl-
official /bin/bash" command --
2025-12-02 23:50:49,650 INFO:      To start a new container, run the /home/rcladmin/
SSSRCL-main/./startRCLContainer command.

```

After this command runs successfully, the Utility Node hosts the RCL service container, but the container is not started.

For a Utility Node without internet access

1. Download the RCL service container image `sss_rcl_7.0.1.x.tar` and the utility package `SSSRCL-main.zip` from IBM Fix central or MyIBM Container software library.

Using IBM Fix central

- a. From a node that is connected to the internet, go to [IBM Fix Central](#).
- b. Select `Scale_System_RCL-7.0.1.x`.
- c. Download the `Scale_System_RCL-7.0.1.x.tgz` file. This file contains the RCL service container image `sss_rcl_7.0.1.x.tar` and the utility package `SSSRCL-main.zip` file.
- d. Transfer the `sss_rcl_7.0.1.x.tar` and `SSSRCL-main.zip` files to `/home/rcladmin` on the Utility Node.
- e. On the Utility Node, extract the `SSSRCL-main` file at `/home/rcladmin`.

```
unzip <ZipFileName>
```

Using MyIBM Container software library

- a. From a node that is connected to the internet, log in to [MyIBM Container Software Library](#) by using the IBMid and password that are associated with your IBM Storage Scale System.
- b. In the **Entitlement keys** section, click **Copy** to get the entitlement key in the clipboard. If no key exists, click **Add new key**, then copy the new key to the clipboard.

Note: To verify that your entitlement key is valid, go to the left pane and click **Container software library**. You can view the list of products that you are entitled to.

- c. Log in to the IBM Cloud container Repository.

```
podman login cp.icr.io --username cp --password <APIKEY>
```

Where,

<APIKEY>

Specifies the entitlement key that is obtained from the **Container software library**.

- d. Pull the image of the RCL service container.

```
podman pull cp.icr.io/cp/scalesystem/sss_rcl:7.0.1.x
```

- e. Export the pulled image as a `.tar` file on the node that is connected to the internet.

```
podman save cp.icr.io/cp/scaletystem/sss_rcl:7.0.1.x -o sss_rcl_7.0.1.x.tar
```

- f. Download the **SSSRCL-main** repository by using the **Download ZIP** option on GitHub on the same node. This repository includes the **startRCLContainer** utility file, which starts the RCL service container.
- g. Move the generated sss_rcl_7.0.1.x.tar file and the downloaded ZIP **SSSRCL-main** file to the /home/rcldadmin on the Utility Node.
- h. On the Utility Node, extract the SSSRCL-main file at /home/rcldadmin.

```
unzip <ZipFileName>
```

2. In the CLI, change the path to the extracted directory.

```
cd </home/rcldadmin/SSSRCL-main>
```

3. Restore the .tar file by using the **-f** option through the **startRCLContainer** utility program.

```
./startRCLContainer -f ../SSSRCL.tar
```

Note: To get help information for the **startRCLContainer** utility, issue the following command:

```
./startRCLContainer --help
```

A sample output is as follows:

```
usage: startRCLContainer [-h] [-v] [-f FILENAME] [-i CAMPUS_INTERFACE] [-vn
IMAGE_VERSION]

optional arguments:
  -h or --help            Display the help message and exit.
  -v or --verbose          Display verbose messages in the console.
  -f FILENAME or --file-name FILENAME
                           Restore the container from a local tar file. The default value
                           is SSSRCL.tar
  -i CAMPUS_INTERFACE or --campus-interface CAMPUS_INTERFACE
                           Name of the interface connected to the campus network. The
                           default value is campus.
  -vn IMAGE_VERSION or --image-version IMAGE_VERSION
                           API Server version to be deployed. The default value is 7.0.1.x.
```

4. Start the **startRCLContainer** utility on the Utility Node and enter the image version.

```
./startRCLContainer
```

A sample output is as follows:

```
Enter the campus interface name (default: campus):
Enter the image version (default: 7.0.1.x):

Detailed logs are located on ./logs directory

2025-12-02 23:50:36,021 INFO:      All 4 endpoints are reachable on port 22
2025-12-02 23:50:36,025 INFO:      All configurable variables were validated successfully.
2025-12-02 23:50:36,129 INFO:      No images are installed. No cleanup is required.
2025-12-02 23:50:36,131 INFO:      The RCL container image installation is about to
begin. No changes are applied if the image is already installed.
-- [INFO] The container image is about to be pulled from the IBM repository. --
-- [WARNING] The existing container image cp.stg.icr.io/cp/scaletystem/sss_rcl:7.0.1.x
is being deleted forcefully --
Trying to pull cp.stg.icr.io/cp/scaletystem/sss_rcl:7.0.1.x...
Getting image source signatures
Copying blob ee7e8b316b25 done |
Copying blob 35d6a5cce6a1 done |
Copying blob f988781302f6 done |
Copying blob 1ecaa84e2377 done |
Copying blob 2c3055ea4e9b done |
Copying blob b5033592c3e0 done |
Copying blob ba279a2362fa done |
Copying blob c4ee0281ec09 done |
Copying blob 03a3b7808ecb done |
Copying blob 9e19a643aa24 done |
```

```

Copying config f4615fa598 done |
Writing manifest to image destination
f4615fa5988fef226ece1661d7e766c9a58901108fe3e0ce01cca42108a798d8
-- [INFO] The container image was pulled successfully. --
2025-12-02 23:50:46,116 INFO: The container image installation completed
successfully.
2025-12-02 23:50:46,118 INFO: The container is about to start. On later runs, use
the 'startRCLContainer' command to manage this container.
-- [INFO] Running the container image: cp.stg.icr.io/cp/scalesystem/sss_rcl:7.0.1.x
-- [WARNING] The '-x' or '--force' option removes containers that are in the EXIT state
--
-- [INFO] Automatic initialization of the container begin shortly --
-- [INFO] Startup can take several minutes. --
ae702692d757acfb9877200e6f1e92275f49893a6f8c8268f72698855ba79167
-- [INFO] The RCL service container is being configured to start as a systemd service. --
-- [INFO] The RCL service container is set to autostart --

DEPRECATED command:
It is recommended to use Quadlets for running containers and pods under systemd.

Please refer to podman-systemd.unit(5) for details.
/home/rcldadmin/container-utilityBareMetal-rcl-official.service
-- [INFO] The container was created successfully in the background. To
start this container, run the "systemctl --user start container-utilityBareMetal-rcl-
official.service" command. --
-- [INFO] To log in to the container, run the "podman exec -it utilityBareMetal-rcl-
official /bin/bash" command --
2025-12-02 23:50:49,650 INFO: To start a new container, run the /home/rcldadmin/
SSSRCL-main/./startRCLContainer command.

```

After this command runs successfully, the Utility Node hosts the RCL service container, but the container is not started.

Start the RCL service container

To start the RCL service container, complete the following steps:

1. If you are not already logged in, log in as the `rcladmin` user on the Utility Node.
2. Start the RCL service container by using the **systemd** service of this container.

```
systemctl --user start container-utilityBareMetal-rcl-official.service
```

3. Verify that the RCL service container is active.

```
systemctl --user status container-utilityBareMetal-rcl-official.service
```

A sample output is as follows:

```

container-utilityBareMetal-rcl-official.service - Podman container-utilityBareMetal-rcl-
official.service
   Loaded: loaded (/home/rcldadmin/.config/systemd/user/container-utilityBareMetal-rcl-
official.service; enabled; preset: disabled)
   Active: active (running) since Wed 2025-06-04 01:08:20 EDT; 10s ago
     Docs: man:podman-generate-systemd(1)
   Process: 317562 ExecStart=/usr/bin/podman start utilityBareMetal-rcl-official
(code=exited, status=0/SUCCESS)
    Main PID: 317603 (common)
      Tasks: 17 (limit: 819932)
     Memory: 16.7M
        CPU: 82ms
    CGroup: /user.slice/user-1004.slice/user@1004.service/app.slice/container-
utilityBareMetal-rcl-official.service
            └─317584 /usr/bin/slrp4netns --disable-host-loopback --mtu=65520 --enable-
sandbox --enable-seccomp --enable-ipv6 -c -r 3 -e 4 --netns-type=path /run/user/1004/netns/
netn>
                └─317587 rootlessport
                    └─317593 rootlessport-child
                        └─317603 /usr/bin/common --api-version 1
-c c29b682341f35e454917b07cc24087dd64386331601374c941027448a8b0f4d4 -u
c29b682341f35e454917b07cc24087dd64386331601374c941027448a8>
lines 1-14/14 (END)

```

After the RCL service container starts, the Utility Node appears in the IBM Service Center portal and is ready to perform RCL services. Next, set up the API server container if it is not already configured.

Limitation

If the Utility Node is not listed on the portal, restart the RCL service container again by issuing the following **systemctl** commands:

```
systemctl --user stop container-utilityBareMetal-rcl-official.service
```

```
systemctl --user start container-utilityBareMetal-rcl-official.services
```

Setting up the IBM Storage Scale System API server container

The API server container is a dedicated container that runs on the IBM Storage Scale System Utility Node and serves REST endpoints for the deployment artifacts of IBM Storage Scale System. This API server container enables IBM Storage Scale System deployment over the REST interface.

Ensure that the required prerequisites are completed on the system before setting up the container. For more information, see [“Prerequisites” on page 169](#).

To set up the API server container, complete the following steps:

1. [Create user and user group](#).
2. [Pull the image of the API server container and **startAPIContainer** utility](#).
3. [Start the API server container on the Utility Node and verify that it is active](#).
4. If required, [replace the autogenerated self-signed certificate with your organization-owned, CA-signed certificates](#).
5. [Verify that the API server container is configured correctly on the Utility Node](#).

Create user and user group

Create the **apiadmin** user and user group to host the API server container. The **apiadmin** user is a superuser that can change any authorization. This user must be familiar with the IBM Storage Scale native REST API feature. For more information about the IBM Storage Scale native REST API, see [IBM Storage Scale native REST API](#).

If required, you can also add the **apiuser** to the API server container. This user has non-privileged access and restricted to change the authorization.

To create the user and user group, complete the following steps:

1. Log in as the root user on the Utility Node.
2. Create the **apiadmin** user group.

```
groupadd apiadmin
```

3. Create the **apiadmin** user and assign it to the **apiadmin** user group.

```
useradd -m -g apiadmin apiadmin
```

4. Set the password for the **apiadmin** user.

```
passwd apiadmin
```

When the **New password:** prompt appears, enter the new password. When the **Retype new password:** prompt appears, re-enter the password to confirm it.

5. Verify that the **apiadmin** user is created.

```
cat /etc/passwd
```

6. Log out as the root user and log in as the **apiadmin** user.

Pull the API sever container image and the startAPIContainer utility

Complete the next steps to pull the image of the API server container from the IBM Cloud container repository and the **startAPIContainer** utility package from the [GitHub](#). Internet access is required on the node to complete this step. If the Utility Node does not have internet access, pull the container image from a node that is connected to internet and move the downloaded files to the Utility Node. Alternatively, you can download these files from the [IBM Fix Central](#).

To pull the image of the API server container and **startAPIContainer** utility package, complete one of the following methods:

For a Utility Node with internet access

1. In a browser, log in to [MyIBM Container Software Library](#) by using the IBMid and password that are associated with your IBM Storage Scale System.
2. In the **Entitlement keys** section, click **Copy** to get the entitlement key in the clipboard. If no key exists, click **Add new key**, then copy the new key to the clipboard.

Note: To verify that your entitlement key is valid, go to the left pane and click **Container software library**. You can view the list of products that you are entitled to.

3. On the Utility Node, log in as the **apiadmin** user and issue the following command to log in to the IBM Cloud container Repository:

```
podman login cp.icr.io --username cp --password <APIKEY>
```

Where,

<APIKEY>

Specifies the entitlement key that is obtained from the **Container software library**.

4. Clone the **SSSAPI** repository from GitHub on the Utility Node at /home/apiadmin. Alternatively, use the **Download ZIP** option on GitHub to manually download the **SSSAPI** repository on the Utility Node at /home/apiadmin. This repository includes the **startAPIContainer** utility file, which starts the API server container.
5. If downloaded manually, extract the downloaded SSSAPI-main.zip file at /home/apiadmin.

```
unzip <SSSAPI-main.zip>
```

6. In the CLI, change the path to the extracted directory.

```
cd </home/apiadmin/SSSAPI-main>
```

7. Start the **startAPIContainer** utility on the Utility Node and enter the image version.

```
./startAPIContainer
```

A sample output is as follows:

```
Enter the campus interface name (default: campus):
Enter the image version (default: 7.0.1.x):
Enter the API port (default: 46443):
Enter the IBM Storage Scale Utility Node password to configure passwordless SSH
(default: cluster):
Specify whether EMSVM-23E exists (default: no): yes
Enter the EMSVM-23E password to configure passwordless SSH (default: cluster):

Detailed logs are located on ./logs directory

2025-12-02 23:46:09,966 INFO:      All configurable variables were validated successfully.
2025-12-02 23:46:10,082 INFO:      No images related to api-official were found for
deletion.
2025-12-02 23:46:10,084 INFO:      The API server container image installation is about
to begin. No changes are applied if the image is already installed.
-- [INFO] The API container image is about to be pulled from the IBM repository. --
-- [WARNING] The existing container image cp.stg.icr.io/cp/scalesystem/sss_api:7.0.1.xis
being deleted forcibly. --
Untagged: cp.stg.icr.io/cp/scalesystem/sss_api:7.0.1.x
Deleted: bc77eb54e68150279d1c1e7afd72109ff33e0b5899bb6c2f0cf4a62cdc69ace3
Trying to pull cp.stg.icr.io/cp/scalesystem/sss_api:7.0.1.x...
```

```

Getting image source signatures
Copying blob 8d6cf32b1d96 done |
Copying blob 35d6a5cce6a1 done |
Copying blob b5033592c3e0 done |
Copying blob 29d4b2abb1d4 done |
Copying blob 354a1969e4a5 done |
Copying blob 0382955305cb done |
Copying blob c8cad13252da done |
Copying blob fbb2a441be54 done |
Copying blob fa7bd7163b6b done |
Copying blob 394966977691 done |
Copying blob f3e63b1012d6 done |
Copying config cf62860064 done |
Writing manifest to image destination
cf628600645cfe7d95fc528504b5406ef5d43f1e55d47717b1f487717784e7c3
-- [INFO] The container image was pulled successfully. --
2025-12-02 23:46:38,901 INFO: The container image installation completed
successfully.
2025-12-02 23:46:38,904 INFO: The container is about to start. On later runs, use
the 'startAPIContainer' command with optional arguments to manage the container.
-- [INFO] Running the container image - cp.stg.icr.io/cp/scalesystem/sss_api:7.0.1.x
-- [WARNING] The '-x' or '--force' option removes containers that are in the EXIT state.
--
utilityBareMetal-api-official
-- [INFO] Automatic initialization of the container will begin shortly. --
-- Startup can take several minutes. --
456cbc4056b214186b2d0ed9a48784857cb8782f30b6c8671ec75576ee1c9b61
-- [WARNING] The /home/apiadmin/SSSAPI-main_7000/credentials.properties file was deleted.
-- [INFO] The API server container is being configured to start as a systemd service. --
-- [INFO] The API server container is set to autostart. --

DEPRECATED command:
It is recommended to use Quadlets for running containers and pods under systemd.

Please refer to podman-systemd.unit(5) for details.
/home/apiadmin/container-utilityBareMetal-api-official.service
-- [INFO] The container was created successfully in the background. To start
the container, run the "systemctl --user start container-utilityBareMetal-api-
official.service" to start using container. --
-- [INFO] To log in to the container, run the "podman exec -it utilityBareMetal-api-
official /bin/bash" command. --
2025-12-02 23:46:45,233 INFO: To start a new container again run the /home/apiadmin/
SSSAPI-main_7000/./startAPIContainer command.

```

After this command runs successfully, the Utility Node hosts the API server container, but the container is not started.

For a Utility Node without internet access

1. Download the API server container image `sss_api_7.0.1.x.tar` and the utility package `SSSAPI-main.zip` from IBM Fix central or MyIBM Container software library.

Using IBM Fix central

- a. From a node that is connected to the internet, go to [IBM Fix Central](#).
- b. Select `Scale_System_RCL-7.0.1.x`.
- c. Download the `Scale_System_RCL-7.0.1.x.tgz` file. This file contains the API server container image `sss_api_7.0.1.x.tar` and the utility package `SSSAPI-main.zip` file.
- d. Transfer the `sss_api_7.0.1.x.tar` and `SSSAPI-main.zip` files to `/home/apiadmin` on the Utility Node.
- e. On the Utility Node, extract the `SSSAPI-main.zip` file at `/home/apiadmin`.

```
unzip <ZipFileName>
```

Using MyIBM Container software library

- a. From a node that is connected to the internet, log in to MyIBM Container Software Library by using the IBMid and password that are associated with your IBM Storage Scale System.
- b. In the **Entitlement keys** section, click **Copy** to get the entitlement key in the clipboard. If no key exists, click **Add new key**, then copy the new key to the clipboard.

Note: To verify that your entitlement key is valid, go to the left pane and click **Container software library**. You can view the list of products that you are entitled to.

- c. Log in to the IBM Cloud container Repository.

```
podman login cp.icr.io --username cp --password <APIKEY>
```

Where,

<APIKEY>

Specifies the entitlement key that is obtained from the **Container software library**.

- d. Pull the image of the API server container.

```
podman pull cp.icr.io/cp/scalesystem/sss_api:7.0.1.x
```

- e. Export the pulled image as a .tar file on the node that is connected to the internet.

```
podman save cp.icr.io/cp/scalesystem/sss_api:7.0.1.x -o sss_api_7.0.1.x.tar
```

- f. Download the **SSSAPI-main** repository by using the **Download ZIP** option on [GitHub](#) on the same node. This repository includes the **startAPIContainer** utility file, which starts the API server container.

- g. Move the generated sss_api_7.0.1.x.tar file and the downloaded ZIP **SSSAPI-main** file to the /home/apiadmin on the Utility Node.

- h. On the Utility Node, extract the SSSAPI-main files at /home/apiadmin.

```
unzip <ZipFileName>
```

2. In the CLI, change the path to the extracted directory.

```
cd </home/apiadmin/SSSAPI-main>
```

3. Restore the .tar file by using the **-f** option through the **startAPIContainer** utility program.

```
./startAPIContainer -f ../sss_api_7.0.1.x.tar
```

Note: To get help information for the **startAPIContainer** utility, issue the following command:

```
./startAPIContainer --help
```

A sample output is as follows:

```
usage: startAPIContainer [-h] [-v] [-f FILENAME] [-p PORT] [-i CAMPUS_INTERFACE]
[-vn IMAGE_VERSION] [-up UTILITYBAREMETAL_PASSWORD] [-e IS_EMSVM23E_EXIST] [-ep
EMSV23E_PASSWORD]

optional arguments:
  -h, --help                Display the help message and exit.
  -v, --verbose             Display verbose messages in the console.
  -f FILENAME, --file-name FILENAME
                           Restore the container from a local tar file. The default value
                           is SSSDEPLOYGUI.tar.
  -p PORT, --port PORT      Port on which the API server listens. The default value is
                           46443.
  -i CAMPUS_INTERFACE, --campus-interface CAMPUS_INTERFACE
                           Name of the interface connected to the campus network. The
                           default value is campus.
  -vn IMAGE_VERSION, --image-version IMAGE_VERSION
                           API server version to deploy. The default value is 7.0.1.x.
  -up UTILITYBAREMETAL_PASSWORD, --utilitybaremetal-password UTILITYBAREMETAL_PASSWORD
                           Password for the root user on the 'utilityBareMetal' host.
                           The default value is cluster.
  -e IS_EMSVM23E_EXIST, --is-emsvm23e-exist IS_EMSVM23E_EXIST
                           Specify whether EMSVM-23E exists. The default value is no.
  -ep EMSVM23E_PASSWORD, --emsvm23e-password EMSVM23E_PASSWORD
                           Password for the root user on EMSVM-23E. The default value is
                           cluster.
```

4. Start the **startAPIContainer** utility on the Utility Node and enter the image version.

```
./startAPIContainer
```

A sample output is as follows:

```
Enter the campus interface name (default: campus):
Enter the image version (default: 7.0.1.x):
Enter the API port (default: 46443):
Enter the IBM Storage Scale Utility Node password to configure passwordless SSH
(default: cluster):
Specify whether EMSVM-23E exists (default: no): yes
Enter the EMSVM-23E password to configure passwordless SSH (default: cluster):

Detailed logs are located on ./logs directory

2025-12-02 23:46:09,966 INFO:      All configurable variables were validated successfully.
2025-12-02 23:46:10,082 INFO:      No images related to api-official were found for
deletion.
2025-12-02 23:46:10,084 INFO:      The API server container image installation is about
to begin. No changes are applied if the image is already installed.
-- [INFO] The API container image is about to be pulled from the IBM repository. --
-- [WARNING] The existing container image cp.stg.icr.io/cp/scalesystem/sss_api:7.0.1.xis
being deleted forcibly. --
Untagged: cp.stg.icr.io/cp/scalesystem/sss_api:7.0.1.x
Deleted: bc77eb54e68150279d1c1e7afd72109ff33e0b5899bb6c2f0cf4a62cdc69ace3
Trying to pull cp.stg.icr.io/cp/scalesystem/sss_api:7.0.1.x...
Getting image source signatures
Copying blob 8d6cf32b1d96 done |
Copying blob 35d6a5cce6a1 done |
Copying blob b5033592c3e0 done |
Copying blob 29d4b2abb1d4 done |
Copying blob 354a1969e4a5 done |
Copying blob 0382955305cb done |
Copying blob c8cad13252da done |
Copying blob fbb2a441be54 done |
Copying blob fa7bd7163b6b done |
Copying blob 394966977691 done |
Copying blob f3e63b1012d6 done |
Copying config cf62860064 done |
Writing manifest to image destination
cf628600645cfe7d95fc528504b5406ef5d43f1e55d47717b1f487717784e7c3
-- [INFO] The container image was pulled successfully. --
2025-12-02 23:46:38,901 INFO:      The container image installation completed
successfully.
2025-12-02 23:46:38,904 INFO:      The container is about to start. On later runs, use
the 'startAPIContainer' command with optional arguments to manage the container.
-- [INFO] Running the container image - cp.stg.icr.io/cp/scalesystem/sss_api:7.0.1.x
-- [WARNING] The '-x' or '--force' option removes containers that are in the EXIT state.
--
utilityBareMetal-api-official
-- [INFO] Automatic initialization of the container will begin shortly. --
-- Startup can take several minutes. --
456cbc4056b214186b2d0ed9a48784857cb8782f30b6c8671ec75576ee1c9b61
-- [WARNING] The /home/apiadmin/SSSAPI-main_7000/credentials.properties file was deleted.
-- [INFO] The API server container is being configured to start as a systemd service. --
-- [INFO] The API server container is set to autostart. --

DEPRECATED command:
It is recommended to use Quadlets for running containers and pods under systemd.

Please refer to podman-systemd.unit(5) for details.
/home/apiadmin/container-utilityBareMetal-api-official.service
-- [INFO] The container was created successfully in the background. To start
the container, run the "systemctl --user start container-utilityBareMetal-api-
official.service" to start using container. --
-- [INFO] To log in to the container, run the "podman exec -it utilityBareMetal-api-
official /bin/bash" command. --
2025-12-02 23:46:45,233 INFO:      To start a new container again run the /home/apiadmin/
SSSAPI-main_7000/./startAPIContainer command.
```

After this command runs successfully, the Utility Node hosts the API server container, but the container is not started.

Start the API server container

To start the API server container, complete the following steps:

1. If you are not already logged in, log in as the **apiadmin** user on the Utility Node.
2. Start the API server container by using the **systemd** service of this container.

```
systemctl --user start container-utilityBareMetal-api-official.service
```

Note: When the API server container is started, a self-signed certificate is autogenerated for the authentication. If required, replace this certificate with your organization-owned, CA-signed certificate. For more information, see [“Replace the autogenerated self-signed certificate with your organization-owned, CA-signed certificate”](#) on page 181.

3. Verify that the API server container is active.

```
systemctl --user status container-utilityBareMetal-api-official.service
```

A sample output is as follows:

```
container-utilityBareMetal-api-official.service - Podman container-utilityBareMetal-api-official.service
Loaded: loaded (/home/apiadmin/.config/systemd/user/container-utilityBareMetal-api-official.service; enabled; preset: disabled)
Active: active (running) since Tue 2025-05-06 04:30:36 EDT; 6s ago
Docs: man:podman-generate-systemd(1)
Process: 156908 ExecStart=/usr/bin/podman start utilityBareMetal-api-official (code=exited, status=0/SUCCESS)
Main PID: 156952 (common)
Tasks: 18 (limit: 820520)
Memory: 17.0M
CPU: 140ms
CGroup: /user.slice/user-1002.slice/user@1002.service/app.slice/container-utilityBareMetal-api-official.service
        └─156928 /usr/bin/slrp4netns --disable-host-loopback --mtu=65520 --enable-sandbox --enable-seccomp --enable-ipv6 -c -r 3 -e 4 --netns-type=path /run/user/1002/netns>
            └─156930 rootlessport
                └─156936 rootlessport-child
                    └─156952 /usr/bin/common --api-version 1
                        -c 9a8d34d2ea684840873a06f7f5d76e26afd1c95c3818625f8cdb877ee46ccab2 -u 9a8d34d2ea684840873a06f7f5d76e26afd1c95c3818625f8cdb8>
```

Replace the autogenerated self-signed certificate with your organization-owned, CA-signed certificate

When the API server container is started, a TLS or SSL self-signed certificate is autogenerated and placed at `/home/apiadmin/keys`. By default, this self-signed certificate is imported to the API server container to start the API server. Replace this self-signed certificate of the API server container with your organization-owned, CA-signed certificates. Alternatively, generate a new self-signed certificate for your organization and import it to the API server container.

To replace the certificate, complete the following steps:

1. If you are not already logged in, log in as the **apiadmin** user on the Utility Node.
2. Get the CA certificate and its keys that are owned by your organization and save it in the `/home/apiadmin/backup` folder on the Utility Node.
3. Log in to the API server container by issuing the following command:

```
podman exec -it utilityBareMetal-api-official /bin/bash
```

4. Go to the `/home/backup` folder and verify that your certificates are listed by issuing the following commands:

```
cd /home/backup
```

```
ls -la
```

A sample output is as follows:

```
total 8
drwxr-xr-x 2 root root 4096 Jun  2 07:18 .
drwxr-xr-x 1 root root 4096 Jun  4 06:57 ..
-rW----- 1 root root 1704 Feb 12 07:56 ca.key
-rW-r--r-- 1 root root 1440 Feb 12 07:56 ca.pem
-rw-r--r-- 1 root root  41 Feb 12 07:56 ca.srl
-rw-r--r-- 1 root root  90 Feb 12 07:56 san.conf
-rw-r--r-- 1 root root 1054 Feb 12 07:56 server.cr
-rW----- 1 root root 1708 Feb 12 07:56 server.key
-rw-r--r-- 1 root root 1509 Feb 12 07:56 server.pem
```

5. Delete the existing TLS or SSL configuration:

```
/usr/lpp/mmfs/bin/essctl node config delete
```

A sample output is as follows:

```
Node configuration deleted successfully
```

6. Import the new TLS or SSL configuration by issuing the following command:

```
/usr/lpp/mmfs/bin/essctl node config set --cert /home/apiadmin/server.pem --key /home/apiadminserver.key --chain /home/backup
```

A sample output is as follows:

```
Node configuration set successfully
```

7. Verify the status of the **deployadmd** service by issuing the following command:

```
systemctl status deployadmd
```

A sample output is as follows:

```
• deployadmd.service - Deployment API Admin Daemon Service
   Loaded: loaded (/usr/lib/systemd/system/deployadmd.service; enabled; preset: disabled)
   Active: active (running) since Wed 2025-02-12 07:22:51 UTC; 39min ago
   Process: 52 ExecStartPre=/bin/sh -c targetLink=$(/bin/readlink -f /var/mmfs/run) ;
  [[ -n "$targetLink" && "$targetLink" == */mmfs ]]>
   Main PID: 55 (scaleadmd)
     Tasks: 18 (limit: 1638)
    Memory: 17.2M
       CPU: 284ms
    CGroup: /system.slice/deployadmd.service
            └─55 /usr/lpp/mmfs/bin/scaleadmd -deployment -serverType ess -hostname
utilityBareMetal-api-official.gpfs.local

Feb 12 07:22:51 utilityBareMetal-api-official.gpfs.local deployadmd.sh[55]:
%home%:70_MMFS_CFG::9:sdNotifyAuthEnabled:yes::::::::::::::::::>
Feb 12 07:22:51 utilityBareMetal-api-official.gpfs.local deployadmd.sh[55]:
-----
Feb 12 07:22:51 utilityBareMetal-api-official.gpfs.local deployadmd.sh[55]: Starting GRPC
server on unix:///var/mmfs/run/scaleadmd.grpc.sock,>
Feb 12 07:22:51 utilityBareMetal-api-official.gpfs.local deployadmd.sh[55]: REST server
stopped with error: unable to start admin daemon rest>
Feb 12 07:22:51 utilityBareMetal-api-official.gpfs.local deployadmd.sh[55]: TCP gRPC server
stopped with error: unable to start admin daemon >
Feb 12 08:00:39 utilityBareMetal-api-official.gpfs.local deployadmd.sh[55]: REST server
stopped with error: http: Server closed
Feb 12 08:01:38 utilityBareMetal-api-official.gpfs.local deployadmd.sh[55]: Starting GRPC
server on https://[:]:50052, boot mode 2
Feb 12 08:01:38 utilityBareMetal-api-official.gpfs.local deployadmd.sh[55]: Starting REST
API server on https://0.0.0.0:46443, boot mode 2
```

Verify that the last two lines of output state that the GRPC and REST API server are started with boot mode 2. This mode indicates that the TLS or SSL certificate is imported successfully into the API server container.

Verifying that the API server container is configured on the Utility Node

If the API server is configured correctly on the Utility Node, you can log in to the OpenAPI specification page and access the API endpoints. To verify that the API server container is configured correctly on the Utility Node, complete the following steps:

1. In a browser, navigate to the following URL:

Replace `<utility_node_IP>` with your system IP address.

```
https://<utility_node_IP>:46443/openapi/#
```

This URL opens the OpenAPI specification page for the IBM Storage Scale native REST API endpoints.

2. If the **This Connection Is Not Private** warning message appears, click the **Visit this website** link in the message. This message appears if the autogenerated self-signed certificate is not replaced with the organization owned CA certificates. For more information, see [“Replace the autogenerated self-signed certificate with your organization-owned, CA-signed certificate”](#) on page 181.
3. After the successful authentication, verify that the IBM Storage Scale native REST API endpoints are listed on the page. If the endpoints are listed, then the API server container is configured correctly on the Utility Node. You can now authorize and access the endpoints.

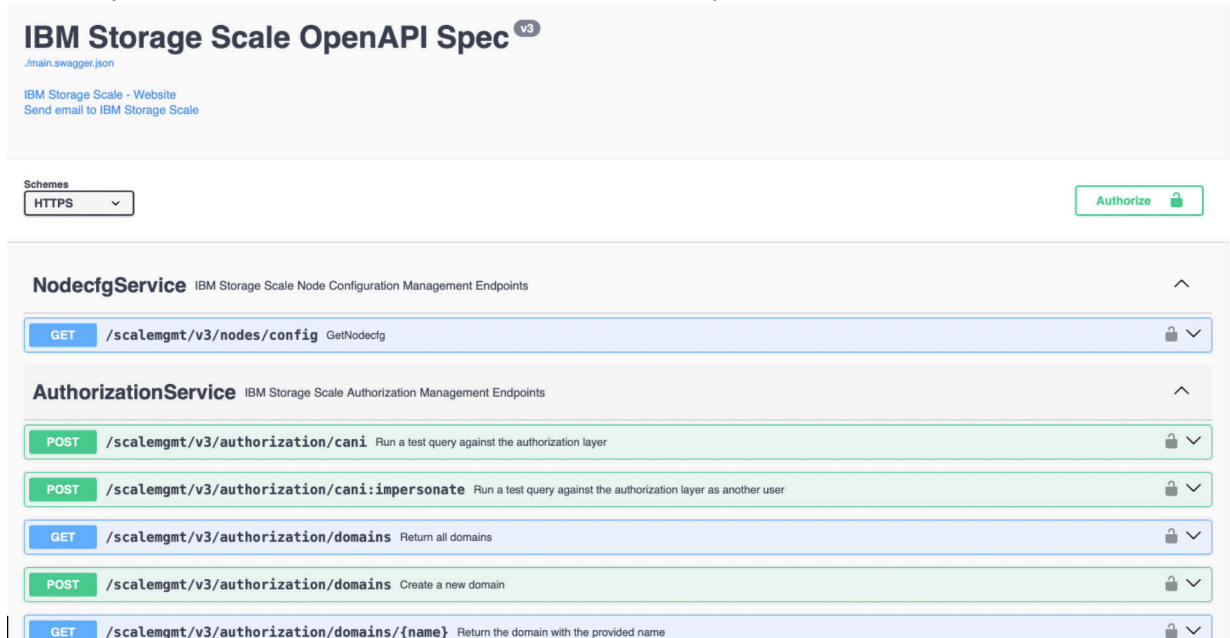


Figure 31. IBM Storage Scale Native REST API endpoints

4. In the OpenAPI specification page, click **Authorize**.
5. On the **Available authorizations** dialog, enter **Username** and **Password**. Then, click **Authorize**. You can use the **apiadmin** or **apiuser** credentials as required.

Note: All authentication in the API server container use the pluggable authentication modules (PAM). To add a new user, create the user in the API server container by using the **useradd** command. Then, set the required authorization by using the **AuthorizationService** endpoints. Any changes to authorization require authentication as the **apiadmin** user. For more information about configuring PAM authentication, see [Authentication for IBM Storage Scale native REST API](#).

6. After a successful authentication, access the endpoints. For example, the following procedure initiates the **VersionService** endpoint to retrieve the version of the API server container:
 - a. In the OpenAPI specification page, navigate to the **VersionService** endpoint.
 - b. Click **GET /scalemgmt/v3/version** to expand it.
 - c. Click **Try it out > Execute**. The version of **scaleadmd** is displayed in the **Responses** section.

GET /scalemgmt/v3/version GetVersion

Parameters

Name

Description

X-StorageScaleDomain

Domain to be authorized against for the request (default 'StorageScaleDomain')

string

X-StorageScaleDomain

(header)

Execute

Clear

Responses

Response content type application/json

Curl

```
curl -X 'GET' \
  'https:///:46443/scalemgmt/v3/version' \
  -H 'accept: application/json' \
  -H 'authorization: Basic YXBpYWRtaW46Y2x1c3Ricg=='
```

Request URL

https:///:46443/scalemgmt/v3/version

Server response

Code

Details

200

Response body

```
{
  "name": "scaleadm",
  "version": "5.2.3-0",
  "build_date": "Built on Jan 29 2025 at 01:05:36",
  "commit": "ee46de4a12e"
}
```

Download

Response headers

Figure 32. VersionService

Chapter 7. Troubleshooting

The following sections describe the known issues in IBM Storage Scale Systems and how to resolve these issues.

The following sections provide troubleshooting and support information. This information contains instructions for using several user interfaces to troubleshoot, recover, or maintain your system.

The `mm1srecoverygroup` might show invalid space state in a mixed environment with IBM FlashCore Modules 4 and IBM Storage Scale System 5149-091 Storage Enclosures

Product:

- IBM Storage Scale System 6000

In an IBM Storage Scale System 6000 environment with IBM FlashCore Modules 4 and MTM 5149-091 storage enclosures, the `mm1srecoverygroup <recovery_group> -L` command might report an invalid space state for the DAs that are associated with the storage enclosure.

Resolution or action

The message can be safely ignored. No workaround or fix is available.

File system creation by using the 16+2p RAID code through the IBM Storage Scale System GUI fails.

Product:

- IBM Storage Scale System 6000

When a user tries to create a 16+2p based file system through the GUI, the task fails due to the error: `mmvdisk: command failed examine previous error messages to determine cause.`

Resolution or action

No workaround or fix is available. To create a 16+2p file system, use the `mmvdisk` or `essrun` command.

During an MES procedures for NVMe drive that involves the IBM Storage Scale System 6000, some of the newly inserted NVMe drives might not be discovered properly.

Product:

- IBM Storage Scale System 6000

During an MES procedure, the hardware abstraction layer does not discover some IBM Storage Scale System 6000 drives properly.

Resolution or action

After you insert all the NVMe drives, use `essfindmissingdisks` from each canister to determine whether you have topology problems.

Use the information from the output to complete the following steps. Start with the least invasive process first.

Note: If any of the preexisting drives are reported as missing, call IBM Support before you attempt any resolution.

1. If the system reports missing drives, run the following command once from each canister to help you determine which drive slots have issues.

```
tslsencslot -ad |grep <system_serial_number> |grep '::::::::'
```

2. Reinsert each missing drive. After you remove a drive, wait 15 seconds before reinserting it. Wait another 15 seconds before you remove the next drive.
3. After you reinsert all missing drives, run **essfindmissingdisks** again.

If this step fails to recover all the missing drives, contact IBM Support.

If the previous steps fail, coordinate with IBM Support to complete the following steps.

1. If some drives are still missing after you complete the previous steps, run the **tslsencslot** command again to identify the current set of missing drives.

Note: Coordinate with the customer before you power cycle canisters or complete an AC power cycle.

If you are familiar with the **tsctlencslot** command, use this command to power cycle the drive slots that are missing. Call IBM Support if you are not familiar with this command.

2. Run **essfindmissingdisks** again.

During the resize operation for the recovery group of an MTM 5149-F26 to MTM 5149-F26 MES, the output might show an error about log vdisks creation.

Product:

- IBM Storage Scale System 6000

When you install many MTM 5149-F26 storage enclosures in an IBM Storage Scale System 6000 in flash hybrid configuration, the MES procedure completes successfully, but you might encounter an error: `mmvdisk: Error creating log vdisks.`

Resolution or action

1. Check the output of `mmvdisk rg list --rg <recovery_group_name>` to see whether the recovery group is properly resized. If it is, continue with the next steps of the MES process.
2. If it was not properly resized, attempt to complete the log format by using the next command.

```
mmvdisk rg create --complete-log-format
```

If these steps do not resolve the issue, call IBM Support before you continue with the MES procedure.

Used and physical capacity of the storage declustered arrays is not displayed for systems that do not use IBM FlashCore Modules 4.

Product:

- IBM Storage Scale System 6000

In the GUI of systems with no IBM FlashCore Modules, the **Storage > Declustered Arrays** pane states that the performance collector cannot return any data.

Resolution or action

This feature is implemented only for systems with IBM FlashCore Modules 4. Ignore this report if your system does not use IBM FlashCore Modules 4.

With certain error conditions, a DPU adapter in an MTM 5149-F26 canister might become unreachable.

Product:

- IBM Storage Scale System 6000

An adapter for a data processing unit (DPU) might become unreachable after the power cycle of an MTM 5149-F26 canister.

Resolution or action

If this condition is encountered, call IBM Support to recover the DPU.

GPFS and protocol packages removed after LEAPP operating system upgrade

Product:

- IBM Storage Scale System 3500 CES VM

After a LEAPP update on an operating system, IBM Storage Scale and protocol packages might be removed from one or more CES VM nodes.

Resolution or action

The resolution depends on your IBM Storage Scale System version:

For version 7.0.0 and later:

1. From the container, reinstall IBM Storage Scale RPMs on the affected CES VM nodes. Replace *emsnode* with your EMS node name and *listofcesvmnode* with the comma-separated list of affected CES VM nodes:

```
essrun -N emsnode cluster --add-cesvm-nodes listofcesvmnode
```

If your environment uses a high-speed network, add the `--suffix=-hs` option:

```
essrun -N emsnode cluster --add-cesvm-nodes listofcesvmnode --suffix=-hs
```

2. Install protocol RPMs on the affected CES VM nodes. Replace *cesvm03* with your CES VM node name.

For version 7.0.0:

```
essrun -N cesvm03 protocol --enable nfs smb s3 --configure
```

For version 7.0.1 and later:

```
essrun -N cesvm03 protocol configure --enable nfs smb s3
```

For more information, see [Adding and configuring protocols in an existing cluster by using the essrun command](#).

For 7.0.0 and earlier:

Use the IBM Storage Scale installation toolkit (Ansible toolkit) to reinstall the IBM Storage Scale and protocol RPMs on the affected CES VM nodes. For detailed instructions, see [Adding and configuring protocols in an existing cluster by using the Storage Scale Installation Toolkit](#).

SSH connections from one node to another might fail when FIPS is enabled.

Product:

- IBM Storage Scale System 6000
- Utility Node Protocol
- Utility Node management server
- IBM Storage Scale System 3000
- IBM Storage Scale System 3200
- IBM Storage Scale System 3500
- IBM Storage Scale System 5000

The following error message is displayed when you try to use SSH from one node to another, or when you run a command that requires SSH connections:

```
sshkey_read: Ed25519 keys are not allowed in FIPS mode
```

Resolution or action

Run the following command in all affected nodes, where FIPS is enabled:

```
sed -i '/ed25519/d' /root/.ssh/known_hosts*
```

An error is displayed during the firmware check.

Product:

- IBM Storage Scale System 5000

When you run the **mmfswfirmware --type storage-enclosure** command, the following error is displayed:

```
/usr/lpp/mmfs/bin/mmremote[8067]: lsFirmware[7565]: mmgetpdisktopologylite: line 2938: /usr/lpp/mmfs/bin/diskman/ljsjbofnet: not found  
mmfswfirmware: Command failed. Examine previous error messages to determine cause.
```

The firmware check incorrectly searches for an IBM Storage Scale System 5149-F26 storage enclosure, which is not applicable to IBM Storage Scale System 5000.

Resolution or action

This error can be ignored. Contact IBM Support for further guidance.

A node might not be updated successfully after an update and reboot.

Product:

- IBM Storage Scale System 6000
- IBM Storage Scale System 3500

Sometimes, when you update a server by using the **essrun** command, the server might not be updated after the server is rebooted.

Resolution or action

Reboot the node and issue the **essrun update --offline** command. Contact IBM Service for guidance, if the problem persists.

The secondary PSU might fail to update during an offline update from both canisters concurrently.

Product:

- IBM Storage Scale System 6000

When you update a 5149-F48 server offline from both canisters, the secondary PSU might not be updated. During the service, LED illuminates because of this issue. The normal operation of the PSU has no impact. The following error message might be displayed:

```
[E] Error sg1 updating firmware for enclosure.
```

Resolution or action

If the cluster is offline, then use **ipmitool power cycle** and run the update offline.

```
essrun update --offline
```

Else, contact IBM Support for guidance on how to complete the PSU firmware update and reset the LED.

Deployment from the GUI might fail because the RAS interface IP address is not 10.23.16.2.

Product:

- IBM Storage Scale System 6000
- Utility Node management server
- IBM Storage Scale System 3500

Resolution or action

1. Stop the existing bridge.

```
virsh net-destroy ess_vm_host_only_network
```

2. Undefine the bridge.

```
virsh net-undefine ess_vm_host_only_network
```

3. Create bridge with the latest DHCP reservation information.

```
virsh net-define /opt/ibm/ess/tools/conf/network_host_only_kvm_dev_esstility.xml
```

4. Edit the XML.

```
virsh net-edit ess_vm_host_only_network
```

5. Add the MAC address from the RAS interface to the EMSVM. An example is as follows:

```
<ip address='10.23.16.1' netmask='255.255.255.248'>
  <dhcp>
    <range start='10.23.16.1' end='10.23.16.6' />
    <host mac='52:54:00:b2:57:2e' name='EMSVM-23E' ip='10.23.16.2' />
    <host mac='52:54:00:b2:57:2d' name='PROTOCOL_F48' ip='10.23.16.3' />
  </dhcp>
</ip>
```

6. Start the bridge.

```
virsh net-start ess_vm_host_only_network
```

7. Set to autostart.

```
virsh net-start ess_vm_host_virsh net-autostart ess_vm_host_only_networkonly_network
```

8. Verify whether the EMSVM RAS IP is 10.23.16.2 and start deployment from the GUI.

During the drive-replacement of an IBM Storage Scale System 5149-F26 storage enclosure, the enclosure indicator LED might not illuminate.

Product:

- IBM Storage Scale System 6000

Resolution or action

Run the following command to illuminate the associated enclosure LEDs. The blue enclosure LED illuminates and flashes for 60 seconds, and then turns off.

```
essjbofctl -i mgmt -j <JB0F S/N> -I -l Bottom
```

The drive LED remains amber. If you have confirmed the enclosure location, use the LED to locate the correct drive for replacement.

The FCM4 drive replacement might fail in the first attempt.

Product:

- IBM Storage Scale System 6000 with FCM4

```
[E] Can not determine slothandle to power drive /dev/nvmeXnY on/off
```

This issue is a race condition between format and reset.

Resolution or action

Replace the FCM4 drive again.

```
mmvdisk pdisk replace --rg <rg> --pdisk <pdisk name>
```

For example:

```
mmvdisk pdisk replace --rg rg1 --pdisk e1s5
```

For more information, see [“FCM4 drive replacement failure in the first attempt” on page 194.](#)

Update failed because of the missing drive connection to the attached IBM Storage Scale System 5149-F26 storage enclosures.

Product:

- IBM Storage Scale System 6000 attached to the IBM Storage Scale System 5149-F26 storage enclosure

Because I/O canisters upgrade, the **mmvdisk topology check** might fail during the vdisk configuration. This issue occurs if a drive connection to the attached IBM Storage Scale System 5149-F26 storage enclosure is missing.

Resolution or action

1. From both I/O canisters check connectivity to the attached IBM Storage Scale System 5149-F26 storage enclosures.

```
essjbofctl -i auto -t JX
```

The JX is the JBOF config; for example, J1, J2.

2. Address errors if any. Reestablish connections to IBM Storage Scale System 5149-F26 storage enclosure that is attached to drives.

```
essjbofctl -I auto -t JX -d -c
```

3. Verify the topology.

```
mmvdisk server list --nc <building block node class> --disk-topology
```

4. Rerun the initial failing step after topology is verified.

The drive path is missing.

Product:

- IBM Storage Scale System 6000 with QLC mixed tiering

Pdisk health check shows that the drive path is missing.

```
mmvdisk pdisk list --rg all --not-ok
```

Resolution or action

1. Verify that corresponding DA is otherwise healthy and no rebuild operation is in progress.

```
mmvdisk rg list --rg <RG name> --da --fault-tolerance
```

2. Power cycle the drive to reinitiate path discovery. Contact IBM Support for guidance on how to properly power cycle the drive.

Online upgrade by using **essrun** changes autoload setting from yes to no

Product:

- IBM Storage Scale System 6000
- Utility Node Protocol
- Utility Node management server
- IBM Storage Scale System 3000
- IBM Storage Scale System 3200
- IBM Storage Scale System 3500
- IBM Storage Scale System 5000

During an online upgrade by using the **essrun** command, the autoload setting must be disabled. The setting must not change after completion of the upgrade. The setting must be disabled before and after the upgrade or enabled before and after the upgrade. Autoload ensures that IBM Storage Scale starts automatically when a node reboots.

Resolution or action

1. Check the autoload setting.

```
mmlsconfig autoload
```

2. After the completion of the upgrade, if you want to enable autoload, issue the following command:

```
mmchconfig autoload=yes
```

A firmware upgrade by using **--serial** on a 5147-092 JBOD fails

Product:

- IBM Storage Scale System 5000

An example of the error when you attempt to upgrade a single JBOD in a 5147-092:

```
mmchfirmware --type storage-enclosure --update-id PREVIOUS --serial-number 789A39W
```

The following error was displayed:

```
essio1-hs: [E] Storage enclosure 789A39W not found.  
[17:20:07.829157999] /usr/lpp/mmfs/bin/mmchfirmware <- chEnclFirmwareVer2 rc:1  
[17:20:07.829914727] /usr/lpp/mmfs/bin/mmchfirmware gpfs is up, putting pdisk paths back in  
service  
mmchfirmware: Command failed. Examine previous error messages to determine cause.
```

Resolution or action

Issue the command with the **--serial** option.

QLC SSDs might run Gen 1 speeds (2.5 GT/s)

Product:

- IBM Storage Scale System 6000

Resolution or action

1. To check whether the system has drives with degraded speed run the following:

```
for i in $(nvme list -v | grep pcie | grep SAMS | awk '{print $NF}'); do nvme list -v |
grep pcie | grep $i |
awk '{printf $2" "}' ; nvme list -v | grep pcie | grep $i | awk '{print $(NF-2)}' | xargs -i
lspci -vvs {}
| grep LnkSta: ; done | grep -v 16GT
```

A sample output is as follows:

```
S7ZDNG0Y100024      LnkSta:      Speed 2.5GT/s (downgraded), Width x2 (ok)
S7ZDNG0Y100038      LnkSta:      Speed 2.5GT/s (downgraded), Width x2 (ok)
S7ZDNG0Y100023      LnkSta:      Speed 2.5GT/s (downgraded), Width x2 (ok)
S7ZDNG0Y100051      LnkSta:      Speed 2.5GT/s (downgraded), Width x2 (ok)
```

2. Make sure that all drives are at 16 GT/s, otherwise, run the recovery steps:

- a. Power cycle the drive to reinitiate the link training.
- b. Reboot the node to reinitiate link training.
- c. Run recovery script to retrain the link. For example,

```
/opt/ibm/ess/tools/samples/setlnkspd.sh 47
```

A sample output is as follows:

```
Slot: 47, target pcie address: d4:00.0, port pcie address: ca:09.0
Link cap: 0a7b7904, Link Speed: 3, Link width 2
Link cap 2: 021e0003
Setting link cap 2 of slot 47 with 021E0004
Trigger link training of slot 47? (y/n) y
Triggering link control with 0060
```

SELinux enablement might cause failures on IBM Storage Scale System Utility Node VM

Product:

- IBM Storage Scale System Utility Node VM

You might see the following error when you enable SELinux on IBM Storage Scale System Utility Node VM:

```
[WARNING]: log file at /var/log/ess/6.2.3.0/essansible.json is not writeable and we cannot
create it, aborting
```

Resolution or action

Create **touch /.autorelabel**, and reboot the node.

Deployment inventory file might be overwritten if config load is run with individual nodes only.

Product:

- IBM Storage Scale System 6000
- Utility Node Protocol
- Utility Node management server
- IBM Storage Scale System 3000
- IBM Storage Scale System 3200
- IBM Storage Scale System 3500
- IBM Storage Scale System 5000

The inventory file is stored in /vpd/Inventory. This file might be incorrect if you are running config load with one node at a time.

Resolution or action

Run the **essrun config load** with all nodes in the same command.

Drive failure is not reported by Callhome due to being marked as a duplicate

Product:

- IBM Storage Scale System 6000
- Utility Node Protocol
- Utility Node management server
- IBM Storage Scale System 3000
- IBM Storage Scale System 3200
- IBM Storage Scale System 3500
- IBM Storage Scale System 5000

Resolution or action

None. Contact IBM Support if a drive failure occurs.

Protocol VM creation fails if a previous protocol VM exists

Product:

- IBM Storage Scale System 3000
- IBM Storage Scale System 3200
- IBM Storage Scale System 3500

When you attempt to create a new protocol VM from the GUI, if a previous VM exists, the creation of the new protocol VM fails. The deployment fails with errors that are related to permanently added utility2 to the list of known hosts.

Build information

ess_7.0.1.0_2414-01-sr-gr

Prerequisites

Make sure that an existing protocol VM must be removed before initiating deployment from the GUI.

Resolution or action

Before a new protocol VM deployment from the GUI:

1. Verify whether any existing protocol VMs are present on the system.
2. If an existing protocol VM is found, remove it completely before you initiate the new deployment.
3. Clean the yum cache if needed by running **yum clean packages**.

Deployment from GUI fails, manual cleanup is required

Product:

- IBM Storage Scale System 6000
- Utility Node management server
- IBM Storage Scale System 3500

Resolution or action

Run manual mm commands such as **mmdelnode**, **mmvdisk rg delete**, **mmvdisk fs delete**, and **mmvdisk vs delete** to cleanup artifacts that were created, attempt the deployment again. Contact IBM Support for any further guidance.

QLC drives fail to enroll (self-encrypting drives)

Product:

- IBM Storage Scale System 6000 with QLC drives

Resolution or action

1. Power off and on the drives.

```
/opt/ibm/ess/tools/samples/drives_off_on
```

2. Enroll QLC drives (on one of the canisters).

If more issues or concerns, contact IBM Support.

Disk replacement with command fails for QLC drives in mixed configurations

Product:

- IBM Storage Scale System 6000 with QLC drives

Resolution or action

1. Use underlying disk replacement commands.
2. Use the GUI to replace drives.

For more issues or concerns, contact IBM Support.

Formatting of 4 K+8 B drives fails during MES operations

Product:

- IBM Storage Scale System 6000
- Utility Node Protocol
- Utility Node management server
- IBM Storage Scale System 3000
- IBM Storage Scale System 3200
- IBM Storage Scale System 3500
- IBM Storage Scale System 5000

```
mmvdisk: (mkrginput) Only the format of 4K+0 or 512+0 is supported.
```

Resolution or action

If more issues or concerns, contact IBM Support.

FCM4 drive replacement failure in the first attempt

Product:

- IBM Storage Scale System 6000
- Utility Node Protocol
- Utility Node management server
- IBM Storage Scale System 3000
- IBM Storage Scale System 3200
- IBM Storage Scale System 3500
- IBM Storage Scale System 5000

Resolution or action

Run the same command again.

Concurrent new deployments through the GUI from multiple browser sessions can overwrite earlier sessions before deployment starts

Product:

- IBM Storage Scale System Utility Node
- IBM Storage Scale System 6000
- IBM Storage Scale System 3500

If two users load the deployment GUI portal for the same system and attempt to deploy without refreshing the page, the second deployment does not detect the ongoing progress of the first. As a result, the second deployment overwrites the first, and the first deployment stops. Any subsequent refresh of the GUI portal displays only the progress of the second deployment. This issue occurs only if both users reach the configuration step and click **Submit** to start deployment without refreshing the page beforehand.

Resolution or action

To avoid this issue:

- Refresh the deployment GUI portal before deployment. A refresh checks for any ongoing deployment and prevents starting a new one if progress is detected.
- Make sure that only one user initiates the deployment at a time.

GUI does not display full `essrun` logs for failed commands

Product:

- IBM Storage Scale System Utility Node
- IBM Storage Scale System 6000
- IBM Storage Scale System 3500

When a deployment fails, the GUI might display only a partial view of the `essrun` command logs.

Resolution or action

To view the full `essrun` logs, manually access the deployment container and check the log file:

1. SSH into the deployment container host.

```
ssh emsvm
```

2. Log in to the deployment container.

```
podman exec -it IBMRCLCONNECT /bin/bash
```

3. Access the full log file.

```
vim /var/log/ess/6.2.3.0/essansible.json
```

utilityBareMetal update fails due to SSH configuration error

Product:

- IBM Storage Scale System Utility Node
- IBM Storage Scale System 6000
- IBM Storage Scale System 3500

```
TASK [/opt/ibm/ess/deploy/ansible/roles/checkenv : Check known hosts | Fail] ***
fatal: [utilityBareMetal]: FAILED! => changed=false
  msg:
    - Please check above output before proceeding to update.
```


Refer to the *Emergency recovery* section under the [IBM Storage Scale with thin provisioned devices](#) section in the IBM Storage Scale documentation for graceful emergency recovery for thin-provisioned storage systems, LUNs, and volumes created from FCM drives:

Step 4 in the emergency recovery procedure says: ‘After enough physical space is free, unmount the file system.’ It is difficult to determine how much free space is ‘enough’. The following example helps determine the free space:

For example:

```
# mmfsadm dump vdisk|grep -A 1 SpaceInfo
```

A sample output is as follows:

```
daSpaceInfoVersion 24:56376 daPhySpaceState Critical
used/total physical space 148086264/439482672 MiB utilization 0.3370
```

Where,

- 33.70% disk is used which means the space is in the `Critical` state. Run the following command to check how much space is available:

```
grep utiliz /var/adm/ras/mmfs.log.latest
```

A sample output is as follows:

```
2024-12-09_20:50:05.911-0500: [C] Reset DA space utilization state threshold: Info:20
Warning:22 Critical:30 OOS:35 OOSBGT:90
```

OOS is at 35%, 1.3% less than OOS, and you can safely mount the file system. If the usage is at .349 or higher, delete more and rerun the emergency reclaim to free some more space.

If the problem persists, call IBM Support.

Correctable hardware errors on many PCI addresses might be noticed in dmesg on a canister

Product:

- IBM Storage Scale System 6000

In *dmesg*, you might see the following errors:

```
[Hardware Error]: Hardware error from APEI Generic
Hardware Error Source: 516
[Hardware Error]: It has been corrected by h/w and
requires no further action
[Hardware Error]: device_id: 0000:e1:00.0
```

Note: The PCI device address or hardware error source might be different.

Resolution or action

These errors might be safely ignored. The threshold for correctable errors is 1 in 2 TB worth of data. This threshold is measured in IBM manufacturing and systems are only shipped if compliant. If you have any client concerns, contact IBM Support.

GUI reconfiguration error

After reconfiguration of the GUI, you might encounter the following error when you attempt to initialize GUI:

```
com.ibm.fssc.cli.CommandException: EFSSA0635A An error occurred while reading cluster
information on host localhost
```

Resolution or action

- Run the following command from the management node:

```
-rm /root/.pgpass
-systemctl restart gpfsGUI
```

- Initialize the GUI again.

A boot drive mirror degraded when you run the `mmhealth node show` command when both boot drives are healthy

Product:

- IBM Storage Scale System 6000

The symptom for this degraded issue is that all but one partition is mirrored successfully. This issue can be seen by running the `cat /proc/mdstat` command on the affected node and looking for a missing 'U'.

Example:

```
# cat /proc/mdstat
```

A sample output is as follows:

```
Personalities : [raid1]
md2 : active raid1 nvmeXn1p3[2] nvmeYn1p3[0]
133613440 blocks super 1.2 [2/2] [UU]
bitmap: 0/1 pages [0KB], 65536KB chunk
md3 : active raid1 nvmeXn1p5[2] nvmeYn1p5[0]
768587904 blocks super 1.2 [2/2] [UU]
bitmap: 6/6 pages [24KB], 65536KB chunk
md1 : active raid1 nvmeXn1p2[2] nvmeYn1p2[0]
8289280 blocks super 1.2 [2/2] [UU]
bitmap: 0/1 pages [0KB], 65536KB chunk
md0 : active raid1 nvmeXn1p1[2]
2047936 blocks super 1.0 [2/1] [U_] << here is the missing mirrored partition.

unused devices: <none>
```



Attention: The partition numbers (p1 for md0, p2 for md1, and so on).

Resolution or action

In the example, you can see that `nvmeYn1p1` is the missing element. To fix this manually, add the `nvmeYn1p1` partition to `/dev/md0` by running:

```
mdadm -a /dev/md0 /dev/nvmeYn1p1
```

The measured drive temperature is lower than the low critical value

Product:

- IBM Storage Scale System 6000

This condition is a false positive and can be ignored.

Example:

```
# mmhealth node eventlog | grep temp_low_critical
```

A sample output is as follows:

```
2024-XX-XX 02:42:11.850411 EST temp_low_critical WARNING Temperature sensor drive_slot13_id94
has measured a temperature is less than the low critical value.
```

Resolution or action

No action.

In a TPM environment, the key-in-use flag does not get reset after a recovery group is deleted.

Product:

- IBM Storage Scale System 6000

Resolution or action

This error blocks a new key from being generated in the same NV slot.

- Option 1: The key is only mistakenly marked as in use; therefore, the same key can be reused. After the key is in use again, the **rekey** command can be used to change to the new key.
- Option 2: Contact IBM Support to reset the key.

Only one path to the drive is active

Product:

- IBM Storage Scale System 6000

After you replaced an FCM4 drive, you might find that only one path to the drive is active.

Check this error by using the **mmhealth** or the **mmvdisk pdisk** command.

Example:

```
# mmvdisk pdisk list --rg s6krw1 --not-ok
```

A sample output is as follows:

recovery group	pdisk	declustered array	paths	capacity	free space	FRU (type)	state
s6krw1	e1s24	DA1	1	52 TiB	51 TiB	03LG010	ok

Resolution or action

This issue might resolve itself within an hour. If the issue is not resolved within an hour, complete the following steps:

1. Verify whether the system is in good condition and no arrays are rebuilding.
2. Reseat the affected drive.

If the issue is still not resolved, contact IBM Support.

Missing drive paths, enclosure LEDs are on, or unknown status when performing SAS topology

Product:

- IBM Storage Scale System 6000

- [ERROR] Unable to determine enclosure order. Check the HBA to enclosure cabling. [ERROR] Location 78F10GH-2 appears only on the sg369 path

- [ERROR] Enclosure 78F10G8 (IBM-ESS 5149-091, number undetermined):

- Adapter in PCI Slot: 2, index: c0 Enclosures connected: Enclosure e6 Serial 78F10GM Expected 91 Found 91 Connected to C3 at x4 22.5 Gb/s x12 Unknown

Resolution or action

Refer to the [Falcon LEDs lit and paths missing before cluster is created](#) and contact IBM Support.

Missing drives or namespaces from one or both canisters

Product:

- IBM Storage Scale System 6000

Example of IBM Storage Scale System 6000 with 48 FCM4 drives missing 8

```
nvme list -v | grep NVMe | wc -l 40
```

```
esstoragequickcheck -N localhost
```

A sample output is as follows:

```
[INFO] Enclosure 78L005D sees 40 disks (40 SSDs, 0 HDDs) [ERROR] GNR server disk topology:
Scale 6000 F48 48 NVMe (match: 83/100)
```

Resolution or action

1. Reboot the canister.

```
systemctl reboot
```

2. Complete AC power cycle the building block.

- a. Completely remove power from the IBM Storage Scale System 6000 chassis (unplug all the 4 PSUs).
- b. Wait for 5 minutes and plug power back to the PSUs.
- c. Wait for the canisters boot and check the drives again.

3. Scan the PCI bus again.

```
echo 1 > /sys/bus/pci/rescan
```

You might need to power off or power on the affected slots. Contact IBM Support.

After a firmware update or a drive replacement, the drive shows more paths

Product:

- IBM Storage Scale System 6000

This error is due to conflicting device namespaces. The system runs without problem while in this condition.

Example:

```
mmvdisk pdisk list --rg s6krq1 --not-ok
```

A sample output is as follows:

recovery group	pdisk	declustered array	paths	capacity	free space	FRU (type)	state
s6krq1	e1s22	DA1	3	52 TiB	12 TiB	03LG010	ok

Resolution or action

The error might resolve itself within an hour. If the error is not resolved, use one of the following options:

- Verify that the system is in good condition and no arrays are rebuilding, then reseal the affected drive.
- Check your mounts, and verify whether you can reboot a canister without losing quorum. If you can, then reboot one canister. Wait until recovery is complete and check for the error. If the error is still unresolved, double check quorum and mounts again, and reboot that partner canister.
- Contact IBM Support, if the error is still not resolved.

CX7 crypto-enabled high-speed adapters to replace CX7 noncrypto adapters

Product:

- IBM Storage Scale System 6000

CX7 crypto-enabled high-speed adapters can be used to replace CX7 noncrypto adapters. Ethernet has crypto-enabled adapters by default. Therefore, if the replacement is for an InfiniBand system, the link layer needs to be changed.

The **essgennetworks** command can be used to change the link layer of each port, then the canister must be rebooted.

Resolution or action

In the following example, both ports of a single adapter are changed to InfiniBand:

1. Verify that you do not lose the quorum after you reboot the canister or a network connectivity does not change.
2. Change the link layer of the ports by using the **essgennetworks** command from the canister.

```
essgennetworks -N localhost --change InfiniBand --devices /dev/mst/mt4129_pciconf1 --port P1
```

```
essgennetworks -N localhost --change InfiniBand --devices /dev/mst/mt4129_pciconf1 --port P2
```

3. Reboot the canister.

```
reboot
```

Resolution or action

Autoload setting might not be as expected after an online update

Product:

- IBM Storage Scale System 6000

After an online update, the autoload configuration might show some nodes set to autoload=no.

Example:

```
mmlsconfig autoload
autoload yes
autoload no
```

Resolution or action

If the autoload setting is not as desired, use the **mmchconfig** command to change to the expected value.

```
mmchconfig autoload=yes
```

Unsuccessful online update of IBM Storage Scale System 5000

Product:

- IBM Storage Scale System 5000

During an online update for IBM Storage Scale System 5000, you might encounter an error similar as follows:

```
TASK [/opt/ibm/ess/deploy/ansible/roles/updatenode : Set partner node as primary | gather
backup] *****
fatal: [essio2]: FAILED! => changed=true
  cmd: |-
    /usr/lpp/mmfs/bin/mmvdisk rg list --rg ess5k_essio1_essio2
    ess5k_essio2_essio1 --server -Y | grep -i backup | awk -F ':' '{print $8}'
  delta: '0:00:01.149195'
  end: '2024-XX-28 01:24:40.658240'
```

```
failed_when_result: true
msg: ''
rc: 0
start: '2024-XX-28 01:24:39.509045'
stderr: '/bin/sh: line 1: ess5k_essio2_essio1: command not found'
stderr_lines: <omitted>
stdout: |2-
```

Resolution or action

Run an offline update (as long as downtime can be scheduled).

1. Shut down GPFS on both I/O nodes.
2. Start the upgrade by using the `--offline` flag,

```
essrun -N <IO nodeList> update --offline
```

If an offline upgrade is unacceptable, run an online upgrade.

1. Check the update.

```
essrun update --precheck
```

2. Upgrade one node only.
3. From the RG group, move both recovery group names to the peer node.
 - a. Select RG names that are hosted by the node and must be updated.

```
mmvdisk rg list
```

- b. Check recovery groups status.

```
mmvdisk rg change --rg <rg name> --active <peer node >
```

- c. Upgrade only one node (the one that is not hosting any RG).

```
essrun -N <io node> update
```

- d. After one node online upgrade is successful, upgrade other nodes by completing all online upgrade steps.

Incorrect FRU type for drives after the recovery group creation

Product:

- IBM Storage Scale System 6000

Sometimes IBM Storage Scale System 6000 drives might show incorrect FRU type after the recovery group creation. This issue might also occur during MES of drives from 24 to 48.

Resolution or action

Run the **mmvdisk pdisk list --rg** and use the output to determine whether an incorrect FRU type is seen.

- Correct FRU type

```
mmvdisk pdisk list --rg rgs6k
```

A sample output is as follows:

recovery group	pdisk	declustered array	paths	capacity	free space	FRU (type)	state
rgs6k	e1s01	DA1	2	52 TiB	12 TiB	3LG010	ok

- Incorrect FRU type

recovery group	pdisk	declustered array	paths	capacity	free space	FRU (type)	state
rgs6k	e1s01	DA1	2	52 TiB	12 TiB	19.2 TBu FCM4 N	ok

The incorrect FRU does not impact on normal operation. However, a drive with incorrect FRU type needs to be replaced in the future, the normal repair operation might fail.

If the repair fails, double check that you have the proper FRU type (03LG010). The FRU can be seen on the new drive label. If the FRU is incorrect, call IBM Support before continuing. If the FRU is correct, retry the repair by using the `--force-fru` option.

```
mmvdisk pdisk replace -rg rgs6k --pdisk e1s01
```


Chapter 8. Enabling or disabling security settings

The following topics describe how to enable or disable security related settings in IBM Storage Scale System.

-
- [“Enabling or disabling SELinux” on page 210](#)
- [“Enabling or disabling sudo” on page 212](#)
- [“Enabling or disabling FIPS” on page 213](#)

Firewall ports

The **essrun** command opens required ports for IBM Storage Scale System components during deployment. For security, close any unused, duplicate, or legacy ports that are not required by your deployment.

Required firewall ports

The **essrun** command opens the required ports for IBM Storage Scale System components during deployment. Ports associated with Cluster Export Services (CES) are opened only when supported IBM Storage Scale System CES MTM protocol nodes are present. User-managed (“roll your own”) CES nodes are not modified. If no protocol nodes exist, the **essrun** command does not open any CES-related ports. When protocols such as NFS are configured, run the **mmnfs config change** command to align the service with the opened ports.

Table 6. Required firewall ports			
Protocol	Port	Node type	Service
TCP	22	All nodes	SSH/SCP
TCP/UDP	53	All nodes	DNS
TCP	80	All nodes	Management GUI/API
TCP/UDP	88	All nodes	Kerberos authentication
TCP/UDP	111	Protocol nodes	NFSv3 and NFSv4, and portmapper
UDP	123	All nodes	Network time protocol (NTP)
TCP/UDP	389	All nodes	LDAP
TCP/UDP	443	All nodes	Management GUI/API (HTTPS), Call Home
TCP	445	Protocol nodes	SMB
TCP/UDP	464	All nodes	Kerberos password change
TCP/UDP	636	All nodes	LDAP
TCP/UDP	657	All nodes	HMC communication
TCP	873*	All nodes	Re-synchronization for file transfer to IBM FlashCore Module
TCP	1191	All nodes	Internal cluster communication
TCP	2049	Protocol nodes	NFS
TCP	4379	Protocol nodes	SMB CTDB

Table 6. Required firewall ports (continued)

Protocol	Port	Node type	Service
TCP	4444	All nodes	Management GUI
TCP	4739	All nodes	Performance monitoring Allows only external requests from clients
TCP	5000	Protocol nodes	Object authentication
TCP/UDP	5353*	All nodes	HAL zeroconf (DNS multicast)
TCP/UDP	5431	Protocol nodes	PostgreSQL for object storage
TCP	6001	Protocol nodes	S3 service
TCP	6443	Protocol nodes	S3 service
TCP	7004	Protocol nodes	S3 metrics collection on protocol nodes during the performance monitoring
TCP	7005	Protocol nodes	S3, IAM
TCP	8080	Protocol nodes	Object storage
TCP	9085	Performance monitoring collector nodes	Performance monitoring (between collectors)
TCP	9980	Performance monitoring collector nodes	Performance monitoring
TCP	10080	All nodes	Installation Toolkit
TCP/UDP	11211	Protocol nodes	Object storage
TCP	20080*	All nodes	Deployment container
TCP/UDP	32765	Protocol nodes	NFS STATD_PORT (NFSv3 only)
TCP/UDP	32767-32769	Protocol nodes	NFS MNT_PORT, RQUOTA_PORT, NLM_PORT (NFSv3 only)
TCP	35357	Protocol nodes	Object authentication
TCP	40443*	All nodes	HAL REST API
TCP	46443	All nodes	Native REST API
TCP	50052	All nodes	Native REST API AD communication
TCP	51000*	All nodes	Interlink communication
TCP	51001*	All nodes	HAL cross-node communication
TCP/UDP	60000-61000	All nodes	GPFS ephemeral port range (configurable)
TCP	49152-65535	Windows 2012 and higher nodes	Dynamic RPC

Note:

- *All these ports are for only IBM Storage Scale System.

- **Protocol nodes:** Serves NFS, SMB, or S3. Before you enable the firewall with the listed NFS settings, run the following command on each protocol node.

```
mmnfs config change
```

A sample output is as follows:

```
STATD_PORT=32768:RQUOTA_PORT=32767:NLM_PORT=32766:MNT_PORT=32765
mmnfs: The NFS configuration was changed successfully.
mmnfs: NFS server restarted on all NFS nodes on which NFS server is running.
```

- **Port 1191:** Internal cluster communication only, do not use for external access.
- **IBM Storage Scale ephemeral range:** Default 60000-61000, configurable.

Unrequired ports

For enhanced security, ports that are not required for any system operation can be safely closed. These include unnecessary services, duplicate protocol ports (where only TCP or UDP is needed), and legacy service ports.

The following TCP and UDP ports are not required for normal system operation and can be closed to enhance security.

Table 7. Unrequired TCP and UDP ports	
Port	Protocol
7	UDP
22	UDP
67	TCP
67	UDP
68	TCP
69	TCP
69	UDP
80	UDP
123	TCP
162	TCP
162	UDP
514	TCP
514	UDP
623	TCP
623	UDP
782	TCP
873	UDP
1500	TCP
2049	UDP
3001	TCP
3001	UDP

Table 7. Unrequired TCP and UDP ports (continued)	
Port	Protocol
3002	TCP
3002	UDP
4011	TCP
4739	UDP
5024	TCP
6200-6203	TCP
8085	TCP
8123-8127	TCP
9084	TCP
9094	TCP
9981	TCP
10080	UDP
20080	UDP

Note:

- Close these ports only after you verify that your deployment does not require these ports.
- Some ports are optional because the service requires only one protocol, TCP or UDP.
- Close port ranges (for example, 6200-6203 and 8123-8127) as complete ranges.
- Test port closures in a non-production environment before you apply changes to production systems.
- Document all closed ports for future reference and troubleshooting.

Enabling or disabling the firewall

You can enable firewall in an IBM Storage Scale System environment for management server, I/O server nodes, and protocol nodes by using the **firewall** subcommand of the **essrun** command.

Before you begin

- Before you enable the firewall on the system, you must identify and validate all required network ports by referring to the [Firewall ports](#) section that lists component-to-component port requirements.

Important: Check the firewall configuration by using the decoupled precheck tool available at [Scale System Precheck Tool](#).

- Ensure that you have downloaded the container image from Fix Central. If you have not downloaded the image, download the from Fix Central. For more information, see [“Preparing a Utility Node” on page 65](#).

About this task

By default, any node in an IBM Storage Scale System cluster has firewall that is disabled. You can run the **firewall** subcommand of the **essrun** command. This command can be run after the deployment of management server node or I/O server nodes is complete.

Procedure

- To access the container, run the following command:

```
cd /home/deploy/<extracted .dir>
```

- If the container is running, connect to the container by running the following command:

```
./essmkym1 --reconnect
```

- If the container is not running, restart the container by running the following:

```
./essmkym1 --restart
```

- Enable the firewall on IBM Storage Scale System nodes.

- a) To enable the firewall on required nodes, run the following command:

```
essrun -N <ems1,io1,io2,proto1,...> firewall enable
```

where:

ems1

Specifies a management node.

io1,io2

Specify I/O nodes.

proto1,...

Specifies protocol nodes.

- b) After the firewall is enabled on the management node, exit and restart the container.

```
exit
cd /home/deploy/<extracted .dir>
./essmkym1 --restart
```

If the following message is displayed, type no.

```
[INFO] You may want to wipe it out if EMS IP changed or there are wrong entries in the
Inventory --
Inventory file exists, do you want to wipe it out? (y/n):
```

- c) To check the firewall status on all nodes, run the following command:

```
essrun -N <ems1,io1,io2,proto1,...> firewall status
```

The system shows the following firewall status:

```
running
```

- d) To check the open ports on specified nodes, run the following command:

```
essrun -N <ems1,io1,io2,proto1,...> firewall list
```

- Add firewall ports to nodes.

- a) To add one or more ports to the firewall, run the **firewall-cmd** command with the **--permanent** and **--add-port** options.

- To add a single port, run the following command:

```
firewall-cmd --permanent --add-port=<port>/tcp
```

If the specified protocol is UDP, use UDP instead of TCP.

- To add multiple ports, run the following command:

```
firewall-cmd --permanent --add-port=<port1>/tcp --add-port=<port2>/tcp --add-
port=<port3>/tcp
```

- b) To reload the firewall configuration to apply the changes, run the following command:

```
firewall-cmd --reload
```

- To remove one or more ports from the firewall, use the **firewall-cmd** command with the **--permanent** and **--remove-port** options.
- To remove a single port, run the following command:

```
firewall-cmd --permanent --remove-port=<port>/tcp
```

If the specified protocol is UDP, use UDP instead of TCP.

- To remove multiple ports with a single command, run the following command:

```
firewall-cmd --permanent --remove-port=<port1>/tcp --remove-port=<port2>/tcp --remove-port=<port3>/tcp
```

- To disable the firewall on nodes, complete the following steps:
 - To disable firewall on the management server node, run the **firewall** subcommand with the **disable** option.

```
essrun -N ems1 firewall disable
```

- To disable firewall on I/O server nodes, run the **firewall** subcommand with the **disable** option.

```
essrun -N nodeList firewall disable
```

Enabling or disabling SELinux

Enabling SELinux in an IBM Storage Scale System environment is a two-step process and it can be enabled for management server and I/O server nodes using the **selinux** sub-command of the **essrun** command.

About this task

By default, any node in an IBM Storage Scale System cluster has SELinux disabled. You can run the **selinux** sub-command of the **essrun** command to enable or disable SELinux on nodes. This command can be run after the deployment of management server node or I/O server nodes is complete.

Procedure

- Enable SELinux on the management server node as follows.
 - a) Run the **selinux** sub-command on the management server node.

```
# essrun -N ems1 selinux permissive
```

Note: Make sure that you reboot the node when the **selinux** sub-command completes.

- b) Reboot the node.

```
# systemctl reboot
```

The node is rebooted and it comes up with SELinux in Permissive mode.

```
# sestatus
SELinux status:                enabled
SELinuxfs mount:              /sys/fs/selinux
SELinux root directory:      /etc/selinux
Loaded policy name:           targeted
Current mode:                 permissive
Mode from config file:       permissive
Policy MLS status:           enabled
Policy deny_unknown status:   allowed
Max kernel policy version:    31
```

c) Rerun the **selinux** sub-command with the **enable** option to enforce SELinux.

```
# essrun -N ems1 selinux enable
```

No reboot is required in this case.

```
# sestatus
SELinux status:                enabled
SELinuxfs mount:              /sys/fs/selinux
SELinux root directory:      /etc/selinux
Loaded policy name:            targeted
Current mode:                  enforcing
Mode from config file:        enforcing
Policy MLS status:             enabled
Policy deny_unknown status:    allowed
Max kernel policy version:     31
```

After SELinux is enabled, kernel logs any activity in the `/var/log/audit/audit.log` file.

- Enable SELinux on I/O server nodes as follows.

a) Run the **selinux** sub-command on the I/O server nodes.

```
# essrun -N ess_x86_64 selinux permissive
```

Note: Make sure that you reboot the node when the **selinux** sub-command completes.

b) Reboot the I/O server nodes.

```
# systemctl reboot
```

The node is rebooted and it comes up with SELinux in Permissive mode.

c) Rerun the **selinux** sub-command with the **enable** option to enforce SELinux.

```
# essrun -N ess_x86_64 selinux enable
```

No reboot is required in this case.

After SELinux is enabled, kernel logs any activity in the `/var/log/audit/audit.log` file.

- Disable SELinux on IBM Storage Scale System nodes as follows.

- To disable SELinux on the management server node, use the following command.

```
# essrun -N ems1 selinux disable
```

Reboot the node after the command completes. When the node comes up after reboots, SELinux is disabled.

You can check the status as follows.

```
# sestatus
SELinux status:                disabled
```

- To disable SELinux on the I/O server nodes, use the following command.

```
# essrun -N ess_x86_64 selinux disable
```

Reboot the node after the command completes. When the node comes up after reboots, SELinux is disabled. Any I/O server node name can also be used instead of the group name.

Additional information: Any mentioned security item is an optional feature and you can enable it on demand for an IBM Storage Scale System cluster. Security commands can be run using the **essrun** command after deployment of the node is done and before creating the GPFS cluster. In upgrade cases, any such security commands must be run after stopping the GPFS cluster. Do not attempt to run any security command while GPFS cluster is up and running.

Container consideration: Make sure that none of the security command is run against the container node. The container has a very light footprint of Red Hat Enterprise Linux 7.x OS on which any security parameters are not supported.

Enabling or disabling sudo

Enabling sudo requires a sudo-capable user (gpfsadmin) to be added to all nodes which are a part of or which are going to be a part of an IBM Storage Scale System cluster. Sudo must be enabled for management server and I/O server nodes by using the **sudo** sub-command of the **essrun** command.

Note: Sudo user across all GPFS nodes must have the same Linux group ID and user ID.

Enabling sudo on IBM Storage Scale System

You can enable sudo configuration on a Linux node by using the enable option of the **sudo** sub-command.

```
# essrun -N nodeList sudo enable
```

Note: To configure a sudo user, see the **essrun sudo** command in the *IBM Storage Scale System: Command Reference*.

This command creates the gpfsadmin Linux user and gpfs Linux group on the node and performs all necessary sudoers set up. For detailed information, see the `/etc/sudoers.d/ess_sudoers` file.

User can now log in to the node server using the gpfsadmin user and they can perform GPFS administration tasks.

Make sure that the **sudo** sub-command is run on all GPFS nodes (management server node, I/O server nodes, and any client nodes) as part of the cluster to be completely compliant with the sudo requirement. Change the node name in the **sudo** sub-command accordingly. Enabling sudo also allows the gpfsadmin user to administer xCAT and the GPFS GUI on the management server node.

Disabling sudo on IBM Storage Scale System

You can disable sudo configuration on a Linux node by using enable option of the **sudo** sub-command.

```
# essrun -N nodeList sudo disable
```

Disabling sudo reverts the xCAT policy table to its previous state, deletes `/etc/sudoers.d/ess_sudoers` file, and deletes the gpfsadmin user from the Linux node. Make sure that you have disabled sudo user configuration on all GPFS nodes (management server node, I/O server nodes, and any client nodes) as part of the cluster to be completely compliant with the sudo requirement. Change the node name in the **sudo** sub-command accordingly.

Important: You must not disable sudo user until the GPFS cluster is set to configure not to use sudo wrapper and sudo user. Failing to do so might result in cluster corruption.

Enabling root access on IBM Storage Scale System

To enable root access, issue the following command:

```
# essrun -N nodeList sudo enable_root --sudo-password YourSudoPassword
```

Disabling root access on IBM Storage Scale System

To disable root access, issue the following command:

```
# essrun -N nodeList sudo disable
```

Note: You can disable root in all IBM Storage Scale System nodes except IBM Storage Scale System 3000.

Enabling or disabling FIPS

IBM Storage Scale System 7.0.1.x supports Federal Information Processing Standards (FIPS)-compliant operations on IBM Storage Scale System nodes. You can enable or disable FIPS on all nodes in cluster by using the **essrun** command.

1. Ensure that the following conditions are met to avoid certain errors:
 - Ensure that all nodes are updated with IBM Storage Scale System 7.0.1.x.
 - Manually shut down IBM Storage Scale.
2. Enable or disable FIPS.



Attention: When you enable or disable the FIPS, the nodes are rebooted automatically. Enter **accept** to enable or **no** to not enable the FIPS on the nodes.

```
essrun fips [-h] {enable | disable}
```

Where:

enable

Enables FIPS on cluster nodes. To enable the FIPS, set the **1** value.

Note: When you want to enable FIPS in the management server, enable FIPS on the management server node and reboot after the management server node. After the management sever becomes available, enable FIPS on the remaining nodes. Do not enable FIPS on the management server and all nodes at the same time. If you enable FIPS on all nodes, the management server node reboots immediately and the FIPS is not enabled on the remaining nodes.

disable

Disables FIPS on cluster nodes. To disable the FIPS, set the **0** value.

The following example shows how to enable FIPS:

```
essrun -N essio1 fips enable
```

The following system output is displayed:

```
[INFO]      2025-12-08T20:02:23.836099      Going to check if can resolve this node: essio1
Enter 'accept' indicating that you want to enable FIPS in the following list of nodes:
essio1 or enter 'no' if you don't want to proceed. Note that the nodes will be rebooted
during the process.
>>> accept
```

```
PLAY [Enable FIPS ESS]
```

3. Check whether FIPS is enabled or disabled on all cluster nodes. If FIPS is not consistently enabled or disabled, a warning message is displayed when you run the **essinstallcheck** command only if the node belongs to a cluster.

```
[WARN]      FIPS check:      FAILED - FIPS is not consistent in all cluster nodes.
```

4. Check whether the FIPS enabled or disabled on each nodes by checking the content of the `/proc/sys/crypto/fips_enabled` file.

```
cat /proc/sys/crypto/fips_enabled
1
```

Where:

1

Shows nodes where the FIPS is enabled.

0

Shows nodes where the FIPS disabled.

Security Technical Implementation Guide

Security Technical Implementation Guide (STIG) standards are developed by the Defense Information Systems Agency (DISA) to secure information systems. STIG defines specific security requirements that reduce vulnerabilities and help organizations meet federal security compliance standards.

The STIG audit is mandatory for Department of Defense (DoD) and federal systems, best practice for security hardening, and required for security audits and compliance checks, such as FISMA, NIST.

Scope

The STIG audit addresses approximately 20 STIG findings for audit logging on IBM Storage Scale System nodes that run RHEL 8 or RHEL 9.

The STIG provides the following capabilities:

- STIG-compliant audit daemon configuration
- Immutable audit rules (prevent tampering)
- Automated log rotation and disk space monitoring
- Comprehensive file and login monitoring

Monitored files

When STIG audit is enabled on a system, the system monitors the following critical files for changes:

File	Purpose
/etc/sudoers	Administrative privilege changes
/etc/passwd, /etc/shadow	User account changes
/etc/group, /etc/gshadow	Group membership changes
/var/log/lastlog, /var/log/faillock	Login attempts

Requirements

The STIG audit meets the STIG requirements that are listed in the following table:

STIG ID	Requirement	Implementation
RHEL-09-653010	Audit service must be running.	Service auto-starts and is monitored
RHEL-09-653020	Audit rules must be active.	Nine rules deploy automatically.
RHEL-09-653030	Configuration must be immutable.	Immutable mode prevents changes.
RHEL-09-653060	Admin actions must be logged.	Sudoers file is monitored.
RHEL-09-653070	User changes must be logged.	Password files are monitored.
RHEL-09-653080	Group changes must be logged.	Group files are monitored.
RHEL-09-653090	Login attempts must be logged.	Login tracking is enabled.

Immutable mode

Immutable mode prevents auditing configuration changes at run time. This security feature prevents attackers from disabling audit logging. System reboot is required to modify audit rules after immutable mode is enabled.

Important: Reboot nodes after you enable STIG audit to activate full immutable mode.

Log management

The STIG audit includes the following automated log management:

- Logs rotate automatically at 6 MB.
- The system retains five log files (deletes the oldest automatically).
- The system sends automatic alerts when disk space is low.

Supported systems

The STIG audit supports the following systems:

- RHEL 8.10 (VM, bare metal, storage nodes)
- RHEL 9.6 (VM, bare metal, storage nodes)
- All IBM Storage Scale System node types (IBM Storage Scale System Utility Node, storage, protocol, management)

Support and references

For more information about the STIG, refer to the following documentation:

- [DISA STIG](#)
- [Red Hat STIG](#)

You can contact support team for any further assistance.

- Security team: security@example.com
- System administration: sysadmin@example.com
- [Red Hat customer support](#)

To enable STIG audit on a node, see [“Enabling STIG audit” on page 215](#).

Enabling STIG audit

Use **essrun** commands to enable, verify, and restore STIG configuration on IBM Storage Scale System nodes.

Before you begin

Verify that packages are installed on nodes where you want to enable STIG audit.

```
dnf install -y audit audispd-plugins
```

About this task

The **essrun** command provides three operations for the STIG audit management:

enable

Deploy STIG audit configuration (approximately 10-15 seconds per node).

verify

Check compliance status (5-10 seconds per node).

rollback

Restore previous configuration (5-10 seconds per node).

Procedure

1. Enable the STIG audit. You can enable STIG audit on a single node, multiple specific nodes, or all IBM Storage Scale System Utility Nodes at once.

- Deploy STIG on a single node.

```
essrun -N node1 audit enable
```

- Deploy STIG on multiple nodes.

```
essrun -N "node1,node2,node3" audit enable
```

- Deploy STIG on all nodes.

```
essrun -N all_nodes audit enable
```

The system displays a similar output as follows:

```
[node1] Enabling STIG on nodes...
[node1] Backing up existing configuration
[node1] Deploying STIG-compliant audit settings
[node1] Loading audit rules
[node1] Restarting audit service
```

Important: Reboot nodes to activate full immutable mode.

2. Verify STIG audit compliance. You can verify STIG audit compliance on a single node, multiple specific nodes, or all IBM Storage Scale System Utility Nodes at once.

- Verify STIG audit compliance on a single node.

```
essrun -N node1 audit verify
```

- Verify STIG audit compliance on multiple nodes.

```
essrun -N "node1,node2,node3" audit verify
```

- Verify STIG audit compliance monthly on all nodes.

```
essrun -N all_nodes audit verify
```

The system displays a similar output as follows:

```
[node1] Running STIG compliance checks...
[node1] ✓ PASS: Auditd service is running
[node1] ✓ PASS: 9 audit rules loaded
[node1] ✓ PASS: Immutable mode active
[node1] ... (15 more checks)
[node1] ✓ All compliance checks passed (18/18)
```

3. Rollback audit configuration, if you want to back up files from a previous deployment. You can rollback audit configuration on a single node or multiple IBM Storage Scale System Utility Node nodes.

- Rollback STIG audit on a single node.

```
essrun -N node1 audit rollback
```

- Rollback STIG audit on multiple nodes.

```
essrun -N "node1,node2,node3" audit rollback
```

The system displays a similar output as follows:

```
[node1] Checking for backup files...
[node1] ✓ Backup files found
[node1] Restoring previous configuration...
[node1] Removing STIG audit rules...
[node1] Restarting auditd service...
[node1] ✓ Rollback completed successfully
```



CAUTION: When you restore configuration of a node, STIG audit protection is removed from the node. Nodes are no longer compliant.

What to do next

To verify audit status, complete the following steps:

1. Verify the audit service status.

```
systemctl status auditd
```

2. Verify the loaded rules.

```
auditctl -l
```

3. Search audit logs.

```
ausearch -k actions      # Admin actions
```

```
ausearch -k identity     # User/group changes
```

```
ausearch -k logins       # Login attempts
```

4. Review audit events.

```
ausearch -ts today
```

```
aureport --summary
```

Federal Information Processing mode

Learn about Federal Information Processing standards (FIPS) mode capabilities and limitations for IBM Storage Scale System to make informed security configuration decisions.

Difference between FIPS mode and compliance

FIPS mode configures the operating system to use approved cryptographic algorithms. This mode does not represent full FIPS compliance or certification.

Important: If you enable FIPS mode by using the **essrun** command, your system is not FIPS 140-2/140-3 compliant or certified.

FIPS mode capabilities and limitations

When you enable FIPS mode on your IBM Storage Scale System, you activate a specific set of cryptographic controls at the operating system level. Understanding what FIPS mode provides and does not provide helps you make informed decisions about your security configuration and compliance requirements.

What does FIPS mode provide?

FIPS mode provides the following cryptographic security capabilities:

System-wide enforcement of approved cryptographic policies

All system components use only FIPS-approved algorithms.

Use of validated kernel cryptographic modules

The Linux kernel loads only FIPS 140-2 validated cryptographic modules.

Restriction to approved algorithms

Only strong algorithms are allowed (AES, SHA-2, RSA, ECDSA) and weak algorithms (MD5, DES, RC4) are blocked.

Generation of compliant SSH keys and certificates

SSH host keys and certificates are automatically regenerated to meet FIPS requirements.

Automatic rejection of weak cryptographic operations

Applications that attempt to use noncompliant algorithms fail immediately.

What does FIPS mode not provide?

FIPS mode is a technical configuration, not a certification. FIPS mode does not provide the following capabilities:

FIPS 140-2 or FIPS 140-3 certification for your system

Certification requires formal testing and validation by accredited laboratories.

Validation of IBM Storage Scale software components

IBM Storage Scale uses operating system cryptography but is not independently FIPS-certified.

Operational or audit compliance coverage

Implement proper key management, access controls, and audit procedures.

Physical security enforcement or tamper protection

Hardware-level security controls require more measures.

Automatic compliance with regulatory requirements

Compliance frameworks (FedRAMP, FISMA, PCI-DSS) require more controls more than what FIPS mode provides.

Protection against all security threats

FIPS mode addresses cryptographic security only, not network security, access control, or other threat vectors.

Comparison between FIPS mode and FIPS compliance

The following table lists the important differences between FIPS mode (a technical configuration) and achieving full FIPS compliance (a comprehensive certification process):

Aspect	FIPS mode essrun-enabled	FIPS 140-2/140-3 compliance
Definition	OS-level cryptographic policy enforcement	Formal NIST certification process
Scope	Operating system cryptographic modules	Entire system and processes
Validation	Pre-validated RHEL cryptography modules	Formal NIST validation and certification
Requirements	Configuration change	Operational procedures, physical security, documentation, audits
Cost	Included with RHEL	Expensive certification process (\$50 K- \$500 K, or more)
Timeline	Minutes	Months or years
Maintenance	OS updates	Regular recertification required

Behavior of IBM Storage Scale in FIPS mode

When you enable FIPS mode on nodes that are running IBM Storage Scale, the file system continues to operate normally by following FIPS cryptographic restrictions. IBM Storage Scale inherits the cryptographic behavior from the underlying RHEL.

File system operations

All standard IBM Storage Scale features remain functional.

Use of approved encryption algorithms

Data encryption and secure communications use only FIPS-approved algorithms, such as AES.

Compliance with system-level cryptographic restrictions

IBM Storage Scale follows the FIPS policy of the operating system.

Note: IBM Storage Scale relies on operating system cryptography and is not independently FIPS-certified. The software operates in the FIPS-enabled environment, which is provided by RHEL.

Operational considerations

Before you enable FIPS mode, consider the following operational impacts on an IBM Storage Scale System:

Not FIPS 140-2/140-3 certified

IBM Storage Scale relies on FIPS-validated cryptographic modules that are provided by RHEL. FIPS mode can disable some legacy protocols.

Increased cryptographic processing overhead

FIPS mode introduces 10–30% cryptographic overhead. This impact varies by workload and is the highest for encryption-intensive operations. Benchmark your workload before production use.

Disabled legacy protocols and algorithms

Applications that use MD5, DES, RC4, or SSL/TLS 1.0/1.1 fail. Make sure that all applications support TLS 1.2 or higher.

Potential third-party compatibility issues

Some third-party tools or integrations might not support FIPS mode. Test thoroughly before production use.

To enable FIPS mode on your nodes, see [“Enabling and managing FIPS mode on nodes” on page 220](#).

Compliance considerations

To achieve compliance, complete the following steps:

1. Enable FIPS mode by using the `essrun` command.
2. Define and maintain documented operational procedures.
3. Implement physical security controls.
4. Perform regular security audits.
5. Complete formal NIST validation if required.
6. Maintain continuous monitoring and periodic recertification.

Support and references

FIPS mode is a technical configuration that enables cryptographic controls. Full FIPS compliance requires more procedures, documentation, and audits. FIPS mode is a foundation for compliance, not compliance itself.

IBM resources

- [IBM Storage Scale Documentation](#)
- [IBM Support](#)

Red Hat resources

- [RHEL 9 Security Guide](#)
- [FIPS mode setup](#)
- [Red Hat Support](#)

NIST resources

- [FIPS 140-2 Standard](#)
- [FIPS 140-3 Standard](#)
- [Cryptographic Module Validation Program](#)

Compliance frameworks

- FISMA: Federal Information Security Management Act
- FedRAMP: Federal Risk and Authorization Management Program
- NIST SP 800-53: Security and Privacy Controls
- NIST SP 800-171: Protecting Controlled Unclassified Information

Related tasks

[Enabling and managing FIPS mode on nodes](#)

Enable FIPS-validated cryptography on your nodes, verify the configuration, and disable FIPS mode when needed by using the **essrun** command.

Related information

[FIPS 140-2 Standard](#)

[FIPS 140-3 Standard](#)

Enabling and managing FIPS mode on nodes

Enable FIPS-validated cryptography on your nodes, verify the configuration, and disable FIPS mode when needed by using the **essrun** command.

Before you begin

Verify that the nodes run supported Red Hat Enterprise Linux (RHEL) versions:

- RHEL 8.10
- RHEL 9.6

About this task

Use the **essrun** command to manage FIPS mode across nodes. FIPS mode configures your nodes to use FIPS-validated cryptographic modules, ensuring that all cryptographic operations use only approved algorithms.

For detailed information about what FIPS mode provides, limitations, and how FIPS mode affects IBM Storage Scale System operations, see [“Federal Information Processing mode” on page 217](#).

The following operations support FIPS mode:

fips enable

Installs FIPS crypto policy, regenerates SSH keys, and configures the system to use FIPS-validated modules (approximately 2-5 minutes and reboot required).

fips verify

Checks kernel status, crypto policy, and loaded modules to confirm FIPS mode is active (approximately 30 seconds).

fips disable

Removes FIPS crypto policy and restores default cryptographic configuration (approximately 2-5 minutes and reboot required).

Important: You must reboot nodes when you enable or disable FIPS mode. The configuration changes take effect only after reboot.

Procedure

1. Enable FIPS mode on nodes. You can enable FIPS mode on a single node, multiple specific nodes, or all IBM Storage Scale System Utility Node nodes at once.

- Enable FIPS mode on a single node.

```
essrun -N node1 fips enable
```

- Enable FIPS mode on multiple nodes.

```
essrun -N "node1,node2,node3" fips enable
```

- Enable FIPS mode on all nodes.

```
essrun -N nodes fips enable
```

The following tasks are completed:

- a. Installed FIPS crypto policy.
- b. Regenerated SSH host keys with FIPS-compliant algorithms.
- c. Updated system crypto configuration.

Reboot nodes to activate FIPS mode. The kernel must restart with FIPS mode enabled. If FIPS mode is not enabled after reboot, complete steps in [“FIPS mode is not enabled.”](#) on page 222.

2. Reboot nodes to activate FIPS mode. The kernel loads FIPS-validated cryptographic modules during the boot process.

```
reboot
```

3. Verify whether FIPS mode is active. You can verify FIPS mode on a single node or multiple specific nodes.

- Verify FIPS mode on a single node.

```
essrun -N node1 fips verify
```

- Verify FIPS mode on multiple nodes.

```
essrun -N "node1,node2,node3" fips verify
```

The following verification is completed:

- a. FIPS mode enabled in kernel.
- b. Crypto policy set to FIPS.
- c. FIPS-validated modules loaded.
- d. System configuration is correct.

If all checks pass, the output displays `FIPS mode is active` and the node is operating in FIPS mode. If verification fails, ensure that the node was rebooted after you enabled FIPS mode. If the problem persists, rerun the enable command and reboot.

4. Disable FIPS mode to remove a FIPS crypto policy and restore the default cryptographic configuration, if needed. You can disable FIPS mode on a single node or multiple specific nodes.

- Disable FIPS mode on a single node.

```
essrun -N node1 fips disable
```

- Disable FIPS mode on multiple nodes.

```
essrun -N "node1,node2,node3" fips disable
```



CAUTION: Disable FIPS only if you do not need FIPS mode. Because if you disable FIPS on a production system, compliance requirements might be violated.

The following tasks are completed:

- a. Removed FIPS crypto policy.
- b. Restored default crypto configuration.

Reboot the node to complete the process and return to standard cryptographic operations.

What to do next

To verify FIPS mode manually, complete the following steps:

1. Check FIPS mode status.

```
fips-mode-setup --check
```

2. View current crypto policy.

```
update-crypto-policies --show
```

3. Verify kernel FIPS mode.

```
cat /proc/sys/crypto/fips_enabled
```

A sample output is as follows:

```
1 = FIPS enabled, 0 = FIPS disabled
```

4. Check the IBM Storage Scale status.

```
mmlsconfig
```

5. Test SSH only with FIPS algorithms.

```
ssh -vv user@host 2>&1 | grep -i fips
```

Related concepts

[FIPS mode and compliance](#)

Learn about Federal Information Processing standards (FIPS) mode capabilities and limitations for IBM Storage Scale System to make informed security configuration decisions.

FIPS mode issues

FIPS mode configuration issue and resolutions are provided, to enable FIPS mode successfully on nodes.

FIPS mode is not enabled.

Symptoms

- The **fips-mode-setup --check** command shows FIPS disabled after reboot.
- The **essrun fips verify** command reports FIPS mode is not active.
- The `/proc/sys/crypto/fips_enabled` file contains 0 instead of 1.

Cause

- System is not rebooted after the enable command is run.
- Boot loader configuration issue prevents FIPS mode activation.
- Kernel has an incompatible version that does not support FIPS mode.

- FIPS module is corrupt or kernel parameters are missing.

Resolution

1. Verify FIPS status.

```
fips-mode-setup --check
```

The FIPS is disabled after reboot.

2. Check boot loader configuration for compatible kernel version.

```
grubby --info=ALL | grep fips
```

3. Rerun FIPS setup.

```
fips-mode-setup --enable
```

4. Reboot the system.

```
reboot
```

Application is not compatible.

Symptoms

Applications fail with cryptographic errors after FIPS mode is enabled.

Cause

- Application uses non-FIPS algorithms (MD5, DES, RC4, 3DES, and so on).
- Legacy SSL/TLS versions (SSLv3, TLS 1.0, TLS 1.1) that are disabled in FIPS mode.
- Hardcoded weak cryptographic algorithms.

Resolution

1. Update applications to use FIPS-approved algorithms.
2. Configure applications to use TLS 1.2 or higher.
3. Contact the application vendor for FIPS-compatible versions if the application cannot be configured to use FIPS-approved algorithms.

Related concepts

[FIPS mode and compliance](#)

Learn about Federal Information Processing standards (FIPS) mode capabilities and limitations for IBM Storage Scale System to make informed security configuration decisions.

Related tasks

[Enabling and managing FIPS mode on nodes](#)

Enable FIPS-validated cryptography on your nodes, verify the configuration, and disable FIPS mode when needed by using the **essrun** command.

Accessibility features for the system

Accessibility features help users who have a disability, such as restricted mobility or limited vision, to use information technology products successfully.

Accessibility features

The following list includes the major accessibility features in IBM Storage Scale RAID:

- Keyboard-only operation
- Interfaces that are commonly used by screen readers
- Keys that are discernible by touch but do not activate just by touching them
- Industry-standard devices for ports and connectors
- The attachment of alternative input and output devices

IBM Documentation, and its related publications, are accessibility-enabled.

Keyboard navigation

This product uses standard Microsoft Windows navigation keys.

IBM and accessibility

See the [IBM Human Ability and Accessibility Center \(www.ibm.com/able\)](http://www.ibm.com/able) for more information about the commitment that IBM has to accessibility.

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Glossary

This glossary provides terms and definitions for the IBM Storage Scale System solution.

The following cross-references are used in this glossary:

- *See* refers you from a non-preferred term to the preferred term or from an abbreviation to the spelled-out form.
- *See also* refers you to a related or contrasting term.

For other terms and definitions, see the [IBM Terminology website](http://www.ibm.com/software/globalization/terminology) (opens in new window):

<http://www.ibm.com/software/globalization/terminology>

B

building block

A pair of servers with shared disk enclosures attached.

BOOTP

See *Bootstrap Protocol (BOOTP)*.

Bootstrap Protocol (BOOTP)

A computer networking protocol that is used in IP networks to automatically assign an IP address to network devices from a configuration server.

Bring-your-own management server (BYOMS)

An x86 management server of your choice. See also *management server*.

C

CEC

See *central processor complex (CPC)*.

central electronic complex (CEC)

See *central processor complex (CPC)*.

central processor complex (CPC)

A physical collection of hardware that consists of channels, timers, main storage, and one or more central processors.

cluster

A loosely coupled collection of independent systems, or *nodes*, organized into a network for the purpose of sharing resources and communicating with each other. See also *GPFS cluster*.

cluster manager

The node that monitors node status using disk leases, detects failures, drives recovery, and selects file system managers. The cluster manager is the node with the lowest node number among the quorum nodes that are operating at a particular time.

compute node

A node with a mounted GPFS file system that is used specifically to run a customer job. ESS disks are not directly visible from and are not managed by this type of node.

CPC

See *central processor complex (CPC)*.

D

DA

See *declustered array (DA)*.

datagram

A basic transfer unit associated with a packet-switched network.

DCM

See *drawer control module (DCM)*.

declustered array (DA)

A disjoint subset of the pdisks in a recovery group.

dependent fileset

A fileset that shares the inode space of an existing independent fileset.

DFM

See *direct FSP management (DFM)*.

DHCP

See *Dynamic Host Configuration Protocol (DHCP)*.

drawer control module (DCM)

Essentially, a SAS expander on a storage enclosure drawer.

Dynamic Host Configuration Protocol (DHCP)

A standardized network protocol that is used on IP networks to dynamically distribute such network configuration parameters as IP addresses for interfaces and services.

E**Elastic Storage System (ESS)**

See IBM Storage Scale System.

EMS virtual machine (EMSVM)

Virtual machine of an IBM Storage Scale System management server. See *MS management server* and IBM Storage Scale System.

ESS management server (EMS)

See *management server* and IBM Storage Scale System.

encryption key

A mathematical value that allows components to verify that they are in communication with the expected server. Encryption keys are based on a public or private key pair that is created during the installation process. See also *file encryption key (FEK)*, *master encryption key (MEK)*.

ESS

See *Elastic Storage System (ESS)*.

environmental service module (ESM)

Essentially, a SAS expander that attaches to the storage enclosure drives. In the case of multiple drawers in a storage enclosure, the ESM attaches to drawer control modules.

ESM

See *environmental service module (ESM)*.

F**failback**

Cluster recovery from failover following repair. See also *failover*.

failover

(1) The assumption of file system duties by another node when a node fails. (2) The process of transferring all control of the ESS to a single cluster in the ESS when the other clusters in the ESS fails. See also *cluster*. (3) The routing of all transactions to a second controller when the first controller fails. See also *cluster*.

failure group

A collection of disks that share common access paths or adapter connection, and could all become unavailable through a single hardware failure.

FEK

See *file encryption key (FEK)*.

file encryption key (FEK)

A key used to encrypt sectors of an individual file. See also *encryption key*.

file system

The methods and data structures used to control how data is stored and retrieved.

file system descriptor

A data structure containing key information about a file system. This information includes the disks assigned to the file system (*stripe group*), the current state of the file system, and pointers to key files such as quota files and log files.

file system descriptor quorum

The number of disks needed in order to write the file system descriptor correctly.

file system manager

The provider of services for all the nodes using a single file system. A file system manager processes changes to the state or description of the file system, controls the regions of disks that are allocated to each node, and controls token management and quota management.

fileset

A hierarchical grouping of files managed as a unit for balancing workload across a cluster. See also *dependent fileset*, *independent fileset*.

fileset snapshot

A snapshot of an independent fileset plus all dependent filesets.

flexible service processor (FSP)

Firmware that provides diagnosis, initialization, configuration, runtime error detection, and correction. Connects to the HMC.

FQDN

See *fully qualified domain name (FQDN)*.

FSP

See *flexible service processor (FSP)*.

fully qualified domain name (FQDN)

The complete domain name for a specific computer, or host, on the Internet. The FQDN consists of two parts: the hostname and the domain name.

G**GPFS cluster**

A cluster of nodes defined as being available for use by GPFS file systems.

GPFS portability layer

The interface module that each installation must build for its specific hardware platform and Linux distribution.

GPFS Storage Server (GSS)

A high-performance, GPFS NSD solution made up of one or more building blocks that runs on System x servers.

GSS

See *GPFS Storage Server (GSS)*.

H**Hardware Management Console (HMC)**

Standard interface for configuring and operating partitioned (LPAR) and SMP systems.

HMC

See *Hardware Management Console (HMC)*.

I

IBM Security Key Lifecycle Manager (ISKLM)

For GPFS encryption, the ISKLM is used as an RKM server to store MEKs.

IBM Storage Scale System

A high-performance, IBM Storage Scale NSD solution made up of one or more building blocks. The IBM Storage Scale System software runs on IBM Storage Scale System nodes (management server nodes and I/O server nodes).

independent fileset

A fileset that has its own inode space.

indirect block

A block that contains pointers to other blocks.

inode

The internal structure that describes the individual files in the file system. There is one inode for each file.

inode space

A collection of inode number ranges reserved for an independent fileset, which enables more efficient per-fileset functions.

Internet Protocol (IP)

The primary communication protocol for relaying datagrams across network boundaries. Its routing function enables internetworking and essentially establishes the Internet.

I/O server node

An ESS node that is attached to the ESS storage enclosures. It is the NSD server for the GPFS cluster.

IP

See *Internet Protocol (IP)*.

IP over InfiniBand (IPoIB)

Provides an IP network emulation layer on top of InfiniBand RDMA networks, which allows existing applications to run over InfiniBand networks unmodified.

IPoIB

See *IP over InfiniBand (IPoIB)*.

ISKLM

See *IBM Security Key Lifecycle Manager (ISKLM)*.

J

JBOD array

The total collection of disks and enclosures over which a recovery group pair is defined.

K

kernel

The part of an operating system that contains programs for such tasks as input/output, management and control of hardware, and the scheduling of user tasks.

L

LACP

See *Link Aggregation Control Protocol (LACP)*.

Link Aggregation Control Protocol (LACP)

Provides a way to control the bundling of several physical ports together to form a single logical channel.

logical partition (LPAR)

A subset of a server's hardware resources virtualized as a separate computer, each with its own operating system. See also *node*.

LPAR

See *logical partition (LPAR)*.

M

management hostname

This is the short name that gets associated with the IP address that is tied to the management (mgmt) interface on a given node. The definition is contained within the `/etc/hosts` file (or DNS), and it is typically a subset of the high-speed interface name.

management network

A network that is primarily responsible for booting and installing the designated server and compute nodes from the management server.

management server (MS)

An IBM Storage Scale System node that hosts the IBM Storage Scale System GUI and is not connected to storage. It must be part of an IBM Storage Scale cluster. From a system management perspective, it is the central coordinator of the cluster. It also serves as a client node in an IBM Storage Scale System building block.

master encryption key (MEK)

A key that is used to encrypt other keys. See also *encryption key*.

maximum transmission unit (MTU)

The largest packet or frame, specified in octets (eight-bit bytes), that can be sent in a packet- or frame-based network, such as the Internet. The TCP uses the MTU to determine the maximum size of each packet in any transmission.

MEK

See *master encryption key (MEK)*.

metadata

A data structure that contains access information about file data. Such structures include inodes, indirect blocks, and directories. These data structures are not accessible to user applications.

MS

See *management server (MS)*.

MS virtual machine (MSVM)

Virtual machine of an IBM Storage Scale System management server. See *management server* and *IBM Storage Scale System*.

MTU

See *maximum transmission unit (MTU)*.

N

Network File System (NFS)

A protocol (developed by Sun Microsystems, Incorporated) that allows any host in a network to gain access to another host or netgroup and their file directories.

Network Shared Disk (NSD)

A component for cluster-wide disk naming and access.

NSD volume ID

A unique 16-digit hexadecimal number that is used to identify and access all NSDs.

node

An individual operating-system image within a cluster. Depending on the way in which the computer system is partitioned, it can contain one or more nodes. In a Power Systems environment, synonymous with *logical partition*.

node descriptor

A definition that indicates how ESS uses a node. Possible functions include: manager node, client node, quorum node, and non-quorum node.

node number

A number that is generated and maintained by ESS as the cluster is created, and as nodes are added to or deleted from the cluster.

node quorum

The minimum number of nodes that must be running in order for the daemon to start.

node quorum with tiebreaker disks

A form of quorum that allows ESS to run with as little as one quorum node available, as long as there is access to a majority of the quorum disks.

non-quorum node

A node in a cluster that is not counted for the purposes of quorum determination.

O**OFED**

See *OpenFabrics Enterprise Distribution (OFED)*.

OpenFabrics Enterprise Distribution (OFED)

An open-source software stack includes software drivers, core kernel code, middleware, and user-level interfaces.

P**pdisk**

A physical disk.

PortFast

A Cisco network function that can be configured to resolve any problems that could be caused by the amount of time STP takes to transition ports to the Forwarding state.

R**RAID**

See *redundant array of independent disks (RAID)*.

RDMA

See *remote direct memory access (RDMA)*.

redundant array of independent disks (RAID)

A collection of two or more disk physical drives that present to the host an image of one or more logical disk drives. In the event of a single physical device failure, the data can be read or regenerated from the other disk drives in the array due to data redundancy.

recovery

The process of restoring access to file system data when a failure has occurred. Recovery can involve reconstructing data or providing alternative routing through a different server.

recovery group (RG)

A collection of disks that is set up by ESS, in which each disk is connected physically to two servers: a primary server and a backup server.

remote direct memory access (RDMA)

A direct memory access from the memory of one computer into that of another without involving either one's operating system. This permits high-throughput, low-latency networking, which is especially useful in massively parallel computer clusters.

RGD

See *recovery group data (RGD)*.

remote key management server (RKM server)

A server that is used to store master encryption keys.

RG

See *recovery group (RG)*.

recovery group data (RGD)

Data that is associated with a recovery group.

RKM server

See *remote key management server (RKM server)*.

S**SAS**

See *Serial Attached SCSI (SAS)*.

secure shell (SSH)

A cryptographic (encrypted) network protocol for initiating text-based shell sessions securely on remote computers.

Serial Attached SCSI (SAS)

A point-to-point serial protocol that moves data to and from such computer storage devices as hard drives and tape drives.

service network

A private network that is dedicated to managing Power8 servers. Provides Ethernet-based connectivity among the FSP, CPC, HMC, and management server.

SMP

See *symmetric multiprocessing (SMP)*.

Spanning Tree Protocol (STP)

A network protocol that ensures a loop-free topology for any bridged Ethernet local-area network. The basic function of STP is to prevent bridge loops and the broadcast radiation that results from them.

SSH

See *secure shell (SSH)*.

STP

See *Spanning Tree Protocol (STP)*.

symmetric multiprocessing (SMP)

A computer architecture that provides fast performance by making multiple processors available to complete individual processes simultaneously.

T**TCP**

See *Transmission Control Protocol (TCP)*.

Transmission Control Protocol (TCP)

A core protocol of the Internet Protocol Suite that provides reliable, ordered, and error-checked delivery of a stream of octets between applications running on hosts communicating over an IP network.

V**VCD**

See *vdisk configuration data (VCD)*.

vdisk

A virtual disk.

vdisk configuration data (VCD)

Configuration data that is associated with a virtual disk.

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